MYP YEAR 2 · STUDENT COURSEBOOK

MYP

MATHEMATICS 2

STUDENT COURSEBOOK



A concept-driven inquiry course

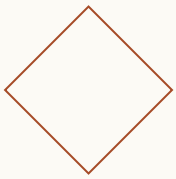
Ratio and proportion · Algebra and pattern · Geometry and space · Probability and data

Complete worked solutions · Investigations · Criterion-mapped tasks

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MIDDLE YEARS PROGRAMME



MYP Mathematics 2

STUDENT COURSEBOOK

SADIKOU MOUKODDIM

FIRST EDITION

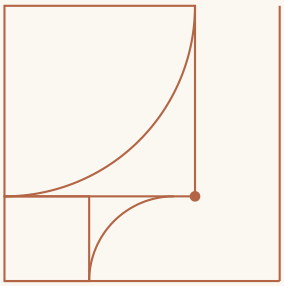
STUDENT COURSEBOOK
FIRST EDITION · 2026

MYP

Mathematics *2*

Student Coursebook

An inquiry-led course in number, algebra, geometry, probability and statistics, built around the concepts, contexts and assessment criteria of the Middle Years Programme.



SADIKOU Moukoddim

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M²

MYP Mathematics 2 — Student Coursebook

FIRST EDITION · 2026

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Data and contexts. Real-world scenarios use rounded, simplified or classroom-generated data. Figures are chosen to support the mathematics and may not reflect the most recent published statistics unless a source is named. Students should verify current data from a reliable source when a task requires it.

Assessment. Criterion tasks, rubrics and level descriptors in this book are teaching tools written for formative classroom use. They paraphrase and interpret the MYP assessment criteria; they are not official IB descriptors and do not replace the current subject guide.

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How to use this coursebook

This is a working book, not a book of answers to memorise. Read with a pencil in hand, test ideas, write down conjectures, and return to your reasoning when it needs repair.

The course is arranged in four parts. Each part gathers chapters that share a way of thinking: comparing quantities; generalising with symbols; describing space and measurement; and making decisions from uncertainty and data. The chapters build on one another, but each is also designed to be entered on its own after the prerequisite check.

Every chapter follows the same inquiry rhythm. You meet a situation that mathematics can clarify. You frame a question. You build the concepts and representations needed to answer it. You practise until the method is reliable. Then you investigate, communicate, apply and reflect. The answer key at the back supports this cycle; it should confirm your reasoning, not replace it.

MATHEMATICAL HABITS THAT THIS COURSE DEVELOPS

- → **Represent.** Move between words, tables, diagrams, graphs, symbols and physical models.
- → **Justify.** Say why a method works, not only what the answer is.
- → **Check.** Estimate first, then test whether the answer is reasonable in its context.
- → **Generalise.** Look for structure that remains true beyond the example in front of you.
- → **Reflect.** Name what you understand, what is still fragile, and what you will do next.

ANATOMY OF A CHAPTER

01 **Chapter opener**
A real context, a visual motif and an inquiry prompt that frames the mathematics.

02 **MYP unit planner**
Key and related concepts, global context, statement of inquiry, inquiry questions and criterion mapping.

03 **Intentions and prerequisites**
Learning intentions, success criteria and a short diagnostic check from earlier learning.

04 **Concept sequence**
Definitions, representations, worked examples, common errors and graduated practice.

05 **Investigate and apply**
An investigation, a real-life application and a criterion-mapped task with clear strands.

06 **Review and mixed practice**
Summary, revision checklist, reflection and a substantial mixed set combining the chapter's ideas.

HOW THE LABELS WORK

Definition boxes state the precise meaning of a term. **Key idea** boxes connect a method to a bigger principle. **Common error** boxes show a mistake worth anticipating. **ATL** boxes make a learning skill explicit and give you a way to practise it. **Criterion** strips name the MYP criterion and strands being developed.

EXERCISE TYPES

F

FLUENCY

Short, deliberate practice to make a method accurate and efficient.

R

REASONING

Explain, compare, justify, find a counterexample or diagnose an error.

P

PROBLEM SOLVING

Multi-step and unfamiliar contexts that require selection and planning.

E

EXTENSION

Deeper structure, generalisation or a more open task.

ATL · USING THE ANSWER KEY WELL

Attempt the question fully before checking. If your answer differs, locate the first line where your reasoning diverges. Mark the correction in a different colour and write one sentence explaining why the alternative was wrong. That sentence is often more valuable than the correct number.

ACADEMIC HONESTY

Copying a solution without understanding it creates the appearance of progress and hides a gap. Use sample answers to compare *methods*, not to replace your attempt. In investigations, report your own data and reasoning, and acknowledge any source you use.

A FOUR-STEP STUDY CYCLE

STEP 01

Read the question

Restate it in your own words and name what is given, what is asked and which units apply.

STEP 02

Attempt first

Cover the worked solution. Commit to a method, even a partial one, before comparing.

STEP 03

Compare lines

If your answer differs, find the first line where the reasoning diverges, not the last.

STEP 04

Name the change

Write one sentence about what you will do differently on the next similar task.

The MYP framework in this course

The Middle Years Programme asks students to learn mathematics as a connected, conceptual and applicable discipline. This course is designed around that expectation: concepts organise the content, contexts give it purpose, and criteria make the quality of thinking visible.

The aims of MYP mathematics

The MYP mathematics framework sets out to help students enjoy mathematics, develop curiosity, build conceptual understanding, apply mathematics to real situations, communicate reasoning, think critically and creatively, and appreciate the international and historical dimensions of the subject. In Year 2, this means strengthening the bridge between informal reasoning from earlier years and the more formal symbolic work that follows.

KEY CONCEPT 01

Form

The shape, structure and organisation of mathematical objects: how a ratio is written, how a graph is arranged, how an expression is built. Form helps us notice what is mathematically the same beneath different appearances.

KEY CONCEPT 02

Logic

The reasoning that connects claims to conclusions. Logic asks not only "Is it true?" but "How do we know, and for which cases?" It is the discipline behind justification, generalisation and proof.

KEY CONCEPT 03

Relationships

Connections between quantities, variables, representations and ideas. A ratio, a linear pattern, a proportional graph and a rate are different ways of expressing relationships.

Related concepts used in this course

EQUIVALENCE
Different forms with the same value or meaning.

JUSTIFICATION
Reasons and evidence that support a claim.

MODELS
Simplified representations of a real situation.

PATTERNS
Regular structure that repeats or grows predictably.

QUANTITY
Number, magnitude, measure and unit.

REPRESENTATION
Words, symbols, tables, diagrams, graphs.

SIMPLIFICATION
Rewriting without changing value or meaning.

SPACE
Position, shape, direction and dimension.

SYSTEMS
Related parts working under shared rules.

VALIDITY
Whether a conclusion follows from its premise.

MEASUREMENT
Assigning numbers and units to attributes.

GENERALISATION
Extending a relationship to a whole class of cases.

Global contexts

Each chapter is anchored in one of the six MYP global contexts. The context is not decoration: it shapes the questions asked and the type of solution that counts as useful.

GLOBAL CONTEXT	HOW MATHEMATICS ENTERS THE INQUIRY
Identities and relationships	Fair comparison, personal data, growth, choices and healthy proportions.
Orientation in space and time	Scale, maps, coordinate systems, symmetry and geometric change.
Personal and cultural expression	Pattern, design, rhythm, tessellation and mathematical beauty.
Scientific and technical innovation	Measurement, rate, precision, modelling and simulation.
Fairness and development	Sharing resources, comparing rates, interpreting claims and data.
Globalization and sustainability	Consumption, density, population, energy and environmental data.

The inquiry pathway

1 CONTEXT
What situation needs mathematical sense-making?

2 QUESTION
What exactly do we want to know?

3 CONCEPT
Which ideas and representations will help?

4 INVESTIGATION
What patterns, relationships or cases can we test?

5 COMMUNICATION
How will we present and justify the reasoning?

6 REFLECTION
What is the quality, scope and limit of the conclusion?

The pathway is an instructional design tool for this book. It is compatible with MYP inquiry but is not an official IB sequence.

ASSESSMENT

Knowing, investigating, communicating, applying

MYP mathematics is assessed against four equally weighted criteria. Each is marked out of 8, giving a total out of 32. Every chapter in this book identifies the strands it develops and provides at least one task designed for criterion-based feedback.

A

KNOWING AND UNDERSTANDING

- i. Select appropriate mathematics when solving problems in familiar and unfamiliar situations.
- ii. Apply the selected mathematics successfully when solving problems.
- iii. Solve problems correctly in a variety of contexts.

B

INVESTIGATING PATTERNS

- i. Apply mathematical problem-solving techniques to recognise patterns.
- ii. Describe patterns as relationships or general rules consistent with correct findings.
- iii. Verify whether the pattern works for other examples.

C

COMMUNICATING

- i. Use appropriate mathematical language, notation, symbols and terminology in oral and written explanations.
- ii. Use different forms of mathematical representation to present information.
- iii. Communicate coherent mathematical lines of reasoning.
- iv. Organise information using a logical structure.

D

APPLYING MATHEMATICS IN REAL-LIFE CONTEXTS

- i. Identify relevant elements of authentic real-life situations.
- ii. Select appropriate mathematical strategies when solving authentic real-life situations.
- iii. Apply the selected strategies successfully to reach a solution.
- iv. Justify the degree of accuracy of a solution.
- v. Justify whether a solution makes sense in the context of the authentic real-life situation.

What Year 2 looks like in practice

By Year 2, students should be moving from recognising a method to selecting it. A student working securely at this level can choose a representation, carry a multi-step procedure accurately, explain why the steps

are valid, and judge whether the result is sensible. In unfamiliar contexts, they may need prompting, but they can recover and justify a route.

Criterion B is not only "spot the pattern". Students need to test cases, express the relationship in words or symbols, and test the general rule against further examples, including a case that could have broken it. Criterion C values clarity: correct notation, labelled diagrams and a logical order. Criterion D requires attention to units, accuracy and meaning—not just calculation.

LEVEL	TYPICAL EVIDENCE IN THIS COURSE
1–2	Uses limited mathematics; needs support to start; communication is fragmented.
3–4	Uses appropriate methods in familiar tasks with some errors; explanation is partial.
5–6	Selects and applies methods accurately in familiar and some unfamiliar tasks; justifies most steps.
7–8	Works independently and flexibly; verifies generalisations; communicates precise, coherent reasoning and evaluates accuracy and meaning.

These are summarised, course-specific descriptions to help students recognise progress. They are not official IB level descriptors and must not be used as a substitute for the current subject guide.

ATL skills in mathematics

Approaches to learning are the deliberate habits that make inquiry possible. This book integrates them where they do real work: organising a solution, testing a conjecture, choosing a representation, reading a data set critically, or collaborating on a fair experiment.

- **Communication** — notation, diagrams, labelled axes, clear explanations.
- **Critical thinking** — counterexamples, error analysis, evaluating claims.
- **Research** — finding and judging data, naming sources.
- **Collaboration** — sharing methods, dividing a task, reaching agreement with evidence.
- **Organisation** — planning an investigation, managing data and units.
- **Reflection** — judging the quality of a solution and setting a next step.

Criterion tasks

Criterion tasks are longer than routine practice. They are marked against named strands, so the goal is not only the answer. You are judged against the descriptors, not ranked on a curve: two students can take different valid routes and both reach high levels. A strong response typically:

- states the mathematics being used and why it fits;
- shows the method with units, labels and intermediate values;
- checks a general rule against further cases;
- comments on accuracy, assumptions and reasonableness;
- uses a structure a reader can follow without guessing.

SEQUENCE

Course map and progression

Year 2 sits between the exploratory mathematics of Year 1 and the more formal algebraic and geometric work of Year 3. The sequence below builds multiplicative reasoning first, then symbolic generalisation, then spatial measurement, and finally probabilistic and statistical reasoning.

PART	CHAPTER	CORE IDEAS	CRITERIA
I Number & proportion	1 Ratios and equivalent ratios	Ratio language, simplifying, ratio tables, three-part ratios, sharing	A, C
	2 Proportional reasoning	**Tape and double-line models, unitary method, scale, direct proportion**	A, B, D
	3 Rates and unit rates	**Rates, unit rates, comparison, speed and best buys**	A, C, D
	4 Integers and operations	**Order, addition, subtraction, multiplication, division, order of operations**	A, B
II Algebra & pattern	5 Algebraic expressions and equations	Variables, substitution, like terms, one- and two-step equations	A, C
	6 Numerical patterns and generalisation	**Sequences, term rules, function machines, linear graphs, conjectures**	B, C
III Geometry & space	7 2D geometry	Angles, triangles, quadrilaterals, perimeter and area	A, C, D
	8 3D geometry	**Solids, nets, surface area, volume, composite solids**	A, B, D
	9 Transformations	**Translation, reflection, rotation, enlargement, symmetry**	B, C
IV Probability & data	10 Probability and simulation	Events, sample spaces, theoretical and experimental probability, simulation, fairness	A, B, C, D
	11 Univariate data	**Tables, dot plots, bar charts, centre and spread, comparing distributions**	A, B, C, D

Progression across the programme

STRAND	YEAR 1 FOUNDATIONS	YEAR 2 THIS COURSE	YEAR 3 NEXT STEPS
Ratio and proportion	Fractions of a whole; simple sharing; informal comparison.	Equivalent ratios; ratio tables; proportional reasoning; scale; rates.	Direct and inverse proportion; proportional graphs; algebraic constant of proportionality.
Number	Number line; four operations; simple negative numbers.	Integer operations; order of operations; reasoning about sign.	Rational and irrational numbers; exponent laws; scientific notation.
Algebra	Variables as unknowns; simple function machines.	Expressions; substitution; like terms; linear equations.	Linear inequalities; simultaneous equations; factorisation and expansion.

Pattern	Continue and describe repeating patterns.	Term rules; nth-term expressions; verifying generalisations.	Quadratic and non-linear patterns; algebraic justification.
Geometry	Shape names; simple perimeter and area by counting.	Angle facts; parallelogram, trapezium and composite areas; surface area and volume.	Circles, Pythagoras, similarity; trigonometric ratios.
Probability & data	Chance language; simple pictograms and bar charts.	Sample spaces; theoretical and experimental probability; mean, median, range; comparing distributions.	Compound events; tree diagrams; standard deviation; bivariate data.

OVERLAP IS DELIBERATE

Some ideas return from Year 1, now with more structure. Fractions, sharing and simple graphs are prerequisites, not repeats: the Year 2 tasks ask for generalisation, comparison and justification. Later chapters revisit earlier representations so that ratio, algebra, geometry and data become connected rather than separate topics.

A NOTE ON SEQUENCE

Publicly listed contents for commercial MYP 2 courses commonly include integers, 2D and 3D shape, univariate data, ratio and proportion, rates, probability and algebraic expressions and equations. This course covers those domains and adds dedicated work on patterns and transformations. The sequence is an independent design, not an official or compulsory MYP order.

Notation, working and habits

Clear notation is part of the mathematics. A solution that cannot be read cannot be credited, even when the calculation is right.

Notation used in this book

$a : b$	the ratio of a to b
$\frac{a}{b}$	a divided by b; also the fraction a of b
$=$	is equal to
$\neq$	is not equal to
$\approx$	is approximately equal to
$<,\ >,\ \leq,\ \geq$	is less than, greater than, at most, at least
x^2	x squared, or x multiplied by x
$3x$	3 multiplied by x, not the two-digit number $3x$
$\angle ABC$	the angle at vertex B in a figure
π	the constant ratio of circumference to diameter
%	per cent, “out of 100”

Working conventions

- **Copy the question** or state the problem in your own words before starting.
- **Show the method.** A correct answer without working cannot demonstrate understanding.
- **Line up equals signs** when solving equations so that each line is equivalent to the one above.
- **Include units** in the statement of an answer and in any interpretation.
- **Round only at the end** of a calculation unless the context requires otherwise. State the degree of accuracy.
- **Check by another route** where possible: substitute, estimate, measure, or use a diagram.
- **Answer the question asked.** If the question says “explain”, a number alone is incomplete.

Technology

Mental estimation and written efficiency come first. A calculator may be used for investigations involving measurement, data and geometry when the task says so. Never round a calculator display before the final step; store intermediate values or keep full precision.

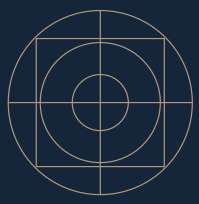
> **THE EQUALS SIGN IS A RELATIONSHIP**
>
> Writing $3 + 5 = 8 + 2 = 10$ is incorrect: it says that 8 and 10 are the same. Write one relationship per line: $3 + 5 = 8$, and $8 + 2 = 10$.

PART I · MULTIPLICATIVE REASONING

I

Number and proportion

Much of the mathematics we use outside the classroom is about comparison: which offer is better, how a quantity scales, how two changing amounts stay in step. This part builds a precise language for comparison and gives you tools to reason with it confidently.

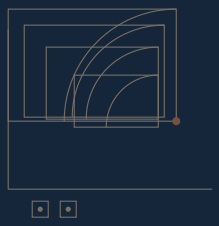


01 Ratios and equivalent ratios

02 Proportional reasoning

03 Rates and unit rates

04 Integers and operations



1

NUMBER AND PROPORTION

Ratios and equivalent ratios

A ratio can describe an entire family of situations: the same relationship at a different size, in different units, or on a different scale.

KEY CONCEPT · FORM

RELATED · EQUIVALENCE

RELATED · REPRESENTATION

CONTEXT · FAIRNESS AND DEVELOPMENT

INQUIRY

How can one comparison describe many different-sized situations without losing what matters?

MYP UNIT PLANNER

Framing the inquiry

UNIT 1 · APPROX. 12 HOURS
ATL FOCUS · CRITICAL THINKING

KEY AND RELATED CONCEPTS

Form. A ratio has a written form, $a : b$, and a simplest form. Its form shows which quantities are compared, and in what order.

Equivalence. Multiplying or dividing both parts by the same non-zero number gives a different-looking ratio with the same relationship.

Representation. Tape diagrams, tables, double number lines and graphs each expose a different feature.

GLOBAL CONTEXT AND EXPLORATION

Fairness and development. Exploration: fair distribution, shared resources and just comparison.

Sharing a harvest or scaling a recipe needs a defensible rule. Ratios make that rule visible, so a decision can be examined rather than merely asserted. The same comparison can look fair in one representation and misleading in another.

STATEMENT OF INQUIRY

Equivalent ratios preserve a multiplicative relationship, so one comparison can organise fair shares and scaled quantities across totals and units.

FACTUAL

What is a ratio, and how is an equivalent ratio formed?

CONCEPTUAL

How can different ratios show the same relationship?

DEBATABLE

When is a fixed-ratio share fair, and when is another comparison needed?

CRITERION	STRANDS DEVELOPED	EVIDENCE IN THIS CHAPTER
A · Knowing and understanding	i, ii, iii	Simplify and interpret ratios; share quantities accurately.
B · Investigating patterns	i, ii, iii	Investigate equivalent ratio chains and state the scaling rule.
C · Communicating	i, ii, iii, iv	Use ratio notation and units; move between representations.
D · Applying mathematics	i, ii, iii, iv, v	Garden task: select quantities, justify a share, evaluate fairness.



ATL · CRITICAL THINKING IN ACTION

Test the claim “adding the same amount to both parts keeps a ratio the same” with three cases. Find a counterexample, explain why it fails, and repair the claim. Use the corrected rule in Exercise 1.2 question 11.

INQUIRY PATHWAY FOR THIS CHAPTER

1	2	3	4	5	6
CONTEXT	QUESTION	CONCEPT	INVESTIGATION	COMMUNICATION	REFLECTION
Community kitchen shares among unequal groups.	How can shares stay comparable when totals change?	Ratio as multiplicative comparison; equivalence.	Lemonade recipes, ratio tables and graphs.	Tape diagrams, tables and precise ratio language.	When might a simplified ratio hide a difference?

Learning intentions and success criteria

YOU ARE LEARNING TO

- **Understand** that a ratio compares quantities multiplicatively, and that order carries meaning.
- **Know** how to simplify a ratio and generate equivalent ratios.
- **Be able to** represent a ratio with a diagram, table, double number line and graph.
- **Be able to** share a total in a given ratio and check the result.
- **Be able to** compare two ratios and justify which is greater.

SUCCESS CRITERIA

- ☐ I can write a ratio from a description and explain what each term represents.
- ☐ I can simplify a ratio using a common factor and state when it is in simplest form.
- ☐ I can generate equivalent ratios and test equivalence without guessing.
- ☐ I can complete a ratio table and use a double number line to find a missing value.
- ☐ I can share a total in a two-part or three-part ratio and verify by recombining.
- ☐ I can compare ratios and communicate the comparison with evidence.

PREREQUISITE CHECK · 10 MINUTES, NO CALCULATOR

1. Write $\frac{12}{18}$ in simplest form.
2. Find $\frac{1}{4}$ of 60.
3. Convert 1.5 kg to grams.
4. Calculate $3 \times 7 + 4 \times 7$.
5. Solve $5x = 45$.
6. Write 0.25 as a fraction in simplest form.

Answers are in the Chapter 1 answer section. If more than two are incorrect, review equivalent fractions and unit conversion before continuing.

YEAR 1
Fractions of a whole; simple sharing; comparing quantities by counting.

YEAR 2 · THIS CHAPTER
Equivalent ratios; ratio tables and graphs; sharing and comparing quantities.

YEAR 3
Direct and inverse proportion; algebraic constants; proportional graphs.

1.1 Understanding ratio language

CRITERION A · I, III ◆ REPRESENTATION ◆ RATIO AS COMPARISON

A ratio is not a pair of numbers that happen to sit beside each other. It is a statement about how quantities are related by multiplication, and the order of the statement is part of its meaning.

DEFINITION

A **ratio** compares two or more quantities multiplicatively. For quantities a and b, the ratio is written $a : b$, read "a to b". The quantities must be measured in the same unit before the ratio is simplified.

There are three common ways to write a two-part ratio: in words, with a colon, or as a fraction. A ratio of 3 parts blue to 5 parts yellow can be written "3 to 5", $3 : 5$, or $\frac{3}{5}$ when the fraction is understood as a comparison, not as a share of the whole. The notation you choose should make the meaning clearer, not merely shorter.



Figure 1.1 A tape diagram for the ratio $3 : 5$. The whole bar has $3 + 5 = 8$ equal parts. Blue is $\frac{3}{8}$ of the whole, while the ratio of blue to yellow is $3 : 5$.

PART TO PART, AND PART TO WHOLE

A **part-to-part** ratio compares two parts of a group. A **part-to-whole** ratio compares one part with the entire group. In Figure 1.1, the part-to-part ratio of blue to yellow is $3 : 5$. The part-to-whole ratio of blue to all counters is $3 : 8$. Confusing these two is a common source of errors, because the second term answers a different question.

KEY IDEA

Order matters. The ratio of blue to yellow is not the same statement as the ratio of yellow to blue. Reversing the terms describes a different comparison, even though both ratios use the same two numbers.

WORKED EXAMPLE 1.1.1 — READING THE TERMS

PROBLEM

A box contains 8 red pencils and 12 blue pencils. Write the ratio of red to blue pencils in simplest form, then write the ratio of red pencils to the total number of pencils.

1. The part-to-part ratio is red : blue $= 8 : 12$.
2. Both terms have common factor 4: $8 \div 4 = 2$ and $12 \div 4 = 3$. So $8 : 12 = 2 : 3$.
3. The total number of pencils is $8 + 12 = 20$. The part-to-whole ratio is red : total $= 8 : 20$.
4. Simplify by dividing both terms by 4: $8 : 20 = 2 : 5$.

ANSWER **Red : blue** $= 2 : 3$**; red : total** $= 2 : 5$**.**

WORKED EXAMPLE 1.1.2 — UNITS BEFORE RATIO

PROBLEM

A student spends 45 minutes on mathematics and 2 hours on a project. Write the ratio of mathematics time to project time in simplest form.

1. The quantities are in different units, so convert to the same unit: 2 hours $= 120$ minutes.
2. Write the ratio $45 : 120$.
3. The highest common factor of 45 and 120 is 15. Divide both terms by 15: $45 \div 15 = 3$, $120 \div 15 = 8$.

ANSWER **Mathematics : project** $= 3 : 8$**.**

Alternatively, write both times in hours: $0.75 : 2$. Multiply both terms by 4 to obtain $3 : 8$.

COMMON ERROR

Writing $45 : 2$ by keeping the numbers as they appear in the question ignores the units. A ratio compares quantities, not numerals. Convert to a common unit first, then simplify.

PRACTICE 1.1 · READING AND WRITING RATIOS

SUPPORT 1–4 · CORE 1–8 · EXTENSION 9–10

Write each ratio in simplest form unless the question says otherwise. Explain your reasoning where the question asks for it.

1. A bag has 6 green and 9 yellow counters. Write the ratio of green to yellow counters. F
2. In a class of 24 students, 10 walk to school and 14 take the bus. Write the ratio of students who walk to students who take the bus, and the ratio of students who walk to the total number of students. F
3. A rectangular card is 18 cm long and 7 cm wide. Write the ratio of length to width. F
4. Write the ratio 45 minutes to 2 hours in simplest form. F
5. The ratio of cats to dogs at a shelter is $4 : 7$. There are 12 cats. How many dogs are there? P
6. The ratio of fiction to non-fiction books in a library is $5 : 8$. There are 320 non-fiction books. How many fiction books are there? P
7. A fruit bowl contains apples and oranges in the ratio $3 : 2$. There are 15 apples. How many pieces of fruit are in the bowl altogether? P

8. True or false: “The ratio $2 : 3$ is the same as $3 : 2$ because they use the same numbers.” Explain your answer with an example. R

9. Which ratio is equivalent to $4 : 6$? F

   A $2 : 4$
   B $6 : 8$
   C $8 : 12$
   D $4 : 8$

10. Sam says, “$5 : 10$ simplifies to $1 : 5$ because $5 \div 5 = 1$ and $10 \div 2 = 5$.” Identify the error and write the correct simplest form. E

1.2 Equivalent ratios and simplest form

CRITERION A · II ◆ EQUIVALENCE ◆ SIMPLIFICATION

Two ratios are equivalent when they describe the same multiplicative relationship. The numbers may change; the comparison does not.

DEFINITION

Ratios $a : b$ and $c : d$ are **equivalent** if there is a non-zero number k such that $c = ka$ and $d = kb$. A useful test is the cross-product condition $ad = bc$.

Multiplying or dividing both terms by the same non-zero number produces an equivalent ratio. This is the same operation that produces equivalent fractions, applied to each part of the comparison. A ratio is in **simplest form** when the terms are whole numbers with no common factor greater than 1.

KEY IDEA

To test whether $a : b = c : d$, check whether $ad = bc$. For example, $6 : 9$ and $10 : 15$ are equivalent because $6 \times 15 = 90$ and $9 \times 10 = 90$.

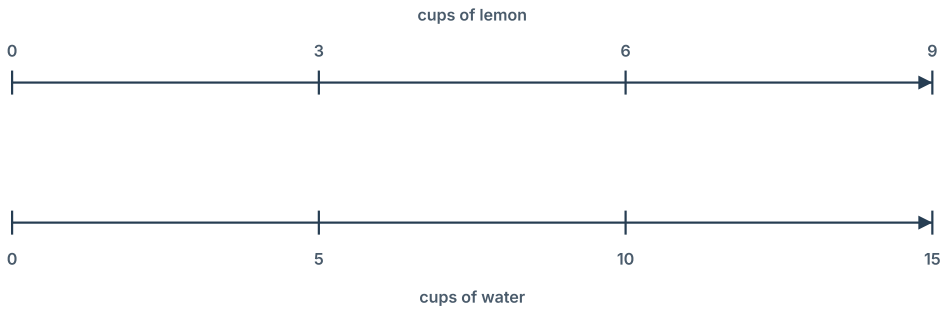


Figure 1.2 A double number line for $3 : 5$. Every mark on the upper line is paired with the mark directly below it. Multiplying both quantities by the same factor moves you to another equivalent pair.

WORKED EXAMPLE 1.2.1 — SIMPLIFY AND JUSTIFY

PROBLEM

Simplify each ratio and state the common factor used.

(a) $42 : 56$ (b) $0.75 : 1.25$ (c) $\frac{3}{4} : \frac{5}{6}$

1. (a) The highest common factor of 42 and 56 is 14: $42 \div 14 = 3$, $56 \div 14 = 4$. So $42 : 56 = 3 : 4$.
2. (b) Multiply both terms by 100 to remove the decimals: $75 : 125$. Divide both by 25: $3 : 5$.
3. (c) Multiply both terms by the lowest common multiple of the denominators, 12: $9 : 10$. The terms are now whole numbers with no common factor.

ANSWER **(a)** $3 : 4$ **(b)** $3 : 5$ **(c)** $9 : 10$

WORKED EXAMPLE 1.2.2 — FINDING A MISSING TERM

PROBLEM

Find the value of x.

(a) $5 : 8 = x : 48$ (b) $4 : x = 6 : 9$

1. (a) The second term has been multiplied by $48 \div 8 = 6$. To keep the ratio equivalent, multiply the first term by the same factor: $x = 5 \times 6 = 30$.
2. (b) Use the cross-product condition: $4 \times 9 = x \times 6$, so $36 = 6x$ and $x = 6$.
3. Check: $4 : 6$ simplifies to $2 : 3$, and $6 : 9$ also simplifies to $2 : 3$.

ANSWER **(a)** $x = 30$ **(b)** $x = 6$

ATL · CRITICAL THINKING

Skill in action. A classmate claims that $3 : 4$ and $4 : 5$ are equivalent because "you add one to each term". Test the claim numerically, find a counterexample, and explain why adding the same number changes the multiplicative relationship. Then write a corrected rule that a younger student could follow.

COMMON ERROR

Adding or subtracting the same amount from both terms does not preserve a ratio: $2 : 3$ becomes $3 : 4$, and $\frac{2}{3} \neq \frac{3}{4}$. Only multiplication or division by the same non-zero factor preserves equivalence.

PRACTICE 1.2 · EQUIVALENT RATIOS

SUPPORT 1–5 · CORE 1–9 · EXTENSION 10–12

Simplify where possible. Show the factor you used, not only the answer.

1. Simplify $12 : 18$. F
2. Simplify $25 : 40$. F
3. Simplify $0.6 : 0.9$. F
4. Simplify $\frac{3}{4} : \frac{5}{6}$. F
5. Copy and complete: $3 : 7 = 15 : ?$. F

6. Copy and complete: $? : 8 = 9 : 24$. F

7. Write three ratios equivalent to $5 : 4$. F

8. Are $8 : 12$ and $12 : 18$ equivalent? Justify your answer using a common simplified form. R

9. Find x: $14 : 21 = x : 3$. F

10. A cordial recipe uses 2 parts concentrate to 5 parts water. Write the ratio of concentrate to water when 6 parts of concentrate are used. P

11. Mia says $6 : 9 = 3 : 4$ because she subtracted 3 from both terms. Explain why this is not valid, then give the correct simplest form. E

12. Write a ratio equivalent to $2 : 3$ whose two terms add to 40. Explain how you found it. E

1.3 Ratio tables and double number lines

CRITERION A · II ◆ CRITERION C · II ◆ REPRESENTATION

A ratio table organises equivalent pairs so that the scaling factor becomes visible. A double number line does the same thing spatially; a graph turns the same relationship into a line.

BUILDING A RATIO TABLE

Each row or column of a ratio table contains one pair of equivalent quantities. To move from one entry to the next, multiply or divide both quantities by the same factor. A useful strategy is to scale through a convenient unit, such as one part, because that makes every other entry easier to find.

TABLE 1.1 · A RATIO TABLE FOR LEMONADE CONCENTRATE AND WATER

Concentrate (cups)	2	4	6	8
Water (cups)	5	10	15	20

The ratio is $2 : 5$ in every column. Multiplying both quantities by 2 gives $4 : 10$; by 3 gives $6 : 15$; by 4 gives $8 : 20$. If a table is built by adding the same amount to each row instead, the ratio changes and the table no longer represents equivalent ratios.

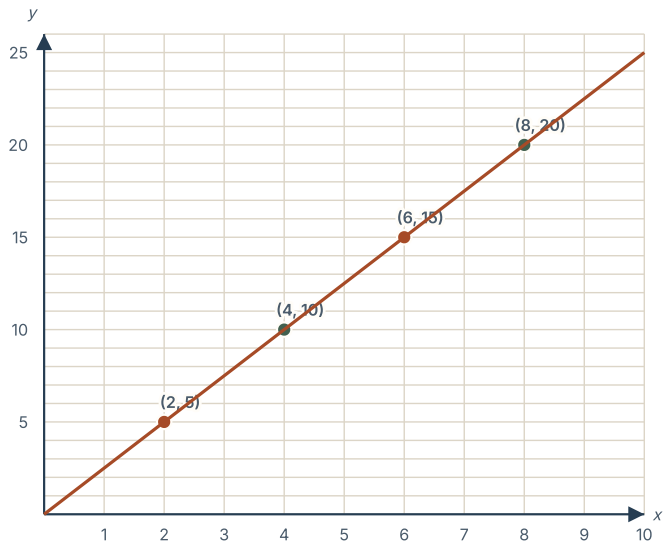


Figure 1.3 Plotting the pairs from Table 1.1 gives points on a straight line through the origin. The line is a visual signature of equivalent ratios: every point on it has the same ratio of y to x.

WORKED EXAMPLE 1.3.1 — TABLE TO MISSING VALUE

PROBLEM

A recipe uses 3 cups of flour for every 4 cups of milk. Complete the table and find the amount of milk needed for 21 cups of flour.

Flour (cups)	3	6	9	21
Milk (cups)	4	?	?	?

1. The scale factor from 3 cups to 6 cups is 2, so multiply 4 by 2: milk $= 8$ cups.
2. The scale factor from 3 to 9 is 3, so milk $= 4 \times 3 = 12$ cups.
3. For 21 cups, the scale factor is $21 \div 3 = 7$, so milk $= 4 \times 7 = 28$ cups.
4. Check using the cross-product condition: $3 \times 28 = 84 = 4 \times 21$.

ANSWER **8 cups, 12 cups and 28 cups of milk.**

WORKED EXAMPLE 1.3.2 — A NON-EXAMPLE

PROBLEM

Explain why the pairs $(2, 5)$, $(4, 9)$ and $(6, 13)$ do not form a ratio table for a single ratio.

1. Compare the ratios: $2 : 5$, $4 : 9$ and $6 : 13$.
2. Write them as fractions: $\frac{2}{5} = 0.4$, $\frac{4}{9} \approx 0.444$, $\frac{6}{13} \approx 0.462$.
3. The values are different, so the quantities do not scale by the same factor. For example, from $2 : 5$ to $4 : 9$, the first term doubles but the second does not.

ANSWER **The pairs are not equivalent; adding 2 to both terms changes the ratio.**

PRACTICE 1.3 · TABLES AND LINES

SUPPORT 1–4 · CORE 1–8 · EXTENSION 9–10

Look for the multiplicative scale factor. State the factor when it helps to justify your answer.

1. Copy and complete the ratio table for $2 : 5$.

First quantity	2	4	6	8
Second quantity	5	?	?	?

F

2. A shop sells pens in packs. The table shows the number of pens and the cost in dollars.

Pens	3	6	9	15
Cost ($)	12	24	36	?

Find the missing cost and describe the ratio of cost to pens. F

3. Use the double number line in Figure 1.2. What amount of water is paired with 12 cups of lemon? F

4. Which table shows equivalent ratios?
A: $(2, 5), (4, 10), (6, 15)$
B: $(2, 5), (4, 9), (6, 13)$
Explain how you decided. R

5. A recipe uses 3 cups of flour to 4 cups of milk. Complete the table for 6, 9 and 12 cups of flour, then state the amount of milk for 12 cups. F

6. A map scale is 2 cm represents 5 km. Complete the table: 4 cm, 6 cm, 9 cm. How many kilometres does 9 cm represent? P

7. A cyclist travels at a constant speed, covering 3 km every 2 minutes. How far does the cyclist travel in 10 minutes? Use a ratio table or double number line. P

8. Do the points $(2, 3)$, $(4, 6)$ and $(6, 9)$ all represent the same ratio? Explain using the graph or the cross-product test. R

9. Which point does **not** belong to the graph of the ratio $3 : 4$? F

   A $(6, 8)$
   B $(9, 12)$
   C $(12, 15)$
   D $(15, 20)$

10. A table contains the pairs $(5, 8)$, $(10, 16)$ and $(15, 23)$. Is this a ratio table for a single ratio? If not, change one number to make it one, and explain your change. E

1.4 Sharing and comparing with ratios

CRITERION A · III ◆ CRITERION C · III ◆ JUSTIFICATION

A ratio becomes a practical tool when a total must be divided into parts that preserve a stated relationship. The central step is to find the value of one part, not to divide by the number of shares.

> METHOD · SHARING IN A RATIO
>
> To share a total T in the ratio $a : b : c$: add the parts to get $a + b + c$; divide T by the total number of parts to find the value of one part; multiply by each term to find the shares; check that the shares add to T.

WORKED EXAMPLE 1.4.1 — TWO-PART SHARING

PROBLEM

Share 72 counters between Ali and Bea in the ratio $5 : 3$.

1. Total number of parts: $5 + 3 = 8$.
2. One part: $72 \div 8 = 9$ counters.
3. Ali: $5 \times 9 = 45$ counters. Bea: $3 \times 9 = 27$ counters.
4. Check: $45 + 27 = 72$, and $45 : 27$ simplifies to $5 : 3$.

ANSWER **Ali receives 45 counters; Bea receives 27.**

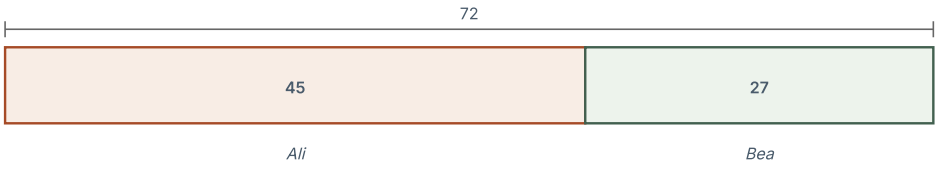


Figure 1.4 The same share represented as a bar. The ratio $45 : 27$ simplifies to $5 : 3$, so both displays describe the same relationship at different scales.

WORKED EXAMPLE 1.4.2 — THREE-PART SHARING

PROBLEM

A school divides 180 exercise books among Year 7, Year 8 and Year 9 in the ratio $2 : 3 : 4$. How many books does each year group receive?

1. Total parts: $2 + 3 + 4 = 9$.
2. One part: $180 \div 9 = 20$ books.
3. Year 7: $2 \times 20 = 40$; Year 8: $3 \times 20 = 60$; Year 9: $4 \times 20 = 80$.
4. Check: $40 + 60 + 80 = 180$.

ANSWER **40, 60 and 80 books.**

WORKED EXAMPLE 1.4.3 — A DIFFERENCE IS GIVEN

PROBLEM

Two amounts are in the ratio $3 : 8$. The larger amount is 45 more than the smaller. Find both amounts.

1. The difference in the ratio is $8 - 3 = 5$ parts.
2. These 5 parts represent 45, so one part is $45 \div 5 = 9$.
3. Smaller amount: $3 \times 9 = 27$. Larger amount: $8 \times 9 = 72$.
4. Check: $72 - 27 = 45$ and $27 : 72 = 3 : 8$.

ANSWER **The amounts are 27 and 72.**

WORKED EXAMPLE 1.4.4 COMPARING TWO RATIOS

PROBLEM

Which is greater, $4 : 5$ or $7 : 9$? Justify your answer.

1. Write each ratio as a fraction of the first quantity to the second: $\frac{4}{5} = 0.8$ and $\frac{7}{9} \approx 0.778$.
2. Since $0.8 > 0.778$, the ratio $4 : 5$ represents a larger first quantity relative to the second.
3. Cross-product check: $4 \times 9 = 36$ and $5 \times 7 = 35$. Because $36 > 35$, $4 : 5 > 7 : 9$.

ANSWER $4 : 5$ **is greater than** $7 : 9$.

Alternatively, write both with a common second term of 45: $4 : 5 = 36 : 45$ and $7 : 9 = 35 : 45$.

COMMON ERROR

To share 72 in the ratio $5 : 3$, dividing $72 \div 2 = 36$ treats the ratio as two equal shares. The ratio tells you there are $5 + 3 = 8$ parts, not two. Always find the total number of parts first.

PRACTICE 1.4 · SHARING AND COMPARING

SUPPORT 1–4 · CORE 1–9 · EXTENSION 10–12

Show the total number of parts and the value of one part. Check each share by recombining.

1. Share 72 in the ratio $5 : 3$. F
2. Share 180 in the ratio $2 : 3 : 4$. F
3. Share 96 in the ratio $3 : 5 : 4$. F
4. Two amounts are in the ratio $4 : 7$. The smaller amount is 36. Find the larger amount and the total. P
5. Two amounts are in the ratio $3 : 8$. Their difference is 45. Find both amounts. P
6. Share 120 in the ratio $7 : 3 : 2$. F
7. A rectangle has perimeter 60 cm. Its length and width are in the ratio $7 : 3$. Find the length and width. P
8. The angles of a triangle are in the ratio $2 : 3 : 4$. Find the three angles. P
9. Which is greater, $4 : 5$ or $7 : 9$? Show your method. R
10. A recipe uses flour and sugar in the ratio $5 : 2$. If 350 g of flour is used, how much sugar is needed, and what is the total mass of these two ingredients? P
11. Three classes share 360 students in the ratio $5 : 6 : 7$. How many students are in each class? P
12. A sum of money is shared in the ratio $3 : 5$. The larger share is \$120 more than the smaller. Find the total sum. E

1.5 Investigation · The lemonade stand

A community fair is selling lemonade. Four teams use different recipes. The recipes are described by the ratio of lemon juice to water. The teams want to know which recipes taste the same and how to scale any recipe to a larger container.

Which recipe is which?

INVESTIGATION

The table gives the ingredients for one jug of each recipe. All measurements are in cups.

TABLE 1.2 · FOUR LEMONADE RECIPES

Recipe	Lemon juice	Water	Total
A	2	5	7
B	4	10	14
C	6	15	21
D	3	8	11

1. Write each recipe as a ratio of lemon juice to water. Simplify each ratio.
2. Which recipes have the same simplified ratio? What does this suggest about their taste?
3. Complete a table showing the amount of lemon juice needed for 30 cups of water for each recipe.
4. Plot the pairs (lemon juice, water) for recipe A for jugs of 1, 2, 3 and 4 times the base recipe. Describe the graph.
5. Write a conjecture: if two lemonade recipes have the same simplified ratio, what is true about the taste? Test your conjecture with a new pair of recipes, $3 : 7$ and $9 : 21$.
6. Recipe D uses 3 cups of lemon and 8 cups of water. Is it more or less lemony than recipe A? Justify with a calculation.
7. Give one reason why the ratio alone might not guarantee that two lemonades taste identical.

Real-life application · Cooking for 45

CRITERION D

A community kitchen has a recipe for 12 people: 600 g rice, 450 g lentils and 90 mL oil. The team needs to cook for 45 people. The ingredients must stay in the same ratio.

1. Find the scale factor from 12 people to 45 people.
2. Calculate the amount of each ingredient needed for 45 people. Give masses to the nearest gram and oil to the nearest millilitre.
3. Explain why tripling the recipe and then adding a quarter of the original recipe gives the same result as multiplying by 3.75.
4. The kitchen has 2.4 kg of rice. Is that enough? Justify with a comparison, and comment on the accuracy of your answer.

Criterion task · The community garden harvest

ASSESSED CRITERIA
A I–III · C I–IV · D I–V

Three classes — 7A, 7B and 8A — look after a community garden. Their volunteering hours are in the ratio 5 : 3 : 2. The first harvest is 120 kg of vegetables. A parent proposes that the harvest be shared equally. The students propose that it be shared in proportion to the hours worked.

1. Calculate the equal share for each class.
2. Calculate the share for each class using the ratio 5 : 3 : 2. Show the total number of parts and the value of one part.
3. Compare the two proposals in a table. Which proposal is fairer? Justify your answer using the ratios, the context and at least one limitation of your judgement.
4. A second harvest of 90 kg is shared in the same ratio. Find each class's share.
5. The committee decides to reserve 15% of the 120 kg harvest for seed. Calculate the amount reserved, then share the remainder in the ratio 5 : 3 : 2. Round each share to the nearest 0.1 kg and check that the rounded shares add to the remainder.
6. Write a short proposal to the garden committee. Include a table of shares, a clear explanation of your method, a statement about accuracy, and one assumption you made.

FORMAT WRITTEN REPORT WITH TABLE **TIME** 45–60 MINUTES **RESOURCES** CALCULATOR PERMITTED

PART	WHAT TO SHOW	STRANDS
1–2	Correct total parts, value of one part, and all shares; units included.	A i–iii
3–5	Comparison organised in a table; a justified recommendation; accuracy and rounding handled explicitly.	C ii–iv, D i–v
6	A coherent proposal with a logical structure, precise language and a limitation.	C i–iv, D iv–v

PLAN BEFORE YOU WRITE

State what the whole represents, the value of one part, and the comparison you will make. Draft one sentence giving your recommendation and one limitation of it.

1.6 Summary, revision and reflection

KEY IDEAS

- A ratio compares quantities multiplicatively; order carries meaning.
- Part-to-part and part-to-whole ratios answer different questions.
- Multiplying or dividing both terms by the same non-zero number gives an equivalent ratio.
- A ratio is in simplest form when the whole-number terms have no common factor greater than 1.
- Ratio tables, double number lines and straight-line graphs all represent equivalent ratios.
- To share a total, divide by the total number of parts, not by the number of shares.

Equivalent ratios: $a : b = ka : kb$ for $k \neq 0$.

Cross-product test: $a : b = c : d$ exactly when $ad = bc$.

Share of one part:

$$\text{value of one part} = \frac{\text{total}}{\text{sum of ratio terms}}.$$

CONNECTIONS

Ratios prepare the ground for proportion (Chapter 2) and rates (Chapter 3). The same scaling relationship will later appear as a linear graph and as an algebraic equation.

REVISION CHECKLIST

- [ ] I can explain the difference between part-to-part and part-to-whole.
- [ ] I can simplify a ratio with whole-number, decimal or fractional terms.
- [ ] I can generate and verify equivalent ratios.
- [ ] I can complete a ratio table and interpret a double number line.
- [ ] I can plot equivalent ratio pairs and describe the graph.
- [ ] I can share a total in a two- or three-part ratio.
- [ ] I can find amounts when one share or a difference is known.
- [ ] I can compare two ratios and justify my conclusion.
- [ ] I can identify and correct a ratio error.

REFLECTION

Which representation made ratio relationships clearest for you, and why?

When might a simplified ratio hide an important difference between two situations?

CHAPTER 1 · MIXED PRACTICE

Consolidate and extend

These questions combine the chapter's ideas. They are not a formal assessment; use them to find what is secure and what still needs attention.

FLUENCY Accurate and efficient

1. Simplify 16 : 24.
2. Simplify 0.8 : 1.2.

3. Copy and complete: $5 : 9 = 20 : ?$.
4. Share 84 in the ratio $3 : 4$.
5. Share 210 in the ratio $2 : 5 : 7$.
6. Write the ratio 40 minutes to 3 hours in simplest form.
7. A mixture uses 2 parts syrup to 7 parts water. If 21 L of water is used, how much syrup is needed?
8. Complete the ratio table for $2 : 5$: first quantity 2, 6, 10; second quantity 5, ?, ?. Find the missing values.
9. Which is larger, $5 : 8$ or $7 : 11$? Show a numerical check.
10. A rectangle has length and width in the ratio $8 : 5$ and perimeter 78 cm. Find its length and width.

REASONING AND PROBLEM SOLVING

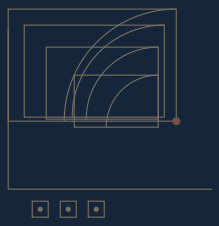
Connect and justify

11. A ratio $12 : 18$ is simplified to $6 : 9$, and then the student stops. Explain why $6 : 9$ is not in simplest form and give the correct answer. R
12. A recipe uses flour to milk in the ratio $3 : 4$. How much milk is needed for 21 cups of flour? What is the total number of cups of these two ingredients? P
13. The ratio of red to blue beads is $5 : 3$. There are 40 red beads. How many beads are there altogether? P
14. The angles of a triangle are in the ratio $3 : 4 : 5$. Find the three angles. P
15. A sum is shared in the ratio $2 : 3 : 5$. The largest share is \$250. Find the total sum and the other two shares. P
16. Which is greater, $3 : 4$ or $5 : 7$? Justify with two different methods. R
17. A map has a scale of 3 cm to 8 km. A path on the map is 12 cm long. What distance does this represent? P
18. A drink is made from 1 part syrup to 4 parts water. If 2.5 L of syrup is used, how much water is needed, and what is the total volume of the drink? P
19. A school has 480 students. The ratio of girls to boys is $7 : 5$. How many girls and boys are there? P
20. A bag contains 72 sweets. The ratio of chocolate to fruit sweets is $5 : 4$. How many chocolate sweets are there? After 8 chocolate sweets are eaten, what is the new ratio of chocolate to fruit sweets? P

EXTENSION AND INQUIRY

Generalise and evaluate

21. A rectangle is 12 cm by 18 cm. It is enlarged so that the shorter side becomes 18 cm. Find the new longer side and the ratio of the new dimensions to the original dimensions. E
22. Given $a : b = 2 : 5$ and $b : c = 3 : 4$, find the ratio $a : b : c$. Explain how you made the two ratios compatible. E
23. A sum is shared in the ratio $4 : 5$. The larger share is \$36 more than the smaller. Find the total sum. E
24. A recipe for 8 people uses flour and milk in the ratio $3 : 2$. For 20 people, how much flour and milk are needed? Explain why the amounts are not whole numbers and whether that matters. E
25. Design a fair-share problem for a context of your choice. Include a table of values, a ratio, and a question that requires a judgement about fairness. Solve it and explain one assumption. E



2

NUMBER AND PROPORTION

Proportional reasoning

Two quantities are proportional when one is always a fixed multiple of the other. That single number can connect a recipe to a banquet, a map to a journey, and a model to the building it represents.

KEY CONCEPT · RELATIONSHIPS

RELATED · MODELS

RELATED · EQUIVALENCE

CONTEXT · ORIENTATION IN SPACE AND TIME

INQUIRY

How can one constant multiplier connect a small model to a much larger reality?

MYP UNIT PLANNER

Framing the inquiry

UNIT 2 · APPROX. 11 HOURS
ATL FOCUS · RESEARCH

KEY AND RELATED CONCEPTS

Relationships. A proportional relationship pairs every input with an output in a fixed multiplicative way. The pairing can be described by a table, a graph, an equation or a scale statement.

Models. A map, plan or scale model is a smaller or simpler version of something real. A model is useful only when its scale is stated and applied consistently.

Equivalence. The ratios $y : x$ are equal for every pair in a proportional relationship; this is what makes one constant multiplier possible.

GLOBAL CONTEXT AND EXPLORATION

Orientation in space and time. Exploration: scale, maps, journeys and representations of place.

A map compresses distance without changing relationships; a plan compresses a room without changing its shape. Proportional reasoning lets us move between the representation and the reality, check that a model is faithful, and recognise when a convenient scale has become misleading.

STATEMENT OF INQUIRY

A proportional relationship preserves a constant multiplier, so one scale factor can translate reliably between a representation and the reality it models.

FACTUAL

What makes a relationship proportional, and how is the constant multiplier found?

CONCEPTUAL

How do tables, graphs, equations and scale statements reveal the same relationship?

DEBATABLE

When does a scale model stop being a trustworthy representation of reality?

ATL · RESEARCH IN ACTION

Find two published maps or scale plans and record their stated scales. Convert one map distance to real distance and one real distance to map distance. Compare how much detail each scale allows, then write a rule for choosing a scale for a given purpose.

INQUIRY PATHWAY FOR THIS CHAPTER

1	2	3	4	5	6
CONTEXT	QUESTION	CONCEPT	INVESTIGATION	COMMUNICATION	REFLECTION
A garden plan and a map of a nature reserve.	How does one scale factor connect model and reality?	Proportional relationships; constant multiplier; scale.	Design a plan that fits the paper and keeps the shape.	Scale statements, tables, equations and labelled plans.	When does a faithful scale still fail to show what matters?

Learning intentions and success criteria

YOU ARE LEARNING TO

- **Understand** what makes a relationship proportional and how to recognise it in a table, graph or equation.
- **Know** how to find and use the constant of proportionality.
- **Be able to** solve proportional problems using scale factors and cross products.
- **Be able to** interpret and calculate with map scales and scale drawings.
- **Be able to** decide whether a model or comparison is proportional and justify the decision.

SUCCESS CRITERIA

- ☐ I can test whether a table or graph represents a proportional relationship and explain my test.
- ☐ I can find the constant of proportionality and write the equation $y = kx$.
- ☐ I can solve a proportion from a context and state the units of the answer.
- ☐ I can choose an efficient method and check my result by substitution.
- ☐ I can convert between map and real distances using a stated scale.
- ☐ I can use a scale factor to find a missing length in a similar figure.

PREREQUISITE CHECK · 10 MINUTES, NO CALCULATOR

1. Simplify $\frac{18}{24}$.
2. Find the value of 7×12.
3. Convert 3.5 m to centimetres.
4. Solve $4x = 52$.
5. Write 0.4 as a fraction in simplest form.
6. Calculate $15 \div 0.5$.

Answers are in the Chapter 2 answer section. If more than two are incorrect, review simplifying fractions and metric conversions before continuing.

YEAR 1
Equivalent fractions; simple scaling; coordinates in the first quadrant.

YEAR 2 · THIS CHAPTER
Proportional relationships; constants; scale drawings and maps.

YEAR 3
Direct and inverse proportion; algebraic constants; linear functions.

CRITERION	STRANDS	EVIDENCE IN THIS CHAPTER
A · Knowing and understanding	i, ii, iii	Identify proportional relationships; calculate constants; solve proportions and scale problems.
B · Investigating patterns	i, ii, iii	Investigation: generalise the relationship between plan length, real length and scale.
C · Communicating	i, ii, iii, iv	Use scale statements, units and labelled diagrams; justify the choice of scale.
D · Applying mathematics	i, ii, iii, iv, v	Nature-reserve task: model a real space, evaluate accuracy, and state a limitation.

2.1 Recognising proportional relationships

CRITERION A · I, II ◆ CRITERION C · II ◆ RELATIONSHIPS

A proportional relationship is more than a table of paired numbers. It is a relationship in which every output is obtained from its input by the same multiplication.

DEFINITION

Two quantities x and y are **directly proportional** when $y = kx$ for some constant $k \neq 0$. The number k is the **constant of proportionality**. Equivalently, $\frac{y}{x} = k$ for every pair of corresponding values.

A proportional relationship has three signatures: equal ratios $y : x$ in every column of a table, a straight-line graph through the origin, and an equation of the form $y = kx$. Any one signature can be used to test the relationship, but the table is usually the quickest place to start.

TABLE 2.1 · A PROPORTIONAL RELATIONSHIP AND A NEAR MISS

x	2	4	6	8
y (proportional)	5	10	15	20
y (not proportional)	5	8	11	14

In the first row, $\frac{5}{2} = \frac{10}{4} = \frac{15}{6} = \frac{20}{8} = 2.5$, so $y = 2.5x$. In the second row, the values increase by 3 each time, but the ratios are 2.5, 2, 1.83 and 1.75. Adding a fixed amount is not proportional reasoning.

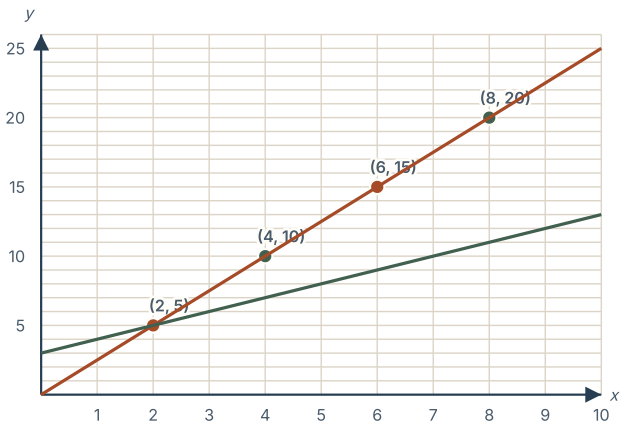


Figure 2.1 Both relationships are linear, but only the copper line passes through the origin. The green line has equation $y = x + 3$; its ratio $y : x$ changes, so it is not proportional.

KEY IDEA

A straight line is not enough evidence for proportionality. The line must also pass through $(0, 0)$, because a proportional relationship sends zero input to zero output.

WORKED EXAMPLE 2.1.1 — TEST AND GENERALISE

PROBLEM

A printer produces 45 pages in 3 minutes, 75 pages in 5 minutes and 120 pages in 8 minutes at a constant rate. Is the number of pages proportional to the time? Find the constant and write an equation.

1. Compare the ratios $\frac{\text{pages}}{\text{minutes}}$: $\frac{45}{3} = 15$, $\frac{75}{5} = 15$, $\frac{120}{8} = 15$.
2. The ratios are equal, so the quantities are directly proportional with constant $k = 15$ pages per minute.
3. Let p be the number of pages and t the time in minutes. Then $p = 15t$.
4. Check with $t = 8$: $15 \times 8 = 120$, which matches the table.

ANSWER **Yes; $k = 15$ pages per minute and $p = 15t$.**

WORKED EXAMPLE 2.1.2 — A NON-PROPORTIONAL MODEL

PROBLEM

A taxi charges a fixed \$4 plus \$2 per kilometre. Copy and complete the table. Explain why the cost is not proportional to the distance.

Distance (km)	1	2	3	5
Cost (\$)	6	8	10	?

1. The cost increases by \$2 for each extra kilometre, so for 5 km the cost is $4 + 5 \times 2 = 14$ dollars.
2. The cost equation is $C = 2d + 4$, not $C = kd$.
3. The ratios $\frac{C}{d}$ are 6, 4, 3.33 and 2.8; they are not constant because the fixed charge is shared over more kilometres.

ANSWER **\$14; the intercept is 4, so the graph is linear, not proportional.**

ATL · CRITICAL THINKING

Skill in action. A classmate claims, "Every straight-line graph shows a proportional relationship." Construct two equations that produce straight lines, one proportional and one not, and sketch both. Then write a precise correction to the claim that names the extra condition.

COMMON ERROR

Checking only that the table increases steadily. A table can increase by equal differences and still be non-proportional. Test the ratios $\frac{y}{x}$, or check that the graph passes through the origin.

PRACTICE 2.1 · PROPORTIONAL OR NOT?

SUPPORT 1–4 · CORE 1–8 · EXTENSION 9–10

Show the test you used, not only the verdict. Where a relationship is proportional, state the constant.

1. Which tables are proportional?

A: x	2	4	6
y	3	6	9

B: x	1	2	3
y	4	6	8

F

2. Find the constant of proportionality for the pairs $(4, 10)$, $(6, 15)$ and $(8, 20)$, then write an equation. F

3. Copy and complete the table for $y = 7x$.

x	1	3	5	?
y	7	21	?	56

F

4. Is $y = 4x + 1$ proportional? Explain using two pairs of values. R

5. A line passes through the origin and the point $(4, 10)$. Find k and the value of y when $x = 7$. F

6. A phone plan costs \$5 per month plus \$0.10 per minute. Is the monthly cost proportional to the minutes used? Explain. R

7. The area A of a square is related to its side length s by $A = s^2$. Is A proportional to s? Justify with a numerical example. R

8. A table shows $x = 3, 5, 8$ and $y = 12, ?, 32$. The relationship is proportional. Find the missing value and explain your method. P

9. A student says, "If x increases by 2 and y increases by 7 each time, then y is proportional to x." Test the claim with a table starting at $x = 1, y = 3$. E

10. Two proportional quantities satisfy $y = 3.5x$. Explain what happens to y when x increases by 2. Use your answer to find y when $x = 11$ given that $y = 21$ when $x = 6$. E

2.2 Solving proportional problems

CRITERION A · II, III ◆ CRITERION D · I, II ◆ QUANTITY

Proportional problems are solved by finding the multiplier that connects the two situations. The multiplier can come from a unit value, from a scale factor between the quantities, or from an equation.

METHOD · THREE ROUTES

Unitary method. Find the value of one unit, then multiply. **Scale factor.** Divide the new quantity by the old quantity, then multiply the paired quantity by the same factor. **Cross products.** From $\frac{a}{b} = \frac{c}{d}$, use $ad = bc$. Choose the route that keeps the numbers simple and the units clear.

WORKED EXAMPLE 2.2.1 — UNIT PRICE

PROBLEM

Four identical notebooks cost \$9.20. Find the cost of seven notebooks and write an equation for the cost C of n notebooks.

1. One notebook costs $9.20 \div 4 = \$2.30$.
2. Seven notebooks cost $7 \times 2.30 = \$16.10$.
3. The constant is $k = 2.30$ dollars per notebook, so $C = 2.30n$.
4. Check: $C = 2.30 \times 7 = 16.10$, and $16.10 \div 7 = 2.30$.

ANSWER **\$16.10; $C = 2.30n$.**

WORKED EXAMPLE 2.2.2 — SCALING WITH A FACTOR

PROBLEM

A recipe for 12 muffins uses 180 g of flour. How much flour is needed for 20 muffins at the same ratio?

1. The scale factor from 12 to 20 muffins is $\frac{20}{12} = \frac{5}{3}$.
2. Multiply the flour by the same factor: $180 \times \frac{5}{3} = 300$.
3. Alternatively, one muffin uses $180 \div 12 = 15$ g, so 20 muffins use $20 \times 15 = 300$ g.
4. The answer is greater than 180 g, as expected, and the ratio $300 : 20 = 15 : 1$ matches $180 : 12 = 15 : 1$.

ANSWER **300 g of flour.**

WORKED EXAMPLE 2.2.3 CROSS PRODUCTS

PROBLEM

A paint mix uses 7 parts colour to 12 parts base. How much colour is needed with 30 cups of base?

1. Write the proportion $\frac{7}{12} = \frac{c}{30}$, where c is the number of cups of colour.
2. Cross multiply: $7 \times 30 = 12 \times c$, so $210 = 12c$.
3. Divide both sides by 12: $c = 17.5$.
4. Check: $\frac{17.5}{30} = 0.583\ldots$ and $\frac{7}{12} = 0.583\ldots$.

ANSWER **17.5 cups of colour.**

COMMON ERROR

Mixing the order of a proportion. If the left ratio compares colour to base, the right ratio must also compare colour to base. Label each term with its quantity before cross multiplying.

PRACTICE 2.2 · PROPORTIONAL PROBLEMS

SUPPORT 1–4 · CORE 1–9 · EXTENSION 10–12

Choose one method, show the proportion or unit value, and include units in your final answer.

1. Five pens cost \$6.25. Find the cost of eight pens. F
2. Three kilograms of potatoes cost \$4.80. Find the cost of seven kilograms. F
3. A paint covers 96 m^2 with 12 L. How many square metres does 5 L cover? F
4. A recipe for 4 servings uses 600 g of rice. How much rice is needed for 10 servings? F
5. Solve $\frac{3}{5} = \frac{x}{20}$. F
6. Solve $\frac{8}{15} = \frac{24}{y}$. F
7. A car travels 150 km in 2 hours at a constant speed. How far does it travel in 5 hours? P
8. A recipe uses 2 cups of flour for every 3 cups of milk. How much flour is needed with 7.5 cups of milk? P
9. Shop A sells 3 notebooks for \$4.20. Shop B sells 5 for \$6.75. Which is better value? Show unit prices. P
10. A student solves $\frac{4}{5} = \frac{x}{12}$ by adding 7 to both terms and obtains $x = 11$. Identify the error and find the correct value. E
11. A 3.5 L tin of paint covers 28 m^2. How much paint is needed for 50 m^2? Give your answer to a sensible accuracy. P
12. The ratio of red to white paint is $3 : 5$. A decorator needs 24 L of the mixture. How much of each colour is needed? E

2.3 Scale drawings, maps and similar figures

CRITERION A · III ◆ CRITERION C · II ◆ MODELS

A scale is a ratio between a representation and the real object. The same ratio applies to every length in the drawing, which is why a plan can preserve shape while changing size.

DEFINITION

A **scale** compares a length in a drawing or model with the corresponding real length. A scale of $1 : 100$ means 1 unit on the drawing represents 100 of the same units in reality. The **scale factor** from drawing to reality is 100; from reality to drawing it is $\frac{1}{100}$.

Map scales compare lengths in the same unit. A scale of $1 : 25\,000$ means that 1 cm on the map represents 25 000 cm, or 250 m. Convert the real length after multiplying, and state the unit in your answer.

WORKED EXAMPLE 2.3.1 MAP SCALE

PROBLEM

A map has scale $1 : 25\,000$. (a) A path measures 6.4 cm on the map. Find its real length in kilometres. (b) A road is 4.5 km long. How long is it on the map?

1. (a) Real length $= 6.4 \times 25\,000 = 160\,000$ cm.
2. Convert: $160\,000$ cm $= 1600$ m $= 1.6$ km.
3. (b) Convert to centimetres first: 4.5 km $= 4500$ m $= 450\,000$ cm.
4. Map length $= 450\,000 \div 25\,000 = 18$ cm.

ANSWER **(a) 1.6 km (b) 18 cm.**

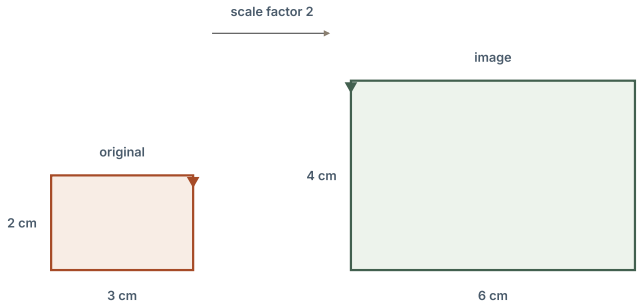


Figure 2.2 The rectangles are similar: corresponding angles are equal and every length has been multiplied by 2. The ratio of corresponding sides, $3 : 6$ and $2 : 4$, is the same.

WORKED EXAMPLE 2.3.2 SIMILAR FIGURES

PROBLEM

A photograph is 6 cm wide and 4 cm high. It is enlarged so that its width becomes 15 cm while its shape is preserved. Find the new height and the scale factor.

1. The scale factor is $15 \div 6 = 2.5$.
2. The height is multiplied by the same factor: $4 \times 2.5 = 10$ cm.
3. Check the ratio of width to height: $\frac{6}{4} = 1.5$ and $\frac{15}{10} = 1.5$.
4. The area increases by $2.5^2 = 6.25$, from 24 cm^2 to 150 cm^2.

ANSWER **Height 10 cm; scale factor 2.5; area factor 6.25.**

COMMON ERROR

Using the length scale factor for area. If every length is multiplied by k, every area is multiplied by k^2. At $1 : 100$, the drawing's lengths are one hundredth of reality, but its areas are one ten-thousandth.

PRACTICE 2.3 · SCALE AND SIMILARITY

SUPPORT 1–4 · CORE 1–8 · EXTENSION 9–10

State the scale you use and convert units before calculating. Draw a labelled sketch where it helps.

1. A plan has scale $1 : 100$. A wall measures 4.5 cm on the plan. Find its real length in metres. F
2. A plan has scale $1 : 200$. A room is 8 m long. How long is it on the plan, in centimetres? F
3. A map has scale $1 : 50\,000$. Two villages are 7.2 cm apart on the map. Find the real distance in kilometres. F
4. On a map, 3 cm represents 12 km. Find the scale in the form $1 : n$. P
5. Two similar triangles have sides 4 cm, 6 cm, 8 cm and 6 cm, 9 cm, 12 cm. Find the scale factor and verify that the ratios of corresponding sides are equal. P
6. A rectangle is 5 cm by 8 cm. It is enlarged by a scale factor of 3. Find the new dimensions, the perimeter and the area. P
7. A model car is 18 cm long. The scale is $1 : 24$. Find the real length in metres. P
8. A photograph 10 cm by 15 cm is enlarged so that its width becomes 25 cm. Find the new height and the area of the enlargement. P
9. A student says, "A scale of $1 : 100$ means the real object is 100 times smaller than the drawing." Explain the error and correct the statement. E
10. Two maps show the same 2 km road at scales $1 : 25\,000$ and $1 : 50\,000$. Find the length of the road on each map and explain which map shows more detail. E

2.4 Investigation · A plan that fits

CRITERION B · I, II, III ◆ CRITERION C · II, III ◆ MODELS

A school garden measures 15 m by 9 m. The garden club needs a scale plan on A4 paper, with a shed, four rectangular beds and a path. The plan must preserve the garden's shape and fit inside a 2 cm border.

Which scale should we use?

INVESTIGATION

A4 paper measures 21.0 cm by 29.7 cm. The usable drawing area is 17 cm by 25.7 cm after the border is removed. Work in centimetres and keep a record of your scale statements.

1. Calculate the plan dimensions of the garden at scales $1 : 50$, $1 : 100$ and $1 : 200$.
2. Which of these scales fit inside the usable area? Explain, using both dimensions.
3. A shed is 3 m by 2 m. Find its plan dimensions at the scale you chose.
4. A bed is 4.5 m by 1.8 m. Find its plan dimensions at the same scale.
5. On your plan, two measured lengths are 7.2 cm and 4.8 cm. Use your scale to find the real lengths.
6. Write a general rule connecting plan length, real length and the scale denominator n for a scale of $1 : n$.
7. Explain one reason a plan at $1 : 200$ might be less useful than a plan at $1 : 100$, even though both fit.

Real-life application · A model sports hall

CRITERION D

A sports hall is 36 m long, 24 m wide and 8 m high. A model is built at a scale of $1 : 200$ for an open day.

1. Find the model's length, width and height in centimetres.
2. The model must fit on a board 50 cm by 40 cm. Does it fit? Justify with a comparison.
3. The model's base area is 216 cm^2. Show that the real floor area is 864 m^2 and verify the area scale factor.
4. Give one reason the model cannot show every feature of the real hall, and explain how the scale affects that limitation.

Criterion task · The nature reserve map

ASSESSED CRITERIA
A i–iii · C i–iv · D i–v

A nature reserve is 2.4 km long and 1.8 km wide. A draft trail map uses a scale of 1 : 10 000. Four trail sections measure 8.4 cm, 6.7 cm, 4.2 cm and 5.1 cm on the draft. The reserve wants a new map on A3 paper, which has a usable area of 38 cm by 26 cm after a border.

1. Convert each trail length on the draft to a real length in metres.
2. Find the total real length of the trail in kilometres.
3. Explain what the scale 1 : 10 000 means, using both map and real units.
4. Choose a scale that fits the reserve on the A3 map and leaves a sensible border. Show the plan dimensions at your scale and justify the choice.
5. At your chosen scale, find the length on the map of the 4.2 cm section. Compare it with the original 4.2 cm and explain the change.
6. Write a short proposal for the reserve manager. Include a scale statement, a table of real trail lengths, a note on accuracy, and one limitation of the map.

FORMAT ANNOTATED MAP AND SHORT REPORT

TIME 45–60 MINUTES

RESOURCES RULER, CALCULATOR PERMITTED

PART	WHAT TO SHOW	STRANDS
1–2	Correct conversions, units and total length; a clear scale statement.	A i–iii
3–5	A labelled plan, a justified scale choice, and a correct comparison of map lengths.	C ii–iv, D i–v
6	A coherent proposal with accurate data, a limitation and a conclusion.	C i–iv, D iv–v

PLAN BEFORE YOU WRITE

Decide what 1 cm represents at your chosen scale, check both reserve dimensions against the paper, and record the trail lengths before writing the proposal.

2.5 Summary, revision and reflection

KEY IDEAS

- Two quantities are directly proportional when $y = kx$ for a constant $k \neq 0$.
- Every pair in a proportional relationship has the same ratio $\frac{y}{x}$.
- A proportional graph is a straight line through the origin; a linear graph with a non-zero intercept is not proportional.
- Proportional problems can be solved with a unit value, a scale factor or cross products.
- A scale $1 : n$ means 1 unit on the representation equals n of the same units in reality.
- If lengths are multiplied by k, areas are multiplied by k^2.

Proportional relationship: $y = kx$.

Constant of proportionality: $k = \frac{y}{x}$.

Proportion: $\frac{a}{b} = \frac{c}{d}$ exactly when $ad = bc$.

Scale: drawing : real $= 1 : n$.

Area scale factor: k^2 when the length factor is k.

CONNECTIONS

Proportional relationships become linear functions in algebra, rate problems in Chapter 3, and similar-shape reasoning in geometry. The constant k is the bridge between the situations.

REVISION CHECKLIST

- ☐ I can test a table or graph for proportionality and explain the test.
- ☐ I can find k and write the equation of a proportional relationship.
- ☐ I can solve proportions using two different methods.
- ☐ I can choose and state units correctly in a proportional problem.
- ☐ I can convert between map and real distances.
- ☐ I can use a scale factor to find a missing length.
- ☐ I can calculate an area scale factor and explain it.
- ☐ I can identify when a linear relationship is not proportional.

REFLECTION

Which representation of a proportional relationship did you find most convincing, and why?

When might a map be mathematically accurate but still misleading?

KEY VOCABULARY

proportional
one quantity is a constant multiple of another

constant of proportionality
the multiplier k in $y = kx$

cross product
the result of multiplying across a proportion

scale
the ratio of a representation to reality

scale factor
the multiplier that changes one length to another

similar
same shape, with every length multiplied by the same factor

CHAPTER 2 · MIXED PRACTICE

Consolidate and extend

These questions combine the chapter's ideas. They are not a formal assessment; use them to find what is secure and what still needs attention.

FLUENCY Accurate and efficient

1. Find the constant of proportionality for the pairs $(5, 12.5)$, $(8, 20)$ and $(12, 30)$.
2. Write an equation for the relationship in question 1 and find y when $x = 15$.
3. Solve $\frac{9}{4} = \frac{x}{10}$.
4. Solve $\frac{6}{11} = \frac{18}{y}$.
5. A 5 kg bag of rice costs \$9.50. Find the cost of 8 kg.
6. A car uses 7 L of fuel for 84 km. How much fuel is needed for 156 km?
7. A plan has scale $1 : 150$. A corridor is 6 m long. Find its length on the plan.
8. A map has scale $1 : 40\,000$. Two points are 9.5 cm apart. Find the real distance in kilometres.
9. Are the pairs $(2, 7)$, $(4, 14)$, $(7, 24.5)$ proportional? Explain.
10. A rectangle 7 cm by 11 cm is enlarged by a factor of 4. Find its new area.

REASONING AND PROBLEM SOLVING Connect and justify

11. A phone contract costs \$8 per month plus 5 cents per minute. Is the cost proportional to the minutes? Justify with an equation and a graph description. R
12. A recipe for 6 people uses 450 g of pasta and 300 g of sauce. How much of each is needed for 15 people? P
13. A 2.5 L bottle of juice fills 10 glasses. How many glasses will 6 L fill at the same rate? P
14. Shop A sells 4 kg of apples for \$7.60. Shop B sells 6 kg for \$11.10. Which is better value? Show both unit prices. P
15. A photograph 8 cm by 12 cm is enlarged so that its height becomes 30 cm. Find the new width and the area scale factor. P
16. A model bridge is 45 cm long and built at a scale of $1 : 80$. Find the real length in metres. P
17. A student says, "If I double the side of a square, the area doubles too." Use a 3 cm square to test and correct the claim. R
18. Map A has scale $1 : 20\,000$ and Map B has scale $1 : 50\,000$. A path is 3 cm on Map A. How long is the same path on Map B? P
19. Two quantities are proportional. When $x = 4$, $y = 10$. Find y when $x = 14$, using two different methods. R
20. A plan of a room is 12 cm by 9 cm at $1 : 50$. Find the real dimensions and the real area. P

EXTENSION AND INQUIRY Generalise and evaluate

21. Two proportional quantities satisfy $y = kx$. When x increases from 5 to 9, y increases from 12.5 to 22.5. Find k and explain your method. E
22. A map scale is $1 : 25\,000$. A rectangular park measures 4.8 cm by 3.6 cm on the map. Find its real dimensions and area in square kilometres. E
23. A photograph is enlarged by a factor of 2.5, then the enlargement is reduced by a factor of 0.8. Is the final image the same size as the original? Justify with the combined scale factor. E

24. A scale model of a building is 1.2 m tall and uses a scale of $1 : 150$. The real building's height is later increased by 6 m. How tall should the model be to keep the same scale? E

25. Design a scale plan for a room of your choice. State the scale, calculate at least four real lengths, and explain one feature the plan cannot show. Include a labelled sketch and one accuracy statement. E

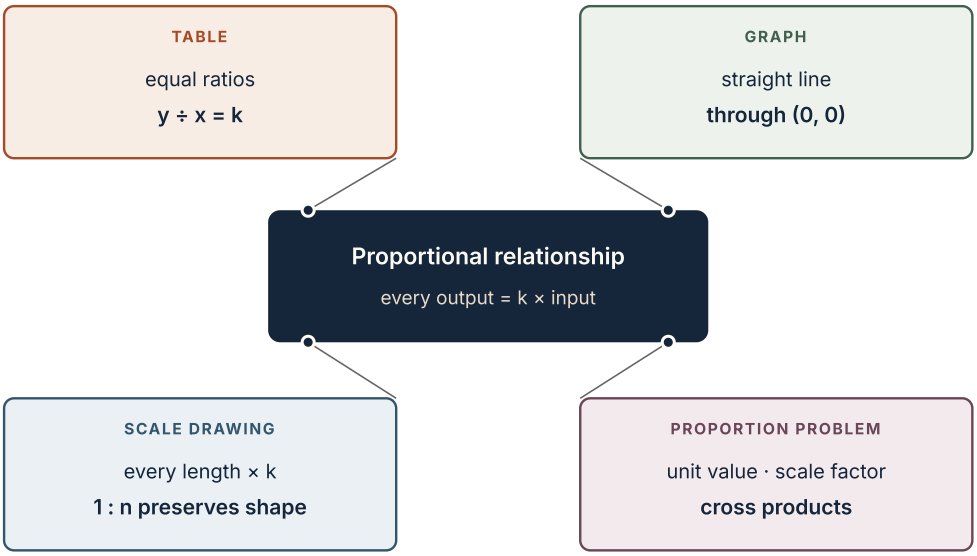
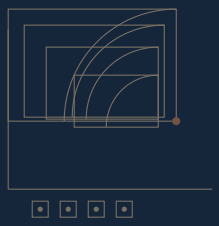


Figure 2.3 The constant multiplier connects four ways of working: a table, a graph, a scale drawing and a proportion problem. Each branch offers a different test.

ATL · REFLECTION

Make the map yours. Add a numerical example to each branch, then answer: which representation would you choose first to test a new relationship, and why?



3

NUMBER AND PROPORTION

Rates and unit rates

A rate compares quantities measured in different units. Reduce it to a single unit and unlike offers, journeys and flows become directly comparable.

KEY CONCEPT · RELATIONSHIPS

RELATED · QUANTITY

RELATED · EQUIVALENCE

CONTEXT · FAIRNESS AND DEVELOPMENT

INQUIRY

How can one common unit make unlike quantities fair to compare?

MYP UNIT PLANNER

Framing the inquiry

UNIT 3 · APPROX. 10 HOURS
ATL FOCUS · CRITICAL THINKING

KEY AND RELATED CONCEPTS

Relationships. A rate is a relationship between two measured quantities. Changing the unit changes the number but not the relationship, provided the conversion is exact.

Quantity. Every rate carries a unit that identifies what is being counted: kilometres per hour, dollars per kilogram, litres per minute.

Equivalence. Two rates are equivalent when they describe the same relationship in different units, for example $60\ \text{km/h} = 1\ \text{km/min}$.

GLOBAL CONTEXT AND EXPLORATION

Fairness and development. Exploration: fair comparison, access to resources and value in everyday decisions.

A unit price makes two different packages comparable; a speed makes two journeys comparable; a flow rate makes two water supplies comparable. The same mathematical move — reduce to one unit — supports fair choices about money, time and resources, provided the comparison is honest about what it leaves out.

STATEMENT OF INQUIRY

Expressing a relationship as a rate per unit creates a fair basis for comparison, revealing both value and the trade-offs hidden behind a single number.

FACTUAL

What is a rate, and how is a unit rate formed from it?

CONCEPTUAL

How does changing the unit change the comparison without changing the relationship?

DEBATABLE

When is the lowest unit price not the fairest or best choice?

ATL · CRITICAL THINKING IN ACTION

Find two multibuy offers for the same kind of product. Calculate the unit price of each, then list the information missing from the labels that could change your decision. Write a short judgement that separates the mathematical comparison from the shopping decision.

INQUIRY PATHWAY FOR THIS CHAPTER

1	2	3	4	5	6
CONTEXT	**QUESTION**	**CONCEPT**	**INVESTIGATION**	**COMMUNICATION**	**REFLECTION**
A supermarket shelf and a garden water tank.	How can unlike quantities be compared fairly?	Rate, unit rate and equivalent measures.	Is the largest package always the best value?	Rate statements, tables, unit-price comparisons and recommendations.	When does a fair calculation still lead to an unfair decision?

Learning intentions and success criteria

YOU ARE LEARNING TO

- **Understand** a rate as a comparison of two quantities with different units.
- **Know** how to calculate and interpret a unit rate.
- **Be able to** compare offers fairly using unit prices and equivalent units.
- **Be able to** solve problems involving speed, density and flow.
- **Be able to** evaluate a rate-based decision and state its limitations.

SUCCESS CRITERIA

- [ ] I can write a rate with correct units and reduce it to a unit rate.
- [ ] I can compare two rates by converting them to the same unit.
- [ ] I can use a unit rate to scale a quantity up or down.
- [ ] I can convert between km/h and m/s and check that the result is sensible.
- [ ] I can solve speed, density and flow problems using the correct relationship.
- [ ] I can explain when a lower unit price does not mean a better choice.

PREREQUISITE CHECK · 10 MINUTES, NO CALCULATOR

1. Calculate $240 \div 8$.
2. Simplify the ratio $45 : 60$.
3. Convert 2.5 hours to minutes.
4. Write 0.75 kg in grams.
5. Calculate 3.6×4.
6. Round 7.846 to two decimal places.

Answers are in the Chapter 3 answer section. If more than two are incorrect, review division, metric conversions and rounding before continuing.

YEAR 1
Ratio language; sharing in a ratio; simple scaling and unit conversion.

YEAR 2 · THIS CHAPTER
Rates, unit rates, best buys, speed, density and flow.

YEAR 3
Direct and inverse proportion; gradients; compound measures and algebraic rates.

CRITERION	STRANDS	EVIDENCE IN THIS CHAPTER
A · Knowing and understanding	i, ii, iii	Calculate unit rates; convert units within a rate; solve speed, density and flow problems.
B · Investigating patterns	i, ii, iii	Investigation: generalise a rule for comparing package prices and test it against an offer.
C · Communicating	i, ii, iii, iv	Use rate statements, units and tables; justify a best-buy recommendation.
D · Applying mathematics	i, ii, iii, iv, v	Breakfast club contract: model a real decision, evaluate assumptions and state a limitation.

3.1 What is a rate?

CRITERION A · I, II ◆ CRITERION C · II ◆ QUANTITY

A rate compares two quantities measured in different units. The comparison becomes a unit rate when the second quantity is reduced to one unit, so that every rate can be read as “this much for each one of those”.

DEFINITION

A **rate** is a comparison of two quantities with different units, such as 180 km in 3 hours. A **unit rate** is the amount of the first quantity for one unit of the second: 60 km per hour. A rate is written with its units, because the units are part of its meaning.

Rates are ratios with a difference: the two quantities measure different things. “240 km to 4 hours” compares distance with time, so it is a rate; “4 hours to 2 hours” compares times, so it is a ratio. Both can be simplified, but only a rate carries two different units.

TABLE 3.1 · EVERYDAY RATES AND THEIR UNIT FORMS

Situation	Rate	Unit rate
A car travels 240 km in 4 hours	240 km : 4 h	60 km/h
Apples cost $12.50 for 5 kg	$12.50 : 5 kg	$2.50/kg
A printer makes 96 pages in 3 min	96 pages : 3 min	32 pages/min
A tap delivers 42 L in 6 min	42 L : 6 min	7 L/min
A heart beats 72 times in 60 s	72 beats : 60 s	72 beats/min

To find a unit rate, divide the first quantity by the second. The units of the answer are “first units per second unit”. Division and the unit labels must agree: $240 \div 4 = 60$, and the answer is 60 km per hour, not 60 hours per kilometre.

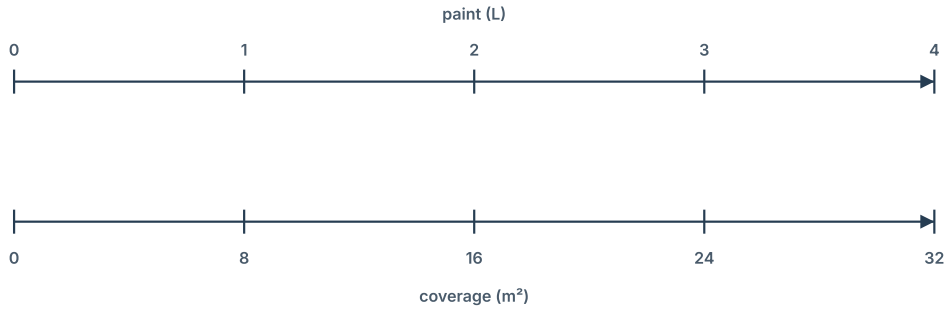


Figure 3.1 Each extra litre of paint covers another 8 m². The unit rate $8\ \mathrm{m^2/L}$ is the constant multiplier linking the two quantities.

KEY IDEA

A unit rate is both a comparison and a multiplier. Once you know the amount per one unit, you can scale to any quantity by multiplying, and scale back by dividing.

WORKED EXAMPLE 3.1.1 — PAGES PER MINUTE

PROBLEM

A printer produces 180 pages in 4 minutes at a constant rate. (a) Find the unit rate in pages per minute. (b) How long does it take to print 585 pages?

1. Divide pages by minutes: $180 \div 4 = 45$. The unit rate is 45 pages per minute.
2. To find the time, divide the number of pages by the unit rate: $585 \div 45 = 13$.
3. Check by multiplying: $45 \times 13 = 585$, so the time is consistent with the rate.

ANSWER **(a) 45 pages/min (b) 13 minutes.**

WORKED EXAMPLE 3.1.2 — PRICE PER KILOGRAM

PROBLEM

A 2.5 kg bag of apples costs \$6.25. (a) Find the price per kilogram. (b) Find the cost of 4.2 kg at the same rate.

1. Price per kilogram: $6.25 \div 2.5 = 2.50$, so the unit price is \$2.50 per kg.
2. Multiply by the new mass: $4.2 \times 2.50 = 10.50$.
3. Check by dividing: $10.50 \div 4.2 = 2.50$, which matches the unit price.

ANSWER **(a) \$2.50/kg (b) \$10.50.**

ATL · CRITICAL THINKING

Skill in action. A 2 L bottle of juice costs \$3.60 and a 1.5 L bottle costs \$2.40. A classmate says, "The 2 L bottle is better value because you get more juice." Calculate both unit prices, then write a correction that separates "more juice" from "better value".

COMMON ERROR

Dividing in the wrong order or losing the units. "12 km in 15 minutes" gives $12 \div 15 = 0.8$ km per minute, not 0.8 km per hour. Decide which unit the question asks for, then convert the time or distance before or after dividing.

PRACTICE 3.1 · RATES AND UNIT RATES

SUPPORT 1–4 · CORE 1–8 · EXTENSION 9–10

Write the rate, show the division, and include the units of the unit rate.

1. Write each as a unit rate. (a) 240 km in 4 h (b) \$18 for 6 kg (c) 96 words in 3 min (d) 150 mL for 5 doses F
2. A tap fills 42 L in 6 minutes. Find the rate in litres per minute and the time to fill 105 L. F
3. Five identical shirts cost \$72.50. Find the cost per shirt and the cost of three shirts. F
4. A car travels 320 km on 40 L of fuel. Find the rate in km/L and in L per 100 km. F
5. A recipe uses 750 g of flour for 15 muffins. Find the flour per muffin and the flour needed for 24 muffins. F

6. A heart beats 72 times in 60 seconds. Find the unit rate in beats per minute and the number of beats in 15 seconds. F

7. A 3.5 kg joint of meat costs $31.50. Find the price per kilogram and the cost of 2.2 kg. P

8. A runner covers 400 m in 80 seconds. Find the speed in m/s and the distance covered in 3 minutes at the same speed. P

9. A student says, "12 km in 15 minutes is 0.8 km per hour because $12 \div 15 = 0.8$." Identify the error and give the speed in km/h. E

10. A pack of four 500 mL bottles of water costs $5.60. Find the cost per litre. Explain why the number of bottles alone does not decide value. E

3.2 Unit prices and best buys

CRITERION A · II, III ◆ CRITERION D · I, II ◆ EQUIVALENCE

A unit price is a rate with money in the numerator and a quantity in the denominator. It lets you compare packages of different sizes without being distracted by the total price on the label.

TABLE 3.2 · THREE BAGS OF RICE, THREE UNIT PRICES

Package	Price	Unit price
1 kg bag	$2.40	$2.40/kg
5 kg bag	$10.50	$2.10/kg
10 kg bag	$21.50	$2.15/kg

The largest bag is not the best value here. The 5 kg bag costs $2.10 per kilogram, while the 10 kg bag costs $2.15 per kilogram. Unit pricing replaces "bigger looks cheaper" with a calculation that can be checked.

METHOD · A FAIR COMPARISON

1. Convert both quantities to the same unit. **2.** Divide price by quantity to find the unit price. **3.** Compare the unit prices, not the totals. **4.** Check whether a multibuy, delivery charge or minimum purchase changes the comparison. State the unit and round money to the nearest cent only at the end.

WORKED EXAMPLE 3.2.1 — CEREAL BOXES

PROBLEM

A 750 g box of cereal costs \$4.80. A 1.2 kg box of the same cereal costs \$7.20. Which box is better value per kilogram?

1. Convert to kilograms: $750\ \text{g} = 0.75\ \text{kg}$.
2. Small box: $4.80 \div 0.75 = 6.40$ dollars per kg.
3. Large box: $7.20 \div 1.2 = 6.00$ dollars per kg.
4. Compare: \$6.00 < \$6.40, so the 1.2 kg box is cheaper per kilogram.

ANSWER **The 1.2 kg box, at \$6.00/kg compared with \$6.40/kg.**

WORKED EXAMPLE 3.2.2 — A MULTIBUY

PROBLEM

Shop A sells 5 cereal bars for \$6.25. Shop B sells 4 of the same bars for \$4.80. Which shop offers the lower price per bar? Find the cost of 20 bars at each shop.

1. Shop A: $6.25 \div 5 = 1.25$ dollars per bar.
2. Shop B: $4.80 \div 4 = 1.20$ dollars per bar, so Shop B is cheaper per bar.
3. For 20 bars, Shop A needs 4 packs: $4 \times 6.25 = 25.00$.
4. Shop B needs 5 packs: $5 \times 4.80 = 24.00$. Shop B saves \$1.00 on 20 bars.

ANSWER **Shop B, at \$1.20 per bar; 20 bars cost \$25.00 at A and \$24.00 at B.**

COMMON ERROR

Comparing totals when the quantities differ. “\$7.50 is more than \$4.80, so the 6-pack is better” ignores how many items each price buys. Convert both offers to the same unit before deciding, and check whether the cheaper unit price requires buying more than you can use.

WORKED EXAMPLE 3.2.3 — LIQUID DETERGENT

PROBLEM

A 2 L bottle of detergent costs \$9.00. A 3 L bottle costs \$12.60. Which is better value per litre, and how much is saved on 6 L at the better unit price?

1. 2 L bottle: $9.00 \div 2 = 4.50$ dollars per litre.
2. 3 L bottle: $12.60 \div 3 = 4.20$ dollars per litre.
3. Six litres need three 2 L bottles or two 3 L bottles.
4. Cost with 2 L bottles: $3 \times 9.00 = 27.00$. Cost with 3 L bottles: $2 \times 12.60 = 25.20$. Saving: $27.00 - 25.20 = 1.80$.

ANSWER **The 3 L bottle is better value; 6 L costs \$25.20 instead of \$27.00, a saving of \$1.80.**

PRACTICE 3.2 · BEST BUYS — SUPPORT 1–4 · CORE 1–9 · EXTENSION 10–12

Show both unit prices, with units, before you choose. Where a decision depends on more than price, say what else matters.

1. 2 kg of potatoes cost \$3.60; 5 kg cost \$8.50. Which is better value per kilogram? F
2. 500 g of cheese costs \$4.20; 750 g costs \$6.00. Which has the lower price per kilogram? F
3. Six 500 mL bottles of water cost \$3.00; twelve cost \$5.40. Find the cost per litre for each. F
4. 250 g of coffee costs \$6.50; 1 kg costs \$23.00. Which is better value? F
5. 3 L of juice costs \$7.20; 5 L costs \$11.50. Which is better value per litre? F
6. A 4-pack of toilet rolls has 200 sheets per roll and costs \$3.20. A 9-pack has 180 sheets per roll and costs \$6.30. Find the cost per 100 sheets for each. P
7. A shop offers "buy 2, get 1 free" on bars priced at \$2.40 each. Another shop sells 3 bars for \$6.60. Compare the cost per bar. P
8. 1.5 kg of chicken costs \$10.50; 2.2 kg costs \$14.96. Which is better value per kilogram? P
9. A 5 kg bag of rice costs \$10.50; a 10 kg bag costs \$21.50. Which is better value? Explain why the larger bag is not automatically cheaper. P
10. Two internet plans cost \$30 per month for 100 Mbps and \$60 per month for 250 Mbps. Find the cost per Mbps and comment on whether the cheaper rate is the better choice. E
11. A student compares 6 pens for \$4.80 with 10 pens for \$7.50 and says, "The 10-pack is worse because \$7.50 is more than \$4.80." Identify the error and correct the comparison. E
12. Choose two real products sold in different sizes. Record the prices, calculate both unit prices, and recommend one. Include one non-price factor that could change your recommendation. E

3.3 Speed, density and flow

CRITERION A · II, III ◆ CRITERION C · II, III ◆ QUANTITY

Speed, density and flow are rates that describe how one quantity changes with another. Each has a defining relationship; the units tell you which quantity is divided by which.

METHOD · THREE DEFINING RELATIONSHIPS

Speed $= \dfrac{\text{distance}}{\text{time}}$, so distance = speed × time. **Density** $= \dfrac{\text{mass}}{\text{volume}}$, so mass = density × volume. **Flow rate** $= \dfrac{\text{volume}}{\text{time}}$, so volume = flow rate × time. Keep the time unit and the volume unit consistent before dividing.

WORKED EXAMPLE 3.3.1 — AVERAGE SPEED

PROBLEM

A train travels 210 km in 2.5 hours. (a) Find its average speed. (b) At the same speed, how long does a 336 km journey take?

1. Speed $= 210 \div 2.5 = 84$ km/h.
2. Time $= 336 \div 84 = 4$ hours.
3. Check: $84 \times 4 = 336$ km, which matches the journey length.

ANSWER **(a) 84 km/h (b) 4 hours.**

WORKED EXAMPLE 3.3.2 — DISTANCE AND UNIT CONVERSION

PROBLEM

A cyclist rides at 18 km/h for 45 minutes. (a) Find the distance travelled. (b) Express the speed in metres per second.

1. Convert the time: $45 \text{ min} = 0.75 \text{ h}$.
2. Distance $= 18 \times 0.75 = 13.5$ km.
3. Convert the speed: $18 \text{ km/h} = 18\,000 \text{ m} \div 3600 \text{ s} = 5 \text{ m/s}$.
4. Check: $5 \text{ m/s} \times 2700 \text{ s} = 13\,500 \text{ m} = 13.5$ km.

ANSWER **(a) 13.5 km (b) 5 m/s.**

WORKED EXAMPLE 3.3.3 — DENSITY

PROBLEM

A metal block has mass 810 g and volume 300 cm^3. (a) Find its density in g/cm^3. (b) Find the mass of a 500 cm^3 block of the same metal.

1. Density $= 810 \div 300 = 2.7 \text{ g/cm}^3$.
2. Mass $= 2.7 \times 500 = 1350$ g.
3. Convert if needed: $1350 \text{ g} = 1.35$ kg.
4. Check: $1350 \div 500 = 2.7$, so the density is unchanged.

ANSWER **(a) 2.7 g/cm^3 (b) 1350 g or 1.35 kg.**

WORKED EXAMPLE 3.3.4 FLOW RATE

PROBLEM

A pump delivers water at 4.5 L/min. (a) How long does it take to deliver 270 L? (b) How much water does it deliver in 1 hour 20 minutes?

1. Time $= 270 \div 4.5 = 60$ minutes, or 1 hour.
2. Convert the time: $1\text{ h }20\text{ min} = 80\text{ min}$.
3. Volume $= 4.5 \times 80 = 360$ L.
4. Check: $360 \div 4.5 = 80$, which matches the time.

ANSWER **(a) 60 minutes (b) 360 L.**

COMMON ERROR

Using incompatible units. A speed of 18 km/h cannot be multiplied by 45 minutes directly; convert 45 minutes to 0.75 hours or the speed to 0.3 km/min first. The same care applies to density: match grams with cm^3, or kilograms with m^3.

PRACTICE 3.3 · SPEED, DENSITY AND FLOW

SUPPORT 1–4 · CORE 1–8 · EXTENSION 9–10

Write the relationship you use, substitute with consistent units, and state the unit of the answer.

1. A car travels 150 km in 2 hours. Find its average speed and the distance it covers in 3.5 hours at that speed. F
2. A runner covers 240 m in 40 seconds. Find the speed in m/s and in km/h. F
3. A train travels at 72 km/h for 45 minutes. Find the distance travelled. F
4. A cyclist rides at 5 m/s for 2 minutes. Find the distance travelled in metres. F
5. A plane flies 1,800 km in 3 hours. Find its average speed and the time for a 2,400 km flight at the same speed. F
6. A block has mass 1.2 kg and volume 150 cm^3. Find its density in g/cm^3 and the mass of an 80 cm^3 piece of the same material. P
7. A liquid has density 0.9 g/cm^3. Find the mass of 250 cm^3 of the liquid. P
8. A tap flows at 3.6 L/min. Find the time to fill a 90 L container and the volume delivered in 12 minutes. P
9. Convert 90 km/h to m/s and 15 m/s to km/h. Show the conversion factor you use. P
10. A student calculates $300 \div 4 = 75$ km/h, then says it takes 75 minutes to travel 100 km. Explain the error and give the correct time. E

ATL · CRITICAL THINKING

Sense-check the size of an answer. At 90 km/h, a car covers about 45 km in half an hour. If a calculation gives 4.5 km or 450 km, the units or the division need checking. Use one benchmark you know to judge whether a rate result is plausible.

3.4 Investigation · The unit-price trap

CRITERION B · I, II, III ◆ CRITERION C · II, III ◆ EQUIVALENCE

A school food drive needs 30 kg of rice. A local shop sells the same rice in three package sizes. The labels invite a simple assumption — the largest bag must be the best value — but a unit-price calculation can overturn it.

PLAN THE INQUIRY

Predict which package you expect to be cheapest per kilogram, and note one reason. **Calculate** all three unit prices before comparing. **Evaluate** whether the cheapest package is automatically the best choice for a food drive that needs exactly 30 kg.

Which package should the food drive buy?

INVESTIGATION

Package	Price	Unit price
1 kg	$2.40	?
5 kg	$10.50	?
10 kg	$21.50	?

1. Calculate the unit price of each package in dollars per kilogram.
2. Which package is cheapest per kilogram? Justify with the three unit prices.
3. Using the cheapest package, find the cost of 30 kg of rice.
4. Is the 10 kg package ever the better choice at these prices? Explain, using the unit prices and one practical consideration.
5. The shop reduces the 10 kg package by 15%. Find the new price, the new unit price, and whether it becomes the cheapest option.
6. Write a general rule for comparing the value of packages. Your rule should mention units, division and the effect of an offer.
7. Give one reason a family might choose a package with a higher unit price. Explain how this reason can coexist with a correct mathematical comparison.

Real-life application · Water for the garden

CRITERION D

A school garden has a rectangular tank measuring 1.2 m by 0.8 m by 0.9 m. A tap supplies water at 12 L/min. A hose can draw water from the tank at 18 L/min while the tap is running.

1. Find the capacity of the tank in litres. Use $1\ \text{m}^3 = 1000\ \text{L}$.
2. How long does the tap take to fill the empty tank when the hose is not being used?
3. Explain why the tank will not fill if the hose is used at 18 L/min while the tap supplies 12 L/min.
4. What is the smallest inlet rate that would allow the tank to fill while the hose is running at 18 L/min? Justify your answer.
5. With an inlet rate of 24 L/min and the hose running at 18 L/min, how long does the tank take to fill from empty?
6. Water costs \$2.10 per 1000 L. Find the cost of filling the tank, and state one assumption behind your answer.

Criterion task · The breakfast club contract

ASSESSED CRITERIA
A I–III · C I–IV · D I–V

A school breakfast club needs 40 L of milk each week. Two suppliers offer delivery.

Supplier A charges a fixed delivery fee of $18 plus $2.10 per litre. **Supplier B** charges $6 delivery plus $2.40 per litre. The club wants a contract that is fair, reliable and affordable.

Milk ordered (L)	Supplier A cost ($)	Supplier B cost ($)
20	?	?
40	?	?
60	?	?

1. Copy and complete the table.
2. At 40 L per week, what does each supplier charge?
3. Calculate the cost per litre at 40 L for each supplier.
4. At 80 L per week, which supplier is cheaper? Show both total costs.
5. Find the quantity at which the two suppliers charge the same amount. Show your method.
6. Write a recommendation for the breakfast club. Include the break-even quantity, one non-price factor and one limitation of your comparison.

FORMAT COMPLETED TABLE AND WRITTEN RECOMMENDATION

TIME 45–60 MINUTES

RESOURCES CALCULATOR PERMITTED

PART	WHAT TO SHOW	STRANDS
1–3	Correct costs, units and unit prices; evidence that the fixed fee is included.	A i–iii
4–5	A correct comparison at 80 L and a valid break-even method.	C ii–iv, D i–iii
6	A clear recommendation that uses the break-even point, a non-price factor and a limitation.	C i–iv, D iv–v

PLAN BEFORE YOU WRITE

Write both cost rules in words before substituting. Check the 40 L case against your table, then use it to explain why the cheaper supplier depends on the quantity ordered.

3.5 Summary, revision and reflection

KEY IDEAS

- A rate compares two quantities with different units; a unit rate gives the amount per one unit.
- To find a unit rate, divide the first quantity by the second and keep the units.
- Unit prices make packages of different sizes comparable when the quantities are in the same unit.
- Speed, density and flow are rates with defining relationships that can be rearranged.
- Convert units before calculating; $1\ \text{m/s} = 3.6\ \text{km/h}$.
- A fixed fee makes a unit rate change with quantity, so the relationship is linear but not proportional.

Unit rate: amount ÷ number of units.

Speed: $s = \frac{d}{t}$, $d = st$, $t = \frac{d}{s}$.

Density: $D = \frac{m}{V}$, so $m = DV$.

Flow rate: $Q = \frac{V}{t}$, so $V = Qt$.

Conversion: $1\ \text{m/s} = 3.6\ \text{km/h}$.

CONNECTIONS

Unit rates are proportional relationships with labelled units. They prepare the linear equations of Chapter 5, the graphs of Chapter 6 and the compound measures used in geometry and science.

REVISION CHECKLIST

- ☐ I can write a rate and reduce it to a unit rate.
- ☐ I can compare two offers using unit prices.
- ☐ I can convert a rate to a different unit.
- ☐ I can solve a speed problem in either direction.
- ☐ I can use density to find a mass or a volume.
- ☐ I can solve a flow-rate problem with consistent units.
- ☐ I can explain why a larger package may not be better value.
- ☐ I can evaluate a rate-based recommendation.

REFLECTION

Which unit rate in this chapter surprised you most, and what made it surprising?

When could a mathematically correct best buy still be a poor decision?

KEY VOCABULARY

rate
a comparison of quantities with different units

unit rate
the amount per one unit

unit price
the cost per unit of quantity

speed
distance travelled per unit of time

density
mass per unit of volume

flow rate
volume delivered per unit of time

CHAPTER 3 · MIXED PRACTICE

Consolidate and extend

These questions combine the chapter's ideas. They are not a formal assessment; use them to find what is secure and what still needs attention.

FLUENCY Accurate and efficient

1. Write 180 km in 3 hours as a unit rate.
2. Write \$12.60 for 4 kg as a unit rate.
3. A tap fills 75 L in 5 minutes. Find the rate in litres per minute.
4. Convert 54 km/h to m/s.
5. Convert 8 m/s to km/h.
6. Find the cost of 7 kg at \$2.80 per kilogram.
7. A car travels at 65 km/h for 4 hours. Find the distance.
8. Find the density of 450 g occupying 50 cm^3.
9. A pipe flows at 2.5 L/min. Find the volume delivered in 8 minutes.
10. Which is better value: 3 kg for \$7.20 or 5 kg for \$11.50? Show unit prices.

REASONING AND PROBLEM SOLVING Connect and justify

11. A 1.5 L bottle costs \$2.70; a 2 L bottle costs \$3.40. Which is better value per litre? R
12. A recipe for 12 servings uses 900 g of rice. How much is needed for 20 servings? P
13. A train travels 240 km in 1 hour 36 minutes. Find its average speed. P
14. A runner's pace is 5 minutes per kilometre. How long does 7.5 km take? P
15. A 2.4 kg chicken costs \$15.36. Find the price per kilogram and the cost of 1.5 kg. P
16. A student says 20 m/s is slower than 70 km/h because $20 < 70$. Convert one rate and decide. R
17. A 750 mL bottle of juice costs \$3.75; a 1.5 L bottle costs \$6.90. Find both unit prices and the better buy. P
18. A car uses 6.5 L of fuel per 100 km. How much fuel is needed for 320 km? P
19. A 5 kg bag costs \$12.50; a 2 kg bag costs \$5.20. Which is better per kilogram, and how much is saved on 10 kg? P
20. A school buys 48 L of milk in 2 L bottles costing \$2.10 each. Find the total cost and the cost per litre. P

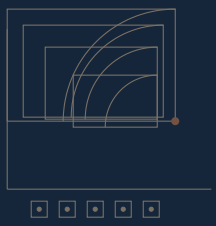
EXTENSION AND INQUIRY Generalise and evaluate

21. A car travels 150 km at 60 km/h and returns along the same road at 40 km/h. Find the total time and the average speed for the whole journey. E
22. Two taps fill a 900 L tank. Tap A delivers 12 L/min and tap B delivers 18 L/min. Find the time when both are open. Then find the time if an outlet drains 6 L/min at the same time. E
23. A shop sells 250 g of coffee for \$8.50 or 1 kg for \$30.00. Which is better value, and what would 2.5 kg cost at the better unit price? E
24. In a 5 km fun run, one runner averages 8 km/h and another averages 12 km/h. How long does each take, and how long after the faster runner finishes does the slower runner finish? E

25. Design a best-buy poster for two sizes of the same product. Include the unit prices, the better buy, and one reason a buyer might choose the other size. E

ATL · REFLECTION

Look back at the inquiry. A unit rate makes a comparison fair only when both offers use the same unit and include the same costs. Identify one comparison in this chapter where a hidden cost or a different unit could change the conclusion.



4

NUMBER AND PROPORTION

Integers and operations

Negative numbers extend the number line in two directions. With one consistent set of sign rules, we can calculate position, change and reversal without losing the meaning of zero.

KEY CONCEPT · FORM

RELATED · QUANTITY

RELATED · EQUIVALENCE

CONTEXT · ORIENTATION IN SPACE AND TIME

INQUIRY

How can a single set of sign rules keep calculations consistent in both directions of the number line?

MYP UNIT PLANNER

Framing the inquiry

UNIT 4 · APPROX. 10 HOURS
ATL FOCUS · TRANSFER

KEY AND RELATED CONCEPTS

Form. An integer is written with a sign and a magnitude. Its position on the number line fixes its order, and its distance from zero fixes its absolute value.

Quantity. Negative integers describe directed quantities: temperatures below zero, positions below sea level, debts and losses. The sign carries meaning that the size alone cannot.

Equivalence. Subtracting an integer gives the same result as adding its opposite, so $a - b = a + (-b)$. An integer and its opposite are equal distances from zero and sum to zero.

GLOBAL CONTEXT AND EXPLORATION

Orientation in space and time.
Exploration: direction, position and scale on a line, including places below sea level and moments before a chosen zero point.

Altitudes around a mountain, depths in a cave and temperatures through a night all use a chosen zero as a reference. Integers let us record how far above or below that reference a value lies, and operations let us calculate the effect of a change.

STATEMENT OF INQUIRY

Directed quantities are represented by integers, whose order and operations follow consistent rules that allow position, change and reversal to be calculated and communicated.

FACTUAL

What is an integer, and how is it positioned and ordered on the number line?

CONCEPTUAL

How do the sign rules for the four operations preserve consistency on the number line?

DEBATABLE

When does a negative value communicate a situation more clearly than a positive one?

ATL · TRANSFER IN ACTION

Choose three contexts where negative numbers appear: temperature, altitude and money, for example. For each, write one statement using an inequality, one addition and one subtraction, then explain what zero means in that context. Compare how the same calculation carries a different meaning in each setting.

INQUIRY PATHWAY FOR THIS CHAPTER

1	2	3	4	5	6
CONTEXT	**QUESTION**	**CONCEPT**	**INVESTIGATION**	**COMMUNICATION**	**REFLECTION**
Mountain heights, sea depths and a cold mountain station.	How can sign rules keep calculations consistent?	Integer, opposite, absolute value and directed change.	Why must a negative times a negative be positive?	Number lines, inequalities, signed statements and general rules.	When is zero a meaningful boundary rather than an absence?

Learning intentions and success criteria

YOU ARE LEARNING TO

- **Understand** integers as directed quantities on an ordered number line.
- **Know** the meaning of opposite and absolute value.
- **Be able to** add and subtract integers using number-line movement and sign rules.
- **Be able to** multiply and divide integers, including powers.
- **Be able to** evaluate expressions using the order of operations.
- **Be able to** apply integer operations to position, temperature and finance problems.

SUCCESS CRITERIA

- ☐ I can order integers and explain the order using the number line.
- ☐ I can find opposites and absolute values, and use them to compare depths or changes.
- ☐ I can add and subtract integers without losing the sign.
- ☐ I can multiply and divide integers and justify the sign of the answer.
- ☐ I can evaluate expressions with brackets, powers and several operations.
- ☐ I can interpret a negative answer in the context of the problem.

PREREQUISITE CHECK · 10 MINUTES, NO CALCULATOR

1. Calculate $47 + 28$.
2. Calculate $90 - 64$.
3. Calculate 7×8.
4. Calculate $72 \div 9$.
5. Evaluate 2^4.
6. Evaluate $5 + 3 \times 4$.

Answers are in the Chapter 4 answer section. If more than two are incorrect, review whole-number operations, powers and the order of operations before continuing.

YEAR 1
Number line to and below zero; negative numbers in temperature and money; simple comparisons.

YEAR 2 · THIS CHAPTER
Order, absolute value, all four operations with integers, powers and order of operations.

YEAR 3
Rational and irrational numbers; exponent laws; algebraic manipulation with signed terms.

CRITERION	STRANDS	EVIDENCE IN THIS CHAPTER
A · Knowing and understanding	i, ii, iii	Order integers; use opposites and absolute value; add, subtract, multiply and divide with signs; evaluate expressions.
B · Investigating patterns	i, ii, iii	Investigate multiplication patterns; generalise the sign rules; justify why the rules must hold.
C · Communicating	i, ii, iii, iv	Use number lines, inequalities, signed statements and clear explanations of each step.
D · Applying mathematics	i, ii, iii, iv, v	Cave research station: model elevations, temperature changes and battery use; evaluate a plan.

4.1 Integers, order and absolute value

CRITERION A · I, II ◆ CRITERION C · II ◆ QUANTITY

Some quantities need a direction as well as a size. A temperature of 7°C below zero, a shore 430 m below sea level and a debt of \$45 are all naturally recorded with negative numbers. The number line gives these values a place and an order.

DEFINITION · INTEGERS

Integers are the whole numbers together with their negatives: $\ldots, -3, -2, -1, 0, 1, 2, 3, \ldots$. The set of integers is written $\mathbb{Z}$. Positive integers are greater than zero, negative integers are less than zero, and zero is neither positive nor negative. Every integer has an **opposite**, the same distance from zero on the other side: the opposite of 7 is -7, and the opposite of -7 is 7.

On a horizontal number line, values increase from left to right. Every negative integer lies to the left of zero, so every negative integer is less than every positive integer. Among negative integers, the one farther from zero is smaller: -9 lies to the left of -2, so $-9 < -2$.



Figure 4.1 The integers -5 and 5 are opposites: each is 5 units from zero, on opposite sides. Their sum is zero.

The absolute value of an integer is its distance from zero on the number line. Distance is never negative, so absolute value is never negative. The absolute value of x is written $|x|$. Thus $|6| = 6$, $|-6| = 6$ and $|0| = 0$. Opposites always have the same absolute value but different signs.

TABLE 4.1 · REFERENCE POINTS ABOVE AND BELOW SEA LEVEL

Place or feature	Elevation (m)	Meaning
Mount Everest summit	+8849	8849 m above sea level
Sea level	0	The chosen reference height
Badwater Basin, Death Valley	−86	86 m below sea level
Dead Sea shore	−430	430 m below sea level

Table 4.1 shows why the sign alone does not tell you which value is larger in size. The Dead Sea shore, at -430 m, has a greater absolute value than Badwater Basin, at -86 m, but it is **lower**. On the number line, $-430 < -86$. Absolute value measures distance from zero; an inequality compares position.

WORKED EXAMPLE 4.1.1 — ORDERING INTEGERS

PROBLEM

Order -7, 4, -2, 0, 6 and -5 from least to greatest. Then write an inequality that places the two negative values correctly.

1. Place the values on a number line: the negative values lie left of zero, the positive values right of zero.
2. The negative values in order are -7, -5, -2.
3. Then come 0, 4 and 6. So the order is $-7, -5, -2, 0, 4, 6$.
4. Check the negatives: -7 is farthest left, so $-7 < -5 < -2 < 0$.

ANSWER $-7 < -5 < -2 < 0 < 4 < 6$. **The farthest negative is the least.**

WORKED EXAMPLE 4.1.2 — ABSOLUTE VALUE

PROBLEM

Evaluate (a) $|-9|$, (b) $|6|$, (c) $-|-4|$, and (d) find all integers x with $|x| = 5$.

1. $|-9| = 9$: the distance from -9 to 0 is 9 units.
2. $|6| = 6$: the distance from 6 to 0 is 6 units.
3. $|-4| = 4$, so $-|-4| = -4$. The minus sign outside the absolute value is applied after evaluating it.
4. $|x| = 5$ means x is 5 units from zero: $x = 5$ or $x = -5$.

ANSWER **(a) 9 (b) 6 (c)** -4 **(d)** $x = 5$ **or** $x = -5$.

WORKED EXAMPLE 4.1.3 — DEPTH AND DISTANCE

PROBLEM

A research submarine is at a depth of -180 m relative to the surface. A diver is at -35 m. (a) Which is deeper? (b) How much deeper is the submarine? (c) Write an inequality using absolute values that supports your answer.

1. Both positions are below the surface, and $-180 < -35$, so the submarine is deeper.
2. The submarine is 180 m below the surface and the diver is 35 m below, so the difference in depth is $180 - 35 = 145$ m.
3. $|-180| = 180$ and $|-35| = 35$, so $|-180| > |-35|$. The larger absolute value corresponds to the deeper position because both values are negative.

ANSWER **(a) the submarine (b) 145 m (c)** $|-180| > |-35|$, **and** $-180 < -35$.

COMMON ERROR

Reading $-8 > -3$ because $8 > 3$. The sign changes the position, not just the size. On the number line -8 is farther left than -3, so $-8 < -3$. Compare position first; use absolute value only when the question asks about distance from zero.

PRACTICE 4.1 · ORDER AND ABSOLUTE VALUE — SUPPORT 1–4 · CORE 1–9 · EXTENSION 10–12

Use a number line when a comparison is not immediate, and state whether a question asks about position or distance.

1. Write an integer for each situation. (a) 5°C below zero (b) 120 m below sea level (c) a gain of \$35 (d) 12 m above sea level (e) a loss of 7 points [F]
2. Order -6, 2, -1, 5, 0 and -4 from least to greatest. [F]
3. Insert $<$ or $>$ to make each statement true. (a) $-4 \;?\; 1$ (b) $-9 \;?\; -2$ (c) $0 \;?\; -3$ (d) $-7 \;?\; -12$ [F]
4. Find the opposite and the absolute value of each integer. (a) -11 (b) 8 (c) 0 (d) -4 [F]
5. Evaluate (a) $|-15|$ (b) $|7|$ (c) $-|-6|$ (d) $|0|$. [F]
6. List all integers whose absolute value is 9. [F]
7. A bank account balance is $-\$45$. Explain what the negative sign means, then state the absolute value and what it represents. [P]
8. The temperature is -7°C at 6 am and 4°C at 2 pm. Which temperature is warmer? Find the difference in degrees. [P]
9. A diver is at -18 m and rises to -5 m. How many metres does the diver ascend? [P]
10. A student says, "-12 is greater than -5 because 12 is greater than 5." Identify the error and correct the comparison. [E]
11. Two integers have absolute value 6. Write both integers, compare them, and find their sum. Explain how the number line shows the result. [E]
12. A weather station records temperatures of -3°C, 0°C, 2°C and -8°C. Order the readings, find the difference between the highest and lowest, and explain what 0°C represents in this context. [E]

4.2 Adding and subtracting integers

CRITERION A · I, II ◆ CRITERION C · III ◆ EQUIVALENCE

A change on the number line is a movement. Adding a positive integer moves right; adding a negative integer moves left. Subtraction asks which movement would undo a given change, which is why subtracting a negative has the same effect as adding a positive.

> **KEY IDEA · MOVEMENT AND OPPOSITES**
>
> Adding a positive integer moves right. Adding a negative integer moves left. Subtracting an integer is the same as adding its opposite: $a - b = a + (-b)$, and in particular $a - (-b) = a + b$. A number and its opposite form a **zero pair**: together they make no net change.

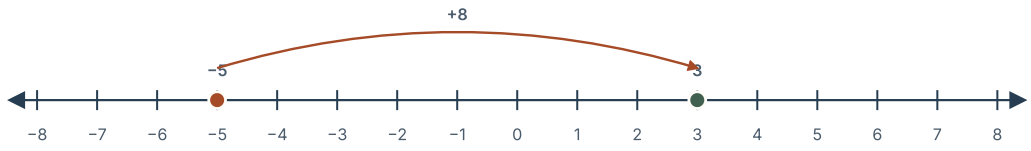


Figure 4.2 Starting at -5 and adding 8 moves 8 units right, ending at 3. The arrow shows both the direction and the size of the change.

METHOD · ADDING AND SUBTRACTING INTEGERS

Same signs: add the absolute values and keep the common sign. For example, $-7 + (-4) = -(7 + 4) = -11$.
Different signs: subtract the smaller absolute value from the larger, and keep the sign of the integer with the larger absolute value. For example, $-9 + 4 = -(9 - 4) = -5$.
Subtraction: rewrite $a - b$ as $a + (-b)$, then use the addition rules. Check the result by estimating its position on the number line.

WORKED EXAMPLE 4.2.1 ADDING WITH SIGNS

PROBLEM

Calculate (a) $-5 + 8$ and (b) $6 + (-9)$. Show the movement on a number line.

1. (a) The signs are different. $8 - 5 = 3$, and 8 has the larger absolute value, so the result is positive: $-5 + 8 = 3$.
2. On the number line, start at -5 and move 8 units right to 3.
3. (b) The signs are different. $9 - 6 = 3$, and -9 has the larger absolute value, so the result is negative: $6 + (-9) = -3$.
4. On the number line, start at 6 and move 9 units left to -3.

ANSWER **(a) 3 (b) -3.**

WORKED EXAMPLE 4.2.2 SUBTRACTING INTEGERS

PROBLEM

Calculate (a) $-7 - 4$, (b) $5 - (-8)$ and (c) $-6 - (-10)$.

1. (a) $-7 - 4 = -7 + (-4)$. Same signs: $-(7 + 4) = -11$.
2. (b) $5 - (-8) = 5 + 8 = 13$. Subtracting -8 is the same as adding 8.
3. (c) $-6 - (-10) = -6 + 10 = 4$. The two negatives make a positive change of 10.
4. Check (c): starting at -6 and moving 10 units right reaches 4.

ANSWER **(a) -11 (b) 13 (c) 4.**

WORKED EXAMPLE 4.2.3 — A MULTI-STEP EXPRESSION

PROBLEM

Evaluate $-12 + 5 - (-7) - 9$.

1. Rewrite every subtraction as adding the opposite: $-12 + 5 + 7 + (-9)$.
2. Group the positive and negative terms: $(5 + 7) - (12 + 9) = 12 - 21$.
3. Subtract: $12 - 21 = -9$.
4. Check step by step: $-12 + 5 = -7$; $-7 + 7 = 0$; $0 - 9 = -9$.

ANSWER -9.

WORKED EXAMPLE 4.2.4 — A SUBMARINE'S MOVEMENT

PROBLEM

A submarine is at -180 m. It rises 65 m, then descends 42 m. (a) Find its final depth. (b) Find its net change in depth.

1. Rising 65 m is a change of $+65$. Descending 42 m is a change of -42.
2. Final position: $-180 + 65 - 42$. First $-180 + 65 = -115$, then $-115 - 42 = -157$.
3. Net change: $65 - 42 = +23$ m. The submarine ends 23 m higher than it started.
4. Check: $-180 + 23 = -157$, which matches the final depth.

ANSWER **(a)** -157 **m** **(b) a rise of 23 m.**

COMMON ERROR

Adding the magnitudes when the signs differ. $-8 + 3$ is not -11: the two changes act in opposite directions. Since 8 is larger, the result is negative, with size $8 - 3 = 5$: $-8 + 3 = -5$. A second common error is treating $4 - (-6)$ as $4 - 6$. Subtracting a negative reverses direction: $4 + 6 = 10$.

PRACTICE 4.2 · ADDING AND SUBTRACTING

SUPPORT 1–4 · CORE 1–9 · EXTENSION 10–12

Rewrite subtractions as additions of the opposite, and check each sign before calculating.

1. Calculate (a) $7 + (-3)$ (b) $-5 + 9$ (c) $-6 + (-4)$ (d) $2 + (-8)$. F
2. Calculate (a) $5 - 9$ (b) $-3 - 7$ (c) $6 - (-2)$ (d) $-4 - (-9)$. F
3. Find the missing integer. (a) $-6 + ? = 2$ (b) $? - 8 = -11$ (c) $5 - ? = -3$ F
4. Evaluate $-9 + 4 - (-6)$. F
5. A temperature of -4°C rises by 9°C, then falls by 6°C. Find the final temperature. P
6. A bank balance is \$80. A withdrawal of \$125 is made, then a deposit of \$60. Find the new balance and interpret the sign. P
7. A submarine at -250 m rises 80 m, then descends 35 m. Find its final depth and its net change. P
8. Start at -3 on a number line. Move 7 units left, then 4 units right. Find the final position. P
9. Explain why $7 - (-3) = 10$ using a number line and the idea of an opposite. E

10. A student writes $-2 - 5 = 3$. Identify the error, give the correct answer, and explain the sign of the result. E

11. Two integers have a sum of -5. One integer is 3. Find the other integer, and explain how you checked your answer. E

12. A football team gains 8 yards, loses 12 yards, gains 5 yards and loses 3 yards. Find the net change and the total distance moved. Explain why these two answers are different. E

4.3 Multiplying and dividing integers

CRITERION A · II, III ◆ CRITERION B · I ◆ FORM

Multiplication extends addition: $3 \times (-4)$ means three groups of -4. The sign pattern that follows from repeated addition, continued through zero, fixes the rule for multiplying and dividing every pair of integers.

TABLE 4.2 · CONTINUING A MULTIPLICATION PATTERN

First factor	Second factor	Product
3	−4	−12
2	−4	−8
1	−4	−4
0	−4	0
−1	−4	4
−2	−4	8

Each time the first factor decreases by 1, the product increases by 4. To keep the pattern consistent through zero, the product of two negative integers must be positive: $(-1) \times (-4) = 4$. Multiplication and division are inverse operations, so the same sign rule applies to division.

KEY IDEA · SIGN RULES FOR PRODUCTS AND QUOTIENTS

Same signs: the product or quotient is positive. $(+) \times (+) = +$, $(-) \times (-) = +$, and the same for division.
Different signs: the product or quotient is negative. $(+) \times (-) = -$, $(-) \times (+) = -$.
Zero: the product of zero and any integer is zero. Zero divided by a non-zero integer is zero; division by zero is undefined.

WORKED EXAMPLE 4.3.1 — PRODUCTS AND QUOTIENTS

PROBLEM

Calculate (a) $(-7) \times 8$, (b) $(-6) \times (-9)$, (c) $72 \div (-8)$ and (d) $-84 \div (-7)$.

1. (a) Different signs: $7 \times 8 = 56$, so the result is -56.
2. (b) Same signs: $6 \times 9 = 54$, so the result is $+54$.
3. (c) Different signs: $72 \div 8 = 9$, so the result is -9. Check: $(-8) \times (-9) = 72$.
4. (d) Same signs: $84 \div 7 = 12$, so the result is $+12$. Check: $(-7) \times 12 = -84$.

ANSWER **(a)** -56 **(b)** 54 **(c)** -9 **(d)** 12.

WORKED EXAMPLE 4.3.2 — POWERS WITH NEGATIVE BASES

PROBLEM

Evaluate (a) $(-3)^4$, (b) -3^4, (c) $(-2)^3$ and (d) $(-1)^{15}$.

1. (a) The brackets include the negative sign in the base: $(-3)^4 = (-3)(-3)(-3)(-3) = 81$. An even number of negative factors gives a positive result.
2. (b) There are no brackets, so the power applies to 3 first: $-3^4 = -(3^4) = -81$.
3. (c) $(-2)^3 = (-2)(-2)(-2) = -8$. Three negative factors give a negative result.
4. (d) 15 is odd, so $(-1)^{15} = -1$.

ANSWER **(a) 81 (b)** -81 **(c)** -8 **(d)** -1**.**

WORKED EXAMPLE 4.3.3 — ORDER OF OPERATIONS

PROBLEM

Evaluate $-4 + 3 \times (-5) - (-2)^3 \div 4$.

1. Evaluate the power: $(-2)^3 = -8$. The expression becomes $-4 + 3 \times (-5) - (-8) \div 4$.
2. Multiplication and division, left to right: $3 \times (-5) = -15$ and $(-8) \div 4 = -2$.
3. Substitute: $-4 + (-15) - (-2) = -4 - 15 + 2$.
4. Add and subtract left to right: $-19 + 2 = -17$.

ANSWER -17**.**

WORKED EXAMPLE 4.3.4 — A FREEZER'S TEMPERATURE

PROBLEM

A freezer starts at -6°C. Its temperature falls by 3°C each hour for 4 hours, then rises by 5°C. (a) Write an expression for the final temperature. (b) Find the final temperature and the overall change.

1. Falling 3°C per hour for 4 hours is a change of $4 \times (-3) = -12$°C. Rising 5°C is $+5$.
2. Final temperature: $-6 + 4 \times (-3) + 5 = -6 - 12 + 5$.
3. $-6 - 12 = -18$, then $-18 + 5 = -13$. The final temperature is -13°C.
4. Overall change: $-13 - (-6) = -13 + 6 = -7$°C, a fall of 7°C.

ANSWER **(a)** $-6 + 4 \times (-3) + 5$ **(b)** -13**°C; a fall of 7°C.**

COMMON ERROR

Ignoring the difference between -3^2 and $(-3)^2$. The first is $-(3 \times 3) = -9$; the second is $(-3) \times (-3) = 9$. Also check division signs: $-24 \div (-6) = 4$, not -4. Work from the innermost brackets outward, evaluate powers before multiplying, and rewrite subtraction of a negative as addition before calculating.

PRACTICE 4.3 · PRODUCTS, QUOTIENTS AND POWERS — SUPPORT 1–4 · CORE 1–9 · EXTENSION 10–12

State the sign first, then calculate the magnitude. Show the order in which you evaluate each part of an expression.

1. Calculate (a) $(-6) \times 7$ (b) $(-8) \times (-5)$ (c) $9 \times (-4)$ (d) $(-1) \times (-12)$. F
2. Calculate (a) $-56 \div 8$ (b) $-72 \div (-9)$ (c) $45 \div (-5)$ (d) $0 \div (-7)$. F
3. Evaluate (a) $(-2)^5$ (b) $(-4)^2$ (c) -4^2 (d) $(-1)^{20}$. F
4. Evaluate (a) $5 + (-3) \times 4$ (b) $-18 \div (-3) + 2$ (c) $(-2)^3 - 5$. F
5. Evaluate $-3 \times (-4) + 20 \div (-5)$. P
6. A temperature of 3°C falls by 2°C each hour for 7 hours. Find the final temperature. P
7. A lift moves 3 floors down per stop for 5 stops, starting at floor 8. Find its final floor. P
8. Find the missing integer. (a) $(-7) \times ? = 56$ (b) $? \div (-4) = -6$ (c) $(-3)^{?} = -27$ P
9. Explain why $(-4) \times (-3) = 12$ using a pattern or the distributive property. E
10. A student writes $(-6)^2 = -36$ and $-6^2 = 36$. Identify both errors and give the correct values. E
11. Evaluate $-2^2 + (-3)^2 \times (-1)^3$. E
12. A bank charges \$4 each month. Find the total change after 6 months. If the balance starts at \$30, find the final balance and explain how the signs are used. E

4.4 Real-life application · A mountain weather station

CRITERION D · I–V ◆ ORIENTATION IN SPACE AND TIME ◆ QUANTITY

Temperature records make signed numbers visible: a reading below zero is negative, and a change can be a rise or a fall. Use the data to calculate, compare and interpret before judging how well a simple model describes the mountain climate.

A day at the mountain station

CRITERION D

A weather station at 2,400 m records the following temperatures during one day.

Time	06:00	09:00	12:00	15:00	18:00	21:00	00:00
Temperature (°C)	−9	−5	0	3	−2	−7	−8

1. Order the seven readings from least to greatest.
2. Find the difference between the highest and lowest readings.
3. Find the change in temperature from 15:00 to 00:00, and state whether it is a rise or a fall.
4. Find the total change from 06:00 to 15:00.
5. Calculate the mean temperature for the day. Use a sum of the signed readings.
6. Explain what 0°C represents in this table, and why it does not mean “no temperature”.
7. The base of the mountain is at 1,600 m and has a temperature of 6°C. A simple model predicts a fall of 6°C per 1000 m of height. Predict the station temperature using this model, then compare it with the recorded mean. Suggest one reason for the difference.

4.5 Investigation · Why is negative times negative positive?

CRITERION B · I, II, III ◆ CRITERION C · II, III ◆ PATTERN

A rule is trustworthy only when it continues a pattern without contradiction. In this investigation you will complete a multiplication table, extend a pattern through zero, and use the distributive property to justify the sign rule rather than simply memorise it.

PLAN THE INQUIRY

Predict: what sign should $(-2) \times (-3)$ have? **Observe:** complete the table and describe how each row changes. **Justify:** use a pattern or the distributive property to show why your prediction must be correct.

The multiplication grid

INVESTIGATION

TABLE 4.3 · COMPLETE THE PRODUCTS

×	−2	−1	0	1	2
−2	?	?	?	?	?
−1	?	?	?	?	?
0	?	?	?	?	?
1	?	?	?	?	?
2	?	?	?	?	?

1. Copy and complete the grid.
2. Describe the pattern in the row for -2 as the column value increases from -2 to 2.
3. Describe the pattern in the column for -2 as the row value increases from -2 to 2.
4. The pattern $3 \times (-4) = -12$, $2 \times (-4) = -8$, $1 \times (-4) = -4$, $0 \times (-4) = 0$ continues. Use it to justify the value of $(-1) \times (-4)$.
5. Use the distributive property to show that $(-1) \times (-1) = 1$. Start with $0 = (-1) \times 0$ and write $0 = 1 + (-1)$.
6. Write a rule for multiplying two integers and a rule for dividing two integers. Test each rule with three examples that include zero.
7. Does the same-sign rule work for powers? Explain the effect of an even exponent and an odd exponent, using $(-2)^2$, $(-2)^3$ and $(-2)^4$.

Criterion task · The cave research station

ASSESSED CRITERIA
A I–III · C I–IV · D I–V

A research team is mapping a cave system. The entrance is at an elevation of +1240 m. Four underground stations lie at depths of 85 m, 160 m, 240 m and 310 m below the entrance. Temperatures have been recorded at each station.

Station	Depth below entrance (m)	Elevation (m)	Temperature (°C)
Entrance	0	+1240	11
Station 1	85	?	8
Station 2	160	?	6
Station 3	240	?	4
Station 4	310	?	3

1. Complete the elevation column by subtracting each station's depth from the entrance elevation.
2. Order the four stations from highest to lowest elevation.
3. Find the elevation difference between the entrance and Station 4, and between Station 1 and Station 3.
4. Find the temperature change from the entrance to Station 4, and from Station 2 to Station 4. State whether each change is a rise or a fall.
5. A sensor at Station 3 records −2°C, then falls by 5°C, then rises by 3°C. Find the final reading.
6. The team's battery starts at 100%. Each trip into the cave uses 12% of the charge, and the solar panel restores 7% per hour. After 3 trips and 5 hours of charging, find the charge. The team wants to make two more trips and keep at least 70% charge. Use integer arithmetic to decide whether this is possible, and write a recommendation that states one assumption and one limitation.

FORMAT COMPLETED TABLE AND WRITTEN RECOMMENDATION

TIME 45–60 MINUTES

RESOURCES CALCULATOR PERMITTED

PART	WHAT TO SHOW	STRANDS
1–2	Correct signed elevations and a valid order from highest to lowest.	A i–iii
3–5	Correct differences and temperature changes, with signs interpreted in context.	C ii–iv, D i–iii
6	A correct charge calculation, a justified decision and a recommendation with an assumption and limitation.	C i–iv, D iv–v

4.6 Summary, revision and reflection

KEY IDEAS

- Integers are ordered on the number line from left to right; absolute value measures distance from zero and is never negative.
- Adding a negative moves left; subtracting a negative moves right.
- For multiplication and division, the same signs give a positive result and different signs give a negative result.
- In a power with a negative base, an even exponent gives a positive result and an odd exponent gives a negative result.
- The order of operations is brackets, powers, multiplication and division from left to right, then addition and subtraction.
- The sign of an answer must be interpreted in context: a negative balance is a debt, a negative depth is below the reference point.

$a - b = a + (-b)$

$-(-a) = a$

$|-a| = |a|$

$(-a)(-b) = ab$

$(-a)^n > 0$ for even n

$(-a)^n < 0$ for odd n

Connections. Signed arithmetic is the grammar of algebra: these rules keep every rearrangement in Chapter 5 valid, and the number line becomes the coordinate grid used for graphs, transformations and data.

REVISION CHECKLIST

- ☐ I can place and order integers on a number line.
- ☐ I can find opposites and absolute values.
- ☐ I can add and subtract integers, including two negatives.
- ☐ I can multiply and divide integers and justify the sign.
- ☐ I can evaluate powers with negative bases.
- ☐ I can apply the order of operations to signed expressions.
- ☐ I can interpret a negative answer in context.

REFLECTION

Which sign rule did you need to check most often, and where did the sign of an answer carry the most important information?

KEY VOCABULARY

integer
a whole number or its negative

opposite
the same distance from zero, on the other side

absolute value
distance from zero, never negative

directed quantity
a quantity with both size and direction

zero pair
an integer and its opposite, whose sum is zero

magnitude
the size of a number, ignoring its sign

evaluate
find the value of an expression

base
the number raised to a power

exponent
the number of repeated factors

CHAPTER 4 · MIXED PRACTICE

Consolidate and extend

These questions mix the chapter's ideas. Routine practice is complete; use these to build fluency, connect concepts and prepare for assessment.

Fluency

Accurate and efficient

1. Order -9, 4, -2, 0, 7 and -6 from least to greatest. F
2. Insert $<$ or $>$. (a) -5 ? -11 (b) 3 ? -4 (c) -8 ? 0 F
3. Evaluate (a) $|-14|$ (b) $-|9|$ (c) $|0|$. F
4. Calculate (a) $-8+13$ (b) $-6+(-9)$ (c) $5-12$ (d) $-3-(-11)$. F
5. Calculate (a) $(-7)\times 6$ (b) $(-8)\times(-4)$ (c) $-63\div 9$ (d) $-45\div(-5)$. F
6. Evaluate (a) $(-3)^3$ (b) $(-2)^4$ (c) -2^4. F
7. Evaluate $8-3\times(-2)$. F
8. Evaluate $-20\div(-4)+7$. F
9. Evaluate $(-4)^2-3\times(-5)$. F
10. Evaluate $-6+2\times(-3)-(-8)\div 4$. F
11. A temperature starts at -5°C, falls 4°C, then rises 7°C. Find the final temperature. P
12. A bank balance is \$60. A withdrawal of \$95 is followed by a deposit of \$40. Find the final balance. P

Reasoning and problem solving

Connect and justify

13. A submarine is at -320 m. It rises 75 m, then descends 40 m. Find its final depth and net change. P
14. A lift starts at floor 6, goes down 9 floors, then up 4 floors. Find its final floor. P
15. Which is colder, -12°C or -5°C? Find the difference in temperature. P
16. A student writes $-7+2=-9$. Identify the error, give the correct answer, and explain how a number line shows the result. E
17. A student writes $(-5)^2=-25$ and $-5^2=25$. Identify both errors and give the correct values. E
18. A diver at -24 m ascends at 3 m per minute for 6 minutes. Find the final depth. E
19. A freezer at -18°C warms by 4°C per hour for 5 hours. Find the final temperature and the overall change. E
20. A hiker at an elevation of $+850$ m descends 1,200 m. Find the final elevation and explain what the sign means. E

Extension and inquiry

Generalise and evaluate

21. The pattern is $4\times(-3)=-12$, $3\times(-3)=-9$, $2\times(-3)=-6$. Continue the pattern for two more terms and write a general rule for multiplying a positive integer by a negative integer. E

22. Explain why $-(-7) = 7$, using the number line and the idea of an opposite. E

23. Find two integers whose product is -24 and whose sum is 2. Show how you searched systematically. E

24. Evaluate $-3^2 + (-2)^3 - (-4) \times 2$. Show each step in the correct order. E

25. A lift starts at floor -2, goes up 7 floors, down 3 floors and up 5 floors. Find its final floor and the total number of floors travelled. Explain why these two numbers are different. E

ATL · TRANSFER

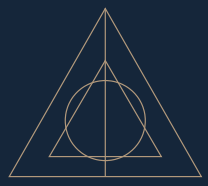
Choose one calculation from this chapter that produced a negative answer. Rewrite the same calculation as a statement about temperature, then as a statement about money, then as a statement about elevation. Explain what stays the same mathematically and what changes in meaning.

PART II · SYMBOLIC GENERALISATION

II

Algebra and pattern

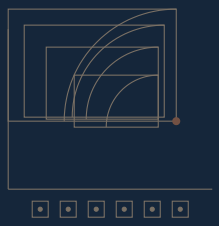
Algebra begins when we stop describing one case and start describing all cases of the same kind. In this part, letters become quantities that vary, expressions become compact descriptions of relationships, and patterns become rules that can be tested, used and justified.



05 Algebraic expressions and equations

06 Numerical patterns and generalisation

5



ALGEBRA AND PATTERN

Algebraic expressions and equations

Letters turn a calculation into a relationship. An expression describes how a quantity is built; an equation asks when two descriptions agree. Equivalent forms let us simplify without changing what is true.

KEY CONCEPT · RELATIONSHIPS

RELATED · PATTERN

RELATED · EQUIVALENCE

CONTEXT · SCIENTIFIC AND TECHNICAL INNOVATION

INQUIRY

How can symbols describe a relationship so that any input can be calculated, compared and reversed?

MYP UNIT PLANNER

Framing the inquiry

UNIT 5 · APPROX. 11 HOURS
ATL FOCUS · CRITICAL THINKING

KEY AND RELATED CONCEPTS

Relationships. An expression links an input variable to an output: change h and the cost $12 + 8h$ changes with it. An equation states the input for which two relationships give the same output.

Pattern. Algebraic notation records structure: a coefficient counts equal variable parts, and a constant marks a fixed part of the relationship.

Equivalence. Simplifying, expanding and factorising change an expression's appearance, not its value; solving preserves equality by treating both sides alike.

GLOBAL CONTEXT AND EXPLORATION

Scientific and technical innovation. Exploration: systems, models and methods. Cost rules, formulas and spreadsheets are models built from variables and operations.

Repair plans, phone contracts and T-shirt orders can each be described by an expression; choosing between them requires equivalent forms or a solved equation.

STATEMENT OF INQUIRY

Algebraic symbols describe relationships between quantities; equivalent expressions and balanced equations allow those relationships to be simplified, evaluated and reversed.

FACTUAL

What are variables, terms, coefficients and constants, and how do we substitute a value into an expression?

CONCEPTUAL

How can different expressions represent the same relationship, and what does it mean to solve an equation?

DEBATABLE

When is a simple formula more useful than a more exact but complicated one?

ATL · CRITICAL THINKING IN ACTION

Do not accept an algebraic claim because it works once. Test zero, a negative number and a fraction. To test equivalence, one match is only evidence while a counterexample is decisive; to test a solution, substitute it into the original equation.

INQUIRY PATHWAY FOR THIS CHAPTER

1	2	3	4	5	6
CONTEXT	**QUESTION**	**CONCEPT**	**INVESTIGATION**	**COMMUNICATION**	**REFLECTION**
A repair workshop pricing its hire by a fixed fee plus an hourly rate.	How can one expression describe every possible bill?	Variable, term, coefficient, equivalence and solution.	Why does a number trick give the same result every time?	Words, tables, graphs, expressions and balanced steps.	Which form of a relationship reveals the most useful information?

Learning intentions and success criteria

YOU ARE LEARNING TO

- **Understand** variables as quantities that can change, and read algebraic notation correctly.
- **Know** the meanings of term, coefficient, constant and like term.
- **Be able to** write and interpret expressions from words and contexts.
- **Be able to** substitute values, including negative numbers, and evaluate expressions.
- **Be able to** simplify, expand and factorise linear expressions.
- **Be able to** solve linear equations and check the solution.

SUCCESS CRITERIA

- ☐ I can explain what each part of an expression represents.
- ☐ I can translate a situation or a sentence into an expression.
- ☐ I can substitute a value using brackets and the order of operations.
- ☐ I can combine like terms and expand or factorise correctly.
- ☐ I can solve one- and two-step equations, including brackets and variables on both sides.
- ☐ I can substitute a solution into the original equation to verify it.

PREREQUISITE CHECK · 10 MINUTES, NO CALCULATOR

1. Calculate 8×7.
2. Calculate $52 - 19$.
3. Calculate $96 \div 6$.
4. Evaluate $4 + 3 \times 5$.
5. Evaluate 2^4.
6. Solve $x + 9 = 15$.

Answers are in the Chapter 5 answer section. If more than two are incorrect, review arithmetic, the order of operations and one-step equations before continuing.

YEAR 1
Letters as unknowns; simple one-step equations; substitution with positive whole numbers.

YEAR 2 · THIS CHAPTER
Expressions, like terms, expanding and factorising, two-step and bracket equations, variables on both sides.

YEAR 3
Rearranging formulas, equations with fractions, inequalities and simultaneous equations.

CRITERION	STRANDS	EVIDENCE IN THIS CHAPTER
A · Knowing and understanding	i, ii, iii	Interpret notation; substitute and evaluate; simplify, expand and factorise; solve linear equations.
B · Investigating patterns	i, ii, iii	Investigate equivalent forms, the number trick and the balance model; justify each general step.
C · Communicating	i, ii, iii, iv	Write expressions from contexts; present balanced steps; define variables and check solutions.
D · Applying mathematics	i, ii, iii, iv, v	Repair plans and a T-shirt order: build models, compare costs, interpret break-even points and evaluate assumptions.

5.1 The language of algebra

CRITERION A · I, II ◆ CRITERION C · I, II ◆ RELATIONSHIPS

A repair workshop rents a bike stand for a fixed booking fee of \$12 plus \$8 for each hour. If the stand is hired for h hours, the cost in dollars is $12 + 8h$. One short expression now describes every possible bill, from one hour to a full day.

TABLE 5.1 · HIRE COST FOR DIFFERENT NUMBERS OF HOURS

Hours, h	0	1	2	3	4
Cost, $12 + 8h$ (\$)	12	20	28	36	44

DEFINITION · THE PARTS OF AN EXPRESSION

A **variable** is a letter or symbol representing a number that can change or is unknown. A **constant** is a value that does not change. A **term** is a part of an expression separated by $+$ or $-$. The **coefficient** is the number multiplying the variable part of a term. An **expression** combines variables, constants and operations without an equals sign; an **equation** states that two expressions are equal.

In $5x^2 - 3x + 7$, the terms are $5x^2$, $-3x$ and 7; the coefficients are 5 and -3; the constant term is 7; the variable is x.

WORKED EXAMPLE 5.1.1 IDENTIFYING STRUCTURE

PROBLEM

For each expression, list the terms, the coefficients and the constant term: (a) $6x + 11$, (b) $4y^2 - 9y + 2$, (c) $15 - 3d$.

1. (a) The terms are $6x$ and 11. The coefficient of x is 6; the constant term is 11.
2. (b) The terms are $4y^2$, $-9y$ and 2. The coefficients are 4 and -9; the constant term is 2. The negative sign belongs to the term $-9y$.
3. (c) The terms are 15 and $-3d$. The coefficient of d is -3; the constant term is 15. Writing the expression as $15 + (-3d)$ makes the signs explicit.

ANSWER **(a)** $6x, 11$**; coefficients** 6**; constant** 11**. (b)** $4y^2, -9y, 2$**; coefficients** $4, -9$**; constant** 2**. (c)** $15, -3d$**; coefficient** -3**; constant** 15**.**

NOTATION · READING SYMBOLS PRECISELY

$3x$ means $3 \times x$, not $3 + x$. ab means $a \times b$. x^2 means $x \times x$, while $2x$ means $x + x$. $\frac{x}{4}$ means $x \div 4$. Brackets group a whole expression: $3(x + 2)$ means $3 \times (x + 2)$. A coefficient of 1 is not written, so $x = 1x$.

WORKED EXAMPLE 5.1.2 — FROM WORDS TO SYMBOLS

PROBLEM

Write an expression for each statement. (a) Seven more than twice a number n. (b) Five less than three times a number x. (c) The cost of h hours at \$9 per hour plus a \$15 call-out fee. (d) The perimeter of a rectangle with length $x + 4$ and width x.

1. (a) "Twice n" is $2n$; "seven more" adds 7: $2n + 7$.
2. (b) "Three times x" is $3x$; "five less than" subtracts 5 from that result: $3x - 5$. The order matters: $5 - 3x$ would mean five minus three times x.
3. (c) The hourly charge is $9h$; adding the fixed fee gives $9h + 15$.
4. (d) Two lengths and two widths give $2(x + 4) + 2x$. This can be simplified later; the unsimplified form shows how the perimeter is built.

ANSWER **(a)** $2n + 7$ **(b)** $3x - 5$ **(c)** $9h + 15$ **(d)** $2(x + 4) + 2x$.

WORKED EXAMPLE 5.1.3 — SUBSTITUTION WITH NEGATIVE VALUES

PROBLEM

Evaluate $3x^2 - 2x + 1$ when (a) $x = 4$ and (b) $x = -2$.

1. (a) Substitute 4 for x: $3(4)^2 - 2(4) + 1$.
2. Evaluate the power first: $3(16) - 8 + 1 = 48 - 8 + 1 = 41$.
3. (b) Substitute -2 inside brackets: $3(-2)^2 - 2(-2) + 1$.
4. Evaluate the power: $(-2)^2 = 4$. Then $3(4) + 4 + 1 = 12 + 4 + 1 = 17$. The product $-2(-2)$ is positive.

ANSWER **(a) 41 (b) 17.**

WORKED EXAMPLE 5.1.4 — EVALUATING A FORMULA

PROBLEM

A delivery company uses the formula $C = 25 + 0.4m$, where C is the cost in dollars and m is the distance in kilometres. (a) Find the cost for 180 km. (b) Find C when $m = 0$ and explain its meaning. (c) What does the 0.4 represent?

1. (a) Substitute $m = 180$: $C = 25 + 0.4(180) = 25 + 72 = 97$.
2. (b) When $m = 0$, $C = 25 + 0 = 25$. This is the fixed charge before any distance is travelled.
3. (c) Each additional kilometre adds 0.4 to the cost, so 0.4 is the charge in dollars per kilometre. The coefficient of m is the rate.

ANSWER **(a) \$97 (b) \$25, the fixed charge (c) \$0.40 per kilometre.**

GUIDED EXAMPLE · A CAFÉ DELIVERY

Problem. A café charges \$3 for each smoothie plus a \$5 delivery fee. Write an expression for the cost of n smoothies and evaluate it for $n = 4$.

Step 1. The variable is n, the number of smoothies. The variable cost is $3n$.

Step 2. Add the fixed delivery fee: the expression is $3n + 5$.

Now complete. Substitute $n = 4$: $3(4) + 5 = ?$. Then find the cost of 10 smoothies, and state what the 5 represents. Compare your reasoning with the answer section.

COMMON ERROR

Reading $3x$ as $3 + x$, or x^2 as $2x$. At $x = 5$, $3x = 15$ but $3 + x = 8$; $x^2 = 25$ but $2x = 10$. When substituting a negative value, always use brackets: $3x$ at $x = -4$ is $3(-4) = -12$, not $3 - 4$. Also check the direction of "less than": five less than x is $x - 5$, not $5 - x$.

PRACTICE 5.1 · NOTATION, EXPRESSIONS AND SUBSTITUTION

SUPPORT 1–5 · CORE 1–9 · EXTENSION 10–12

Write expressions before substituting. Show the substitution in brackets, then evaluate in the order of operations.

1. For each expression, list the terms, the coefficients and the constant term: (a) $6x + 11$ (b) $4y^2 - 9y + 2$ (c) $15 - 3d$. F
2. Write an expression for: (a) seven more than n (b) four times the sum of x and 3 (c) ten less than twice m (d) the quotient of p and 5, increased by 2. F
3. Evaluate $5x + 3$ when (a) $x = 6$ (b) $x = 0.5$ (c) $x = -4$. F
4. Evaluate $2a^2 - 3a$ when (a) $a = 3$ (b) $a = -2$. F
5. Evaluate $4(b + 1) - c$ when $b = 5$ and $c = 7$. F
6. Which expression means "five less than three times x"? A $5 - 3x$ B $3x - 5$ C $3(x - 5)$ D $5x - 3$. P
7. A plumber charges a \$45 call-out fee plus \$30 per hour. (a) Write an expression for the cost of h hours. (b) Find the cost for 3.5 hours. P
8. A rectangle has length $2x + 1$ and width x. Write an expression for its perimeter. P
9. A student evaluates $4x$ at $x = -3$ as 7 and x^2 at $x = -3$ as -9. Identify both errors and give the correct values. R
10. Explain why $2n$ and $n + 2$ are not the same expression. Use $n = 5$ to demonstrate the difference. R
11. The cost of hiring a hall is \$150 plus \$12 per person. (a) Write an expression for n people. (b) Find the cost for 40 people. (c) Explain what the 150 represents. P
12. The expressions $3x + 1$, $3(x + 1)$ and $3x + 3$ are compared. Which two are equivalent? Test $x = 2$ and explain your conclusion. E

ATL · CRITICAL THINKING

Test a claim with a counterexample. A student claims that x^2 is always greater than $2x$. Test $x = 2$: both equal 4, so the claim fails immediately. Test $x = 1$: $1 < 2$, another counterexample. One counterexample is enough to disprove an "always" statement, while many examples never prove one. Use a small table of values to look for a pattern, then look for a reason that explains it.

5.2 Equivalent expressions

CRITERION A · II, III ◆ CRITERION C · III ◆ EQUIVALENCE

Two expressions are equivalent when they give the same value for every value of the variable. Simplifying an expression does not change the relationship it describes; it changes the form so that the structure becomes easier to see and use.

TABLE 5.2 · COMBINING EQUAL VARIABLE PARTS

Expression	Meaning	Simplified
$3x + 2x$	three x's plus two x's	$5x$
$4x + 3 - 2x + 5$	four x's minus two x's, plus 3 and 5	$2x + 8$
$7a + 4 - 2a + 9$	seven a's minus two a's, plus 4 and 9	$5a + 13$

KEY IDEA · LIKE TERMS AND THE DISTRIBUTIVE PROPERTY

Like terms have exactly the same variable part. $3x$ and $-8x$ are like terms; $3x$ and $3x^2$ are not, because x and x^2 are different variable parts. Add or subtract the coefficients and keep the variable part: $3x + 2x = (3 + 2)x = 5x$.

The **distributive property** connects multiplication and addition: $a(b + c) = ab + ac$. Expanding reverses a product; factorising rewrites a sum as a product by taking out a common factor.

DIAGRAM 5.1 · AREA MODEL FOR $4(X + 3)$

	x	3
4	$4x$	12

The rectangle has height 4 and width $x + 3$. Its two parts have areas $4x$ and 12, so $4(x + 3) = 4x + 12$.

WORKED EXAMPLE 5.2.1 — SIMPLIFYING LIKE TERMS

PROBLEM

Simplify $7a + 4 - 2a + 9$.

1. Identify the like terms: $7a$ and $-2a$ are like terms; 4 and 9 are like terms.
2. Combine the variable terms: $7a - 2a = (7 - 2)a = 5a$.
3. Combine the constants: $4 + 9 = 13$.
4. Write the simplified expression: $5a + 13$. Check with $a = 2$: original $14 + 4 - 4 + 9 = 23$; simplified $10 + 13 = 23$.

ANSWER $5a + 13$.

WORKED EXAMPLE 5.2.2 — EXPANDING PRODUCTS

PROBLEM

Expand (a) $3(2x + 5)$ and (b) $-2(4x - 3)$.

1. (a) Multiply each term inside the brackets by 3: $3(2x) + 3(5) = 6x + 15$.
2. (b) Multiply each term by -2: $-2(4x) + (-2)(-3)$. The second product is positive because the signs are the same.
3. Simplify: $-8x + 6$.
4. Check with $x = 1$: original $-2(4 - 3) = -2$; expanded $-8 + 6 = -2$.

ANSWER **(a)** $6x + 15$ **(b)** $-8x + 6$.

WORKED EXAMPLE 5.2.3 — EXPANDING AND SIMPLIFYING

PROBLEM

Expand and simplify $4(2x - 3) - 2(x + 5)$.

1. Expand the first product: $4(2x - 3) = 8x - 12$.
2. Expand the second product, including the negative sign: $-2(x + 5) = -2x - 10$.
3. Subtract the whole second expression: $8x - 12 - 2x - 10$.
4. Combine like terms: $8x - 2x = 6x$ and $-12 - 10 = -22$, so the result is $6x - 22$.
5. Check with $x = 2$: original $4(1) - 2(7) = 4 - 14 = -10$; simplified $12 - 22 = -10$.

ANSWER $6x - 22$.

WORKED EXAMPLE 5.2.4 FACTORISING

PROBLEM

Factorise (a) $12x - 18$ and (b) $8a + 12$.

1. (a) Find the largest common factor of 12 and 18, which is 6. Write each term as a product: $12x = 6(2x)$ and $18 = 6(3)$.
2. Take out the common factor: $12x - 18 = 6(2x - 3)$.
3. (b) The largest common factor of 8 and 12 is 4: $8a + 12 = 4(2a + 3)$.
4. Check by expanding: $6(2x - 3) = 12x - 18$ and $4(2a + 3) = 8a + 12$.

ANSWER **(a)** $6(2x - 3)$ **(b)** $4(2a + 3)$.

GUIDED EXAMPLE · EXPANDING WITH A NEGATIVE COEFFICIENT

Problem. Expand $-5(2x - 4)$.

Step 1. Multiply the first term: $-5(2x) = -10x$.

Step 2. Multiply the second term: $-5(-4) = ?$

Now complete. Write the expanded expression, then check it with $x = 3$. Compare the signs in your two products with the rule that the product of two negatives is positive.

COMMON ERROR

Multiplying only the first term inside the brackets: $3(x + 4) = 3x + 4$ instead of $3x + 12$. A negative sign outside the brackets must multiply every term: $-(x - 5) = -x + 5$, not $-x - 5$. Also, $3x + 4$ cannot be simplified to $7x$, because 3 and 4 count different things; only like terms combine.

CONJECTURE · IS ONE MATCHING VALUE ENOUGH?

Are $4(x + 3)$ and $4x + 12$ always equal? Test $x = 0$, $x = 2$ and $x = -5$. The area model suggests why the two forms match. A single matching value is not proof of equivalence, because two different expressions can happen to agree at one point. The distributive property explains why these two agree at every value.

PRACTICE 5.2 · SIMPLIFY, EXPAND AND FACTORISE

SUPPORT 1–5 · CORE 1–8 · EXTENSION 9–12

Underline like terms before combining them. When expanding, draw arrows from the outside factor to every term inside the brackets.

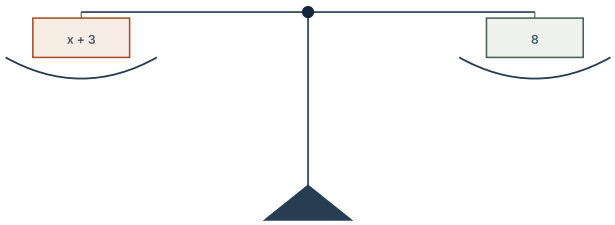
1. Simplify: (a) $6x + 4x$ (b) $9a - 3a$ (c) $5m + 2 - m$ (d) $7 + 3y - y + 4$. F
2. Simplify: (a) $3x + 5 + 2x - 8$ (b) $8a - 2 - 5a + 9$ (c) $4b + 7 - 7b - 2$. F
3. Expand: (a) $2(x + 6)$ (b) $5(2y - 3)$ (c) $-(x + 4)$ (d) $-3(2a - 5)$. F
4. Expand and simplify: (a) $3(x + 2) + 4(x + 1)$ (b) $5(2x - 1) - 2(x + 3)$. P
5. Factorise: (a) $6x + 9$ (b) $10y - 15$ (c) $8a + 12b$. F
6. Which expression is equivalent to $5(2x - 3)$? A $10x - 3$ B $10x - 15$ C $5x - 15$ D $2x - 15$. P

7. A student writes $4(x+3) = 4x+3$ and $7x - 2x = 5x^2$. Identify both errors and give the correct results. R
8. A rectangle has length $2x+5$ and width x. Write two equivalent expressions for its perimeter and show that they are equivalent. P
9. Explain why $3x$ and $3x^2$ are not like terms. Give a value of x for which their values differ. R
10. Factorise $12x - 18$ in two ways: $6(2x-3)$ and $-6(-2x+3)$. Are both correct? Explain. E
11. A student says $2(x+4)$ and $2x+4$ are equivalent because both give 12 when $x = 2$. Test another value and explain what this shows about checking equivalence with one value. E
12. The expressions $4(2x+1)$ and $8x+4$ describe the area of the same rectangle. Show this with an area model and explain using the distributive property. E

5.3 Solving linear equations

CRITERION A · III ◆ CRITERION C · III, IV ◆ RELATIONSHIPS

An equation is a question: for which value of the variable do the two sides have the same value? The solution is that value. Solving is the process of undoing the operations around the variable while keeping the two sides balanced.



A balance model for x + 3 = 8. Removing 3 from each pan leaves x = 5.

Figure 5.1 The pans stay level when the same operation is applied to both sides. Remove 3 from each side to isolate x.

DEFINITION · SOLUTION AND INVERSE OPERATIONS

A **solution** of an equation is a value of the variable that makes the equation true. To **solve** an equation, apply inverse operations to both sides until the variable is alone. Addition and subtraction are inverse operations; multiplication and division are inverse operations. An equation has the same solutions when the same operation is applied to both sides.

METHOD · SOLVING A LINEAR EQUATION

1. Simplify each side: expand brackets and combine like terms.
2. Collect the variable terms on one side and the constants on the other, using inverse operations on both sides.
3. Divide by the coefficient of the variable.
4. Check by substituting the solution into the *original* equation: both sides must be equal.

WORKED EXAMPLE 5.3.1 — ONE-STEP EQUATIONS

PROBLEM

Solve (a) $x + 7 = 12$, (b) $4x = 28$ and (c) $\frac{x}{3} = 5$.

1. (a) Subtract 7 from both sides: $x + 7 - 7 = 12 - 7$, so $x = 5$. Check: $5 + 7 = 12$.
2. (b) Divide both sides by 4: $\frac{4x}{4} = \frac{28}{4}$, so $x = 7$. Check: $4(7) = 28$.
3. (c) Multiply both sides by 3: $3 \cdot \frac{x}{3} = 3(5)$, so $x = 15$. Check: $15 \div 3 = 5$.

ANSWER **(a)** $x = 5$ **(b)** $x = 7$ **(c)** $x = 15$.

WORKED EXAMPLE 5.3.2 — TWO-STEP EQUATIONS

PROBLEM

Solve (a) $3x + 5 = 20$ and (b) $7 - 2x = 1$.

1. (a) Subtract 5 from both sides: $3x = 15$. Then divide by 3: $x = 5$. Check: $3(5) + 5 = 20$.
2. (b) Subtract 7 from both sides: $-2x = -6$. Then divide by -2: $x = 3$. Check: $7 - 2(3) = 1$.
3. In (b), the coefficient of x is negative. Dividing a negative by a negative gives a positive result, so the solution is positive.

ANSWER **(a)** $x = 5$ **(b)** $x = 3$.

WORKED EXAMPLE 5.3.3 — EQUATIONS WITH BRACKETS

PROBLEM

Solve (a) $4(x - 3) = 20$ and (b) $2(3x + 1) = 5x + 12$.

1. (a) Divide both sides by 4 first: $x - 3 = 5$, so $x = 8$. Alternatively, expand: $4x - 12 = 20$, then $4x = 32$, so $x = 8$. Check: $4(8 - 3) = 4(5) = 20$.
2. (b) Expand the left side: $6x + 2 = 5x + 12$.
3. Subtract $5x$ from both sides: $x + 2 = 12$. Then subtract 2: $x = 10$.
4. Check: left $2(30 + 1) = 62$; right $5(10) + 12 = 62$. Both sides agree.

ANSWER **(a)** $x = 8$ **(b)** $x = 10$.

WORKED EXAMPLE 5.3.4 — VARIABLES ON BOTH SIDES

PROBLEM

Solve (a) $5x + 2 = 3x + 10$ and (b) $12 + 8h = 44$.

1. (a) Subtract $3x$ from both sides: $2x + 2 = 10$. Then subtract 2: $2x = 8$, so $x = 4$.
2. Check: left $5(4) + 2 = 22$; right $3(4) + 10 = 22$.
3. (b) Subtract 12: $8h = 32$. Divide by 8: $h = 4$. Check: $12 + 8(4) = 44$.
4. In context, (b) answers "how many hours give a cost of \$44?" The solution $h = 4$ means four hours.

ANSWER **(a)** $x = 4$ **(b)** $h = 4$.

GUIDED EXAMPLE · KEEPING THE BALANCE

Problem. Solve $4x - 3 = 17$.

Step 1. Add 3 to both sides: $4x - 3 + 3 = 17 + 3$, so $4x = 20$.

Now complete. Divide both sides by 4 to find x. Then substitute your value into $4x - 3$ and check that it equals 17. Write the check as a separate line, not as part of the solving steps.

COMMON ERROR

Applying an operation to only one side breaks the balance. A second common error is distributing a negative incorrectly: $2(x - 3) = 10$ becomes $2x - 6 = 10$, not $2x - 3 = 10$. When dividing by a negative coefficient, keep the sign attached: from $-2x = -6$, divide by -2 to get $x = 3$. Always substitute back into the original equation; a solution that works in a later line but not the original is not a solution.

PRACTICE 5.3 · SOLVING LINEAR EQUATIONS

SUPPORT 1–4 · CORE 1–11 · EXTENSION 12–14

Show one balanced step per line. Check every solution by substitution, and state the answer in context where a context is given.

1. Solve and check: (a) $x + 9 = 17$ (b) $y - 6 = 11$ (c) $5m = 45$ (d) $\frac{n}{4} = 7$. F
2. Solve: (a) $2x + 3 = 13$ (b) $4x - 5 = 19$ (c) $7 + 3y = 22$ (d) $10 - 2a = 4$. F
3. Solve: (a) $3(x + 2) = 18$ (b) $5(x - 1) = 25$ (c) $2(2x + 1) = 14$ (d) $4(3x - 2) = 40$. P
4. Solve: (a) $4x + 1 = 2x + 11$ (b) $6x - 5 = 3x + 7$ (c) $8 + 5y = 2y + 29$. P
5. Solve $2(x + 3) = 3(x - 1)$. P
6. A taxi charges \$4 plus \$2.50 per kilometre. Write and solve an equation to find the distance travelled for a \$34 fare. P
7. Maya is 3 years older than twice her brother's age. Maya is 15. Write and solve an equation to find her brother's age. P
8. Which value is the solution of $5 - 2x = 11$? A $x = 3$ B $x = -3$ C $x = 8$ D $x = -8$. P
9. A student solves $3x + 4 = 19$ by adding 4 to both sides, obtaining $3x = 23$. Identify the error and give the correct solution. R
10. A student solves $2(x - 3) = 10$ by writing $2x - 3 = 10$. Identify the error, complete the correct solution, and check it. R
11. Explain why substituting a proposed solution into the original equation is a reliable check. Demonstrate with $x = 4$ in $5x + 2 = 3x + 10$. R
12. Solve $3(2x - 1) = 4x + 9$. E
13. A phone plan costs \$18 per month plus \$0.10 per minute. A bill is \$32. Write and solve an equation to find the number of minutes used. E
14. Solve $2(x + 3) = 2x + 6$. Then solve $2(x + 3) = 2x + 1$. Explain why one equation has infinitely many solutions and the other has none. E

ATL · REFLECTION

Check the original, not the last line. Substitute your value into the equation as it was first written. If both sides agree, it is a solution; if not, the error may lie in an earlier line. Record the check in a different colour.

5.4 Real-life application · Choosing a repair plan

CRITERION D · I–V ◆ SCIENTIFIC AND TECHNICAL INNOVATION ◆ MODELS

A community bike workshop needs to hire a repair stand. Two suppliers advertise different fixed fees and hourly rates. The cheaper plan depends on how long the stand is needed, so the decision requires a table, two expressions and an equation.

Plan A: \$12 booking fee plus \$8 per hour. **Plan B:** \$20 booking fee plus \$6 per hour.

TABLE 5.3 · COST OF EACH PLAN FOR DIFFERENT HIRE TIMES

Hours, h	0	2	4	6	8
Plan A cost (\$)	12	28	44	60	76
Plan B cost (\$)	20	32	44	56	68

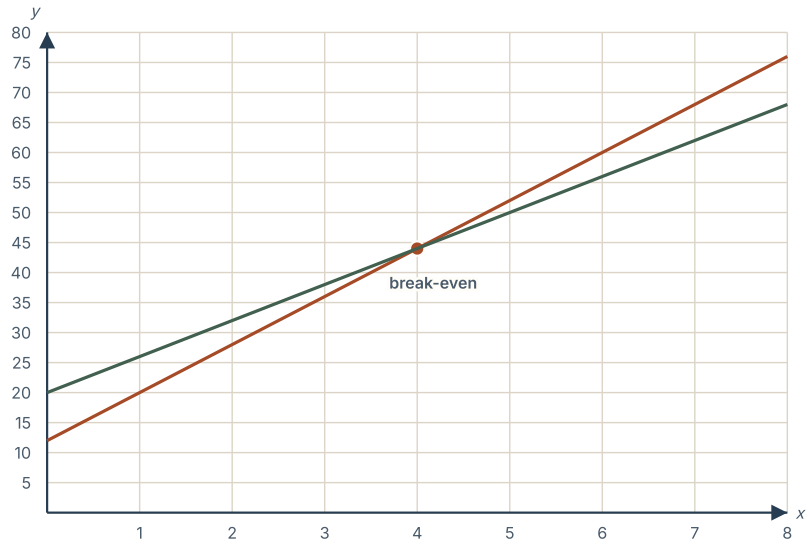


Figure 5.2 Cost in dollars against hire time in hours. The lines cross at $h = 4$, where both plans cost \$44; after that, Plan B rises more slowly.

Comparing the plans

CRITERION D

1. Check the table for 0, 2, 4, 6 and 8 hours against each plan's rule. Then extend it for 1, 3, 5 and 10 hours.
2. Write an expression for the cost of each plan when the stand is hired for h hours. Identify the fixed fee and the hourly rate in each expression.
3. Find the cost of each plan for 3 hours. Which plan is cheaper?
4. Solve $12 + 8h = 20 + 6h$. Interpret both the solution and the cost at that point.
5. For which hire times is Plan A cheaper? For which times is Plan B cheaper? Explain using your table and equation.
6. The workshop expects to use the stand for 12 hours. Which plan should it choose? Justify with calculations.
7. State one assumption and one limitation of the two-plan model. How could each affect the recommendation?

5.5 Investigation · The number trick

CRITERION B · I, II, III ◆ CRITERION C · II, III ◆ PATTERN

PLAN THE INQUIRY

A number trick seems magical because the starting number disappears. **Try** three starting numbers, **represent** each step with n, then **explain** why n cancels.

Think of a number

INVESTIGATION

Follow these steps: **1.** Think of a number. **2.** Add 5. **3.** Double the result. **4.** Subtract 4. **5.** Halve the result. **6.** Subtract your original number.

1. Carry out the trick with starting numbers 4, 10 and -3. Record the result at each step.
2. Write an expression for the result after each of the six steps, starting with n.
3. Simplify the expression after step 5. Then subtract n and explain why the final result is always 3.
4. Test the trick with $n = -2.5$. Does it still work? Explain what this shows about the rule.
5. Design your own trick that always gives 7. Use at least three operations, write the expression after each step, and simplify to prove the result.
6. Why is an algebraic simplification more convincing than testing 20 different starting numbers? Refer to the idea of a general case.

Criterion task · The fun-run T-shirt order

ASSESSED CRITERIA
A I–III · C I–IV · D I–V

The student council is ordering T-shirts for a fun run. A local print shop charges a \$45 setup fee plus \$8 per shirt. An online supplier charges \$12 per shirt with no setup fee, but adds a fixed \$15 delivery charge.

Number of shirts, n	5	10	20	30
Print shop cost (\$)	?	?	?	?
Online supplier cost (\$)	?	?	?	?

1. Complete the table.
2. Write an expression for the total cost of n shirts from each supplier. Identify the fixed and variable parts.
3. The council needs 10 shirts. Which supplier is cheaper, and by how much?
4. Solve $45 + 8n = 12n + 15$. Interpret the solution in the context of the number of shirts, and explain why the practical answer must be adjusted to a whole number.
5. The council sells the shirts for \$15 each. Write an expression for the profit from an order of n shirts from the print shop. Find the smallest whole-number order that makes a positive profit.
6. Write a recommendation for the council. Include one assumption and one limitation, and use your calculations to justify the choice.

FORMAT COMPLETED TABLE AND WRITTEN RECOMMENDATION

TIME 45–60 MINUTES

RESOURCES CALCULATOR PERMITTED

PART	WHAT TO SHOW	STRANDS
1–2	Correct table values and two correctly structured expressions with fixed and variable parts identified.	A i–ii, C i–ii
3–4	A valid cost comparison and a correct solution with the break-even point interpreted in context.	A iii, C iii–iv, D i–iii
5–6	A correct profit expression, a justified minimum order, and a recommendation with an assumption and limitation.	C i–iv, D iii–v

5.6 Summary, revision and reflection

KEY IDEAS

- A variable represents a quantity that can change; a coefficient multiplies the variable part, and a constant is fixed.
- Substitute with brackets, then follow the order of operations; a negative value inside brackets keeps its sign attached.
- Only like terms combine. Add or subtract coefficients, never the variable parts.
- Expanding uses the distributive property; factorising reverses it by taking out a common factor.
- An equation is solved by applying inverse operations to both sides; the solution is checked in the original equation.
- Equivalent expressions have the same value for every variable value; one matching value is not enough to prove equivalence.

$$a(b+c) = ab + ac$$

$$ax + bx = (a+b)x$$

$$ax + b = c \;\Rightarrow\; x = \frac{c-b}{a}$$

$$a(x+b) = c \;\Rightarrow\; x = \frac{c}{a} - b$$

Connections. This notation becomes the language of Chapter 6, where patterns generalise. Perimeter, area and volume formulas in Part III are expressions, and solving equations turns a known measurement into an unknown one.

REVISION CHECKLIST

- ☐ I can identify variables, terms, coefficients and constants.
- ☐ I can write an expression from words or a context.
- ☐ I can substitute positive and negative values correctly.
- ☐ I can simplify expressions by combining like terms.
- ☐ I can expand brackets and factorise using a common factor.
- ☐ I can solve one- and two-step equations, including brackets and variables on both sides.
- ☐ I can check a solution and interpret it in context.

REFLECTION

Which form of a relationship — words, a table, a graph or an expression — helped you see the structure most clearly this chapter, and why?

KEY VOCABULARY

variable
a symbol for a quantity that can change

constant
a fixed value in an expression

term
a part separated by plus or minus

coefficient
the number multiplying a variable part

like terms
terms with exactly the same variable part

expand
rewrite a product as a sum

factorise
rewrite a sum as a product

equation
a statement that two expressions are equal

solution
a value that makes an equation true

Mixed practice · Consolidate and extend

These questions combine the chapter's ideas. Choose the representation that makes the structure clearest, and show enough working for a reader to follow your reasoning. Questions are not grouped by difficulty within each section.

Fluency and accuracy

1. Simplify: (a) $8x + 3x$ (b) $12a - 5a$ (c) $4m + 9 - m$. F
2. Expand: (a) $3(x + 7)$ (b) $6(2y - 1)$ (c) $-4(a + 3)$. F
3. Solve: (a) $x + 8 = 21$ (b) $5x = 35$ (c) $\frac{x}{6} = 4$. F
4. Solve: (a) $2x + 7 = 19$ (b) $5x - 4 = 21$. F
5. Evaluate $4x - 3$ when (a) $x = 6$ (b) $x = -2$. F
6. Which expression is equivalent to $3(2x + 5)$? A $6x + 5$ B $6x + 15$ C $5x + 15$ D $6x - 15$. F
7. Which value solves $4x - 3 = 2x + 9$? A $x = 3$ B $x = 4$ C $x = 6$ D $x = 9$. F

Reasoning and problem solving

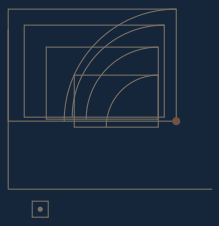
8. A student writes $5(x + 2) = 5x + 2$. Identify the error and give the correct expansion. R
9. A student solves $2x - 3 = 11$ by writing $2x = 8$, so $x = 4$. Identify the error and give the correct solution. R
10. Explain why $4x$ and x^4 are not like terms. Use $x = 2$ to show that their values differ. R
11. A rectangle has length $3x + 1$ and width $x - 2$. Write and simplify an expression for its perimeter. P
12. The sum of three consecutive integers is 42. Write an equation and find the integers. P
13. A phone plan costs \$15 per month plus \$0.08 per minute. A bill is \$27. Find the number of minutes used. P
14. Solve $3(x - 2) = 2(x + 4)$. P
15. Solve $5 - 2x = 13$. P
16. A taxi charges \$5 plus \$2.20 per kilometre. A journey costs \$38. Find the distance travelled. P
17. Two plans cost $20 + 3h$ and $35 + 1.5h$ dollars for h hours. Find the time at which the costs are equal, and state which plan is cheaper for a longer hire. P
18. Show that $2(3x + 4) - 3(x - 2)$ simplifies to $3x + 14$. P
19. A student says $x = 2$ is a solution of $3x + 5 = 2x + 7$ because both sides equal 11. Verify the claim and solve the equation to check whether there are other solutions. P

Extension and inquiry

20. Solve $2(x + 3) = 2x + 6$. Explain why this equation has infinitely many solutions. E
21. Solve $2(x + 3) = 2x + 1$. Explain why this equation has no solution. E
22. Create an equation with variables on both sides whose solution is $x = 5$. Show that your equation works. E
23. Factorise: (a) $10x - 25$ (b) $8a + 12b$. Check each result by expanding. E
24. Show that $3(2n + 1) - 6n + 4$ is always equal to 7, whatever the value of n. E
25. A rectangle has perimeter 30 cm. Its length is $x + 3$ cm and its width is x cm. Write and solve an equation to find its dimensions, then find its area. E

ATL · CRITICAL THINKING

Choose a representation deliberately. A table is efficient for a few values; a graph reveals a break-even point; an equation gives an exact solution. Before starting a mixed question, ask which representation will answer the question with the least chance of losing the meaning. Then use a second representation to check that your answer is plausible.



6

ALGEBRA AND PATTERN

Numerical patterns and generalisation

A sequence is evidence; a general rule is an explanation. The rule does not merely continue a list — it tells us what any term must be, including terms we could never draw or count one by one.

KEY CONCEPT · LOGIC

RELATED · PATTERN

RELATED · GENERALISATION

CONTEXT · PERSONAL AND CULTURAL EXPRESSION

INQUIRY

How can a handful of terms reveal a rule that predicts terms we have not yet seen?

MYP UNIT PLANNER

Framing the inquiry

UNIT 6 · APPROX. 11 HOURS
ATL FOCUS · CRITICAL THINKING

KEY AND RELATED CONCEPTS

Logic. A pattern is not yet a rule. Moving from several cases to a general claim requires a reason that covers every case; a single counterexample is enough to reject a false claim.

Pattern. Constant first differences produce linear sequences. A growing staircase behaves differently: its differences increase, so its rule has a different structure.

Generalisation. The nth term, a function machine and a graph are three representations of one relationship. Each can answer different questions about the same rule.

GLOBAL CONTEXT AND EXPLORATION

Personal and cultural expression. Exploration: artistry, craft, creation, beauty. Repeat patterns in textiles, tiling and lantern displays are designed with counting rules that makers must predict before they cut materials.

A border with a growing number of tiles and an arch with a growing number of lanterns both require a general rule: a maker cannot build every size in order to count it.

STATEMENT OF INQUIRY

Patterns in number and shape can be generalised into algebraic rules; representing those rules as expressions, machines or graphs makes prediction possible and justification necessary.

FACTUAL

What is a sequence, and how do we describe its term-to-term rule and its nth term?

CONCEPTUAL

How can one algebraic rule represent every term of a sequence, including terms we cannot draw?

DEBATABLE

When is a pattern evidence for a rule, and when is it merely a coincidence?

ATL · CRITICAL THINKING IN ACTION

Look for structure before looking for a formula. Compare consecutive terms, then test the proposed rule at the first and last known positions. A rule that works for two terms may fail at the third; a counterexample is decisive, while agreement is only evidence.

INQUIRY PATHWAY FOR THIS CHAPTER

1	2	3	4	5	6
CONTEXT	**QUESTION**	**CONCEPT**	**INVESTIGATION**	**COMMUNICATION**	**REFLECTION**
A workshop building a tiled border and a lantern arch to a growing design.	How many tiles or lanterns are needed for a design of any size?	Sequence, term, common difference, nth term and function.	Does a growing staircase follow a linear rule?	Tables, words, expressions, function machines and graphs.	Which representation makes the structure of a pattern easiest to justify?

Learning intentions and success criteria

YOU ARE LEARNING TO

- **Understand** a sequence as an ordered list of terms, each with a position number.
- **Know** the difference between a term-to-term rule and an nth-term rule.
- **Be able to** find the nth term of a linear sequence and use it to predict.
- **Be able to** use function machines forwards and backwards.
- **Be able to** plot and interpret the graph of a linear rule.
- **Be able to** generalise a context and justify a conjecture with algebra.

SUCCESS CRITERIA

- ☐ I can find the next terms of a sequence and describe the term-to-term rule.
- ☐ I can identify a linear sequence from its constant first difference.
- ☐ I can find and check an expression for the nth term of a linear sequence.
- ☐ I can decide whether a given number is a term of a sequence.
- ☐ I can find the input of a function machine from its output.
- ☐ I can explain the difference between a pattern that is tested and a rule that is justified.

PREREQUISITE CHECK · 10 MINUTES, NO CALCULATOR

1. Calculate 7×6.
2. Calculate $4 \times 9 + 3$.
3. Evaluate $50 - 6 \times 7$.
4. Continue the sequence $3, 6, 9, 12, \ldots$ by one term.
5. In the coordinate pair $(4, 11)$, which number is the x-coordinate?
6. Solve $2n + 1 = 17$.

Answers are in the Chapter 6 answer section. If more than two are incorrect, review multiplication, the order of operations, coordinates and one-step equations before continuing.

YEAR 1
Continue simple shape and number patterns; describe a rule in words; plot coordinates in the first quadrant.

YEAR 2 · THIS CHAPTER
The nth term of a linear sequence, function machines, linear graphs and algebraic justification.

YEAR 3
Quadratic sequences, gradients and intercepts, inverse functions and algebraic proof.

CRITERION	STRANDS	EVIDENCE IN THIS CHAPTER
A · Knowing and understanding	i, ii, iii	Generate terms; find nth terms; use function machines; read values from graphs and coordinate pairs.
B · Investigating patterns	i, ii, iii	Describe, extend and justify patterns; distinguish linear from non-linear growth; test conjectures and find counterexamples.
C · Communicating	i, ii, iii, iv	Move between words, tables, expressions, machines and graphs; define variables and justify a general rule.
D · Applying mathematics	i, ii, iii, iv, v	Model border and lantern counts; compare designs; use a budget; state assumptions and limitations.

6.1 Sequences and term-to-term rules

CRITERION A · I ◆ CRITERION B · I ◆ PATTERN

A craft workshop is making a stepped frieze from L-shaped tiles. The first motif uses 3 tiles. Each new motif adds one tile to each arm, so the sequence of tile counts begins $3, 5, 7, 9, \ldots$. The design is visual, but the workshop needs a numerical description before it can order materials.

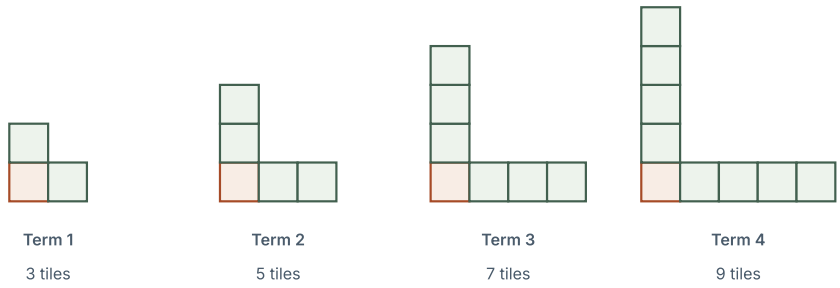


Figure 6.1 Each motif adds one square to each arm, so the tile count increases by 2. The copper square marks the corner that stays fixed.

DEFINITION · SEQUENCE, TERM AND POSITION

A **sequence** is an ordered list of numbers. Each number in the list is a **term**. The **position** of a term, usually written n, counts where it appears: the first term is at position $n = 1$, the second at $n = 2$, and so on. A **term-to-term rule** describes how to get from one term to the next.

The sequence $3, 5, 7, 9, \ldots$ has first term 3, third term 7 and term-to-term rule "add 2". A sequence that continues without end is **infinite**; a sequence with a last term is **finite**.

TABLE 6.1 · POSITION AND TERM FOR THE FRIEZE PATTERN

Position, n	1	2	3	4	5
Term (tiles)	3	5	7	9	11
Change from previous term	—	+2	+2	+2	+2

KEY IDEA · TWO KINDS OF RULE

A **term-to-term rule** says how to step from one term to the next: "add 2" or "multiply by 3". It is efficient for continuing a sequence, but to reach the 100th term you would still need 99 steps.

An **nth-term rule** gives the value directly from the position n. It answers "what is the term at position 100?" in one calculation. This chapter moves from the first kind of rule to the second.

WORKED EXAMPLE 6.1.1 — CONTINUING SEQUENCES

PROBLEM

For each sequence, find the term-to-term rule and the next three terms. (a) $6, 10, 14, 18, \ldots$ (b) $45, 40, 35, 30, \ldots$ (c) $3, 6, 12, 24, \ldots$ (d) $2, 3, 5, 8, 12, \ldots$

1. (a) Each term is 4 more than the previous term: the rule is "add 4". The next terms are $22, 26, 30$.
2. (b) Each term is 5 less than the previous term: "subtract 5". The next terms are $25, 20, 15$.
3. (c) Each term is double the previous term: "multiply by 2". The next terms are $48, 96, 192$.
4. (d) The differences are $1, 2, 3, 4$, so the rule is "add successive whole numbers". The next increases are $+5$ and $+6$: $17, 23, 29$. There is no single constant difference.

ANSWER **(a)** $+4$**;** $22, 26, 30$ **(b)** -5**;** $25, 20, 15$ **(c)** $\times 2$**;** $48, 96, 192$ **(d) add** $1, 2, 3, \ldots$**;** $17, 23, 29$**.**

WORKED EXAMPLE 6.1.2 — GENERATING FROM A FIRST TERM

PROBLEM

Write the first five terms of each sequence. (a) First term 2, rule "add 6". (b) First term 80, rule "subtract 7". (c) First term 1, rule "multiply by 3".

1. (a) $2,\ 2+6=8,\ 8+6=14,\ 14+6=20,\ 20+6=26$.
2. (b) $80,\ 73,\ 66,\ 59,\ 52$. The terms decrease because the rule subtracts.
3. (c) $1,\ 3,\ 9,\ 27,\ 81$. Multiplication by 3 makes the terms grow much faster than repeated addition.

ANSWER **(a)** $2, 8, 14, 20, 26$ **(b)** $80, 73, 66, 59, 52$ **(c)** $1, 3, 9, 27, 81$**.**

WORKED EXAMPLE 6.1.3 — A SEQUENCE IN CONTEXT

PROBLEM

Amara saves \$15 in the first week of a project and adds \$8 each week after that. (a) Write the first five weekly totals. (b) Describe the term-to-term rule. (c) How much has she saved after 10 weeks? (d) In which week does her total reach \$100?

1. (a) Week 1: 15. Each later week adds 8: $23, 31, 39, 47$.
2. (b) The terms increase by 8, so the term-to-term rule is "add 8".
3. (c) Week 10 is nine steps after week 1: $15 + 9(8) = 15 + 72 = 87$. She has saved \$87.
4. (d) Continue or calculate: week 11 gives $15 + 10(8) = 95$; week 12 gives $15 + 11(8) = 103$. She first reaches \$100 in week 12.

ANSWER **(a)** $15, 23, 31, 39, 47$ **(b) add 8 (c) \$87 (d) week 12.**

WORKED EXAMPLE 6.1.4 DIAGNOSING AN OFF-BY-ONE ERROR

PROBLEM

A student says the 10th term of $4, 9, 14, 19, \ldots$ is $4 + 10(5) = 54$. Identify the error and find the correct 10th term.

1. The rule "add 5" is correct, but moving from term 1 to term 10 requires only 9 steps, not 10.
2. Term 10 is $4 + 9(5) = 4 + 45 = 49$.
3. Equivalently, term 1 is $4 + 0(5)$, term 2 is $4 + 1(5)$, and term n is $4 + (n - 1)(5)$.
4. Check by continuing: $24, 29, 34, 39, 44, 49$, which is indeed the 10th term.

ANSWER **The student counted 10 steps instead of 9; the 10th term is 49.**

GUIDED EXAMPLE · FROM TERMS TO A RULE

Problem. A sequence begins $11, 15, 19, 23, \ldots$. Find the term-to-term rule and the next two terms. Then find the 7th term by continuing the sequence.

Step 1. Subtract consecutive terms: $15 - 11 = 4$, $19 - 15 = 4$, $23 - 19 = 4$. The rule is "add 4".

Now complete. Write the next two terms. Then count how many steps separate term 1 from term 7. Use that number of steps to calculate the 7th term, and check it by listing all seven terms.

COMMON ERROR

Do not assume that every sequence has a constant difference. The differences in $2, 4, 8, 16$ are $2, 4, 8$, not constant; the rule is "multiply by 2". Also, a term-to-term rule is not the same as an nth-term rule: "add 4" tells you how to step, not where the 10th term sits. Finally, the number of steps from term 1 to term n is $n - 1$, not n.

PRACTICE 6.1 · SEQUENCES AND TERM-TO-TERM RULES

SUPPORT 1–5 · CORE 1–10 · EXTENSION 11–12

Write the differences between consecutive terms before deciding on a rule. If the differences change, describe how they change.

1. Find the next three terms: (a) $5, 8, 11, 14$ (b) $30, 26, 22, 18$ (c) $2, 4, 8, 16$ (d) $100, 90, 80, 70$. F
2. Find the term-to-term rule and the next two terms: (a) $7, 12, 17, 22$ (b) $60, 54, 48, 42$ (c) $1, 5, 25, 125$. F
3. Write the first five terms: (a) first term 4, rule "add 9" (b) first term 3, rule "multiply by 2" (c) first term 100, rule "subtract 12". F
4. The sequence $1, 4, 9, 16, 25, \ldots$ is made of square numbers. Find the differences between consecutive terms, the next two terms, and explain why the differences are not constant. P
5. Liam starts with \$40 in a savings jar and adds \$15 each week. (a) Write the totals for weeks 1 to 5. (b) In which week does the total first reach \$100? P
6. Which rule generates $2, 6, 18, 54, \ldots$? A add 4 B multiply by 3 C multiply by 2 then add 2 D $n^2 + 1$. P

7. The sequence $12, 9, 6, 3, \ldots$ continues. (a) Write the next two terms. (b) Which position has the term -30? P

8. A student writes the next term of $2, 5, 9, 14$ as 18, saying "the rule is add 4". Identify the error and give the correct next term. R

9. Explain why $2, 4, 8, 16, \ldots$ is not a linear sequence. Refer to the differences in your answer. R

10. Write a sequence with first term 6 and constant difference -4. Continue it to the 8th term and state the term-to-term rule. P

11. Sequence A is $5, 8, 11, 14, \ldots$ and sequence B is $2, 6, 10, 14, \ldots$. They share the term 14. Find the next term that appears in both sequences. Explain how you searched. E

12. The sequence $1, 1, 2, 3, 5, 8, \ldots$ is called the Fibonacci sequence. (a) Describe its term-to-term rule in words. (b) Write the next three terms. (c) Explain why this sequence does not have a constant difference. E

6.2 The nth term of a linear sequence

CRITERION A · II ◆ CRITERION B · II ◆ GENERALISATION

The term-to-term rule says "add 3". The nth-term rule says where you land: term n is $3n + 2$. The second rule is more powerful because it calculates any term directly. For a linear sequence, the nth term always has the form $dn + c$, where d is the constant difference.

TABLE 6.2 · TESTING THE RULE $3N + 2$

Position, n	1	2	3	4	5
Term, $3n + 2$	5	8	11	14	17
First difference	—	+3	+3	+3	+3

METHOD · FINDING THE NTH TERM OF A LINEAR SEQUENCE

1. Find the constant difference d between consecutive terms. This is the coefficient of n.
2. Find the *zeroth term* c: subtract d from the first term. Equivalently, subtract dn from the value of term n and check that the result is the same for two terms.
3. Write the rule as $dn + c$.
4. Check it at $n = 1$ and at another position. For $5, 8, 11, 14$: $d = 3$, $c = 5 - 3 = 2$, so the rule is $3n + 2$.

WORKED EXAMPLE 6.2.1 — FINDING AND USING AN NTH TERM

PROBLEM

The sequence $6, 10, 14, 18, \ldots$ is linear. (a) Find its nth term. (b) Find the 20th term.

1. The difference is $10 - 6 = 4$, so the rule contains $4n$.
2. The zeroth term is $6 - 4 = 2$, so the nth term is $4n + 2$.
3. Check: $n = 1$ gives 6; $n = 3$ gives 14. Both match the sequence.
4. Substitute $n = 20$: $4(20) + 2 = 82$.

ANSWER **(a)** $4n + 2$ **(b)** 82.

WORKED EXAMPLE 6.2.2 — A DECREASING SEQUENCE

PROBLEM

The sequence $20, 17, 14, 11, \ldots$ is linear. (a) Find its nth term. (b) Find the 30th term. (c) Which position has term -10?

1. The difference is $17 - 20 = -3$, so the rule contains $-3n$.
2. The zeroth term is $20 - (-3) = 23$. The nth term is $-3n + 23$, which may also be written $23 - 3n$.
3. Check: $n = 1$ gives 20; $n = 4$ gives $23 - 12 = 11$.
4. (b) $n = 30$: $23 - 3(30) = 23 - 90 = -67$.
5. (c) Solve $23 - 3n = -10$: subtract 23 from both sides, then divide by -3. $23 - 3n = -10$ gives $-3n = -33$, so $n = 11$. The 11th term is -10.

ANSWER **(a)** $-3n + 23$ **(b)** -67 **(c) position 11.**

WORKED EXAMPLE 6.2.3 — PREDICTING FROM A PATTERN

PROBLEM

The L-shaped motifs in Figure 6.1 have $3, 5, 7, 9, \ldots$ tiles. (a) Find the nth term. (b) How many tiles are in motif 30? (c) A motif uses 61 tiles. Which motif number is it? (d) Is there a motif with exactly 50 tiles?

1. (a) The difference is 2. The zeroth term is $3 - 2 = 1$, so the nth term is $2n + 1$.
2. (b) Motif 30: $2(30) + 1 = 61$ tiles.
3. (c) Solve $2n + 1 = 61$: $2n = 60$, so $n = 30$. It is the 30th motif.
4. (d) Solve $2n + 1 = 50$: $2n = 49$, so $n = 24.5$. A position must be a whole number, so no motif has exactly 50 tiles.

ANSWER **(a)** $2n + 1$ **(b) 61** **(c) motif 30** **(d) no; it would be motif 24.5.**

WORKED EXAMPLE 6.2.4 — WORKING FROM A GIVEN RULE

PROBLEM

A sequence has nth term $5n - 3$. (a) Write its first four terms. (b) Is 92 a term of the sequence? (c) Is 100?

1. (a) Substitute $n = 1, 2, 3, 4$: $2, 7, 12, 17$. The difference is 5.
2. (b) Solve $5n - 3 = 92$: $5n = 95$, so $n = 19$. Since 19 is a whole number, 92 is the 19th term.
3. (c) Solve $5n - 3 = 100$: $5n = 103$, so $n = 20.6$. This is not a whole number, so 100 is not a term. The terms around 100 are $5(20) - 3 = 97$ and $5(21) - 3 = 102$.

ANSWER **(a)** $2, 7, 12, 17$ **(b) yes, term 19 (c) no; it lies between terms 20 and 21.**

GUIDED EXAMPLE · FIND AND TEST THE RULE

Problem. The sequence $4, 11, 18, 25, \ldots$ is linear. Find the nth term, the 15th term, and decide whether 95 is a term.

Step 1. The difference is $11 - 4 = 7$, so the rule contains $7n$. The zeroth term is $4 - 7 = -3$, so the nth term is $7n - 3$.

Step 2. Check: $n = 2$ gives $14 - 3 = 11$.

Now complete. Substitute $n = 15$. Then solve $7n - 3 = 95$ and decide whether the solution is a valid position. Compare with the answer section.

COMMON ERROR

For a decreasing sequence, the coefficient of n is negative. The sequence $20, 17, 14, \ldots$ has nth term $-3n + 23$, not $3n + 23$: at $n = 1$, $3n + 23$ gives 26, which is not the first term. A second error is to use the term value as the position. To test whether 92 is a term, solve for n; do not substitute $n = 92$.

PRACTICE 6.2 · FINDING AND USING THE NTH TERM

SUPPORT 1–5 · CORE 1–12 · EXTENSION 13–14

Always check a proposed rule at the first term and at one later term. To test membership, solve the rule equal to the given value and check that the position is a whole number.

1. Find the nth term: (a) $4, 7, 10, 13$ (b) $6, 11, 16, 21$ (c) $9, 7, 5, 3$ (d) $1, 5, 9, 13$. F
2. Find the nth term: (a) $2, 9, 16, 23$ (b) $30, 25, 20, 15$ (c) $0.5, 1.5, 2.5, 3.5$. P
3. A sequence has nth term $4n + 1$. Find the 10th and 50th terms. F
4. A sequence has nth term $7n - 2$. (a) Find the 12th term. (b) Which term is 61? P
5. Is 100 a term of the sequence with nth term $3n + 2$? Is 98? Explain both answers. R
6. The sequence $80, 74, 68, \ldots$ decreases by 6. Find (a) the nth term (b) the 10th term (c) which term equals 2. P
7. Which expression gives the nth term of $7, 12, 17, 22$? A $5n + 2$ B $7n + 5$ C $5n - 2$ D $n + 5$. P
8. A student says the nth term of $5, 9, 13, 17$ is $4n + 5$. Test $n = 1$, identify the error and give the correct rule. R

9. A student writes the nth term of $25, 21, 17, 13$ as $4n + 21$. Show that this fails at $n = 2$, then find the correct rule. R

10. A tiling pattern uses $6, 10, 14, 18, \ldots$ tiles. (a) Find the nth term. (b) How many tiles are in pattern 25? P

11. A border pattern uses $5, 8, 11, 14, \ldots$ tiles. (a) Find the 40th term. (b) A border uses 92 tiles. Which pattern number is it? P

12. Explain why the nth term of $7, 10, 13, 16$ is $3n + 4$ rather than $3n + 7$. Use $n = 2$ in your explanation. R

13. Sequence A has nth term $2n + 5$. Sequence B has nth term $3n + 2$. (a) For which n are the terms equal? (b) Which sequence is larger at $n = 10$? E

14. Show that every term of the sequence with nth term $2n + 1$ is odd. Explain why the result is true for every whole-number position n. E

> **CONJECTURE · EVIDENCE IS NOT YET PROOF**
>
> A rule that fits the first three terms may fail at the fourth. The sequence $2, 4, 8, \ldots$ could be "double each time" or "add 2 then add 4". Before using a rule, check it at every known term and explain why its structure must continue. In a linear sequence, the constant difference is that explanation.

6.3 Function machines and linear graphs

CRITERION A · III ◆ CRITERION C · III ◆ REPRESENTATION

The rule $3n + 2$ can be drawn as a machine: input n, multiply by 3, add 2, output. It can also be drawn as a graph: plot each input with its output. The machine shows the operations; the graph shows the relationship as a whole. A linear rule produces a straight line because every increase of 1 in the input produces the same increase in the output.

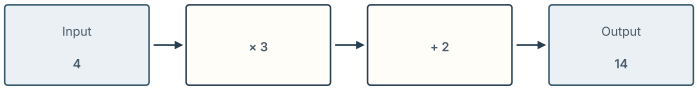


Figure 6.2 A function machine for the rule $3n + 2$. The input 4 produces $3(4) + 2 = 14$.

TABLE 6.3 · INPUTS AND OUTPUTS FOR THE RULE $2N + 1$

Input, n	0	1	2	3	4
Output, $2n + 1$	1	3	5	7	9
Change in output	—	+2	+2	+2	+2

DEFINITION · FUNCTION, INPUT AND OUTPUT

A **function** is a rule that assigns exactly one output to each input. In a function machine, the input enters, the operations are applied in order, and the output leaves. For the rule $y = 2x + 1$, x is the input and y is the output. A **linear graph** is a straight line; its rule can be written $y = mx + c$, where m is the constant change in y for each increase of 1 in x, and c is the value of y when $x = 0$.

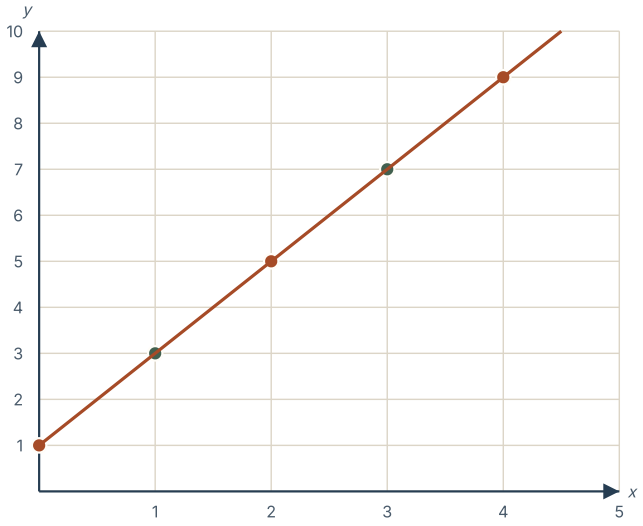


Figure 6.3 The graph of $y = 2x + 1$. Each step of 1 to the right raises the line by 2. The line crosses the y-axis at 1.

WORKED EXAMPLE 6.3.1 FORWARDS THROUGH A MACHINE

PROBLEM

A machine multiplies the input by 3 and then subtracts 2. (a) Find the outputs for inputs $1, 2, 3, 4, 5$. (b) Write the rule as an expression. (c) Which input gives output 25?

1. (a) Apply the operations in order: $3(1) - 2 = 1$, $3(2) - 2 = 4$, $3(3) - 2 = 7$, $3(4) - 2 = 10$, $3(5) - 2 = 13$.
2. (b) For input n, the output is $3n - 2$.
3. (c) Solve $3n - 2 = 25$: add 2 to both sides, $3n = 27$, so $n = 9$. Check: $3(9) - 2 = 25$.

ANSWER **(a)** $1, 4, 7, 10, 13$ **(b)** $3n - 2$ **(c) input 9.**

WORKED EXAMPLE 6.3.2 — FROM TABLE TO GRAPH

PROBLEM

For the rule $y = 2x + 1$, complete a table for $x = 0, 1, 2, 3, 4$, plot the points and describe the graph.

1. Substitute each input: $x = 0$ gives $y = 1$; $x = 1$ gives 3; $x = 2$ gives 5; $x = 3$ gives 7; $x = 4$ gives 9.
2. Write each pair as (x, y): $(0, 1), (1, 3), (2, 5), (3, 7), (4, 9)$.
3. Plot the points with the input on the horizontal axis and the output on the vertical axis. They lie on a straight line.
4. Each time x increases by 1, y increases by 2. The line crosses the vertical axis at $y = 1$, because $2(0) + 1 = 1$.

ANSWER **The points lie on the straight line $y = 2x + 1$; the line rises 2 units for every 1 unit to the right and crosses the y-axis at 1.**

WORKED EXAMPLE 6.3.3 — REVERSING A MACHINE

PROBLEM

A machine uses the rule "multiply by 2, then add 1". Its output is 17. Find the input by reversing the machine, then check.

1. Reverse the last operation first: undo "add 1" by subtracting 1. $17 - 1 = 16$.
2. Undo "multiply by 2" by dividing by 2: $16 \div 2 = 8$.
3. The input is 8. Check forwards: $2(8) + 1 = 17$.
4. In symbols, solve $2n + 1 = 17$: subtract 1, then divide by 2. The reverse machine and the equation are the same reasoning.

ANSWER **Input $= 8$.**

WORKED EXAMPLE 6.3.4 — COMPARING TWO RULES

PROBLEM

Machine A uses $4n - 1$ and machine B uses $2n + 7$. (a) Compare their outputs for $n = 3, 4, 5$. (b) For which input are the outputs equal?

1. (a) At $n = 3$: A gives $12 - 1 = 11$; B gives $6 + 7 = 13$. At $n = 4$: A gives 15; B gives 15. At $n = 5$: A gives 19; B gives 17.
2. (b) Set the rules equal: $4n - 1 = 2n + 7$. Subtract $2n$: $2n - 1 = 7$. Add 1: $2n = 8$, so $n = 4$.
3. The common output is $4(4) - 1 = 15$, which matches $2(4) + 7 = 15$.
4. A grows faster because its coefficient of n is larger; after $n = 4$, A overtakes B.

ANSWER **(a) A: $11, 15, 19$; B: $13, 15, 17$ (b) $n = 4$, output 15.**

GUIDED EXAMPLE · REVERSE THE OPERATIONS

Problem. A machine uses $5n - 3$. (a) Find the output for input 4. (b) Find the input that gives output 42.

Step 1. For input 4: $5(4) - 3 = 20 - 3 = 17$.

Step 2. To reverse, start from 42 and undo "subtract 3" by adding 3: $42 + 3 = 45$.

Now complete. Undo "multiply by 5" by dividing by 5 to find the input. Check your answer forwards.

COMMON ERROR

Apply operations in the order the machine states. The rule $3n - 2$ means "multiply by 3, then subtract 2"; at $n = 5$ it gives 13, not $3(5 - 2) = 9$. When plotting, the input is the first coordinate and the output is the second: the point for input 2 and output 5 is $(2, 5)$, not $(5, 2)$. When reversing, undo operations in reverse order: subtract before dividing, not after.

PRACTICE 6.3 · MACHINES AND GRAPHS

SUPPORT 1–5 · CORE 1–11 · EXTENSION 12–13

For each machine, write the rule before substituting. For each graph, label the axes with the input and output, not merely x and y.

1. A machine multiplies by 4 and adds 1. Find the outputs for inputs $0, 1, 2, 3, 4$. F
2. A machine uses $5n - 2$. (a) Complete a table for $n = 1, 2, 3, 4, 5$. (b) Find the output for $n = 12$. F
3. A machine uses $3n + 4$. Which input gives output 31? P
4. A machine uses $6n - 5$. Which input gives output 43? P
5. For $y = 3x - 1$, complete a table for $x = 1, 2, 3, 4, 5$, then describe how the output changes as x increases by 1. P
6. Which point lies on the graph of $y = 2x + 3$? A $(2, 5)$ B $(2, 7)$ C $(3, 7)$ D $(0, 2)$. P
7. For $y = 4x - 2$, find (a) y when $x = 0$ (b) y when $x = 3$ (c) x when $y = 18$. P
8. A taxi fare is $C = 3.5 + 2d$ dollars for d kilometres. (a) Find the fare for $d = 0, 2, 4, 6$. (b) Find the fare for 10 km. (c) Explain what the 3.5 and the 2 represent. P
9. A machine uses $3n + 5$ and gives output 26. Find the input by reversing the operations. F
10. Machine A uses $2n + 9$; machine B uses $4n + 1$. (a) For which input are the outputs equal? (b) Compare the outputs at $n = 10$. P
11. A student calculates the output of "multiply by 2, then add 3" for input 5 as 16, by computing $2(5 + 3)$. Identify the error and give the correct output. R
12. The points $(0, 2), (1, 5), (2, 8), (3, 11)$ lie on a straight line. (a) Find the rule. (b) Predict y when $x = 10$. (c) Does $(5, 17)$ lie on the line? E
13. A line has rule $y = 5x - 4$. (a) Which term has value 81? (b) Does the line pass through $(10, 46)$? Justify with a calculation. E

6.4 Real-life application · The lantern arch

CRITERION D · I–V ◆ PERSONAL AND CULTURAL EXPRESSION ◆ MODELS

A lantern festival workshop is designing a lighted arch for a courtyard. The arch is assembled from identical curved panels, with one larger lantern at the crown. Design A puts 3 lanterns on each panel; design B puts 2 lanterns on each panel but adds 5 lanterns along the base and crown. The workshop needs to predict the number of lanterns, the cost and the largest design that fits its budget.

TABLE 6.4 · LANTERN COUNTS FOR EACH DESIGN

Number of panels, n	1	2	3	4	5	6
Design A: $3n + 1$	4	7	10	13	16	19
Design B: $2n + 5$	7	9	11	13	15	17

WORKED EXAMPLE 6.4.1 BUILDING THE COST MODEL

PROBLEM

Design A uses $3n + 1$ lanterns for n panels. Each lantern costs \$6, and the frame and wiring cost \$45. (a) Write an expression for the total cost. (b) Find the cost for 10 panels. (c) Explain what each part of the expression represents.

1. (a) The lanterns cost $6(3n + 1)$ dollars. Adding the fixed \$45 gives $6(3n + 1) + 45$.
2. Expand and simplify: $18n + 6 + 45 = 18n + 51$.
3. (b) Substitute $n = 10$: $18(10) + 51 = 180 + 51 = 231$. The design costs \$231.
4. (c) The term $18n$ is the cost of the three lanterns on each of the n panels. The 51 combines the \$6 crown lantern with the \$45 frame and wiring charge. Both parts of the model are needed for the total.

ANSWER **(a)** $6(3n + 1) + 45 = 18n + 51$ **(b) \$231** **(c)** $18n$**: panel lanterns; 51: crown lantern and frame.**

Comparing the lantern designs

CRITERION D

1. Check the table values against each design rule. Then extend both rows for 8 and 10 panels.
2. Identify the coefficient and the constant in each rule. Explain what each number represents in the design.
3. For an arch with 6 panels, which design uses fewer lanterns? By how many?
4. Solve $3n + 1 = 2n + 5$. Interpret the solution and the number of lanterns at that point.
5. Each lantern costs \$6, and the frame and wiring cost a fixed \$45. Write an expression for the total cost of design A with n panels. Find the largest number of panels the workshop can afford with a budget of \$300.
6. The workshop wants exactly 61 lanterns. Find the number of panels each design would need, and the cost of each. Which design should it choose if the courtyard can fit either? Justify using the number of panels and the appearance of the arch.
7. State one assumption and one limitation of the cost model. How could each affect the workshop's decision?

6.5 Investigation · The growing staircase

CRITERION B · I, II, III ◆ CRITERION C · II, III ◆ PATTERN

PLAN THE INQUIRY

A staircase is built from square tiles. Staircase 1 has one tile. Each new staircase adds one row, one tile longer than the row below. **Build** the first four, **record** the totals, then **explain** why the totals are not linear.



Figure 6.4 Staircases 1 to 4. Each new staircase adds a row containing one more tile than the previous row.

Counting the tiles

INVESTIGATION

1. Copy and complete the table for staircases 1 to 6. Include a row for the change from the previous total.
   Steps n: 1, 2, 3, 4, 5, 6; tiles T_n: 1, 3, 6, 10, ?, ?
2. Describe the term-to-term rule for the totals. Is it constant? How does the number of tiles added at step n relate to n?
3. Predict the total for staircase 10 and staircase 20. Explain how you could calculate each without drawing the staircase.
4. Is the sequence of totals linear? Justify your answer using first differences.
5. A student claims the rule is $2n - 1$. Test the claim for $n = 4$ and explain what your result shows. Why does the claim appear to work for the first two staircases?
6. Staircase n contains $1 + 2 + 3 + \cdots + n$ tiles. Use the pairing method to show that this sum equals $\dfrac{n(n+1)}{2}$. Check the formula for $n = 4$ and $n = 10$.
7. A staircase uses exactly 78 tiles. Find its number of steps. Show your method and check your answer.
8. Compare the staircase totals with the linear sequence $4n - 1$. For which n does the staircase first have more tiles? Explain why the staircase eventually overtakes any linear sequence.

Criterion task · The mosaic tile border

ASSESSED CRITERIA
B I–III · C I–IV

A community art project is making square mosaic panels. Each panel has a one-tile-wide border around a square centre. The side length of the whole panel is n tiles. The border is made from copper tiles; the centre is left for a painted design.

Panel side length, n	3	5	8	10
Border tiles, B	?	?	?	?

1. Complete the table. Then write an expression for the number of border tiles in a panel of side length n.
2. Explain why the expression is $4n - 4$ rather than $4n$. Use the four corner tiles in your explanation.
3. A panel uses 60 border tiles. Find its side length. Is a panel with 100 border tiles possible? Justify both answers.
4. Border tiles cost \$2.50 each, and a fixed \$12 is charged for grout. Write an expression for the total cost of the border for a panel of side length n.
5. The project has a budget of \$200. Find the largest whole-number side length it can afford. Show the cost of that panel and the cost of the next larger one.
6. A second design uses $3n + 6$ border tiles. Solve $4n - 4 = 3n + 6$ and interpret the solution. For panels larger than that, which design uses fewer border tiles? Justify with a calculation.
7. Write a recommendation to the project. Include one assumption and one limitation, and use your expressions to support the choice.

FORMAT COMPLETED TABLE AND WRITTEN RECOMMENDATION

TIME 45–60 MINUTES

RESOURCES CALCULATOR PERMITTED

PART	WHAT TO SHOW	STRANDS
1–3	Correct table, a justified expression, and correct solutions with whole-number checks.	B i–ii, C i–ii
4–5	A correctly composed cost expression and a budget conclusion checked at consecutive whole numbers.	A ii, B iii, C iii, D ii–iii
6–7	A solved comparison interpreted in context and a recommendation with a relevant assumption and limitation.	B ii–iii, C iii–iv, D iv–v

6.6 Summary, revision and reflection

KEY IDEAS

- A sequence is ordered by position n. A term-to-term rule steps from one term to the next; an nth-term rule calculates directly from n.
- A linear sequence has a constant first difference d. Its nth term is $dn + c$, where c is the zeroth term.
- For a decreasing sequence, d is negative. Check any rule by substituting $n = 1$ and one other position.
- To test whether a value is a term, solve for n; the position must be a whole number.
- A function machine, a table, an expression and a straight-line graph can all represent the same linear relationship.
- The staircase sequence $1, 3, 6, 10, \ldots$ is not linear: its differences increase, and its nth term is $\frac{n(n+1)}{2}$.

nth term of a linear sequence: $dn + c$

term n = first term $+(n-1)d$

linear graph: $y = mx + c$

triangular numbers: $T_n = \frac{n(n+1)}{2}$

Connections. The expressions and equations of Chapter 5 are the tools for generalising here. The coordinate grid reappears in Chapter 9, where transformations move shapes across it; the triangular numbers return as a model for counting arrangements in later chapters.

REVISION CHECKLIST

- ☐ I can continue a sequence and state its term-to-term rule.
- ☐ I can identify a linear sequence from its differences.
- ☐ I can find the nth term of a linear sequence and check it.
- ☐ I can use an nth term to find a term or test whether a value belongs.
- ☐ I can use a function machine forwards and in reverse.
- ☐ I can plot and interpret a linear graph.
- ☐ I can distinguish a tested pattern from a justified general rule.

REFLECTION

Which representation — words, a table, a machine or a graph — made the structure of a pattern clearest for you, and what did it reveal that the others hid?

KEY VOCABULARY

sequence
an ordered list of terms

term
one number in a sequence, at position n

term-to-term rule
how to get from one term to the next

nth term
a rule for the term at position n

linear sequence
a sequence with constant first difference

common difference
the constant change between consecutive terms

function machine
a rule that turns an input into an output

conjecture
a general claim that still needs justification

counterexample
a case that disproves a general claim

Mixed practice · Consolidate and extend

These questions combine the chapter's ideas. Decide first whether the question asks you to continue a pattern, find a general rule, or reverse a rule; then choose the representation that answers it most directly.

Fluency and accuracy

1. Find the next three terms of $7, 12, 17, 22, \ldots$. F
2. Find the nth term of $8, 13, 18, 23, \ldots$. F
3. Find the nth term of $40, 36, 32, 28, \ldots$. F
4. A sequence has nth term $6n - 5$. Find the 10th term and the position of 79. F
5. A machine uses $4n - 3$. Find the output for input 6 and the input for output 41. F
6. Complete a table for $y = 3x + 1$ using $x = 0, 1, 2, 3, 4$. F
7. Which expression gives the nth term of $2, 7, 12, 17$? A $5n - 3$ B $2n + 5$ C $5n + 2$ D $n + 5$. F

Reasoning and problem solving

8. A student writes the nth term of $9, 14, 19, 24$ as $5n + 9$. Show that this fails at $n = 1$ and give the correct rule. R
9. A student continues $3, 7, 12, 18, \ldots$ with 24, saying "add 6 each time". Identify the error and give the correct next term. R
10. Explain why $3n + 1$ and $3(n + 1)$ are not equivalent. Evaluate both at $n = 2$. R
11. A border pattern has $4, 7, 10, 13, \ldots$ tiles. How many tiles are in the 30th pattern? P
12. Is 80 a term of the sequence with nth term $4n + 2$? Is 82? Justify both answers. P
13. Sequence A has nth term $3n + 5$. Sequence B has nth term $5n - 1$. Find the input at which their terms are equal, and compare them at $n = 10$. P
14. A student claims that every sequence beginning with a difference of 3 has nth term $3n + 2$. Give two counterexamples and explain what they show. P
15. The sequence $2, 6, 12, 20, 30, \ldots$ has differences $4, 6, 8, 10$. Find the next term and explain why the sequence is not linear. P
16. Nia starts with \$25 and adds \$12 each week. Write the total after n weeks, where week 1 is \$25. Find the first week in which her total is at least \$150. P
17. The points $(0, 4), (1, 7), (2, 10)$ lie on a straight line. Find the rule, predict y when $x = 12$, and find x when $y = 61$. P
18. Machine A uses $5n + 2$ and machine B uses $3n + 12$. For which input are their outputs equal? Compare their outputs at $n = 10$ and at $n = 0$. P
19. A staircase with n steps contains $\dfrac{n(n+1)}{2}$ tiles. Staircase 10 has 55 tiles and staircase 11 has 66. Which step adds exactly 12 tiles? Explain. E

Extension and inquiry

20. Write a linear sequence whose 5th term is 23 and whose common difference is 4. Find its nth term. E

21. A sequence has nth term $6n - 1$. Is 100 a term? Is 101? Justify both answers. E

22. Sequence A has nth term $4n + 3$; sequence B has nth term $7n - 6$. Show that they have a common term, and find it. E

23. Create a real-life context that could be modelled by $5n + 2$. Define the variable, explain both numbers, and evaluate the expression for $n = 8$. E

24. Explain, using the staircase diagram, why the sum of the first n odd numbers is n^2. Use $n = 4$ as an example. E

ATL · CRITICAL THINKING

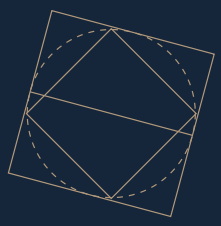
Ask what would prove you wrong. Before trusting a pattern, look for a case that could break it. For a proposed nth term, test the first term and the last known term; for a claimed general rule, try zero, a negative input or a fractional one. A single counterexample ends a conjecture, but no number of matching examples is a proof. The explanation must come from the structure of the rule.

PART III · SHAPE, MEASURE AND CHANGE

III

Geometry and space

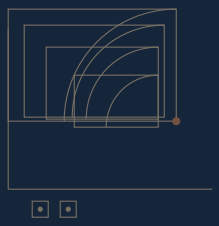
Geometry is the mathematics of the space we live in: the angles that hold a structure together, the surface that must be covered, the volume that must be filled, and the movements that leave a shape unchanged in essential ways. Measurement turns these ideas into numbers that can be checked.



07 2D geometry: angles, perimeter and area

08 3D geometry: solids, surface area and volume

09 Transformations and symmetry



7

GEOMETRY AND MEASURE

2D geometry: angles, perimeter and area

A courtyard is a set of promises: straight edges, true corners, a surface that must be covered and a border that must be bounded. Geometry turns those promises into measurements that can be checked before anything is built.

KEY CONCEPT · FORM

RELATED · SPACE

RELATED · MEASUREMENT

CONTEXT · ORIENTATION IN SPACE AND TIME

INQUIRY

How can breaking a shape into simpler shapes make a difficult measurement possible?

MYP UNIT PLANNER

Framing the inquiry

UNIT 7 · APPROX. 12 HOURS
ATL FOCUS · COMMUNICATION

KEY AND RELATED CONCEPTS

Form. A shape is identified by equal sides, right angles and parallel edges. These properties decide which measurements can be calculated.

Space. Angles describe how lines meet; polygons divide the plane into regions that can be decomposed into simpler ones.

Measurement. Perimeter is a length; area is a surface. Choosing the correct unit and formula is part of the argument.

GLOBAL CONTEXT AND EXPLORATION

Orientation in space and time. Exploration: natural and human landscapes. A community courtyard is shaped by the land, the people who use it and the budget available.

Designers move between a drawing and a quantity: a length becomes metres of edging, a region becomes square metres of paving.

STATEMENT OF INQUIRY

The form of a two-dimensional shape determines its measurable properties; decomposing a complex region into simpler shapes allows us to calculate its perimeter and area and justify a design decision.

FACTUAL

What angle relationships arise when lines intersect, and when a transversal crosses parallel lines?

CONCEPTUAL

Why is a polygon's angle sum fixed by its number of sides, and why does cutting up a region preserve its area?

DEBATABLE

When a courtyard must meet a budget, a safety rule and an aesthetic vision, which measurement should take priority?

ATL · COMMUNICATION IN ACTION

Label every vertex and given length, mark parallel lines and right angles, and give a reason for each calculated angle. State units; a correct number with an unclear diagram is difficult to trust.

INQUIRY PATHWAY FOR THIS CHAPTER

1	2	3	4	5	6
CONTEXT	**QUESTION**	**CONCEPT**	**INVESTIGATION**	**COMMUNICATION**	**REFLECTION**
A neighbourhood converting a bare courtyard into a paved garden.	How much paving, edging and shade can the budget allow?	Angle, polygon, perimeter, area and composite shape.	How does the number of sides control a polygon's angle sum?	Labelled diagrams, formulas, units and a recommendation.	Where did precision change the answer?

Learning intentions and success criteria

YOU ARE LEARNING TO

- **Understand** the angle relationships formed by intersecting and parallel lines.
- **Know** the angle and side properties of triangles and quadrilaterals.
- **Be able to** calculate angles using reasons, not just numerical guesses.
- **Be able to** calculate the perimeter and area of standard 2D shapes.
- **Be able to** use the circumference and area formulas for a circle.
- **Be able to** decompose composite shapes and model a cost.

SUCCESS CRITERIA

- ☐ I can name and classify angles, and use the straight-line and point facts.
- ☐ I can identify corresponding, alternate and co-interior angles.
- ☐ I can find missing angles in triangles and quadrilaterals and give a reason.
- ☐ I can choose the correct measurement formula and include units.
- ☐ I can calculate the area of a composite shape by adding or subtracting parts.
- ☐ I can use a measurement model to compare a design with a budget.

PREREQUISITE CHECK · 10 MINUTES, NO CALCULATOR

1. What is the special name of an angle of $90°$?
2. Calculate $180 - 47$.
3. Calculate 14×6.
4. Calculate $\frac{1}{2} \times 9 \times 8$.
5. Calculate 3.14×5^2.
6. Solve $2x + 30 = 180$.

Answers are in the Chapter 7 answer section. If more than two are incorrect, review angle types, multiplication facts, square numbers and one-step equations before continuing.

YEAR 1
Classify angles; find the perimeter and area of rectangles; recognise basic triangles and quadrilaterals.

YEAR 2 · THIS CHAPTER
Angle relationships, angle sums of polygons, area formulas, circles and composite shapes.

YEAR 3
Pythagoras' theorem, congruence and similarity, arcs and sectors, surface area and volume.

CRITERION	STRANDS	EVIDENCE IN THIS CHAPTER
A · Knowing and understanding	i, ii, iii	Recall angle facts; classify triangles and quadrilaterals; select and use perimeter, area and circle formulas.
C · Communicating	i, ii, iii, iv	Use correct angle notation and units; draw labelled diagrams; write reasons; present a cost comparison and recommendation.
D · Applying mathematics	i, ii, iii, iv, v	Model a courtyard; calculate materials; test a budget; compare designs; state assumptions and limitations.

7.1 Angle relationships

CRITERION A · I ◆ CRITERION C · I ◆ SPACE

At the corner of a courtyard, two paved paths cross. The angle between them is not arbitrary: once one angle is known, the other three are fixed. The same is true wherever straight lines meet. Geometry begins by noticing which measurements constrain the others.

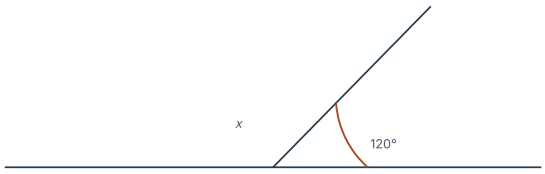


Figure 7.1 The two angles sit on a straight line. Together they form a half-turn, so their measures add to $180°$.

DEFINITION · ANGLE, VERTEX AND NOTATION

An **angle** is formed by two rays that meet at a common point called the **vertex**. The size of an angle measures the turn from one ray to the other and is written in degrees.

An angle of $90°$ is a **right angle**; less than $90°$ is **acute**; between $90°$ and $180°$ is **obtuse**; exactly $180°$ is **straight**; between $180°$ and $360°$ is **reflex**. The angle at vertex B, with arms BA and BC, is written $\angle ABC$. The middle letter always names the vertex.

KEY IDEA · THREE ANGLE FACTS

Angles on a straight line add to $180°$. Two **adjacent** angles share a vertex and an arm; if their outer arms form a straight line, they are supplementary.

Angles at a point add to $360°$, a complete turn.

Vertically opposite angles are formed by two intersecting straight lines and are equal. In Figure 7.1, the angle vertically opposite $120°$ is also $120°$; the adjacent angle is $60°$.

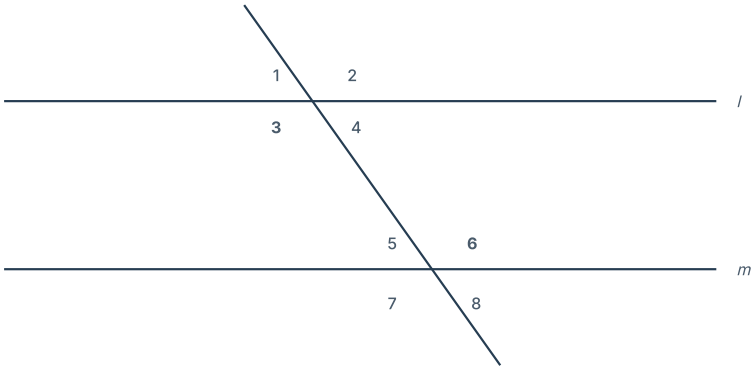


Figure 7.2 A transversal cuts two parallel lines. Angles 3 and 6 are **alternate interior** angles and are equal. Angles 3 and 5 are **co-interior** angles and add to $180°$. Angles 1 and 5 are **corresponding** angles and are equal.

KEY IDEA · PARALLEL LINES AND A TRANSVERSAL

A **transversal** is a line that crosses two or more lines. When the two lines are parallel:

Corresponding angles are in matching positions at each intersection and are equal.
Alternate angles are on opposite sides of the transversal, between the parallel lines, and are equal.
Co-interior angles are on the same side of the transversal, between the parallel lines, and add to $180°$.

WORKED EXAMPLE 7.1.1 ANGLES ON LINES AND AT A POINT

PROBLEM

Find x and the marked angles. (a) $(3x + 5)°$ and $(2x + 15)°$ lie on a straight line. (b) Angles $2x$, $3x$, $4x$ and $90°$ meet at a point. (c) Angles $(3x + 10)°$ and $(5x - 30)°$ are vertically opposite.

1. (a) The angles are supplementary: $(3x + 5) + (2x + 15) = 180$. So $5x + 20 = 180$, $5x = 160$ and $x = 32$. The angles are $3(32) + 5 = 101°$ and $2(32) + 15 = 79°$. Check: $101 + 79 = 180$.
2. (b) The angles complete a full turn: $2x + 3x + 4x + 90 = 360$. So $9x = 270$ and $x = 30$. The angles are $60°, 90°, 120°$ and $90°$, which sum to $360°$.
3. (c) Vertically opposite angles are equal: $3x + 10 = 5x - 30$. So $40 = 2x$ and $x = 20$. Each angle is $3(20) + 10 = 70°$. The adjacent angle is $180 - 70 = 110°$.

ANSWER **(a)** $x = 32$; $101°, 79°$ **(b)** $x = 30$; $60°, 90°, 120°, 90°$ **(c)** $x = 20$; $70°$.

WORKED EXAMPLE 7.1.2 PARALLEL-LINE ALGEBRA

PROBLEM

In each part, the two marked angles lie between parallel lines. Find x and both angles. (a) Corresponding angles $(4x - 5)^\circ$ and $(3x + 15)^\circ$. (b) Alternate angles $(5x + 10)^\circ$ and $(3x + 50)^\circ$. (c) Co-interior angles $(2x + 30)^\circ$ and $(3x + 20)^\circ$.

1. (a) Corresponding angles are equal: $4x - 5 = 3x + 15$, so $x = 20$. Both angles are $4(20) - 5 = 75^\circ$.
2. (b) Alternate angles are equal: $5x + 10 = 3x + 50$, so $2x = 40$ and $x = 20$. Both angles are $5(20) + 10 = 110^\circ$.
3. (c) Co-interior angles are supplementary: $(2x + 30) + (3x + 20) = 180$. So $5x + 50 = 180$, $5x = 130$ and $x = 26$. The angles are $2(26) + 30 = 82^\circ$ and $3(26) + 20 = 98^\circ$. Check: $82 + 98 = 180$.

ANSWER **(a)** $x = 20$; $75^\circ, 75^\circ$ **(b)** $x = 20$; $110^\circ, 110^\circ$ **(c)** $x = 26$; $82^\circ, 98^\circ$.

WORKED EXAMPLE 7.1.3 REASONING ALONG A TRANSVERSAL

PROBLEM

In Figure 7.2, $\angle 1 = 68^\circ$. Find $\angle 2$, $\angle 3$, $\angle 4$ and $\angle 5$, giving a reason for each.

1. $\angle 2 = 180 - 68 = 112^\circ$, because angles 1 and 2 lie on a straight line.
2. $\angle 3 = 112^\circ$, because angles 2 and 3 are vertically opposite.
3. $\angle 4 = 68^\circ$, because angles 1 and 4 are vertically opposite.
4. $\angle 5 = 68^\circ$, because angles 1 and 5 are corresponding angles between parallel lines.
5. As a check, $\angle 3$ and $\angle 5$ are co-interior: $112 + 68 = 180^\circ$, as required.

ANSWER $\angle 2 = 112^\circ$ **(straight line),** $\angle 3 = 112^\circ$ **(vertically opposite),** $\angle 4 = 68^\circ$ **(vertically opposite),** $\angle 5 = 68^\circ$ **(corresponding).**

GUIDED EXAMPLE · ALTERNATE ANGLES WITH ALGEBRA

Problem. Two parallel roads are crossed by a straight path. The alternate angles marked are $(6x - 10)^\circ$ and $(4x + 30)^\circ$. Find x, the size of each alternate angle, and the adjacent angle on the straight line.

Step 1. Alternate angles between parallel lines are equal, so write $6x - 10 = 4x + 30$.

Step 2. Solve: subtract $4x$ from both sides, then add 10. You should obtain $2x = 40$, so $x = 20$.

Now complete. Substitute $x = 20$ into both expressions to check that they agree. Then use the straight-line fact to find the angle adjacent to either one.

COMMON ERROR

Co-interior angles are **supplementary**, not equal: they add to 180°. Alternate and corresponding angles are equal only when the lines are parallel; if the lines are not parallel, no equality can be assumed. Also check which angle you are naming: $\angle ABC$ and $\angle ACB$ share the same letters but have different vertices and usually different sizes.

PRACTICE 7.1 · ANGLE RELATIONSHIPS

SUPPORT 1–5 · CORE 1–11 · EXTENSION 12–13

For every calculated angle, write the fact you used: straight line, point, vertically opposite, corresponding, alternate or co-interior. A correct number with no reason is only half an answer.

1. Classify each angle: (a) $37°$ (b) $90°$ (c) $128°$ (d) $212°$ (e) $180°$. F
2. Find the missing angle on each straight line: (a) $124°$ and x (b) $38°$, $67°$ and y. F
3. Find the missing angle at each point: (a) $120°, 85°, 95°, x$ (b) two right angles, $48°$ and y. F
4. Two straight lines cross. One angle is $68°$. Find the other three angles. F
5. Two parallel lines are cut by a transversal. Find the marked angle: (a) a corresponding angle to $73°$ (b) a co-interior angle to $108°$ (c) an alternate angle to $41°$. P
6. Angles $(3x + 15)°$ and $(2x + 10)°$ lie on a straight line. Find x and both angles. P
7. Angles $2x, 3x, 4x$ and $90°$ meet at a point. Find x and all four angles. P
8. Alternate angles between parallel lines are $(5x - 10)°$ and $(3x + 40)°$. Find x and the size of each angle. P
9. Co-interior angles are $(2x + 20)°$ and $(3x + 30)°$. Find x and both angles. P
10. A student writes: "The co-interior angles are $108°$ and $72°$, so they are equal." Identify the error and correct the statement. R
11. Which pair of angles is always equal? A adjacent angles on a straight line B vertically opposite angles C co-interior angles D angles at a point. P
12. Two straight roads cross at a junction. One of the four angles is $63°$. Find the other three and explain how a road painter could check the layout using only one measured angle. R
13. Prove that vertically opposite angles are equal. Let the two angles be a and c, with an adjacent angle b. Use the straight-line fact twice. E

7.2 Triangles

CRITERION A · II ◆ CRITERION C · II ◆ FORM

A roof truss is a triangle for a reason. Its three sides brace one another, and its angles cannot change without changing a side. Triangles are the simplest polygons, yet their angle and side relationships underpin every structure that must stay rigid.

> **DEFINITION · CLASSIFYING TRIANGLES**
>
> By sides: an **equilateral** triangle has three equal sides; an **isosceles** triangle has two equal sides; a **scalene** triangle has no equal sides.
>
> By angles: an **acute** triangle has three acute angles; a **right** triangle has one right angle; an **obtuse** triangle has one obtuse angle.
>
> In an isosceles triangle, the angles opposite the equal sides are equal. These are called the **base angles**. The remaining angle is the **apex angle**.

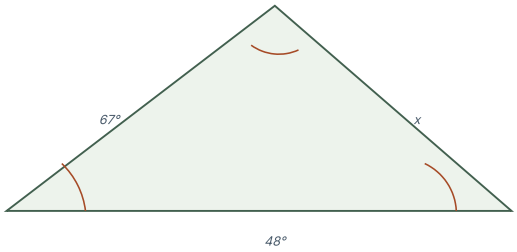


Figure 7.3 The three interior angles of any triangle add to $180°$, so here $x = 180 - 48 - 67 = 65°$.

KEY IDEA · THE TRIANGLE ANGLE SUM

The interior angles of a triangle add to $180°$.

Why? Draw a line through the top vertex parallel to the base. The two angles between this line and the sloping sides are alternate angles to the base angles. Together with the top angle they form a straight line, so the three interior angles add to $180°$. The result is a consequence of the parallel-line facts in 7.1, not a separate coincidence.

DEFINITION · EXTERIOR ANGLE

An **exterior angle** of a triangle is formed by extending one side. It is supplementary to the interior angle at that vertex. The exterior angle equals the sum of the two **opposite interior angles**:

$$\text{exterior angle} = \text{opposite interior angle}_1 + \text{opposite interior angle}_2.$$

This follows from the angle sum: the exterior and adjacent interior angles add to $180°$, and the three interior angles also add to $180°$.

WORKED EXAMPLE 7.2.1 MISSING ANGLES IN TRIANGLES

PROBLEM

Find the missing angle. (a) A triangle has angles $47°$ and $68°$. (b) A right triangle has one acute angle $31°$. (c) An isosceles triangle has apex angle $46°$.

1. (a) The three angles add to $180°$: $180 - 47 - 68 = 65°$.
2. (b) The right angle is $90°$, so the third angle is $180 - 90 - 31 = 59°$.
3. (c) The two base angles are equal. Their sum is $180 - 46 = 134°$, so each is $134 \div 2 = 67°$.

ANSWER **(a)** $65°$ **(b)** $59°$ **(c)** $67°$ **and** $67°$**.**

WORKED EXAMPLE 7.2.2 TRIANGLE ALGEBRA

PROBLEM

(a) The angles of a triangle are $2x$, $3x$ and $4x$. Find x and the angles. (b) The angles are $(x + 20)°$, $(2x - 10)°$ and $(x + 10)°$. Find x and the angles.

1. (a) $2x + 3x + 4x = 180$, so $9x = 180$ and $x = 20$. The angles are $40°$, $60°$ and $80°$.
2. (b) $(x + 20) + (2x - 10) + (x + 10) = 180$. The constants simplify: $20 - 10 + 10 = 20$, so $4x + 20 = 180$, $4x = 160$ and $x = 40$.
3. The angles are $40 + 20 = 60°$, $2(40) - 10 = 70°$ and $40 + 10 = 50°$. Check: $60 + 70 + 50 = 180$.

ANSWER **(a)** $x = 20$; $40°, 60°, 80°$ **(b)** $x = 40$; $60°, 70°, 50°$.

WORKED EXAMPLE 7.2.3 USING AN EXTERIOR ANGLE

PROBLEM

(a) An exterior angle of a triangle is $125°$, and one opposite interior angle is $48°$. Find the other opposite interior angle. (b) An exterior angle is $(5x)°$, and the two opposite interior angles are $(2x)°$ and $60°$. Find x and the exterior angle.

1. (a) The exterior angle equals the sum of the two opposite interior angles: $125 = 48 + y$, so $y = 77°$. The adjacent interior angle is $180 - 125 = 55°$, and $55 + 48 + 77 = 180$, as required.
2. (b) $5x = 2x + 60$, so $3x = 60$ and $x = 20$. The exterior angle is $5(20) = 100°$. The opposite interior angles are $40°$ and $60°$; their sum is $100°$.

ANSWER **(a)** $77°$ **(b)** $x = 20$; **exterior angle** $100°$.

WORKED EXAMPLE 7.2.4 ISOSCELES TRIANGLES AND AN EXTERIOR ANGLE

PROBLEM

An isosceles triangle has an exterior angle of $130°$ at its apex. Find the two base angles. Then find the apex angle if an exterior angle at a base is $115°$.

1. The interior apex angle is $180 - 130 = 50°$, because the exterior and adjacent interior angles lie on a straight line.
2. The base angles are equal: $(180 - 50) \div 2 = 65°$ each.
3. If an exterior angle at a base is $115°$, the interior base angle is $180 - 115 = 65°$. The other base angle is also $65°$, so the apex angle is $180 - 65 - 65 = 50°$.

ANSWER **Base angles** $65°$ **each; with a base exterior angle of** $115°$**, the apex angle is** $50°$.

GUIDED EXAMPLE · THREE EXPRESSIONS, ONE TRIANGLE

Problem. The angles of a triangle are $(2x + 10)°$, $(3x - 20)°$ and $(x + 40)°$. Find x and the three angles.

Step 1. Write the angle-sum equation: $(2x + 10) + (3x - 20) + (x + 40) = 180$.

Step 2. Collect like terms. The x-terms give $6x$; the constants give $10 - 20 + 40 = 30$. So $6x + 30 = 180$.

Now complete. Solve for x, substitute into all three expressions, and check that the three angles add to $180°$.

COMMON ERROR

The angles of a triangle add to $180°$, not $360°$; $360°$ belongs to a quadrilateral or to angles at a point. In an isosceles triangle, the two equal angles are opposite the equal sides — they are not necessarily the apex angle. When an exterior angle is given, subtract from $180°$ only if you need the adjacent interior angle; to find an opposite interior angle, use the exterior-angle rule directly.

PRACTICE 7.2 · TRIANGLES

SUPPORT 1–6 · CORE 1–12 · EXTENSION 13

Sketch the triangle and mark what is given before calculating. If the triangle is isosceles, mark the two equal base angles explicitly.

1. Classify each triangle by its sides and by its angles: (a) sides $5, 5, 5$, angles $60°, 60°, 60°$ (b) sides $6, 8, 10$, one angle $90°$ (c) sides $4, 4, 7$, one angle $110°$ (d) sides $7, 8, 9$, angles $48°, 61°, 71°$. F
2. Find the third angle: (a) $52°$ and $71°$ (b) $90°$ and $34°$ (c) $25°$ and $25°$. F
3. An isosceles triangle has (a) apex angle $84°$ (b) base angle $63°$. Find the other angles. F
4. The angles of a triangle are x, $(x + 20)°$ and $2x$. Find x and the angles. P
5. Find the missing angle: (a) an exterior angle is $115°$ and one opposite interior angle is $40°$ (b) an exterior angle is $3x$, and the opposite interior angles are x and $80°$. P
6. A right triangle has one acute angle of $29°$. Find the other acute angle. F
7. A student claims a triangle can have angles $70°$, $60°$ and $60°$. Without drawing it, decide whether this is possible and explain. R
8. The exterior angle of a triangle is equal to: A the adjacent interior angle B the sum of all three interior angles C the sum of the two opposite interior angles D $360°$ minus the adjacent angle. P
9. An isosceles triangle has an exterior angle of $130°$ at its apex. Find its base angles. P
10. A wheelchair ramp rises from the ground at $12°$. The ramp meets the door at a right angle to the vertical wall. Find the third angle of the triangular side view. P
11. Explain why a triangle cannot have two right angles. Use the angle sum in your answer. R
12. The interior angles of a triangle are $50°$, $60°$ and $70°$. Find each exterior angle and their sum. Explain why the sum is $360°$. E
13. The angles of a triangle are $(2x - 10)°$, $(x + 30)°$ and $(3x - 20)°$. Find x, the angles, and classify the triangle. E

7.3 Quadrilaterals

CRITERION A · II ◆ CRITERION C · II ◆ FORM

A shade canopy is built from parallelogram panels; a window is braced by a rectangle; a garden bed is edged with a trapezium. Each four-sided shape obeys a small set of rules, and those rules decide which angles and lengths can be calculated.

DEFINITION · THE QUADRILATERAL FAMILY

A **quadrilateral** is a polygon with four straight sides. Special quadrilaterals are defined by their sides, angles or parallel pairs. A shape may belong to more than one family: a square is also a rectangle, a rhombus and a parallelogram.

TABLE 7.1 · PROPERTIES OF SPECIAL QUADRILATERALS

Shape	Sides	Angles	Diagonals
Parallelogram	Two pairs of parallel sides; opposite sides equal.	Opposite angles equal; adjacent angles add to $180°$.	Bisect each other.
Rectangle	Parallelogram with four right angles; opposite sides equal.	All four angles $90°$.	Equal in length; bisect each other.
Rhombus	Parallelogram with four equal sides.	Opposite angles equal; adjacent angles add to $180°$.	Perpendicular; bisect each other and the angles.
Square	Four equal sides; both pairs of opposite sides parallel.	All four angles $90°$.	Equal, perpendicular and bisect each other.
Trapezium (trapezoid)	Exactly one pair of parallel sides.	Co-interior angles between the parallel sides add to $180°$.	Not generally equal or perpendicular.
Kite	Two pairs of adjacent equal sides.	One pair of opposite angles equal.	Perpendicular.



Figure 7.4 In a parallelogram, opposite sides are equal and parallel. The *height* is measured perpendicular to the base, not along the slant side.

KEY IDEA · THE QUADRILATERAL ANGLE SUM

The interior angles of any quadrilateral add to $360°$. A diagonal divides the quadrilateral into two triangles, each with angle sum $180°$, so the total is $2 \times 180 = 360°$. This argument works for every quadrilateral, including concave ones.

In a parallelogram, adjacent angles are co-interior angles between parallel sides, so they add to $180°$. Opposite angles are equal because they are made from the same pair of parallel directions.

WORKED EXAMPLE 7.3.1 — MISSING ANGLES IN QUADRILATERALS

PROBLEM

(a) A quadrilateral has angles $85°, 95°$ and $100°$. Find the fourth angle. (b) A kite has angles $x, x, 120°$ and $80°$, where the two x angles are opposite. Find x.

1. (a) The four angles add to $360°$: $360 - 85 - 95 - 100 = 80°$.
2. (b) The four angles add to $360°$: $2x + 120 + 80 = 360$. So $2x + 200 = 360$, $2x = 160$ and $x = 80$.
3. The angles are $80°, 80°, 120°, 80°$. The equal pair is opposite, as required for a kite.

ANSWER **(a)** $80°$ **(b)** $x = 80°$**.**

WORKED EXAMPLE 7.3.2 — ANGLES IN A PARALLELOGRAM

PROBLEM

Two adjacent angles of a parallelogram are $(3x + 20)°$ and $(2x + 30)°$. Find x and all four angles.

1. Adjacent angles in a parallelogram are supplementary: $(3x + 20) + (2x + 30) = 180$.
2. So $5x + 50 = 180$, $5x = 130$ and $x = 26$.
3. The angles are $3(26) + 20 = 98°$ and $2(26) + 30 = 82°$. Check: $98 + 82 = 180$.
4. Opposite angles are equal, so the four angles are $98°, 82°, 98°, 82°$.

ANSWER $x = 26$**; angles** $98°, 82°, 98°, 82°$**.**

WORKED EXAMPLE 7.3.3 — A TRAPEZIUM AND A KITE

PROBLEM

(a) A trapezium has an angle of $72°$ between its parallel sides. Find the angle beside it on the same side. (b) Another trapezium has angles $72°, 108°, 105°$ and y. Find y.

1. (a) The two angles between the parallel sides are co-interior, so they add to $180°$: $180 - 72 = 108°$.
2. (b) The four angles add to $360°$: $y = 360 - 72 - 108 - 105 = 75°$.
3. Check the co-interior pair: $105 + 75 = 180°$, so the remaining pair also works.

ANSWER **(a)** $108°$ **(b)** $y = 75°$**.**

GUIDED EXAMPLE · OPPOSITE ANGLES IN A PARALLELOGRAM

Problem. A parallelogram has opposite angles $(4x - 20)^\circ$ and $(2x + 40)^\circ$. Find x, both angles and the adjacent angle.

Step 1. Opposite angles of a parallelogram are equal, so $4x - 20 = 2x + 40$.

Step 2. Solve: subtract $2x$, then add 20 to both sides. You should obtain $2x = 60$, so $x = 30$.

Now complete. Substitute into both expressions to confirm they are equal. Then use the fact that adjacent angles are supplementary to find the other angle, and state all four angles.

COMMON ERROR

A quadrilateral's angles add to 360°, not 180°. In a parallelogram, it is the **opposite** angles that are equal and the **adjacent** angles that are supplementary; do not assume all four angles are equal. A rhombus is not necessarily a square, and a kite is not necessarily a rhombus: check the definitions rather than the appearance of a sketch.

PRACTICE 7.3 · QUADRILATERALS

SUPPORT 1–5 · CORE 1–11 · EXTENSION 12–13

Mark the parallel sides and equal angles on a sketch before you calculate. State which property you are using in each step.

1. Find the fourth angle of a quadrilateral with angles: (a) $95^\circ, 85^\circ, 100^\circ$ (b) $120^\circ, 60^\circ, 60^\circ$. F
2. A parallelogram has one angle of 68°. Find its other three angles. F
3. Two adjacent angles of a parallelogram are $(3x + 20)^\circ$ and $(2x + 30)^\circ$. Find x and all four angles. P
4. A kite has angles $x, x, 120^\circ$ and 80°, where the two equal angles are opposite. Find x. P
5. A trapezium has an angle of 72° between its parallel sides. Find the co-interior angle beside it. P
6. The angles of a quadrilateral are $x, 2x, 3x$ and $4x$. Find x and all four angles. P
7. The angles of a quadrilateral are $(x + 20)^\circ, (2x - 10)^\circ, (x + 40)^\circ$ and $(2x + 10)^\circ$. Find x and the angles. P
8. A student says the angles of a quadrilateral add to 180°, so a quadrilateral with angles $100^\circ, 80^\circ, 90^\circ$ and 90° is impossible. Identify the error and correct the conclusion. R
9. Which quadrilateral has four equal sides and four right angles? A rhombus B rectangle C square D kite. P
10. Which statement is always true for a parallelogram? A all sides are equal B all angles are right angles C opposite angles are equal D diagonals are perpendicular. P
11. A rectangular window is braced by a diagonal. The diagonal makes an angle of 58° with the bottom edge. Find the angle it makes with the side edge, and explain your reasoning. R
12. Explain why a quadrilateral cannot have four acute angles. Use the angle sum in your answer. E
13. Explain why the angle sum of any quadrilateral is 360°. Draw a diagonal and refer to the two triangles in your explanation. E

7.4 Perimeter and area

CRITERION A · III ◆ CRITERION C · III ◆ MEASUREMENT

The courtyard needs edging around its boundary and paving across its surface. These are different measurements: edging is a length, paving is an area. Confusing the two is one of the most expensive mistakes a designer can make.

DEFINITION · PERIMETER AND AREA

Perimeter is the distance around the outside of a shape. It is measured in units of length: millimetres (mm), centimetres (cm), metres (m) or kilometres (km).

Area is the amount of surface a shape covers. It is measured in square units: mm^2, cm^2, m^2 or km^2. One square metre is the area of a square with side 1 m, so $1\ \text{m}^2 = 100\ \text{cm} \times 100\ \text{cm} = 10\,000\ \text{cm}^2$.

A **composite shape** is made from simpler shapes joined together. Its area can be found by adding the parts or by subtracting a cut-out from a larger shape. Its perimeter is the length of the *outer boundary only*; internal dividing lines are not included.

TABLE 7.2 · PERIMETER AND AREA FORMULAS

Shape	Perimeter	Area
Rectangle	$P = 2(l + w)$	$A = lw$
Square	$P = 4s$	$A = s^2$
Triangle	$P = a + b + c$	$A = \frac{1}{2}bh$
Parallelogram	$P = 2(a + b)$	$A = bh$
Trapezium	$P = a + b + c + d$	$A = \frac{1}{2}(a + b)h$
Circle	$C = 2\pi r = \pi d$	$A = \pi r^2$

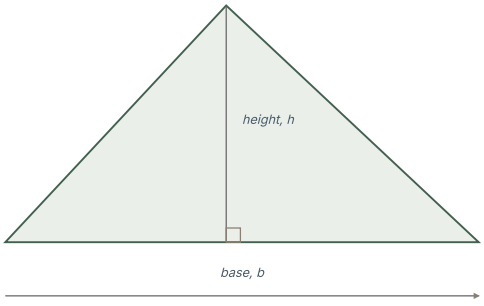


Figure 7.5 The area of a triangle is half the area of the rectangle with the same base and height.

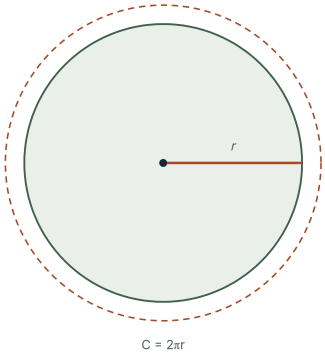


Figure 7.6 The circumference is the distance around the circle; th surface inside it.

KEY IDEA · HEIGHT MUST BE PERPENDICULAR

In the formulas for a triangle, parallelogram and trapezium, h is the **perpendicular height** — the shortest distance between the base and the opposite side or vertex. It is not the length of a sloping side.

Use $\pi \approx 3.14$ unless a question gives another value. Keep the full calculator value until the final rounding step, then round money to the nearest cent and measurements to a sensible precision.

WORKED EXAMPLE 7.4.1

PERIMETER OF RECTANGLES AND COMPOSITE SHAPES

PROBLEM

(a) Find the perimeter of a rectangle measuring 12.5 cm by 7 cm. (b) An L-shaped courtyard is 10 m wide and 8 m high, with a 4 m by 3 m rectangle cut from the top-right corner. Find its perimeter.

1. (a) $P = 2(l + w) = 2(12.5 + 7) = 2(19.5) = 39$ cm.
2. (b) Follow the outer boundary. The bottom is 10 m and the left side is 8 m. The top is $10 - 4 = 6$ m. The notch goes down 3 m, across 4 m, and down the remaining $8 - 3 = 5$ m.
3. $P = 10 + 5 + 4 + 3 + 6 + 8 = 36$ m. Check that every outer edge has been counted once and no inner edge has been added.

ANSWER **(a) 39 cm (b) 36 m.**

WORKED EXAMPLE 7.4.2

AREAS OF TRIANGLES AND QUADRILATERALS

PROBLEM

Find the area. (a) A triangle with base 12 cm and height 7 cm. (b) A parallelogram with base 9 m and perpendicular height 6 m. (c) A trapezium with parallel sides 6 cm and 10 cm and height 5 cm.

1. (a) $A = \frac{1}{2}bh = \frac{1}{2}(12)(7) = 42$ cm^2.
2. (b) $A = bh = 9(6) = 54$ m^2. The slant side is not used.
3. (c) $A = \frac{1}{2}(a + b)h = \frac{1}{2}(6 + 10)(5) = \frac{1}{2}(16)(5) = 40$ cm^2.

ANSWER **(a) 42 cm^2 (b) 54 m^2 (c) 40 cm^2.**

WORKED EXAMPLE 7.4.3

AREA OF A COMPOSITE SHAPE AND A COST

PROBLEM

The L-shaped courtyard in Worked example 7.4.1 is paved at \$18 per square metre. (a) Find its area. (b) Find the cost of the paving.

1. (a) Subtract the notch from the full rectangle: $10 \times 8 = 80$ m^2, and the notch is $4 \times 3 = 12$ m^2. Area $= 80 - 12 = 68$ m^2.
2. (b) Cost $= 68 \times 18 = \$1\,224$.
3. Check by adding the two remaining rectangles: $10 \times 5 = 50$ and $6 \times 3 = 18$; $50 + 18 = 68$ m^2.

ANSWER **(a) 68 m^2 (b) \$1,224.**

WORKED EXAMPLE 7.4.4 CIRCLES AND SEMICIRCLES

PROBLEM

Use $\pi = 3.14$. (a) Find the circumference and area of a circle with radius 7 cm. (b) Find the perimeter and area of a semicircle with radius 5 m.

1. (a) $C = 2\pi r = 2(3.14)(7) = 43.96$ cm. $A = \pi r^2 = 3.14(7^2) = 3.14(49) = 153.86$ cm^2.
2. (b) The curved edge of a semicircle is half the circumference: $\frac{1}{2}(2\pi r) = \pi r = 3.14(5) = 15.7$ m.
3. The perimeter includes the straight diameter: $P = 15.7 + 10 = 25.7$ m.
4. The area is half the circle: $A = \frac{1}{2}\pi r^2 = \frac{1}{2}(3.14)(25) = 39.25$ m^2.

ANSWER **(a) 43.96 cm; 153.86 cm^2 (b) 25.7 m; 39.25 m^2.**

GUIDED EXAMPLE · A TRAPEZIUM WITH FOUR GIVEN SIDES

Problem. A trapezium has parallel sides 8 cm and 14 cm, height 6 cm, and sloping sides 7 cm and 9 cm. Find its perimeter and area.

Step 1. Perimeter is the distance around all four sides: $8 + 14 + 7 + 9$.

Step 2. For the area, use the parallel sides and the perpendicular height: $A = \frac{1}{2}(8 + 14)(6)$.

Now complete. Calculate both values, include units, and check that the area is between the areas of the two rectangles with the same height and each parallel side as base.

COMMON ERROR

Using a slant side as the height overestimates the area of a parallelogram or trapezium. In a triangle and a trapezium, forgetting the factor $\frac{1}{2}$ doubles the answer. For a circle, the diameter is twice the radius: $d = 18$ cm means $r = 9$ cm, not 18 cm. For a composite shape, the perimeter is the outer boundary only — do not add the line where two parts meet.

PRACTICE 7.4 · PERIMETER AND AREA

SUPPORT 1–8 · CORE 1–14 · EXTENSION 15–16

Write the formula before substituting. Check the unit of every answer: a length in cm or m, an area in cm^2 or m^2. Use $\pi = 3.14$.

1. Find the perimeter: (a) a rectangle 12 cm by 7 cm (b) a square of side 9 cm (c) a triangle with sides 6 cm, 8 cm and 10 cm. F
2. An L-shaped region is 10 m by 8 m with a 4 m by 3 m rectangle cut from a corner. Find its perimeter and area. P
3. Find the area: (a) a rectangle 14 cm by 9 cm (b) a square of side 11 m. F
4. Find the area of a triangle with (a) base 12 cm and height 7 cm (b) base 8.5 m and height 4 m. F
5. Find the area of a parallelogram with (a) base 9 cm and height 6 cm (b) base 12.5 m and height 4.8 m. F
6. Find the area of a trapezium with (a) parallel sides 6 cm and 10 cm, height 5 cm (b) parallel sides 3.5 m and 7.5 m, height 4 m. F
7. Find the circumference and area of a circle with (a) radius 7 cm (b) diameter 18 m. P

8. Find the perimeter and area of a semicircle with radius 5 cm. [P]
9. A rectangular courtyard is 15 m by 10 m. A 6 m by 4 m rectangular shed sits in one corner. Find the paved area and the length of the outer boundary of the paving. [P]
10. Paving costs \$35 per square metre. Find the cost of paving 24 m^2. [P]
11. A student calculates the area of a parallelogram with base 9 cm, slant side 7 cm and height 5 cm as $9 \times 7 = 63$ cm^2. Identify the error and give the correct area. [R]
12. Which expression gives the area of a triangle with base b and height h? A bh B $2bh$ C $\frac{1}{2}bh$ D $b + h$. [P]
13. A circular tabletop has diameter 1.2 m. Find its circumference and area, and the cost of edging it at \$18 per metre. [P]
14. Convert (a) 2.5 m^2 to cm^2 (b) 4\,500 cm^2 to m^2. [P]
15. A rectangle has perimeter 36 cm and length 12 cm. Find its width and area. [E]
16. Two rectangles both have perimeter 20 cm. One measures 1 cm by 9 cm; another measures 5 cm by 5 cm. Find both areas and explain what this shows about perimeter and area. [E]

7.5 Real-life application · The courtyard budget

CRITERION D · II, III ◆ CRITERION C · III ◆ MEASUREMENT

A neighbourhood group has a rectangular courtyard 12 m long and 8 m wide. They will keep a rectangular lawn 6 m by 3.5 m and install a circular fountain of radius 1 m. Everything else will be paved. The paving costs \$45 per square metre, and the outer edge is finished with timber edging at \$22 per metre.

TABLE 7.3 · COURTYARD QUANTITIES

Feature	Dimensions	Quantity to calculate
Courtyard	12 m by 8 m	Total area
Lawn	6 m by 3.5 m	Area (not paved)
Fountain	radius 1 m	Area (not paved); rim length
Paving	remaining region	Area; cost at \$45/m^2
Outer edging	perimeter of courtyard	Length; cost at \$22/m

WORKED EXAMPLE 7.5.1 FROM PLAN TO QUANTITY

PROBLEM

Use Table 7.3. (a) Find the paved area. (b) Find the cost of paving and outer edging before wastage.

1. (a) Courtyard area $= 12 \times 8 = 96\ \text{m}^2$. Lawn area $= 6 \times 3.5 = 21\ \text{m}^2$. Fountain area $= 3.14(1^2) = 3.14\ \text{m}^2$.
2. Paved area $= 96 - 21 - 3.14 = 71.86\ \text{m}^2$.
3. (b) Paving cost $= 71.86 \times 45 = \$3\,233.70$.
4. Outer perimeter $= 2(12 + 8) = 40$ m, so edging costs $40 \times 22 = \$880$.
5. Subtotal $= 3\,233.70 + 880 = \$4\,113.70$.

ANSWER **(a) 71.86 m² (b) paving $3,233.70; edging $880; subtotal $4,113.70.**

Testing the budget

CRITERION D

1. Check the three areas in Worked example 7.5.1 and confirm the paved area by subtracting from the courtyard area.
2. The supplier recommends ordering 8% extra paving for cuts and breakages. Calculate the area to order, rounded to 2 decimal places, and the new paving cost.
3. Using the extra paving, find the new subtotal for paving and outer edging. How much remains from a $4,500 budget?
4. The fountain rim costs $30 per metre. Find the rim length and cost. Does the project still meet the $4,500 budget? By how much is it over or under?
5. Suggest one change that would bring the project within budget. Calculate the effect of your change and explain whether it preserves the design's purpose.
6. State two assumptions and one limitation of the cost model. How could a real supplier's quote change the conclusion?

PRACTICE 7.5 · BUDGET DECISIONS

SUPPORT 1–2 · CORE 1–4 · EXTENSION 5

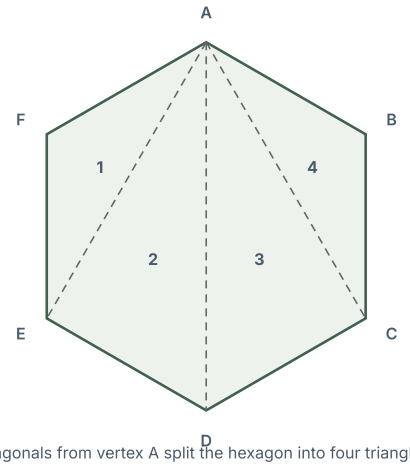
1. A 10 m by 7 m courtyard contains a circular fountain of radius 1.5 m. Find the paved area. [P]
2. Paving costs $40/m². Find the cost for question 1. [P]
3. The 34 m boundary is edged at $25/m. Find the edging cost. [P]
4. With 10% extra paving, find the area to order, the new paving cost and the total including edging. [P]
5. The budget is $3,800. Is it met? Give the difference and one assumption. [E]

7.6 Investigation · Angle sums of polygons

CRITERION A · II ◆ CRITERION C · III ◆ PATTERN

PLAN THE INQUIRY

You already know that a triangle has angle sum $180°$ and a quadrilateral has angle sum $360°$. Does a pentagon have $540°$? **Draw** diagonals from one vertex to split each polygon into triangles. **Record** how many triangles appear. **Generalise** the pattern and **justify** why it must continue for every polygon.



Diagonals from vertex A split the hexagon into four triangles.

Figure 7.7 A hexagon divided into four triangles from one vertex. Each triangle contributes $180°$, so the hexagon's angle sum is $4 \times 180 = 720°$.

WORKED EXAMPLE 7.6.1 USING THE POLYGON FORMULA

PROBLEM

(a) Find the interior angle sum of a 12-sided polygon. (b) Find each interior angle of a regular 12-sided polygon. (c) A polygon has an interior angle sum of $1260°$. How many sides does it have?

1. (a) $S = 180(n - 2) = 180(12 - 2) = 180(10) = 1800°$.
2. (b) A regular polygon has equal angles, so each is $1800 \div 12 = 150°$.
3. (c) Solve $180(n - 2) = 1260$: divide by 180, $n - 2 = 7$, so $n = 9$. The polygon is a nonagon.
4. Check: $180(9 - 2) = 180(7) = 1260°$.

ANSWER **(a)** $1800°$ **(b)** $150°$ **(c) 9 sides.**

From triangles to a general rule

INVESTIGATION

1. Copy and complete the table for polygons with 3 to 8 sides. For each, draw diagonals from one vertex and count the triangles.
   Sides n: 3, 4, 5, 6, 7, 8. Triangles: 1, 2, ?, ?, ?, ?.
2. Describe in words how the number of triangles is related to the number of sides. Write a formula for the number of triangles in an n-gon.
3. Use your formula to write an expression for the angle sum S of an n-gon. Check it for a triangle, a quadrilateral and the hexagon in Figure 7.7.
4. Explain why the diagonals from one vertex divide the polygon into triangles with no gaps and no overlaps. Your explanation should work for any n, not just the examples you drew.
5. Find the angle sum of a 12-sided polygon and a 20-sided polygon. Show the substitution.
6. A **regular** polygon has equal sides and equal angles. Explain why each interior angle of a regular n-gon is $180(n-2) \div n$. Find the interior angle of a regular pentagon, hexagon and decagon.
7. A regular polygon has an interior angle of $150°$. Find its number of sides. Then show that a regular polygon cannot have an interior angle of exactly $100°$.
8. The exterior angle at each vertex is $180°$ minus the interior angle. Find the exterior angle of a regular pentagon, and verify that the five exterior angles add to $360°$. Explain why the exterior angles of any convex polygon add to $360°$.
9. Which regular polygons can tile a flat surface with no gaps? Test the triangle, square, pentagon and hexagon. Explain your answer using the interior angles and the $360°$ around a point.

PRACTICE 7.6 · GENERALISING POLYGON ANGLES

SUPPORT 1–3 · CORE 1–5 · EXTENSION 6

Use $S = 180(n-2)°$ for the interior angle sum. For a regular polygon, divide S by n to find one interior angle, or use $360 \div n$ for one exterior angle.

1. Find the interior angle sum of a 15-sided polygon. F
2. Find each interior angle of a regular octagon. F
3. Find each exterior angle of a regular 12-sided polygon. P
4. A polygon has an interior angle sum of $1620°$. Find its number of sides. P
5. A regular polygon has an exterior angle of $40°$. Find its number of sides and each interior angle. P
6. A student claims that a regular polygon can have an interior angle of $172°$. Find the number of sides this would require and decide whether the claim is possible. E

Criterion task · The community courtyard

ASSESSED CRITERIA
C I–IV · D I–V

A community group is designing a courtyard measuring 14 m by 9 m. The plan includes a right-triangular stage with perpendicular sides 4 m and 3 m and a third side of 5 m, and three circular tree beds each with radius 1.2 m. The rest of the courtyard is paved. Paving costs \$38 per square metre. Edging costs \$15 per metre and runs around the outer boundary, the stage and each tree bed. A 5% contingency is added to the total cost. The budget is \$5,500. Use $\pi = 3.14$.

1. Calculate the area of the courtyard, the stage and all three tree beds. Hence find the paved area, to 2 decimal places. Use $\pi = 3.14$.
2. Calculate the total length of edging: the outer boundary, the stage perimeter and the three tree-bed circumferences. Round to 2 decimal places.
3. Calculate the paving cost and the edging cost. Find the subtotal.
4. Add the 5% contingency to the subtotal. Does the design meet the \$5,500 budget? State the difference.
5. The group considers increasing each tree-bed radius to 1.5 m. Recalculate the paved area, edging length, subtotal and total including contingency. Does this version meet the budget? Explain why enlarging the beds lowers the cost.
6. Write a recommendation to the group. Use your calculations to justify a choice, and state one assumption and one limitation of the model.
7. Check your communication: are all lengths in metres, all areas in square metres, all money rounded to the nearest cent, and every formula visible in your working?

FORMAT STRUCTURED CALCULATIONS AND WRITTEN RECOMMENDATION

TIME 50–60 MINUTES

RESOURCES CALCULATOR PERMITTED

PART	WHAT TO SHOW	STRANDS
1–2	Correct area and perimeter calculations for circles, a triangle and a composite region, with units and reasoned decomposition.	C i–ii, D i–ii
3–4	A correctly composed cost model, contingency applied to the subtotal, and an accurate budget comparison.	C iii–iv, D ii–iii
5–6	A complete alternative calculation, a justified recommendation, and a relevant assumption and limitation.	C iii–iv, D iii–v

7.7 Summary, revision and reflection

KEY IDEAS

— Angles on a straight line add to $180°$; angles at a point add to $360°$; vertically opposite angles are equal.

— When parallel lines are cut by a transversal, corresponding and alternate angles are equal, and co-interior angles add to $180°$.

— The angles of a triangle add to $180°$; an exterior angle equals the sum of the two opposite interior angles. A quadrilateral's angles add to $360°$.

— Perimeter is a length; area is a surface. Composite shapes are handled by adding or subtracting simple parts.

— The angle sum of an n-gon is $180(n-2)°$, because diagonals from one vertex create $n-2$ triangles.

angles on a line: $180°$ · at a point: $360°$

triangle: $A = \frac{1}{2}bh$ · parallelogram: $A = bh$

trapezium: $A = \frac{1}{2}(a+b)h$

circle: $C = 2\pi r$, $A = \pi r^2$

polygon interior sum: $180(n-2)°$

Connections. The angle facts here are used again in Chapter 9, where rotations and reflections preserve angles. The area formulas become surface-area formulas in Chapter 8, and the cost modelling returns whenever a design must be tested against a budget.

REVISION CHECKLIST

- ☐ I can classify angles and use the straight-line and point facts.
- ☐ I can identify corresponding, alternate and co-interior angles.
- ☐ I can find missing angles in triangles and quadrilaterals with reasons.
- ☐ I can use the perimeter and area formulas for standard shapes.
- ☐ I can calculate circumference and area of a circle and a semicircle.
- ☐ I can find the area and perimeter of a composite shape.
- ☐ I can use a measurement model to test a budget.

REFLECTION

Which error would be more costly in the courtyard project — using a slant side as a height, or forgetting the factor $\frac{1}{2}$ in a triangle? Estimate both errors for a 10 m by 6 m triangle and explain your reasoning.

KEY VOCABULARY

vertically opposite
equal angles formed by two intersecting lines

transversal
a line crossing two or more lines

corresponding angles
matching angles at two intersections

alternate angles
equal angles on opposite sides of a transversal

co-interior angles
same-side interior angles; sum to $180°$

polygon
a closed plane shape with straight sides

perimeter
the distance around a shape

area
the surface enclosed by a shape

composite shape
a shape made from simpler shapes

Mixed practice · Consolidate and extend

These questions combine angle reasoning with measurement. Before calculating, decide whether the question asks for an angle, a length or an area; then choose the fact or formula that matches, and keep the units visible.

Fluency and accuracy

1. Classify each angle: $118°$, $90°$, $247°$. F
2. Find the missing angle on a straight line: $133°$ and x. F
3. Find the fourth angle of a quadrilateral with angles $85°, 100°, 95°$. F
4. Find the third angle of a triangle with angles $42°$ and $63°$. F
5. A parallelogram has one angle of $74°$. Find its other three angles. F
6. Find the area of a triangle with base 11 cm and height 6 cm. F
7. Find the circumference and area of a circle with radius 4 m. Use $\pi = 3.14$. F
8. Find the area of a trapezium with parallel sides 5 cm and 9 cm and height 4 cm. F

Reasoning and problem solving

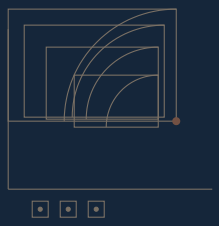
9. Angles $(3x + 10)°$ and $(2x - 5)°$ lie on a straight line. Find x and both angles. P
10. Alternate angles between parallel lines are $(7x - 20)°$ and $(4x + 40)°$. Find x and the angle. P
11. A quadrilateral has angles $(2x + 10)°, (3x - 10)°, (x + 20)°$ and $(2x + 20)°$. Find x and all four angles. P
12. An exterior angle of a triangle is $110°$, and one opposite interior angle is $45°$. Find the other opposite interior angle. P
13. A student says a triangle with angles $80°, 70°$ and $30°$ is impossible because the angles do not add to $360°$. Identify the error and correct the conclusion. R
14. A rectangle is 12 m by 9 m, with a 5 m by 4 m rectangle cut from one corner. Find the area and perimeter of the remaining shape. P
15. A circular garden has diameter 10 m. Find its circumference and area. P
16. Paving 18 m^2 costs \$42.50 per square metre. Find the total cost. P
17. A rectangular fence surrounds a plot 15 m by 9 m. Find the length of fencing and its cost at \$14 per metre. P
18. Two rectangles both have perimeter 20 cm: one is 6 cm by 4 cm, the other 5 cm by 5 cm. Find both areas and explain what the comparison shows. R

Extension and inquiry

19. Find the interior angle sum of a regular 12-sided polygon, and the size of each interior angle. E
20. A regular polygon has an exterior angle of $24°$. Find its number of sides and its interior angle. E
21. The angles of a triangle are in the ratio $2 : 3 : 4$. Find the three angles. E
22. The angles of a quadrilateral are in the ratio $1 : 2 : 3 : 4$. Find the four angles. E
23. A courtyard is 20 m by 15 m with a rectangular lawn 8 m by 6 m. The rest is paved at \$35/m^2, and the outer boundary is edged at \$20/m. Find the paved area, the paving cost, the edging cost and the total. E
24. A student claims that doubling every side of a rectangle doubles its area. Test the claim with a 2 cm by 3 cm rectangle, then explain what really happens to the area and the perimeter. E

ATL · COMMUNICATION

Make the diagram do some of the work. A reader who was absent should be able to rebuild your solution from your diagram and your reasons. Label the vertices, mark the parallel lines and the right angles, write the angle fact beside each step, and carry the units through every calculation. Clear communication is not a finishing touch; it is how a measurement becomes checkable.



8

GEOMETRY AND MEASURE

3D geometry: solids, surface area and volume

A carton protects what it holds; a tank stores what a town needs. The skin of a solid is measured in square units, the space inside in cubic units, and a flat net lets a designer see both before anything is cut, folded or filled.

KEY CONCEPT · FORM

RELATED · SPACE

RELATED · MEASUREMENT

CONTEXT · SCIENTIFIC AND TECHNICAL INNOVATION

INQUIRY

How can a flat sheet of card describe the surface of a solid, and how much space will that solid hold?

MYP UNIT PLANNER

Framing the inquiry

UNIT 8 · APPROX. 12 HOURS
ATL FOCUS · CRITICAL THINKING

KEY AND RELATED CONCEPTS

Form. A solid is described by its faces, edges and vertices, and by the shape that is repeated through its height.

Space. A net unfolds a surface into a plane without stretching it; volume measures the space a solid occupies.

Measurement. Surface area is counted in square units, volume in cubic units; capacity connects the two to litres and millilitres.

GLOBAL CONTEXT AND EXPLORATION

Scientific and technical innovation. Exploration: engineering and design. Packaging and storage ask a designer to protect a product, use material efficiently and hold a stated capacity.

Every carton is a compromise: more material may add strength, but it also adds cost and waste.

STATEMENT OF INQUIRY

The form of a solid determines both the material needed to cover its surface and the space it can enclose; unfolding it into a net or slicing it into a base and height makes those measurements visible and comparable.

FACTUAL

How do faces, edges and vertices describe a solid, and how does a net show its surface?

CONCEPTUAL

Why can two solids with the same volume need different amounts of material, and why is surface area measured in square units but volume in cubic units?

DEBATABLE

When packaging a product, should a designer prioritise the least material, the greatest capacity or the strongest protection?

ATL · CRITICAL THINKING IN ACTION

Unfold the solid, label every edge and check that each face appears once. Before calculating, decide whether the question asks for a length, an area or a volume; after calculating, test the size of the answer against the diagram.

INQUIRY PATHWAY FOR THIS CHAPTER

1	2	3	4	5	6
CONTEXT	**QUESTION**	**CONCEPT**	**INVESTIGATION**	**COMMUNICATION**	**REFLECTION**
A drinks company redesigns its carton.	How much card and how much drink?	Face, net, surface area, volume and capacity.	How much of a carton is empty space?	Labelled net, table of measurements and recommendation.	Which measurement error would change a design decision?

Learning intentions and success criteria

YOU ARE LEARNING TO

- **Understand** the faces, edges and vertices of common solids.
- **Know** how a net unfolds to show surface area.
- **Be able to** calculate the surface area of prisms and cylinders.
- **Be able to** calculate volume and convert between cubic units and capacity.
- **Be able to** decompose a composite solid into simpler parts.
- **Be able to** compare packaging designs using material and capacity.

SUCCESS CRITERIA

- ☐ I can name a solid and count its faces, edges and vertices.
- ☐ I can match a net to its solid and label its dimensions.
- ☐ I can use the formulas for the surface area of a prism and a cylinder.
- ☐ I can use the formulas for the volume of a prism and a cylinder.
- ☐ I can find the volume and surface area of a composite solid.
- ☐ I can justify a packaging decision with measurements and units.

PREREQUISITE CHECK · 10 MINUTES, NO CALCULATOR

1. Calculate the area of a rectangle $7\text{ cm} \times 4\text{ cm}$.
2. Calculate the area of a triangle with base 6 cm and height 5 cm.
3. Calculate the circumference of a circle with radius 3 cm.
4. Calculate 2.5^2.
5. Calculate 4^3.
6. Calculate $3 \times 4 \times 5$.
7. How many square centimetres are in 1 m^2?

Answers are in the Chapter 8 answer section. If more than two are incorrect, review area formulas, square and cube numbers, and metric unit conversions before continuing.

YEAR 1
Name common solids; count faces, edges and vertices; find volume by counting unit cubes.

YEAR 2 · THIS CHAPTER
Nets, surface area and volume of prisms and cylinders, composite solids and capacity.

YEAR 3
Cones, spheres and pyramids; Pythagoras in three dimensions; similarity and scaling.

CRITERION	STRANDS	EVIDENCE IN THIS CHAPTER
A · Knowing and understanding	i, ii, iii	Name solids and their properties; select and use surface area and volume formulas; convert units.
C · Communicating	i, ii, iii, iv	Draw and label nets; use correct square and cubic units; present a measurement table and recommendation.
D · Applying mathematics	i, ii, iii, iv, v	Model a packaging problem; compare designs; evaluate a criterion; state assumptions and limitations.

8.1 Solids, faces and nets

CRITERION A · I ◆ CRITERION C · I ◆ FORM

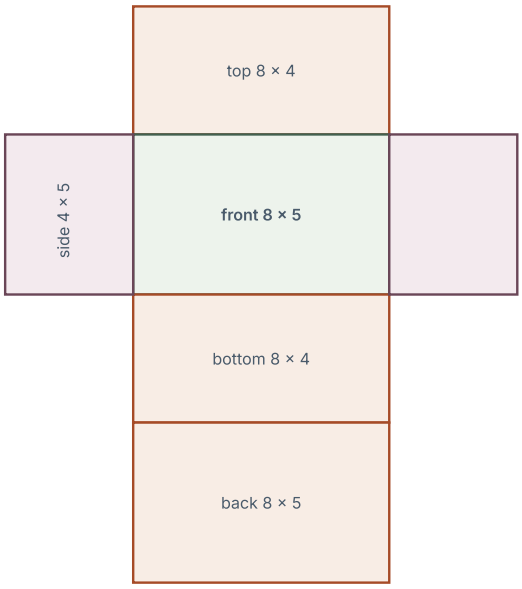
A carton begins life as a flat sheet. Before it is folded, the designer must know which faces will meet, which edges will join and where the tabs will go. A net is the drawing that makes those decisions visible.

DEFINITION · SOLIDS AND THEIR FEATURES

A **solid** is a three-dimensional shape. Its flat surfaces are **faces**; a face meets another face along an **edge**; three or more edges meet at a **vertex**. A **polyhedron** is a solid whose faces are all polygons.

A **prism** has the same polygonal cross-section along its length, so its two end faces are identical and parallel. A **pyramid** has a polygonal base and triangular faces that meet at an apex. A **cylinder** has two parallel circular faces and one curved surface.

A **net** is a flat pattern that folds to make a solid. The **surface area** of a solid is the total area of its faces, measured in square units.



Dashed lines are not shown; all outlines are fold lines.

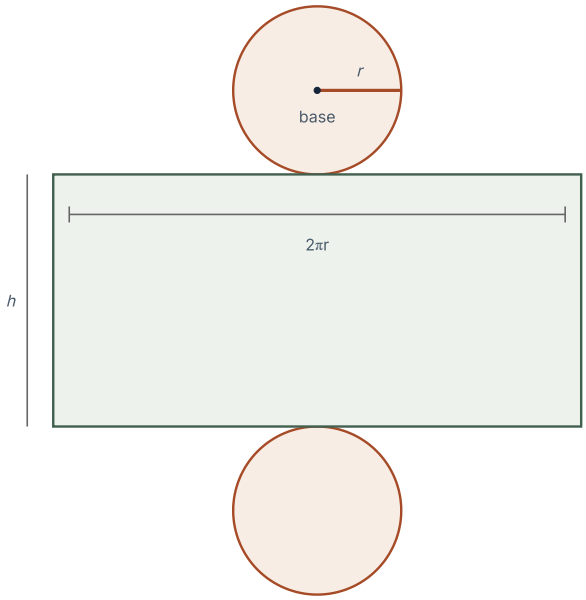
Figure 8.1 A cuboid unfolds into six rectangles. Opposite faces are equal, so the net can be used to calculate the surface area without measuring the solid itself.

TABLE 8.1 · FEATURES OF COMMON SOLIDS

Solid	Faces	Edges	Vertices	Faces in the net
Cuboid	6	12	8	3 pairs of rectangles
Cube	6	12	8	6 equal squares
Triangular prism	5	9	6	3 rectangles and 2 triangles
Square-based pyramid	5	8	5	1 square and 4 triangles
Cylinder	3	2	0	1 rectangle and 2 circles

KEY IDEA · FROM NET TO SURFACE AREA

A net must contain every face exactly once, with no gaps and no overlaps. Folding the net joins some edges, but it does not change any area. Therefore the surface area of a solid equals the total area of its net. For a prism, this means two identical bases plus the rectangles that wrap around them.



The curved surface unrolls to a rectangle 2πr long and h high.

Figure 8.2 The curved surface of a cylinder unrolls into a rectangle. Its length is the circumference $2\pi r$, and its height is the height of the cylinder.

WORKED EXAMPLE 8.1.1 — AREA OF EVERY FACE

PROBLEM

A cuboid measures 8 cm by 4 cm by 5 cm. Use its net to find the area of each face and the total surface area.

1. The two 8×4 faces each have area $32\ \text{cm}^2$, so together they have area $64\ \text{cm}^2$.
2. The two 8×5 faces each have area $40\ \text{cm}^2$, so together they have area $80\ \text{cm}^2$.
3. The two 4×5 faces each have area $20\ \text{cm}^2$, so together they have area $40\ \text{cm}^2$.
4. Total surface area $= 64 + 80 + 40 = 184\ \text{cm}^2$.

ANSWER $184\ \text{cm}^2$**, which is the area of the net.**

WORKED EXAMPLE 8.1.2 — FROM FEATURES TO SOLID

PROBLEM

A polyhedron has 5 faces, 9 edges and 6 vertices. Two of its faces are triangles and three are rectangles. Name the solid and describe its net.

1. The two triangular faces are identical and parallel, so they are the bases of a prism.
2. The three rectangular faces join corresponding edges of the triangles, so the solid is a triangular prism.
3. The net contains a row of three rectangles with one triangle attached above the row and the other below it, on different rectangles. Each rectangle has one side equal to a side of the triangle.
4. Check the counts: faces $= 2 + 3 = 5$, edges $= 3 + 3 + 3 = 9$, vertices $= 3 + 3 = 6$.

ANSWER **A triangular prism; its net has 3 rectangles in a row with a triangle attached to the top and bottom of one rectangle.**

GUIDED EXAMPLE · READING A NET

Problem. A net consists of one rectangle $18.84\ \text{cm}$ long and $10\ \text{cm}$ high, with a circle attached above and below. Find the radius of each circle and identify the solid.

Step 1. The length of the rectangle is the circumference of the circular base, so $2\pi r = 18.84$.

Step 2. Using $\pi = 3.14$, $6.28r = 18.84$, so $r = 3\ \text{cm}$.

Now complete. The net folds into a cylinder with radius 3 cm and height 10 cm. Calculate the area of the two circles and the rectangle, then add them to find the surface area.

COMMON ERROR

The curved surface of a cylinder is one face, not a rectangle in the solid; it only becomes a rectangle when the cylinder is unrolled. A cylinder has two circular faces and one curved face, two circular edges and no vertices. When counting edges, do not count the joins in the net: a join is not an edge. Always check that a net contains every face exactly once; a cube net with five squares, or with two squares covering the same position, cannot close.

PRACTICE 8.1 · SOLIDS AND NETS

SUPPORT 1–4 · CORE 1–8 · EXTENSION 9–10

Use $\pi = 3.14$ where needed. When a net is described in words, sketch it and label the dimensions before calculating.

1. Name the solid: (a) 6 rectangular faces, 12 edges and 8 vertices; (b) 2 circular faces and 1 curved surface; (c) 5 faces, 2 triangular and 3 rectangular. F
2. Count the faces, edges and vertices of a cube. F
3. Count the faces, edges and vertices of a square-based pyramid. F
4. A prism has a pentagonal base. Find its number of faces, edges and vertices. P
5. Explain why six squares in a single straight row cannot fold into a cube. R
6. A net has 4 identical rectangles and 2 identical squares. Name a solid it could make. F
7. Sketch a net of a cuboid measuring 6 cm by 3 cm by 2 cm. Label each rectangle with its dimensions. P
8. A cube has a net with total area $54\ \text{cm}^2$. Find the side length of the cube and its surface area. P
9. A cylinder net has a rectangle $31.4\ \text{cm}$ by $12\ \text{cm}$ and two circles with diameter 10 cm. Find the radius, the height and the total area of the net. P
10. A student says a cylinder has 2 faces because "the curved part is not flat". Explain why this is not the standard count, and state the number of faces, edges and vertices. E

8.2 Surface area of prisms

CRITERION A · II ◆ CRITERION C · II ◆ MEASUREMENT

A prism has two identical bases and a set of rectangular faces joining them. Its surface area is therefore the area of two bases plus the area of the wrap that goes around the sides.

> **KEY IDEA · THE PRISM FORMULA**
>
> For a prism with base area B, base perimeter P and length ℓ between the bases, the surface area is $SA = 2B + P\ell$. The $P\ell$ term is the area of the rectangular faces: unrolled, they form one rectangle whose length is the perimeter of the base.
>
> For a cuboid with length l, width w and height h, $SA = 2(lw + lh + wh)$. For a cube with side s, $SA = 6s^2$.

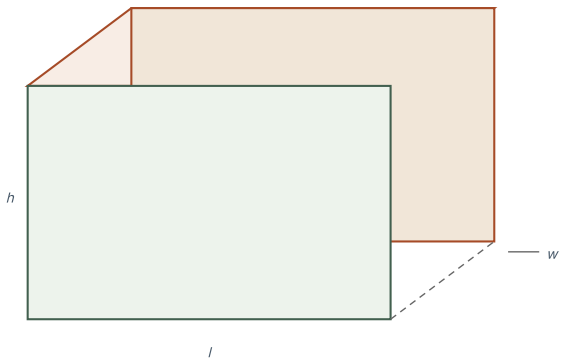


Figure 8.3 A cuboid has three pairs of identical rectangular faces. The formula counts each pair once.

WORKED EXAMPLE 8.2.1 — A CLOSED CUBOID

PROBLEM

Find the surface area of a closed cuboid measuring 12 cm by 7 cm by 5 cm.

1. The three different face areas are $12 \times 7 = 84$, $12 \times 5 = 60$ and $7 \times 5 = 35$ cm^2.
2. Each area occurs twice: $SA = 2(84 + 60 + 35)$.
3. $SA = 2(179) = 358$ cm^2.

ANSWER 358 cm^2.

WORKED EXAMPLE 8.2.2 — A TRIANGULAR PRISM

PROBLEM

A triangular prism has a right-angled triangular base with sides 6 cm, 8 cm and 10 cm. The length of the prism is 15 cm. Find its surface area.

1. Base area $B = \frac{1}{2}(6)(8) = 24$ cm^2.
2. Base perimeter $P = 6 + 8 + 10 = 24$ cm.
3. The two bases contribute $2B = 48$ cm^2. The rectangular faces contribute $P\ell = 24 \times 15 = 360$ cm^2.
4. $SA = 48 + 360 = 408$ cm^2.

ANSWER 408 cm^2.

WORKED EXAMPLE 8.2.3 CUBE: AREA AND EDGING

PROBLEM

A display block is a cube with edge 9 cm. (a) Find its surface area. (b) Find the total length of edging needed to cover all 12 edges.

1. (a) $SA = 6s^2 = 6(9^2) = 6(81) = 486\ \text{cm}^2$.
2. (b) A cube has 12 edges, each 9 cm: $12 \times 9 = 108\ \text{cm}$.
3. The answers have different units because surface area measures a surface and edging measures a length.

ANSWER **(a)** $486\ \text{cm}^2$ **(b)** $108\ \text{cm}$.

COMMON ERROR

Do not confuse the perimeter of a face with its area. The $P\ell$ term uses the perimeter of the base, not the area. Forgetting the factor 2 leaves a prism open at both ends. When a box is open at the top, subtract the area of the missing face from the closed surface area.

PRACTICE 8.2 · SURFACE AREA OF PRISMS

SUPPORT 1–4 · CORE 1–8 · EXTENSION 9–10

Draw the net or list the faces before substituting. State whether the solid is closed or open, and give every area in square units.

1. Find the surface area of a cuboid measuring 10 cm by 4 cm by 6 cm. F
2. Find the surface area of a cube with edge 7 cm. F
3. A triangular prism has base sides 5 cm, 12 cm and 13 cm and length 20 cm. The base is right-angled. Find its surface area. P
4. Find the surface area of a cuboid measuring 2.5 m by 1.2 m by 0.8 m. P
5. A closed box measures 30 cm by 20 cm by 15 cm. Cardboard is ordered with 10% extra for waste. Find the area of cardboard to order. P
6. A tent is a triangular prism with a floor. Its triangular ends have base 2.4 m and height 1.6 m, and the two sloping sides are 2 m each. The tent is 3.5 m long. Find the area of canvas used if the floor is not canvas. R
7. An open-top box measures 40 cm by 25 cm by 20 cm. Find the area of card needed. How much more card would a closed box need? P
8. A cuboid measures 1.5 m by 0.8 m by 0.6 m. Paint covers $12\ \text{m}^2$ per litre. How many litres are needed for one coat, and what is the cost at \$18 per litre? P
9. Two cuboids both have volume $160\ \text{cm}^3$: one is 4 cm by 4 cm by 10 cm, the other 8 cm by 2 cm by 10 cm. Find both surface areas and explain what the comparison shows. R
10. A student calculates the surface area of a 5 cm by 3 cm by 2 cm box as $5 \times 3 \times 2 = 30\ \text{cm}^2$. Identify the error, calculate the correct surface area, and explain what the student's calculation actually found. R

8.3 Surface area of cylinders

A can is a cylinder. To make its label, a printer needs the area of the curved surface only; to make the can itself, the manufacturer needs the two circular ends as well. The net separates these decisions.

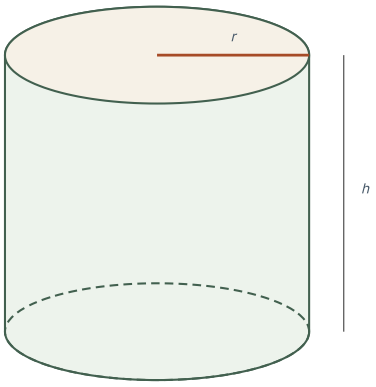


Figure 8.4 A cylinder has two circular faces and one curved surface. The curved surface becomes a rectangle when unrolled.

KEY IDEA · THE CYLINDER FORMULA

The two circular bases have total area $2\pi r^2$. The curved surface unrolls into a rectangle of length $2\pi r$ and height h, so its area is $2\pi rh$. For a closed cylinder, $SA = 2\pi r^2 + 2\pi rh$.

For an open-top cylinder, use one base: $SA = \pi r^2 + 2\pi rh$. For a pipe with no ends, use the curved surface only: $SA = 2\pi rh$.

WORKED EXAMPLE 8.3.1 A CLOSED CYLINDER

PROBLEM

Find the surface area of a closed cylinder with radius 4 cm and height 10 cm. Use $\pi = 3.14$.

1. Area of two bases $= 2\pi r^2 = 2(3.14)(4^2) = 2(3.14)(16) = 100.48$ cm^2.
2. Curved area $= 2\pi rh = 2(3.14)(4)(10) = 251.2$ cm^2.
3. $SA = 100.48 + 251.2 = 351.68$ cm^2.

ANSWER 351.68 cm^2.

WORKED EXAMPLE 8.3.2 — AN OPEN-TOP DRUM

PROBLEM

An open-top cylindrical drum has radius 1.5 m and height 2 m. (a) Find the area to paint on the outside, including the base but not the missing top. (b) Paint costs \$8.50 per square metre. Find the cost.

1. (a) Curved area $= 2(3.14)(1.5)(2) = 18.84\ \text{m}^2$.
2. Base area $= (3.14)(1.5^2) = (3.14)(2.25) = 7.065\ \text{m}^2$.
3. Area to paint $= 18.84 + 7.065 = 25.905\ \text{m}^2 \approx 25.91\ \text{m}^2$.
4. (b) Cost $= 25.905 \times 8.50 = 220.1925$, which rounds to \$220.19.

ANSWER **(a)** $25.91\ \text{m}^2$ **(to 2 d.p.)** **(b) \$220.19.**

WORKED EXAMPLE 8.3.3 — WORKING BACKWARDS

PROBLEM

A closed cylinder has radius 5 cm and surface area $471\ \text{cm}^2$. Find its height. Use $\pi = 3.14$.

1. The two bases have area $2(3.14)(5^2) = 157\ \text{cm}^2$.
2. The curved area is therefore $471 - 157 = 314\ \text{cm}^2$.
3. Solve $2\pi rh = 314$: $2(3.14)(5)h = 31.4h = 314$, so $h = 10$ cm.

ANSWER 10 cm.

COMMON ERROR

Use the radius, not the diameter, in $2\pi rh$. If a question gives the diameter, halve it first. The curved area is $2\pi rh$, not $\pi r^2 h$: the second expression is the volume. Check whether the cylinder is closed, open at one end, or a pipe with no ends before choosing the formula.

PRACTICE 8.3 · SURFACE AREA OF CYLINDERS

SUPPORT 1–4 · CORE 1–8 · EXTENSION 9

Use $\pi = 3.14$. State whether you are including 0, 1 or 2 circular ends, and keep units consistent.

1. Find the surface area of a closed cylinder with radius 3 cm and height 7 cm. F
2. A closed cylinder has diameter 10 cm and height 12 cm. Find its surface area. P
3. An open-top can has radius 4 cm and height 9 cm. Find the area of metal needed. P
4. A metal pipe has radius 0.5 m and length 8 m. Find the area of its curved surface only. P
5. A label covers the curved surface of a can with radius 3.5 cm and height 11 cm. Find the area of the label. P
6. A closed cylinder has surface area $628\ \text{cm}^2$ and radius 5 cm. Find its height. P
7. Cylinder A has radius 3 cm and height 4 cm. Cylinder B has radius 6 cm and height 1 cm. Find both volumes and both surface areas, then explain what the comparison shows. R
8. A closed storage tank has radius 1.2 m and height 2 m. Its outside is painted at \$7 per square metre. Find the paint cost. P

9. A student writes $SA = 2\pi r^2 + 2\pi dh$ for a closed cylinder of diameter d. Explain what is wrong and write the correct formula in terms of d. E

8.4 Volume of prisms and cylinders

CRITERION A · II ◆ CRITERION D · I ◆ MEASUREMENT

Surface area asks how much material covers a solid; volume asks how much space is inside it. A carton can use little cardboard and still hold a lot of drink, which is exactly why designers care about both measurements.

KEY IDEA · VOLUME AS BASE × HEIGHT

Any prism or cylinder can be sliced into thin identical layers. Each layer has the area of the base, so the volume is $V = B\ell$, where B is the base area and ℓ is the distance between the bases.

For a cuboid, $V = lwh$. For a cylinder, $V = \pi r^2 h$. Volume is measured in cubic units. Capacity is the volume a container can hold: $1\ \text{cm}^3 = 1\ \text{mL}$, $1000\ \text{cm}^3 = 1\ \text{L}$, and $1\ \text{m}^3 = 1000\ \text{L}$.

WORKED EXAMPLE 8.4.1 — A CUBOID

PROBLEM

Find the volume of a cuboid measuring 12 cm by 7 cm by 5 cm.

1. The base area is $12 \times 7 = 84\ \text{cm}^2$.
2. The height is 5 cm, so $V = 84 \times 5 = 420\ \text{cm}^3$.
3. Equivalently, $V = 12 \times 7 \times 5 = 420\ \text{cm}^3$.

ANSWER $420\ \text{cm}^3$.

WORKED EXAMPLE 8.4.2 — A TRIANGULAR PRISM

PROBLEM

A triangular prism has base area $24\ \text{cm}^2$ and length 15 cm. Find its volume.

1. The cross-section is the same along the length, so use $V = B\ell$.
2. $V = 24 \times 15 = 360\ \text{cm}^3$.
3. The base's side lengths are not needed once its area is known.

ANSWER $360\ \text{cm}^3$.

WORKED EXAMPLE 8.4.3 — A CYLINDER

PROBLEM

Find the volume of a cylinder with radius 4 cm and height 10 cm. Use $\pi = 3.14$.

1. Base area $= \pi r^2 = 3.14(16) = 50.24$ cm^2.
2. $V = 50.24 \times 10 = 502.4$ cm^3.

ANSWER 502.4 cm^3.

WORKED EXAMPLE 8.4.4 — CAPACITY IN LITRES

PROBLEM

A cylindrical water tank has radius 1.2 m and height 2.5 m. Find its capacity in litres. Use $\pi = 3.14$.

1. $V = \pi r^2 h = 3.14(1.2^2)(2.5) = 3.14(1.44)(2.5)$.
2. $1.44 \times 2.5 = 3.6$, so $V = 3.14 \times 3.6 = 11.304$ m^3.
3. 1 m$^3 = 1000$ L, so the capacity is $11\,304$ L.

ANSWER 11.304 m$^3 = 11\,304$ L.

COMMON ERROR

Surface area and volume are not interchangeable. A formula with r^2 and no height gives an area, not a volume; a formula with r^3 would be needed for a sphere, which is not part of this chapter. Volume units are cubed: writing 420 cm^2 for a volume is wrong. When converting, square the length factor for area but cube it for volume: 1 m $= 100$ cm, so 1 m$^3 = 1\,000\,000$ cm^3.

PRACTICE 8.4 · VOLUME AND CAPACITY

SUPPORT 1–4 · CORE 1–8 · EXTENSION 9–10

Use $\pi = 3.14$. Give every volume in cubic units and every capacity in litres or millilitres where asked.

1. Find the volume of a cuboid measuring 9 cm by 4 cm by 6 cm. [F]
2. Find the volume of a cube with edge 5 cm. [F]
3. A prism has base area 30 cm^2 and length 12 cm. Find its volume. [F]
4. Find the volume of a cylinder with radius 3 cm and height 8 cm. [P]
5. A cylinder has diameter 10 cm and height 14 cm. Find its volume. [P]
6. A carton measures 20 cm by 8 cm by 6 cm. It is labelled “1 litre”. Find its volume in millilitres and decide whether the label is accurate. [R]
7. A rectangular tank measures 2 m by 1.5 m by 1.2 m. Find its capacity in litres. [P]
8. A cylindrical tank has radius 1.5 m and height 3 m. Find its capacity in litres. [P]
9. A prism has volume 240 cm^3 and base area 30 cm^2. Find its length. [P]
10. A box measures 20 cm by 10 cm by 10 cm. A cylinder has radius 6 cm and height 18 cm. Find both volumes and both surface areas. Which container holds more, and which uses less material? [E]

8.5 Composite solids and packaging

CRITERION D · II ◆ CRITERION C · III ◆ FORM

Few products are packed in a single neat cuboid. A composite solid is made from two or more simpler solids joined together; the designer must add the volumes but subtract the faces that are hidden where the parts meet.

KEY IDEA · JOINING SOLIDS

To find the volume of a composite solid, split it into non-overlapping prisms or cylinders, calculate each volume and add. To find the surface area, add the surface areas of the parts and subtract twice the area of every contact face, because each contact hides one face on each part.

When a solid is open, or when one face sits on the ground, that face is not part of the exposed surface.

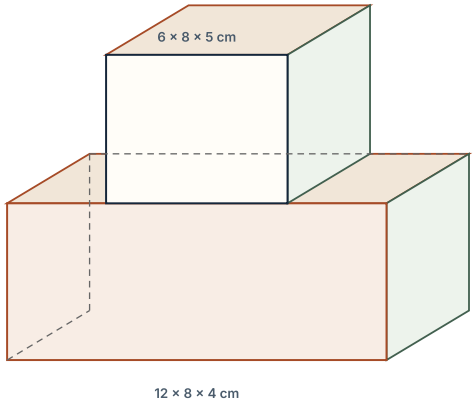


Figure 8.5 The top prism covers part of the top face of the base prism. The hidden contact area is not part of the exposed surface.

WORKED EXAMPLE 8.5.1 A STEP SOLID

PROBLEM

A solid is made from a base prism $12\text{ cm} \times 8\text{ cm} \times 4\text{ cm}$ and a top prism $6\text{ cm} \times 8\text{ cm} \times 5\text{ cm}$ placed on the base. Find (a) the volume and (b) the exposed surface area.

1. (a) Base volume $= 12 \times 8 \times 4 = 384\text{ cm}^3$. Top volume $= 6 \times 8 \times 5 = 240\text{ cm}^3$. Total $= 624\text{ cm}^3$.
2. (b) Base surface area $= 2(96 + 48 + 32) = 352\text{ cm}^2$. Top surface area $= 2(48 + 30 + 40) = 236\text{ cm}^2$.
3. The contact rectangle is $6 \times 8 = 48\text{ cm}^2$. It is hidden on both solids, so subtract $2 \times 48 = 96\text{ cm}^2$.
4. Exposed surface area $= 352 + 236 - 96 = 492\text{ cm}^2$.

ANSWER **(a)** 624 cm^3 **(b)** 492 cm^2.

CONTEXT · A CARTON OF CANS

A drinks company packs 12 cylindrical cans, each with radius 3.5 cm and height 11 cm, in a rectangular carton. The cans stand upright in a 4×3 arrangement, so the carton measures 28 cm by 21 cm by 11 cm. The company wants to know how much cardboard it uses and how much of the carton is empty space.

TABLE 8.2 · CARTON AND CAN MEASUREMENTS

Item	Dimensions	What is needed
One can	radius 3.5 cm, height 11 cm	Volume; 12 cans
Carton	28 cm by 21 cm by 11 cm	Volume; surface area
Empty space	carton volume minus can volume	Volume; percentage of carton
Cardboard	surface area of carton	Cost at \$2.80/m^2

WORKED EXAMPLE 8.5.2 CARDBOARD COST

PROBLEM

Find the surface area of the carton in Table 8.2, and the cardboard cost for 1000 cartons at \$2.80 per square metre.

1. $SA = 2(28 \times 21 + 28 \times 11 + 21 \times 11) = 2(588 + 308 + 231) = 2(1127) = 2254$ cm^2.
2. Convert: 2254 cm$^2 = 0.2254$ m^2.
3. One carton costs $0.2254 \times 2.80 = 0.63112$, or about 63 cents.
4. 1000 cartons cost $0.63112 \times 1000 = \$631.12$.

ANSWER 2254 cm^2; **\$0.63 per carton (to the nearest cent); \$631.12 for 1000 cartons.**

How much of the carton is air?

CRITERION D

1. Calculate the volume of one can and the volume of 12 cans. Use $\pi = 3.14$.
2. Calculate the volume of the carton and the volume of empty space. What percentage of the carton is empty, to 1 decimal place?
3. Calculate the surface area of the carton and the cardboard cost for 1000 cartons.
4. A second layout packs the cans in a 6×2 arrangement. Find the new carton dimensions if each can still has diameter 7 cm and height 11 cm, and calculate its surface area.
5. Which layout uses less cardboard? Find the difference in surface area and in cost for 1000 cartons.
6. Give one reason a manufacturer might not choose the layout with the least cardboard.
7. Suggest one change that would reduce the empty space without changing the can size. Explain its effect on the carton.
8. State two assumptions and one limitation of this packaging model.

PRACTICE 8.5 · COMPOSITE SOLIDS AND PACKAGING

SUPPORT 1–3 · CORE 1–6 · EXTENSION 7–8

Sketch the parts separately, mark the contact faces, and decide which faces are exposed before you calculate.

1. A solid is made from a $10 \times 6 \times 3$ cm base prism and a $4 \times 6 \times 4$ cm top prism. Find its volume and exposed surface area. [P]
2. A closed rectangular tank measures 2 m by 1.5 m by 1 m. Find its surface area and the cost of painting it at \$9.50 per square metre. [P]
3. A closed cylinder has radius 0.8 m and height 1.5 m. Find its volume in litres and its surface area. [P]
4. Six cans of radius 3.5 cm and height 11 cm are packed in a 3×2 layer in a carton 21 cm by 14 cm by 11 cm. Find the volume of the six cans, the carton volume and the percentage of empty space. [R]
5. A composite solid is made from two prisms. The first has base area $24\ \text{cm}^2$ and height 5 cm. The total volume is $500\ \text{cm}^3$. Find the volume of the second prism. [P]
6. Explain why the surface area of a composite solid is not simply the sum of the surface areas of its parts. Use a numerical example. [R]
7. A cube has edge 3 cm. A second cube has edge 6 cm. Find both surface areas and both volumes, and describe how each measurement has changed. [E]
8. A carton has surface area $2400\ \text{cm}^2$. A designer reduces every dimension by 10%. Estimate the new surface area and explain whether the reduction is also 10%. [E]

Criterion task · The water tank

ASSESSED CRITERIA
C I–IV · D I–V

A community needs a water tank. Design A is a closed cylinder with radius 1.2 m and height 2.5 m, standing on a square concrete base 3 m by 3 m by 0.15 m. The tank's curved surface and top are painted, but its base is not. The concrete base is painted on its top and four sides, but not underneath. Paint costs \$6.50 per square metre. Concrete costs \$180 per cubic metre. The budget for paint and concrete is \$500. Use $\pi = 3.14$.

1. Find the area of the tank that is painted: the curved surface and the top.
2. Find the exposed area of the concrete base and the total area to paint. Add the tank and base areas.
3. Calculate the paint cost.
4. Calculate the tank's capacity in litres. Find the volume of concrete and its cost, then the total cost of paint and concrete.
5. Design B is a closed cuboid 2.4 m by 2.4 m by 2 m on the same concrete base. Calculate its capacity in litres, its painted area (top and four sides), the exposed area of the base, and its total paint and concrete cost.
6. Compare the two designs. Which holds more water, which costs less, and which would you recommend? Justify your choice using your calculations.
7. State one assumption and one limitation of the model. Check that every length is in metres, every area in square metres, every volume in cubic metres or litres, and every cost is rounded to the nearest cent.

FORMAT STRUCTURED CALCULATIONS AND WRITTEN RECOMMENDATION

TIME 50–60 MINUTES

RESOURCES CALCULATOR PERMITTED

PART	WHAT TO SHOW	STRANDS
1–3	Correct cylinder and composite surface areas, with the unpainted base and hidden contact excluded, and accurate costs.	C i–ii, D i–ii
4	Accurate capacity conversion and concrete volume, with units carried through.	C iii–iv, D ii–iii
5–7	A complete alternative design, a justified comparison, a relevant assumption and limitation.	C iii–iv, D iii–v

8.6 Summary, revision and reflection

KEY IDEAS

- A solid has faces, edges and vertices. A prism has identical parallel bases; a cylinder has two circular faces and one curved surface.
- A net is a flat pattern that folds into a solid. Its area equals the surface area, so every face must appear exactly once.
- For a prism, $SA = 2B + P\ell$. For a cuboid, $SA = 2(lw + lh + wh)$; for a cube, $SA = 6s^2$.
- For a cylinder, $SA = 2\pi r^2 + 2\pi rh$, with fewer bases if the cylinder is open or a pipe.
- Volume is base area times height: $V = B\ell$, $V = lwh$ or $V = \pi r^2 h$. Capacity converts to litres.
- For composite solids, add volumes and subtract twice each contact area from the surface area.

cuboid: $SA = 2(lw + lh + wh) \cdot V = lwh$

prism: $SA = 2B + P\ell \cdot V = B\ell$

cylinder: $SA = 2\pi r^2 + 2\pi rh$

cylinder volume: $V = \pi r^2 h$

capacity: $1000\ \text{cm}^3 = 1\ \text{L} \cdot 1\ \text{m}^3 = 1000\ \text{L}$

Connections. The area formulas from Chapter 7 become the bases of the volume formulas here. Chapter 9 uses coordinates and transformations to describe shape, while Year 3 extends these formulas to cones, spheres and pyramids.

REVISION CHECKLIST

- ☐ I can name a solid and count faces, edges and vertices.
- ☐ I can match a net to a solid and label its dimensions.
- ☐ I can calculate the surface area of a cuboid, prism and cylinder.
- ☐ I can calculate volume and convert to litres or millilitres.
- ☐ I can decide whether a problem needs area, surface area or volume.
- ☐ I can find the volume and exposed surface area of a composite solid.
- ☐ I can compare packaging designs and justify a choice.

REFLECTION

A box doubles in length, width and height. Does its surface area double? Does its volume double? Explain both answers with a 2 cm cube and a 4 cm cube, and describe one design decision this makes possible.

KEY VOCABULARY

face
a flat surface of a solid

edge
where two faces meet

vertex
where three or more edges meet

prism
a solid with identical parallel bases

pyramid
a solid with a polygonal base and an apex

net
a flat pattern that folds into a solid

surface area
the total area of all faces

volume
the space inside a solid

capacity
the volume a container can hold

Mixed practice · Consolidate and extend

Decide first whether the question asks for a length, an area, a volume or a capacity. Draw the net or the separate parts, keep the units visible, and check each answer against the size of the object.

Fluency and accuracy

1. Name the solid with 6 square faces. F
2. Count the faces, edges and vertices of a triangular prism. F
3. Find the surface area of a cube with edge 8 cm. F
4. Find the surface area of a cuboid measuring 9 cm by 5 cm by 4 cm. F
5. Find the volume of a cuboid measuring 7 cm by 6 cm by 3 cm. F
6. Find the volume of a cylinder with radius 2 cm and height 9 cm. F
7. Find the surface area of a closed cylinder with radius 3 cm and height 5 cm. F
8. Convert 2.5 litres to cubic centimetres. F

Reasoning and problem solving

9. A cuboid has volume $240\ \text{cm}^3$, length 8 cm and width 5 cm. Find its height and surface area. P
10. A closed cylinder has volume $628\ \text{cm}^3$ and radius 5 cm. Find its height and surface area. P
11. An open-top box measures 50 cm by 30 cm by 20 cm. Find the area of card needed. P
12. A cube has surface area $294\ \text{cm}^2$. Find its edge length and volume. P
13. A cylindrical column has radius 1.5 m and height 4 m. Its curved surface is painted at \$6 per square metre. Find the cost. P
14. Two cuboids have volume $48\ \text{cm}^3$: one is 6 cm by 4 cm by 2 cm, the other 8 cm by 3 cm by 2 cm. Find both surface areas and compare them. R
15. A rectangular tank measures 1.2 m by 0.8 m by 0.5 m. Find its capacity in litres. P
16. A cylindrical tank has diameter 2 m and height 3 m. Find its capacity in litres. P
17. A student says that doubling every dimension of a solid doubles its volume. Identify the error and correct it using a cube of edge 3 cm. R
18. A composite solid is made from a $5 \times 4 \times 3$ cm prism and a $2 \times 4 \times 2$ cm prism. Find its volume and exposed surface area. P
19. A cylinder net has two circles of radius 4 cm and a rectangle 25.12 cm by 10 cm. Name the solid and find the total area of the net. P
20. Two cuboids both have volume $1000\ \text{cm}^3$: one is $10 \times 10 \times 10$ cm, the other $20 \times 5 \times 10$ cm. Find both surface areas and explain what the comparison shows. R

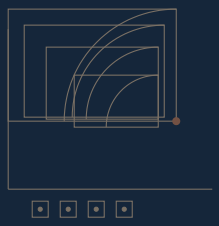
Extension and inquiry

21. A box has volume $720\ \text{cm}^3$, length 12 cm and width 10 cm. Find its height and surface area. E
22. A designer claims that a box with a larger volume always needs more cardboard. Test the claim with a $10 \times 10 \times 10$ cm cube and a $20 \times 20 \times 2$ cm box. Find both volumes and both surface areas, then explain your conclusion. E
23. A cube's volume increases from $27\ \text{cm}^3$ to $216\ \text{cm}^3$. Find both side lengths and both surface areas, and describe the change in each measurement. E
24. A cylinder's radius doubles while its height halves. Find the change in volume and the change in surface area if the original radius is 3 cm and the original height is 8 cm. E

25. A water tank is a cylinder of radius 1 m and height 3 m standing on a cuboid base 2 m by 2 m by 1 m. Find the total volume in litres and explain why the surface area cannot be found by simply adding the two surface areas. E

26. Return to the carton investigation. The company wants to reduce the cardboard used by 10% without changing the number of cans. Propose one change, calculate its effect, and state one reason it might fail in practice. E

> **ATL · CRITICAL THINKING**
>
> A diagram is a claim about which faces are exposed. Before you add areas, mark every hidden face; before you compare volumes, check that the units match. Explaining why a design wins is more convincing than reporting a single number.



9

GEOMETRY AND MEASURE

Transformations and symmetry

A motif is repeated across a banner, a dance formation shifts across a stage and a tile turns to fill a wall. Each move can be described exactly, and that description is what lets a design be reproduced, checked and explained.

KEY CONCEPT · FORM

RELATED · SPACE

RELATED · CHANGE

CONTEXT · PERSONAL AND CULTURAL EXPRESSION

INQUIRY

How can a shape be moved, flipped, turned or resized so that the result can be described exactly and reproduced by someone else?

MYP UNIT PLANNER

Framing the inquiry

UNIT 9 · APPROX. 12 HOURS
ATL FOCUS · COMMUNICATION

KEY AND RELATED CONCEPTS

Form. A transformation is defined by its rule: a vector, a mirror line, a centre and angle, or a centre and scale factor. The rule determines the image.

Space. Coordinates locate a shape in the plane, and a transformation moves it while preserving selected properties.

Change. Position, size and orientation can change; length, angle or shape may stay the same.

GLOBAL CONTEXT AND EXPLORATION

Personal and cultural expression. Exploration: design, pattern and aesthetic expression. Banners, textiles, logos and building façades repeat motifs to create rhythm and balance.

A designer chooses a transformation for its effect: a reflection may create balance, a rotation may suggest movement, and an enlargement may create emphasis.

STATEMENT OF INQUIRY

Describing a transformation precisely — by vector, mirror line, centre and angle, or scale factor — makes a design reproducible and shows which properties have been preserved, changed or lost.

FACTUAL

What information is needed to describe a translation, reflection, rotation or enlargement completely?

CONCEPTUAL

Which properties are preserved by each transformation, and how does the description determine the image?

DEBATABLE

When does symmetry strengthen a design, and when does breaking symmetry communicate more?

ATL · COMMUNICATION IN ACTION

Write each transformation as an instruction that another student can follow without seeing your diagram. Name the original points and their images, state the vector, mirror line, centre and angle, or centre and scale factor, and check that the instruction produces the same image twice.

INQUIRY PATHWAY FOR THIS CHAPTER

1	2	3	4	5	6
CONTEXT	**QUESTION**	**CONCEPT**	**INVESTIGATION**	**COMMUNICATION**	**REFLECTION**
A community arts centre designs a banner.	How can a motif be moved and repeated exactly?	Form, congruence, similarity and symmetry.	Which shapes tile a surface without gaps?	Precise instructions, labelled diagrams and coordinates.	Which properties survive each transformation?

Learning intentions and success criteria

YOU ARE LEARNING TO

- **Understand** coordinates as a precise description of position in the plane.
- **Know** how to translate, reflect, rotate and enlarge a shape on a grid.
- **Be able to** describe a transformation completely and unambiguously.
- **Be able to** identify line symmetry and rotational symmetry.
- **Be able to** decide whether two shapes are congruent or similar.
- **Be able to** explain how transformations are used in a repeating pattern.

SUCCESS CRITERIA

- ☐ I can plot and read coordinates in all four quadrants.
- ☐ I can translate a shape using a vector and name the image coordinates.
- ☐ I can reflect a shape in a given horizontal, vertical or diagonal line.
- ☐ I can rotate a shape about a given centre by 90°, 180° or 270°.
- ☐ I can enlarge a shape by a positive scale factor and find the area change.
- ☐ I can justify whether a pair of shapes is congruent or similar.

PREREQUISITE CHECK · 10 MINUTES, NO CALCULATOR

1. Plot the point $(3, -2)$. Which quadrant contains it?
2. Start at $(1, 3)$. Move 4 right and 2 down. What are the new coordinates?
3. Reflect the point $(4, 2)$ in the x-axis.
4. Multiply each coordinate of $(2, -3)$ by 3.
5. How many degrees are in a quarter turn?
6. A square has side 5 cm. Find its area.
7. Solve $2x = 18$.

Answers are in the Chapter 9 answer section. If more than two are incorrect, review plotting points and the four quadrants before continuing.

YEAR 1
Plot points in the first quadrant; recognise line symmetry in simple shapes and letters.

YEAR 2 · THIS CHAPTER
Translations, reflections, rotations and enlargements on the coordinate plane; symmetry and tessellation.

YEAR 3
Negative scale factors; combined transformations; similarity and trigonometric ratios.

CRITERION	STRANDS	EVIDENCE IN THIS CHAPTER
A · Knowing and understanding	i, ii, iii	Identify and perform transformations; use coordinates; apply symmetry and scale-factor facts.
B · Investigating patterns	i, ii, iii	Investigate which regular polygons tessellate and generalise the angle condition.
C · Communicating	i, ii, iii, iv	Describe transformations precisely; label images; present a design with coordinates and justification.
D · Applying mathematics	i, ii, iii, iv, v	Design a banner from a motif; evaluate congruence, symmetry and cost; state limitations.

9.1 Coordinates and translations

CRITERION A · I ◆ CRITERION C · I ◆ SPACE

A coordinate pair is an address. It fixes one point exactly, so a movement can be recorded as a change to that address rather than as a vague instruction to “move it across”.

DEFINITION · COORDINATES AND TRANSFORMATIONS

The first coordinate gives the horizontal position and the second gives the vertical position: (x, y). The four regions formed by the axes are the **quadrants**, numbered anticlockwise from the top right.

A **transformation** moves or changes a shape. The original shape is the **object**; the result is the **image**. A **translation** slides every point of a shape the same distance in the same direction. A **vector** records that movement: $\begin{pmatrix} 3 \\ -4 \end{pmatrix}$ means 3 right and 4 down.

A translation does not turn, flip or resize a shape. Every side keeps its length and direction, every angle keeps its size, and the image is congruent to the object. Only the position changes.

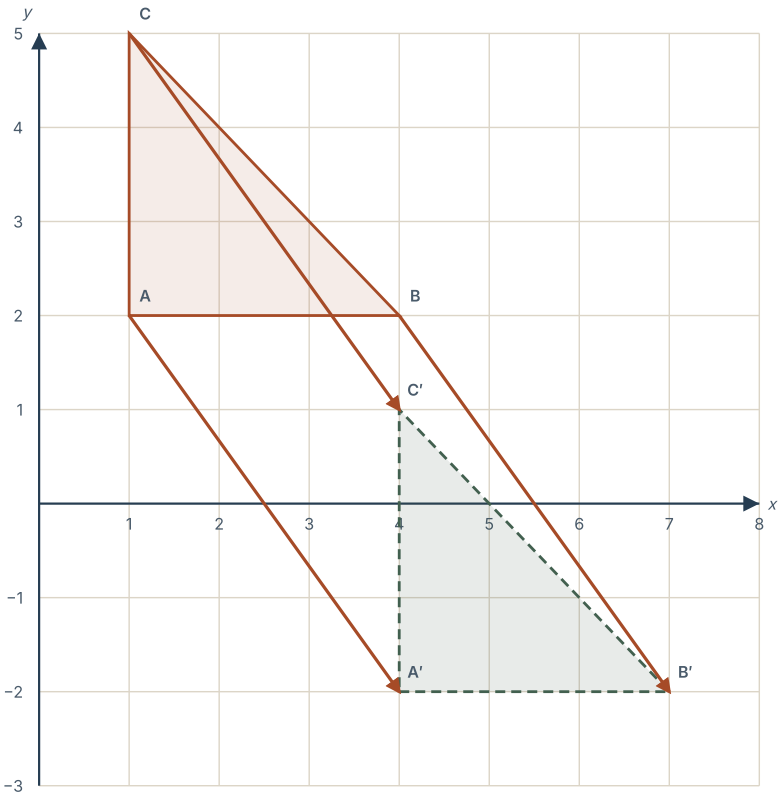


Figure 9.1 Translating triangle ABC by $\begin{pmatrix} 3 \\ -4 \end{pmatrix}$. Each vertex moves 3 right and 4 down, so the image has the same size and orientation.

METHOD · TRANSLATING A SHAPE

Step 1. Write the vector as a column: top number right or left, bottom number up or down.

Step 2. Add the top number to every x-coordinate and the bottom number to every y-coordinate.

Step 3. Plot the image vertices, join them in the same order, and label them with primes: A', B', C'.

Step 4. Check one pair: the arrow from a vertex to its image must equal the vector.

WORKED EXAMPLE 9.1.1

FROM VERTICES TO IMAGE

PROBLEM

Triangle ABC has vertices $A(1, 2)$, $B(4, 2)$ and $C(1, 5)$. Translate it by $\begin{pmatrix} 3 \\ -4 \end{pmatrix}$ and state the image coordinates.

1. Add 3 to each x-coordinate: $1 + 3 = 4$, $4 + 3 = 7$, $1 + 3 = 4$.
2. Add -4 to each y-coordinate: $2 - 4 = -2$, $2 - 4 = -2$, $5 - 4 = 1$.
3. Combine the results: $A'(4, -2)$, $B'(7, -2)$, $C'(4, 1)$.
4. Check: from $A(1, 2)$ to $A'(4, -2)$ the movement is 3 right and 4 down, which matches $\begin{pmatrix} 3 \\ -4 \end{pmatrix}$.

ANSWER $A'(4, -2)$, $B'(7, -2)$, $C'(4, 1)$. **The image is congruent to ABC.**

WORKED EXAMPLE 9.1.2

FINDING THE VECTOR

PROBLEM

A rectangle has vertices $P(-3, 1)$, $Q(1, 1)$, $R(1, 3)$ and $S(-3, 3)$. Its image is $P'(2, 5)$, $Q'(6, 5)$, $R'(6, 7)$, $S'(2, 7)$. Describe the translation.

1. Subtract the coordinates of a vertex from its image: $2 - (-3) = 5$ and $5 - 1 = 4$.
2. The vector is $\begin{pmatrix} 5 \\ 4 \end{pmatrix}$: 5 right and 4 up.
3. Check with a second vertex: $Q(1, 1) \to Q'(6, 5)$, and $6 - 1 = 5$, $5 - 1 = 4$.
4. Every vertex gives the same vector, which confirms a translation rather than a rotation or reflection.

ANSWER **The translation is $\begin{pmatrix} 5 \\ 4 \end{pmatrix}$.**

WORKED EXAMPLE 9.1.3 — TWO TRANSLATIONS

PROBLEM

A point is translated by $\begin{pmatrix} -2 \\ 5 \end{pmatrix}$ and then by $\begin{pmatrix} 6 \\ -1 \end{pmatrix}$. Find a single vector that has the same effect, and test it on the point $(3, -2)$.

1. Add the horizontal movements: $-2 + 6 = 4$.
2. Add the vertical movements: $5 + (-1) = 4$.
3. The combined vector is $\begin{pmatrix} 4 \\ 4 \end{pmatrix}$.
4. Test: $(3, -2) \to (1, 3) \to (7, 2)$. Using the single vector: $(3, -2) + (4, 4) = (7, 2)$.

ANSWER $\begin{pmatrix} 4 \\ 4 \end{pmatrix}$**; both routes give** $(7, 2)$**.**

GUIDED EXAMPLE · TRANSLATING A TRIANGLE

Problem. Triangle DEF has vertices $D(0, 0)$, $E(2, 0)$ and $F(0, 3)$. Translate it by $\begin{pmatrix} 2 \\ -1 \end{pmatrix}$.

Step 1. Add 2 to each x-coordinate: $0 + 2 = 2$, $2 + 2 = 4$, $0 + 2 = 2$.

Step 2. Add -1 to each y-coordinate: $0 - 1 = -1$, $0 - 1 = -1$, $3 - 1 = 2$.

Now complete. Write the image coordinates, calculate the area of the object and the area of the image, and explain why the two areas are equal.

COMMON ERROR

Do not change only one coordinate, and do not subtract the vector automatically. The vector $\begin{pmatrix} -2 \\ 3 \end{pmatrix}$ means 2 *left* and 3 *up*: $(x, y) \to (x - 2, y + 3)$. A quick check with one vertex catches most sign errors before you plot the whole image.

PRACTICE 9.1 · COORDINATES AND TRANSLATIONS

SUPPORT 1–4 · CORE 1–8 · EXTENSION 9–10

Write the vector before you calculate. Check one vertex against the vector after plotting.

1. Translate $A(2, 5)$ by $\begin{pmatrix} 3 \\ -7 \end{pmatrix}$. F
2. Translate triangle $(1, 1)$, $(4, 1)$, $(1, 2)$ by $\begin{pmatrix} -2 \\ 3 \end{pmatrix}$. F
3. Find the image of $(5, -3)$ under the translation $\begin{pmatrix} -4 \\ -2 \end{pmatrix}$. F
4. $A(-2, 4)$ maps to $A'(3, -1)$. Find the translation vector. P
5. A shape has image $(2, 1)$ after translation by $\begin{pmatrix} 3 \\ -4 \end{pmatrix}$. Find the object point. P
6. Translate the point $(4, -1)$ first by $\begin{pmatrix} 1 \\ 2 \end{pmatrix}$, then by $\begin{pmatrix} -3 \\ 5 \end{pmatrix}$. State the final point and the single equivalent vector. P
7. A warehouse robot moves from $(2, 3)$ to $(8, -1)$. Find the vector and describe the movement in words. P

8. A student translates $(3, 4)$ by $\begin{pmatrix} 2 \\ -5 \end{pmatrix}$ and writes $(5, 9)$. Identify the error and give the correct image. R

9. Two translations $\begin{pmatrix} a \\ b \end{pmatrix}$ and $\begin{pmatrix} c \\ d \end{pmatrix}$ are performed in either order. Show with a point that the result is always $\begin{pmatrix} a+c \\ b+d \end{pmatrix}$. E

10. A shape returns to its starting position after two translations. What is the relationship between the two vectors? Give an example and explain. E

9.2 Reflection and line symmetry

CRITERION A · I, II ◆ CRITERION C · II ◆ FORM

A reflection flips a shape across a line. The line is the mirror: every point and its image are the same distance from it, and the segment joining them crosses the mirror at right angles.

DEFINITION · REFLECTION

A **reflection** maps every point to an image on the opposite side of a **mirror line**, so that the mirror line is the perpendicular bisector of the segment joining the point to its image. A reflection preserves lengths and angles but reverses orientation, so the image is congruent to the object.

A shape has **line symmetry** if a mirror line divides it into two halves that coincide when folded. That line is an **axis of symmetry**.

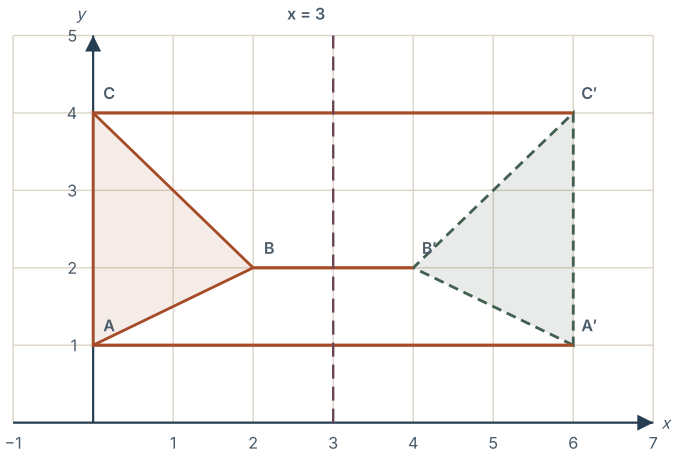


Figure 9.2 Reflecting ABC in the line $x = 3$. Each pair of corresponding vertices is the same distance from the mirror, and the joining segment is perpendicular to it.

KEY IDEA · REFLECTION RULES

For a point (x, y):

in the x-axis: $(x, -y)$ · in the y-axis: $(-x, y)$ · in $y = x$: (y, x) · in $y = -x$: $(-y, -x)$.

In a vertical line $x = a$: $(2a - x, y)$. In a horizontal line $y = b$: $(x, 2b - y)$. The rule works because the mirror line is exactly halfway between the point and its image.

WORKED EXAMPLE 9.2.1 — REFLECTING IN THE AXES

PROBLEM

Triangle PQR has vertices $P(2, 1)$, $Q(5, 1)$ and $R(2, 4)$. Reflect it in the x-axis, then in the y-axis.

1. In the x-axis, keep x and change the sign of y: $P'(2, -1)$, $Q'(5, -1)$, $R'(2, -4)$.
2. In the y-axis, change the sign of x and keep y: $P''(-2, 1)$, $Q''(-5, 1)$, $R''(-2, 4)$.
3. Both images are congruent to PQR. The first image is below the x-axis; the second is to the left of the y-axis.

ANSWER **x-axis:** $(2, -1), (5, -1), (2, -4)$. **$y$-axis:** $(-2, 1), (-5, 1), (-2, 4)$.

WORKED EXAMPLE 9.2.2 — REFLECTING IN X = −2

PROBLEM

Quadrilateral $ABCD$ has vertices $A(-4, 1)$, $B(-1, 1)$, $C(-1, 3)$ and $D(-4, 3)$. Reflect it in the line $x = -2$.

1. Use $x' = 2a - x$ with $a = -2$: $x' = -4 - x$.
2. A: $-4 - (-4) = 0$, so $A'(0, 1)$. B: $-4 - (-1) = -3$, so $B'(-3, 1)$.
3. C: $-4 - (-1) = -3$, so $C'(-3, 3)$. D: $-4 - (-4) = 0$, so $D'(0, 3)$.
4. The y-coordinates do not change because the mirror is vertical. The image lies on the other side of $x = -2$, at the same distances.

ANSWER $(0, 1), (-3, 1), (-3, 3), (0, 3)$.

WORKED EXAMPLE 9.2.3 — REFLECTING IN Y = X

PROBLEM

Reflect the triangle with vertices $A(1, 3)$, $B(4, 0)$ and $C(2, -2)$ in the line $y = x$.

1. In $y = x$, swap the coordinates: $(x, y) \to (y, x)$.
2. $A(1, 3) \to A'(3, 1)$. $B(4, 0) \to B'(0, 4)$. $C(2, -2) \to C'(-2, 2)$.
3. Points on the line $y = x$, such as $(0, 0)$ and $(2, 2)$, do not move. They are the fixed points of the reflection.
4. Check one distance: $A(1, 3)$ and $A'(3, 1)$ are both $\sqrt{2}$ units from the line, on opposite sides.

ANSWER $A'(3, 1)$, $B'(0, 4)$, $C'(-2, 2)$.

WORKED EXAMPLE 9.2.4 — FINDING THE MIRROR LINE

PROBLEM

$A(1, 2)$ reflects to $A'(5, 2)$. Find the mirror line and explain how you know.

1. The joining segment is horizontal because both y-coordinates are 2.
2. The mirror line is perpendicular to it, so the mirror is vertical.
3. The midpoint is $\left(\frac{1+5}{2}, 2\right) = (3, 2)$. A vertical mirror through this midpoint is $x = 3$.
4. Check: $2(3) - 1 = 5$, so the rule produces A'.

ANSWER **The mirror line is $x = 3$.**

TABLE 9.1 · AXES OF SYMMETRY OF COMMON SHAPES

Shape	Number of axes	Where they lie
Isosceles triangle	1	Through the apex and the midpoint of the base
Equilateral triangle	3	Through each vertex and the midpoint of the opposite side
Rectangle	2	Through the midpoints of opposite sides; the diagonals are not axes
Square	4	Two through midpoints of opposite sides, two along the diagonals
Regular pentagon	5	Through each vertex and the midpoint of the opposite side
Circle	Infinite	Every line through the centre

COMMON ERROR

Reflecting in the x-axis does *not* mean "change the x-coordinate". The x-axis is horizontal, so it flips the vertical direction: $(x, y) \to (x, -y)$. A second error is placing the image at the same distance from the axis but on the same side; measure from the mirror, not from the other shape.

PRACTICE 9.2 · REFLECTIONS AND SYMMETRY

SUPPORT 1–4 · CORE 1–8 · EXTENSION 9–12

State the rule you are using before substituting coordinates.

1. Reflect $(3, 4)$ in the x-axis. F
2. Reflect $(-2, 5)$ in the y-axis. F
3. Reflect $(4, -1)$ in the line $y = x$. F
4. Reflect $(6, -2)$ in the line $y = -x$. F
5. Reflect $(3, 1)$ in the line $x = 2$. P
6. Reflect $(-1, 4)$ in the line $y = 3$. P
7. Reflect the triangle $(1, 1), (4, 1), (1, 3)$ in the y-axis and state the image coordinates. P
8. $P(2, 1)$ reflects to $P'(6, 1)$. Find the mirror line. R

9. A student reflects $(2, 5)$ in the x-axis and writes $(-2, 5)$. Identify the error and give the correct image. R

10. How many axes of symmetry does each shape have: a square, a rectangle, an equilateral triangle and a regular hexagon? F

11. A motif is 5 cm wide and its inner edge lies 3 cm from a vertical mirror. How wide is the finished logo? Explain your calculation. P

12. Reflect a point in $x = 1$, then reflect the image in $x = 4$. Show that the combined effect is a translation. Find its vector. E

9.3 Rotation and rotational symmetry

CRITERION A · I ◆ CRITERION C · III ◆ CHANGE

A rotation turns a shape about a fixed point. The point matters as much as the angle: turning a shape about its corner and turning it about the origin produce different images.

DEFINITION · ROTATION

A **rotation** turns a shape about a **centre of rotation** through a given angle, either clockwise or anticlockwise. Every point moves along a circular arc centred on the fixed point, and every point stays the same distance from the centre.

A rotation preserves lengths and angles and does not reverse orientation, so the image is congruent to the object. A shape has **rotational symmetry** if a rotation of less than $360°$ about its centre maps it onto itself. The **order** is the number of positions that fit; the angle is $360° \div$ order.

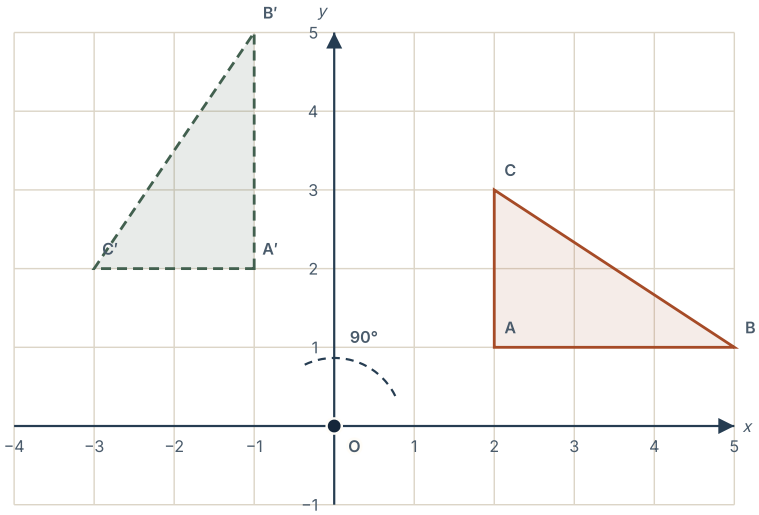


Figure 9.3 A $90°$ anticlockwise rotation about the origin. The dashed arc marks the $90°$ angle at the centre, and A and A' are the same distance from O.

KEY IDEA · ROTATIONS ABOUT THE ORIGIN

For a point (x, y):

$90°$ anticlockwise: $(-y, x)$ · $90°$ clockwise: $(y, -x)$ · $180°$: $(-x, -y)$.

A $270°$ clockwise turn is the same as $90°$ anticlockwise. For a centre that is not the origin, translate the centre to the origin, apply the rule, then translate back; on a grid, tracing paper does the same job.

WORKED EXAMPLE 9.3.1 — 90° ANTICLOCKWISE

PROBLEM

Rotate triangle ABC with $A(2, 1)$, $B(5, 1)$ and $C(2, 3)$ through $90°$ anticlockwise about the origin.

1. Apply $(x, y) \to (-y, x)$ to each vertex.
2. $A(2, 1) \to A'(-1, 2)$. $B(5, 1) \to B'(-1, 5)$. $C(2, 3) \to C'(-3, 2)$.
3. Check distances from the origin: $OA = \sqrt{2^2 + 1^2} = \sqrt{5}$ and $OA' = \sqrt{(-1)^2 + 2^2} = \sqrt{5}$.
4. The image has the same side lengths as the object; only its orientation in the plane has changed.

ANSWER $A'(-1, 2)$, $B'(-1, 5)$, $C'(-3, 2)$.

WORKED EXAMPLE 9.3.2 — 90° CLOCKWISE

PROBLEM

Rotate the quadrilateral $P(-2, 1)$, $Q(1, 1)$, $R(1, 3)$, $S(-2, 3)$ through $90°$ clockwise about the origin.

1. Apply $(x, y) \to (y, -x)$.
2. $P(-2, 1) \to P'(1, 2)$. $Q(1, 1) \to Q'(1, -1)$.
3. $R(1, 3) \to R'(3, -1)$. $S(-2, 3) \to S'(3, 2)$.
4. The rectangle is now taller than it is wide: its 3-unit side lies vertically. A rotation changes orientation but not side lengths.

ANSWER $P'(1, 2)$, $Q'(1, -1)$, $R'(3, -1)$, $S'(3, 2)$.

WORKED EXAMPLE 9.3.3 — 180° ABOUT (2, 2)

PROBLEM

Rotate the triangle with vertices $(1, 1)$, $(4, 1)$ and $(1, 2)$ through $180°$ about the point $(2, 2)$.

1. A $180°$ rotation about (a, b) sends (x, y) to $(2a - x, 2b - y)$.
2. With $(a, b) = (2, 2)$: $(x, y) \to (4 - x, 4 - y)$.
3. $(1, 1) \to (3, 3)$; $(4, 1) \to (0, 3)$; $(1, 2) \to (3, 2)$.
4. Check the midpoint of each point and its image is $(2, 2)$, as a half-turn requires.

ANSWER $(3, 3)$, $(0, 3)$, $(3, 2)$.

TABLE 9.2 · ROTATIONAL SYMMETRY

Shape	Order	Angle
Rectangle (not a square)	2	180°
Equilateral triangle	3	120°
Square	4	90°
Regular pentagon	5	72°
Regular hexagon	6	60°

COMMON ERROR

Check the direction: $(y, -x)$ is 90° *clockwise*, while $(-y, x)$ is 90° *anticlockwise*. A 90° turn is a quarter turn, not a half turn. When the centre is not the origin, do not rotate the coordinates about $(0, 0)$; measure every point from the stated centre.

PRACTICE 9.3 · ROTATIONS

SUPPORT 1–4 · CORE 1–8 · EXTENSION 9–10

Write the centre, angle and direction before applying a rule.

1. Rotate $(3, 1)$ through 90° anticlockwise about the origin. F
2. Rotate $(-2, 4)$ through 90° clockwise about the origin. F
3. Rotate $(5, -2)$ through 180° about the origin. F
4. Rotate the triangle $(1, 0), (3, 0), (1, 2)$ through 90° clockwise about the origin. P
5. Rotate $(4, 2)$ through 90° anticlockwise about $(1, 1)$. P
6. Rotate $(2, 3)$ through 180° about $(4, 1)$. P
7. State the order and angle of rotational symmetry of a regular octagon. F
8. A shape has rotational symmetry of order 5. Find its angle and explain whether it can also have line symmetry. R
9. A student rotates a point 90° clockwise and uses $(-y, x)$. Identify the error and give the correct image of $(3, -1)$. R
10. Two 90° anticlockwise rotations about the origin are performed in succession. Show that the result is a 180° rotation, using the point $(2, 1)$. E

9.4 Enlargement and similarity

CRITERION A · II, III ◆ CRITERION C · IV ◆ CHANGE

A photograph can be printed larger without distortion because every length is multiplied by the same factor. An enlargement does the same thing to a shape: angles stay the same, and every length changes by one scale factor.

DEFINITION · ENLARGEMENT

An **enlargement** maps a shape to a similar image using a **centre of enlargement** and a **scale factor** k. Each image point lies on the ray from the centre through its object point, at k times the distance from the centre.

If $k > 1$ the image is larger; if k lies between 0 and 1 it is smaller. Angles are unchanged, lengths are multiplied by k, and areas are multiplied by k^2. The image is **similar** to the object, not congruent unless $k = 1$.

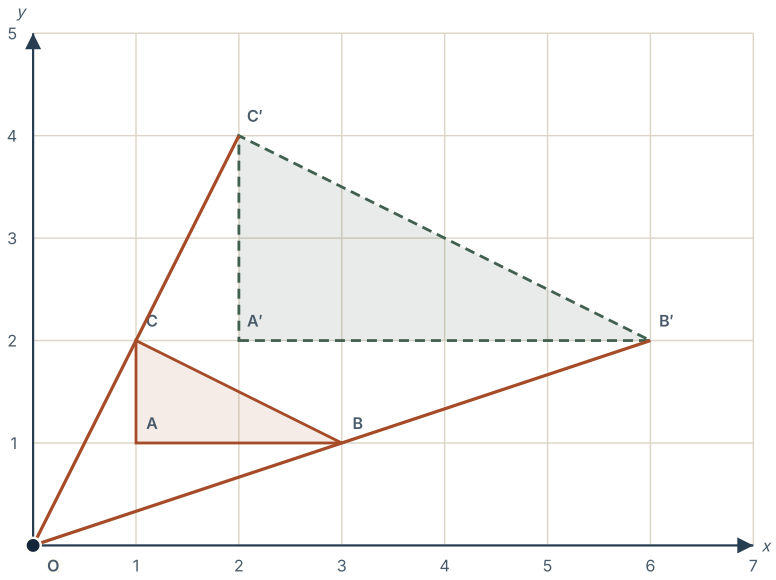


Figure 9.4 Enlarging ABC by scale factor 2 about the origin. The rays from O pass through each object point and its image, and every image length is twice the corresponding object length.

KEY IDEA · COORDINATES AND AREA

For a centre at the origin, $(x, y) \to (kx, ky)$. For a centre $C(a, b)$, subtract C, multiply by k, then add C: $P' = C + k(P - C)$.

Lengths and perimeters scale by k; areas scale by k^2. A scale factor of 3 makes a rectangle nine times larger in area, not three times.

WORKED EXAMPLE 9.4.1 SCALE FACTOR 3 FROM O

PROBLEM

Enlarge the triangle $A(1, 1)$, $B(3, 1)$, $C(1, 2)$ by scale factor 3 about the origin. Find the image coordinates and compare the areas.

1. Multiply each coordinate by 3: $A'(3, 3)$, $B'(9, 3)$, $C'(3, 6)$.
2. Object area: $\frac{1}{2} \times 2 \times 1 = 1$ square unit.
3. Image area: $\frac{1}{2} \times 6 \times 3 = 9$ square units.
4. The ratio of areas is $9 = 3^2$, which matches the square of the scale factor.

ANSWER $A'(3, 3)$, $B'(9, 3)$, $C'(3, 6)$; **area multiplied by 9.**

WORKED EXAMPLE 9.4.2 — A FRACTIONAL SCALE FACTOR

PROBLEM

Enlarge the rectangle with vertices $(4,2)$, $(8,2)$, $(8,6)$, $(4,6)$ by scale factor $\frac{1}{2}$ about the origin.

1. Multiply each coordinate by $\frac{1}{2}$: $(2,1)$, $(4,1)$, $(4,3)$, $(2,3)$.
2. The object is 4 by 4; the image is 2 by 2.
3. The image area is one quarter of the object area because $\left(\frac{1}{2}\right)^2 = \frac{1}{4}$.
4. The image is similar to the object: all angles remain 90° and all lengths are halved.

ANSWER $(2,1)$, $(4,1)$, $(4,3)$, $(2,3)$; **area multiplied by** $\frac{1}{4}$.

WORKED EXAMPLE 9.4.3 — CENTRE (1, 1)

PROBLEM

Enlarge the triangle $P(3,1)$, $Q(3,3)$, $R(1,3)$ by scale factor 2 about the centre $(1,1)$.

1. For P: the vector from the centre is $(2,0)$. Multiplying by 2 gives $(4,0)$, so $P' = (1+4, 1+0) = (5,1)$.
2. For Q: $(2,2) \to (4,4)$, so $Q' = (5,5)$.
3. For R: $(0,2) \to (0,4)$, so $R' = (1,5)$.
4. The side PR lay on the vertical line $x = 1$, which passes through the centre, so its image also lies on $x = 1$.

ANSWER $P'(5,1)$, $Q'(5,5)$, $R'(1,5)$.

WORKED EXAMPLE 9.4.4 — AREA AFTER ENLARGEMENT

PROBLEM

A logo has area $6\ \text{cm}^2$. It is enlarged by scale factor 3. Find the new area, and the new perimeter if the original perimeter is 10 cm.

1. Lengths scale by $k = 3$, so the perimeter becomes $10 \times 3 = 30$ cm.
2. Areas scale by $k^2 = 9$, so the area becomes $6 \times 9 = 54\ \text{cm}^2$.
3. Check: the area is not $18\ \text{cm}^2$. Multiplying the area by 3 would only enlarge one dimension.

ANSWER **Area** $54\ \text{cm}^2$; **perimeter** 30 cm.

COMMON ERROR

When the centre is not the origin, multiplying the coordinates directly gives the wrong image. For centre $(1,1)$, the point $(3,1)$ enlarges to $(5,1)$, not $(6,2)$. Always measure the distance from the stated centre. Also check whether the question asks for a length (scale by k) or an area (scale by k^2).

PRACTICE 9.4 · ENLARGEMENT

SUPPORT 1–4 · CORE 1–8 · EXTENSION 9–11

For each question, state the centre and scale factor before calculating.

1. Enlarge $(2, 3)$ by scale factor 4 about the origin. F
2. Enlarge $(-3, 2)$ by scale factor 2 about the origin. F
3. Enlarge $(6, -4)$ by scale factor $\frac{1}{2}$ about the origin. F
4. Enlarge the triangle $(1, 1), (2, 1), (1, 3)$ by scale factor 2 about the origin. P
5. Enlarge the rectangle $(2, 2), (6, 2), (6, 4), (2, 4)$ by scale factor $\frac{1}{2}$ about the origin. P
6. Enlarge $(3, 4)$ by scale factor 3 about the centre $(2, 2)$. P
7. A shape with area $5\ \text{cm}^2$ is enlarged by scale factor 4. Find the image area. P
8. $(2, 3)$ enlarges to $(6, 9)$ about the origin. Find the scale factor. R
9. A student says the area triples when the scale factor is 3. Identify the error and give the correct area factor. R
10. A photograph measuring 6 cm by 4 cm is enlarged to 18 cm by 12 cm. Find the scale factor and the ratio of the areas. P
11. A logo is enlarged by scale factor 1.5. Its original perimeter is 24 cm and its area is $32\ \text{cm}^2$. Find the new perimeter and area, and explain why the area does not increase by 50%. E

9.5 Investigation · Symmetry and tessellation

CRITERION B · I, II, III ◆ CRITERION D · II ◆ PATTERN

A tiled floor, a honeycomb and a brick wall all cover a surface without gaps or overlaps. The mathematics behind these patterns is an angle calculation: the angles that meet at each point must add to $360°$.

DEFINITION · TESSELLATION

A **tessellation** is a repeating pattern of shapes that covers a surface with no gaps and no overlaps. The shapes are **tiles**. A tessellation is **regular** when it uses congruent regular polygons of one kind; it is **semi-regular** when more than one kind of regular polygon is used but the arrangement at every vertex is the same.

TABLE 9.3 · CAN A REGULAR POLYGON TILE BY ITSELF?

Regular polygon	Interior angle	$360° \div$ angle	Tessellates?
Triangle	$60°$	6	Yes
Square	$90°$	4	Yes
Pentagon	$108°$	$3.33\ldots$	No
Hexagon	$120°$	3	Yes
Octagon	$135°$	$2.67\ldots$	No
Decagon	$144°$	2.5	No

The table suggests a rule: a regular polygon tiles by itself exactly when $360°$ is a whole-number multiple of its interior angle. Three hexagons meet at each vertex because $3 \times 120° = 360°$; three pentagons give only $324°$, leaving a $36°$ gap.

Investigation · Which shapes tile, and how can a motif be repeated?

Pairs · 30 minutes

Plan. Test the angle condition, then build a tiling from a motif using transformations. Record every instruction so that another pair can reproduce your pattern.

1. Copy and complete Table 9.3 for a regular nonagon (9 sides) and a regular dodecagon (12 sides). Use the interior-angle formula $\frac{(n-2)\times 180^\circ}{n}$.
2. Which of the eight polygons in your table tessellate by themselves? Explain using the 360° condition.
3. Three regular pentagons meet at a point. Calculate the gap and explain why a fourth pentagon cannot fit without overlapping.
4. Two regular octagons and one square meet at a point. Show that the angles fit exactly, and explain why this is a semi-regular tessellation.
5. Draw a scalene triangle on squared paper. Rotate a copy through 180° about the midpoint of one side and join it to the original. Repeat around the boundary. Does the pattern tessellate? Explain.
6. A honeycomb uses regular hexagons. Show that three hexagons meet at each vertex, and explain why the pattern has no gaps or overlaps.
7. A student claims that only regular polygons can tessellate. Test the claim with a scalene triangle and a parallelogram, then write a corrected statement.
8. Design a motif with at least one line of symmetry and one rotational symmetry. Describe three transformations that repeat it across a strip, and state the order of rotational symmetry of the finished strip (if it has one).
9. Evaluate your design: does it cover the strip without gaps, is the repetition exact, and which transformation is hardest to describe precisely? Suggest one improvement.

Criterion task · The community banner

ASSESSED CRITERIA
A I–III · C I–IV · D I–V

A community arts centre is making a rectangular banner for a festival. The design grid is 12 units wide and 8 units high; each unit represents 15 cm, so the banner is 1.8 m by 1.2 m. The original motif is a right triangle with vertices $A(1, 1)$, $B(3, 1)$ and $C(1, 3)$. The designer will keep the original motif and produce four more copies, each by a different transformation; the fourth copy is an enlargement.

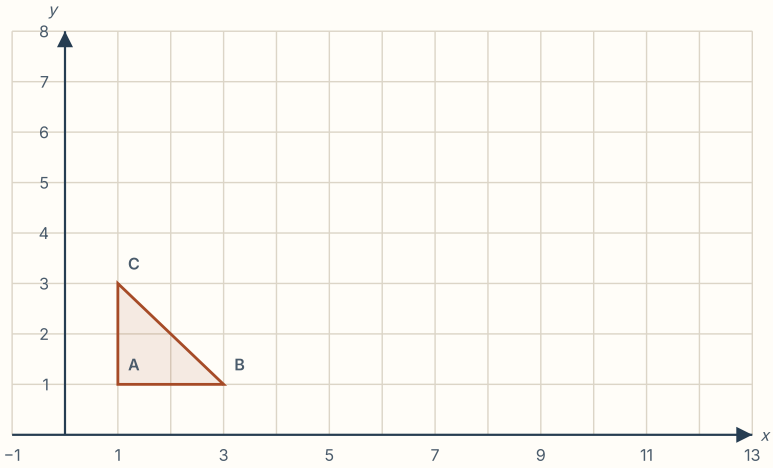


Figure 9.5 The banner grid, with the original motif. One grid unit is 15 cm.

1. Plot the motif. Calculate its area in square units and in square centimetres.
2. Copy 1 is a translation by $\begin{pmatrix} 6 \\ 0 \end{pmatrix}$. Find its vertices and plot it.
3. Copy 2 is a reflection in the line $x = 6$. Find its vertices and plot it.
4. Copy 3 is a rotation of $180°$ about $(6, 4)$. Find its vertices and plot it.
5. Copy 4 is an enlargement by scale factor 2 about the origin. Find its vertices, plot it, and state its area in square units.
6. Which copies are congruent to the original? Which is similar but not congruent? Justify using lengths and angles.
7. Find the total area of the five motifs in square units and in square metres.
8. The fabric costs \$12 per square metre. Calculate the cost of the motifs with 10% extra for waste, rounded to the nearest cent.
9. Write a design note of 60–90 words. Explain how the transformations create balance and emphasis, state one assumption and one limitation of your cost model, and suggest one change that would make the banner easier to reproduce.

CRITERION MAPPING Parts 1–5 · A i–iii, C i–iii | Parts 6–8 · C iii–iv, D i–iii | Part 9 · D iii–v, C iv

FORMAT LABELLED GRID, CALCULATIONS AND DESIGN NOTE

TIME 50–60 MINUTES

RESOURCES RULER, GRID PAPER, CALCULATOR

9.6 Summary, revision and reflection

KEY IDEAS

- A translation slides a shape by a vector. Every point moves the same distance in the same direction, and the image is congruent.
- A reflection flips a shape across a mirror line. The mirror is the perpendicular bisector of each segment joining a point to its image, and orientation is reversed.
- A rotation turns a shape about a centre through a given angle and direction. Distances from the centre are unchanged.
- An enlargement uses a centre and scale factor k. Angles stay the same, lengths multiply by k, and areas multiply by k^2.
- Line symmetry counts mirror lines; rotational symmetry has an order, and its angle is $360° \div$ order.
- A regular polygon tessellates exactly when $360°$ is a whole-number multiple of its interior angle.

translation by $\begin{pmatrix} a \\ b \end{pmatrix}$: $(x, y) \to (x + a, y + b)$

x-axis: $(x, -y)$ · y-axis: $(-x, y)$ · $y = x$: (y, x)

$90°$ anticlockwise: $(-y, x)$ · $180°$: $(-x, -y)$

enlargement about O: $(x, y) \to (kx, ky)$

area factor: k^2 · rotational angle: $360° \div$ order

Connections. The coordinate grid from Chapter 6 is the stage for every transformation here. Angle facts from Chapter 7 explain why rotations preserve shape, and area formulas become the scale-factor rules. Year 3 extends these ideas to negative scale factors and combined transformations.

REVISION CHECKLIST

- ☐ I can plot and read coordinates in all four quadrants.
- ☐ I can translate a shape using a vector and describe the reverse movement.
- ☐ I can reflect a shape in a horizontal, vertical or diagonal mirror line.
- ☐ I can rotate a shape about a centre by $90°$, $180°$ or $270°$.
- ☐ I can enlarge a shape by a positive scale factor and calculate the area change.
- ☐ I can identify lines of symmetry and the order of rotational symmetry.
- ☐ I can decide whether shapes are congruent or similar.
- ☐ I can explain why a regular polygon does or does not tessellate.

REFLECTION

A design uses a reflection on one side and a rotation on the other. From the finished pattern alone, which transformation produced each copy, and what would remain uncertain?

KEY VOCABULARY

transformation — a rule that moves or changes a shape

object / image — the original shape and its result

translation — a slide by a vector

vector — a column showing horizontal and vertical movement

reflection — a flip across a mirror line

line symmetry — a shape folds onto itself

rotation — a turn about a centre

order — the number of positions matching a shape

enlargement — a resize by a scale factor

congruent — same shape and same size

similar — same shape, different size

tessellation — a tiling with no gaps or overlaps

Mixed practice · Consolidate and extend

Name the transformation before you calculate. Write coordinates as pairs, use primes for images, and check one point against the rule you have chosen.

Fluency and accuracy

1. Translate $(-3, 4)$ by $\begin{pmatrix} 5 \\ -2 \end{pmatrix}$. F
2. Reflect $(2, -5)$ in the y-axis. F
3. Rotate $(4, 1)$ $90°$ clockwise about the origin. F
4. Enlarge $(3, -2)$ by scale factor 2 about the origin. F
5. How many axes of symmetry does a regular hexagon have? F
6. State the order and angle of rotational symmetry of an equilateral triangle. F
7. Reflect $(5, 3)$ in the line $x = 2$. F
8. A shape has area $7\ \text{cm}^2$. Its image after enlargement by scale factor 2 has what area? F

Reasoning and problem solving

9. Triangle ABC has vertices $A(0, 1)$, $B(3, 1)$, $C(0, 4)$. Translate it by $\begin{pmatrix} -2 \\ 3 \end{pmatrix}$ and state the image coordinates. P
10. Reflect the quadrilateral $(-2, 1), (1, 1), (1, 4), (-2, 4)$ in the line $y = x$. P
11. Rotate the triangle $(1, 1), (4, 1), (1, 3)$ through $180°$ about $(2, 2)$. P
12. A point $(2, 1)$ enlarges to $(5, 1)$ about the centre $(1, 1)$. Find the scale factor. P
13. Find the mirror line that maps $(1, 3)$ to $(5, 3)$. R
14. A shape is rotated $90°$ clockwise and then $90°$ anticlockwise about the same centre. Describe the final result. R
15. Two rectangles have sides 3 cm by 4 cm and 6 cm by 8 cm. Are they congruent or similar? Find the ratio of their areas. R
16. A pentagon has interior angle $108°$. Show that three pentagons cannot meet at a point without a gap, and calculate the gap. P
17. A designer reflects a motif in a vertical line and then translates it 4 units right. Does the order of the two transformations change the final image? Test with the point $(2, 1)$, using the mirror $x = 0$. E
18. A shape has rotational symmetry of order 4 and exactly 4 lines of symmetry. Name one shape that fits, and explain whether every shape with order 4 has four mirror lines. R

Extension and inquiry

19. A square has vertices $(1, 1), (4, 1), (4, 4), (1, 4)$. Rotate it $90°$ anticlockwise about its centre and describe how the image compares with the original. E
20. Enlarge a 2×3 rectangle by scale factor 2, then by scale factor 3. Find the total area factor compared with the original and explain why it is 6^2. E
21. A regular polygon tessellates with four copies meeting at each vertex. Find its interior angle and the number of sides. E

22. A motif 5 units wide is repeated across a strip by translations of 5 units. The strip is 60 units long. How many copies fit, and where does the last copy end if the first starts at $x = 0$? P

23. A design uses a reflection in $x = 3$ followed by a reflection in $x = 7$. Show that the result is a translation and find its vector. E

24. A student claims that every shape with line symmetry also has rotational symmetry. Test the claim with an isosceles triangle and a rectangle, and write a correct statement. E

25. Return to the banner task. The designer wants the enlarged motif to occupy no more than one quarter of the banner area. The banner is 96 square units. Is the scale factor 2 acceptable? Justify with areas. E

26. Write a set of instructions that would let another student reproduce a pattern of your choice using at least three different transformations. Include one check that would reveal a mistake. E

PART IV · UNCERTAINTY AND EVIDENCE

IV

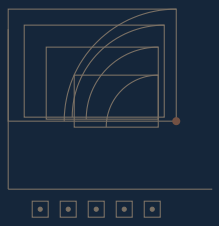
Probability and data

We cannot know everything in advance, and we rarely observe everything. Probability gives us a disciplined way to describe uncertainty; statistics gives us disciplined ways to learn from limited evidence. Together they help us judge claims, design fair experiments and make decisions we can defend.



10 Probability and simulation

11 Univariate data



10

PROBABILITY AND DATA

Probability and simulation

A weather app gives a 70% chance of rain, a hospital screens thousands of test results and a game at a fair promises a prize. Each claim rests on a model of what could happen, and on evidence from what did happen.

KEY CONCEPT · LOGIC | RELATED · CHANGE | RELATED · MODELS

CONTEXT · SCIENTIFIC AND TECHNICAL INNOVATION

INQUIRY

How can we measure uncertainty, test a prediction against evidence, and decide whether a game or a process is fair?

MYP UNIT PLANNER

Framing the inquiry

UNIT 10 · APPROX. 11 HOURS
ATL FOCUS · CRITICAL THINKING

KEY AND RELATED CONCEPTS

Logic. Probability reasons from a defined set of possible outcomes to a numerical measure of uncertainty. Its conclusions are only as reliable as the assumptions behind the model.

Change. Relative frequency changes from one set of trials to the next, yet it tends to settle as the number of trials increases.

Models. A sample space and a simulation are models. Comparing a model with data shows whether the model is useful, incomplete or wrong.

GLOBAL CONTEXT AND EXPLORATION

Scientific and technical innovation. Exploration: mathematical models and simulations. Forecasters, engineers, insurers and medical researchers use probability models to plan under uncertainty.

A simulation replaces a costly or slow real process with many rapid random trials. Its predictions are only trustworthy when the model captures the important features of the real situation.

STATEMENT OF INQUIRY

Probability measures uncertainty by comparing favourable outcomes with a well-defined sample space, while simulation tests whether a model's predictions hold when the same trial is repeated many times.

FACTUAL

How is theoretical probability calculated from a sample space, and how is relative frequency calculated from data?

CONCEPTUAL

Why can experimental probability differ from theoretical probability, and what does the size of that difference reveal about the model?

DEBATABLE

When should a simulation be trusted to inform a real decision, and when is evidence from real trials essential?

ATL · CRITICAL THINKING IN ACTION

Before calculating, state the assumptions: are all outcomes equally likely, are trials independent, and is the sample space complete? After calculating, ask what would change if an assumption failed. A probability is an argument, not just a number.

INQUIRY PATHWAY FOR THIS CHAPTER

1	2	3	4	5	6
CONTEXT	**QUESTION**	**CONCEPT**	**INVESTIGATION**	**COMMUNICATION**	**REFLECTION**
A school fair needs a game that is exciting and financially sustainable.	How can we measure uncertainty and judge whether a game is fair?	Logic, models, change and fairness.	Do two dice really favour one player in a prime-sum game?	Sample spaces, probability notation, tables and simulation reports.	When does a close match between model and data support a conclusion?

Learning intentions and success criteria

YOU ARE LEARNING TO

- **Understand** a sample space as the complete list of equally likely outcomes of a trial.
- **Know** how to calculate theoretical probability and the probability of a complement.
- **Be able to** collect and interpret relative frequencies from data.
- **Be able to** design and run a simulation using random digits or a random device.
- **Be able to** calculate expected frequencies and test whether a game is fair.
- **Be able to** communicate a probability argument with clear assumptions and limitations.

SUCCESS CRITERIA

- ☐ I can list a sample space systematically using a list, table or tree diagram.
- ☐ I can count favourable outcomes and calculate a theoretical probability.
- ☐ I can use $P(A') = 1 - P(A)$ to find the probability that an event does not happen.
- ☐ I can calculate relative frequency and expected frequency.
- ☐ I can design a simulation whose digit assignments match a given probability.
- ☐ I can compare experimental results with a theoretical model and evaluate the evidence.
- ☐ I can decide whether a game is fair and justify a change to its rules.

PREREQUISITE CHECK · 10 MINUTES, NO CALCULATOR

1. Simplify $\frac{6}{36}$.
2. Write 0.35 as a fraction in simplest form.
3. Calculate $1 - \frac{3}{8}$.
4. Find 15% of 240.
5. Two coins are tossed. How many different outcomes are possible?
6. How many faces does a fair six-sided die have?
7. List the prime numbers between 1 and 12.
8. Write 28 out of 80 as a decimal.

Answers are in the Chapter 10 answer section. If more than two are incorrect, review fractions, decimals and percentages before continuing.

YEAR 1

Use chance language such as likely and unlikely; find simple probabilities with dice and spinners; read bar charts.

YEAR 2 · THIS CHAPTER

Sample spaces, theoretical and experimental probability, complements, simulation, expected frequency and fairness.

YEAR 3

Combined events, tree diagrams for dependent events, conditional probability and standard deviation.

CRITERION	STRANDS	EVIDENCE IN THIS CHAPTER
A · Knowing and understanding	i, ii, iii	Construct sample spaces; apply probability rules; calculate expected frequencies.
B · Investigating patterns	i, ii, iii	Investigate the prime-sum game; compare simulation with theory; generalise fairness.
C · Communicating	i, ii, iii, iv	Use probability notation, tables, tree diagrams and reports; explain assumptions.
D · Applying mathematics	i, ii, iii, iv, v	Model a fair game; calculate expected profit; evaluate assumptions and recommend a design.

10.1 Outcomes, sample spaces and events

CRITERION A · I ◆ CRITERION C · I ◆ LOGIC

Before asking how likely something is, we must be precise about what could happen. A complete list of possible results turns a vague question into a counting problem.

DEFINITION · TRIAL, OUTCOME, SAMPLE SPACE AND EVENT

A **trial** is one performance of a chance experiment, such as one roll of a die. An **outcome** is one possible result of a trial. The **sample space** S is the set of all possible outcomes. An **event** is a set of outcomes, often written A or B.

Outcomes are **equally likely** when each has the same chance of occurring. A fair die has six equally likely outcomes; a fair coin has two.

For one roll of a fair six-sided die, $S = \{1, 2, 3, 4, 5, 6\}$. The event "the score is even" is $E = \{2, 4, 6\}$, so E contains three outcomes. We write $n(S) = 6$ and $n(E) = 3$.

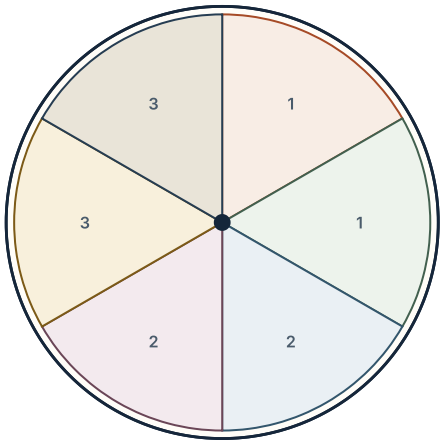


Figure 10.1 Six equal sectors, with two sectors showing each score. The distinct scores are 1, 2 and 3, but the sample space has six equally likely sector outcomes.

The spinner in Figure 10.1 warns us about a common shortcut. The sample space of *sectors* is $\{1, 1, 2, 2, 3, 3\}$, with six equally likely outcomes. The set of distinct scores $\{1, 2, 3\}$ is useful for describing results, but its three members are not equally likely: score 2 can occur in two ways, while score 1 can also occur in two ways and score 3 in two ways. Here the scores happen to be equally likely, but that is a conclusion to check, not an assumption to make.

When a trial has two stages, a tree diagram shows each choice as a branch. The sample space is read along the paths from the start to the end. Figure 10.2 shows two tosses of a coin.

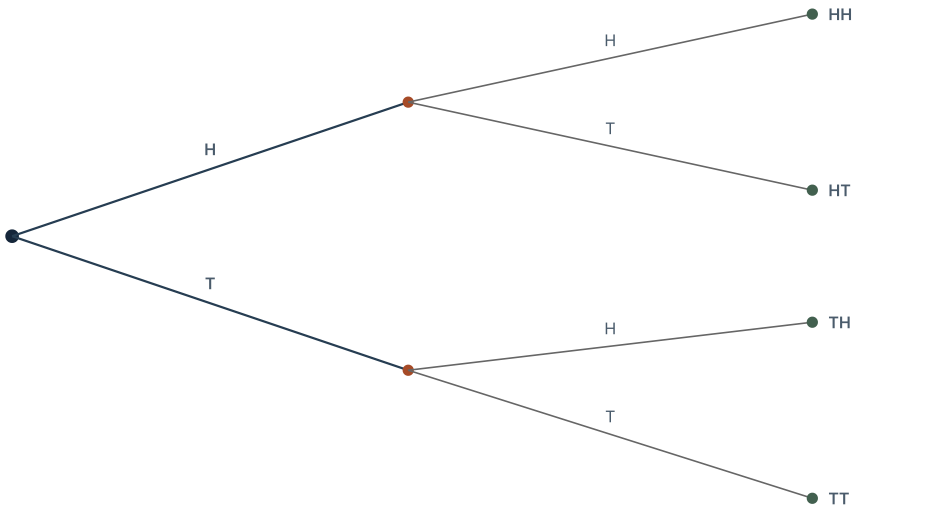


Figure 10.2 A tree diagram for two coin tosses. The four paths give the sample space $\{HH, HT, TH, TT\}$. Order matters: HT means head then tail, while TH means tail then head.

For two dice, a table is more compact than a tree. Rows record the first die and columns record the second. Each cell is one ordered pair, and there are $6 \times 6 = 36$ cells. Table 10.1 shows the *sum* in each cell; the sum 7 appears six times, while the sum 12 appears once.

TABLE 10.1 · SUMS FROM TWO FAIR DICE. ROWS SHOW THE FIRST DIE; COLUMNS SHOW THE SECOND.

+	1	2	3	4	5	6
1	2	3	4	5	6	7
2	3	4	5	6	7	8
3	4	5	6	7	8	9
4	5	6	7	8	9	10
5	6	7	8	9	10	11
6	7	8	9	10	11	12

METHOD · CONSTRUCTING A SAMPLE SPACE

Step 1. Identify the stages of the trial: one roll, two coins, a spinner and a die.

Step 2. List outcomes systematically, using an ordered list, a table or a tree. Do not stop when you think you have "enough".

Step 3. Count using multiplication: if there are m choices at the first stage and n at the second, the sample space has $m \times n$ outcomes.

Step 4. Check that each listed outcome is equally likely, and that no outcome is repeated or missing.

Step 5. Define each event in words first, then list its outcomes and count them.

WORKED EXAMPLE 10.1.1 — TWO COINS

PROBLEM

Two fair coins are tossed. List the sample space. Then find the number of outcomes in each event: A = at least one head, B = exactly one head.

1. The first coin has two possible results, H or T. For each, the second coin has two results.
2. Reading the paths in a tree gives HH, HT, TH and TT. So $S = \{HH, HT, TH, TT\}$ and $n(S) = 4$.
3. At least one head means one head or two: $A = \{HH, HT, TH\}$, so $n(A) = 3$.
4. Exactly one head excludes HH: $B = \{HT, TH\}$, so $n(B) = 2$.

ANSWER $n(S) = 4$; $n(A) = 3$; $n(B) = 2$. **HT and TH are different outcomes because the coins are distinct.**

WORKED EXAMPLE 10.1.2 — COUNTING WITH A TABLE

PROBLEM

Two fair dice are rolled. Using Table 10.1, count the outcomes in each event: C = the sum is 7, D = the sum is 12, E = both dice show the same number.

1. The sample space has $6 \times 6 = 36$ ordered outcomes.
2. For C, locate every cell containing 7. They form a diagonal of six cells, so $n(C) = 6$.
3. For D, only the cell (6, 6) contains 12, so $n(D) = 1$.
4. For E, the matching pairs are (1,1), (2,2), (3,3), (4,4), (5,5) and (6,6), so $n(E) = 6$.

ANSWER $n(C) = 6$, $n(D) = 1$, $n(E) = 6$. **The sum 7 is six times as likely as the sum 12, even though both are possible sums.**

GUIDED EXAMPLE · A SPINNER AND A DIE

Problem. A fair four-sector spinner is numbered 1 to 4, and a fair three-sector spinner is numbered 1 to 3. One spin of each is made.

Step 1. The first spinner has 4 outcomes and the second has 3, so there are $4 \times 3 = 12$ ordered outcomes.

Step 2. The outcome $(2, 3)$ means 2 on the first spinner and 3 on the second.

Now complete. State $n(S)$. Count the outcomes for which the sum of the two scores is 5, and list them. Explain why the sum 5 is not the only event with more than one outcome.

COMMON ERROR

Do not treat HT and TH as the same outcome when the coins are distinct, and do not add the numbers of choices when stages are combined: two dice give $6 \times 6 = 36$ outcomes, not $6 + 6 = 12$. Also, the eleven possible sums from 2 to 12 are not equally likely. Count the cells, not the labels.

PRACTICE 10.1 · OUTCOMES AND SAMPLE SPACES — SUPPORT 1–4 · CORE 1–8 · EXTENSION 9–10

List outcomes systematically. Before you count an event, write it in words and then as a set.

1. A bag contains four tickets labelled A, B, C and D. One ticket is drawn. List the sample space. F
2. A fair six-sided die is rolled once. List the sample space. Write the event "the score is even" as a set and count its outcomes. F
3. Two fair coins are tossed. List the sample space using H and T. Count the outcomes with at least one tail. F
4. A spinner has eight equal sectors labelled $1, 1, 2, 2, 2, 3, 3, 4$. State the number of outcomes in the sample space and list the distinct scores. F
5. Two fair dice are rolled. How many outcomes are in the sample space? List the outcomes with sum 9 and count them. P
6. Three fair coins are tossed. List all eight outcomes. Count the outcomes with exactly two heads, and the outcomes with at least two heads. P
7. A café offers three mains — pasta, curry, fish — and two desserts — fruit, cake. Draw a table or tree to list all six meal combinations. P
8. A student lists the sample space for tossing two coins as $\{HH, HT, TT\}$. Explain the error and write the correct sample space. R
9. Two dice are rolled. A student says that sum 7 and sum 12 are equally likely "because both are sums". Use counts from Table 10.1 to explain why the student is wrong. R
10. A code uses two different letters chosen from A, B and C, and order matters. List the sample space and count its outcomes. E

10.2 Theoretical probability and complements

CRITERION A · II ◆ CRITERION C · II ◆ LOGIC

Once the sample space is complete and its outcomes are equally likely, probability becomes a ratio: the part we are interested in divided by the whole.

DEFINITION · THEORETICAL PROBABILITY

If all outcomes in a sample space S are equally likely, the **theoretical probability** of an event A is

$$P(A) = \frac{n(A)}{n(S)}.$$

An impossible event has probability 0, and a certain event has probability 1. For any event, $0 \le P(A) \le 1$. The probabilities of all the outcomes in S add to 1.

The **complement** of A, written A', is the event that A does not happen. Since A and A' together contain every outcome exactly once, $P(A') = 1 - P(A)$.

Probability is a number between 0 and 1, and it can be written as a fraction, a decimal or a percentage. $\frac{1}{4} = 0.25 = 25\%$. A probability of 0.25 does not mean that the event happens exactly once in every four trials; it describes the long-run proportion under the model.

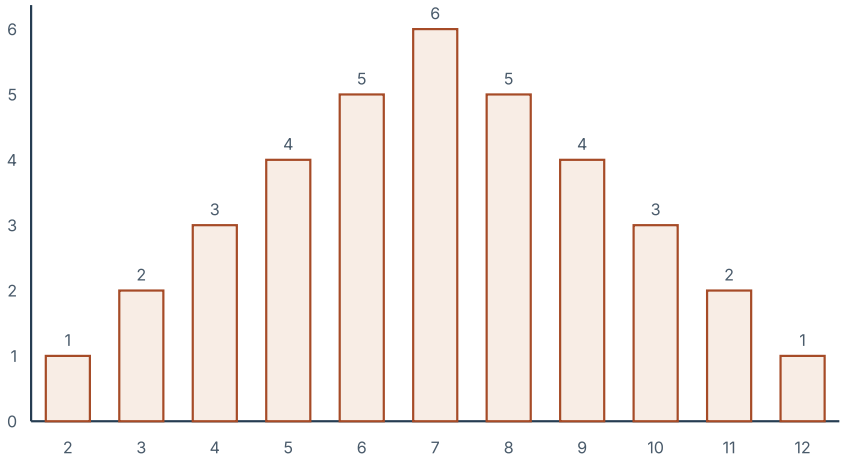


Figure 10.3 The 36 equally likely outcomes are not spread evenly across the eleven possible sums. Sum 7 occurs in 6 outcomes, while sums 2 and 12 occur once each.

KEY IDEA · COUNTING A PROBABILITY

1. List the sample space, or count its outcomes using a table or tree. **2.** Count the outcomes in the event. **3.** Divide the second count by the first. **4.** Simplify the fraction and check that it lies between 0 and 1. **5.** If the event is described with "not", use $P(A') = 1 - P(A)$.

WORKED EXAMPLE 10.2.1 ONE DIE AND A COMPLEMENT

PROBLEM

A fair six-sided die is rolled. Find the probability that the score is greater than 4, and the probability that it is not greater than 4.

1. The sample space is $\{1, 2, 3, 4, 5, 6\}$, so $n(S) = 6$.
2. The scores greater than 4 are 5 and 6, so the event has $n(A) = 2$.
3. $P(A) = \frac{2}{6} = \frac{1}{3}$.
4. The complement is "not greater than 4", so $P(A') = 1 - \frac{1}{3} = \frac{2}{3}$. Check: the scores 1 to 4 give 4 outcomes, and $\frac{4}{6} = \frac{2}{3}$.

ANSWER $P(\text{score} > 4) = \frac{1}{3}$; $P(\text{score not} > 4) = \frac{2}{3}$.

WORKED EXAMPLE 10.2.2 — TWO DICE

PROBLEM

Two fair dice are rolled. Find the probability that (a) the sum is 7, (b) the sum is at least 10, (c) the sum is less than 4.

1. The sample space has 36 equally likely ordered outcomes.
2. For a sum of 7, the cells are (1,6), (2,5), (3,4), (4,3), (5,2) and (6,1): 6 outcomes. So $P(\text{sum} = 7) = \frac{6}{36} = \frac{1}{6}$.
3. For a sum of at least 10, include sums 10, 11 and 12: $3 + 2 + 1 = 6$ outcomes. So $P(\text{sum} \geq 10) = \frac{6}{36} = \frac{1}{6}$.
4. For a sum less than 4, only sums 2 and 3 qualify: $1 + 2 = 3$ outcomes. So $P(\text{sum} < 4) = \frac{3}{36} = \frac{1}{12}$.

ANSWER **(a)** $\frac{1}{6}$ **(b)** $\frac{1}{6}$ **(c)** $\frac{1}{12}$.

WORKED EXAMPLE 10.2.3 — A BAG OF COUNTERS

PROBLEM

A bag contains 5 red, 3 blue and 2 green counters. One counter is taken at random. Find $P(\text{red})$, $P(\text{not red})$, $P(\text{blue or green})$ and $P(\text{red or blue})$.

1. The bag contains $5 + 3 + 2 = 10$ counters, and each is equally likely to be chosen.
2. There are 5 red counters, so $P(\text{red}) = \frac{5}{10} = \frac{1}{2}$.
3. The complement of red is not red: $P(\text{not red}) = 1 - \frac{1}{2} = \frac{1}{2}$. This agrees with the $3 + 2 = 5$ non-red counters.
4. Blue and green cannot happen at the same time, so add their counts: $P(\text{blue or green}) = \frac{3+2}{10} = \frac{1}{2}$.
5. Red and blue are also mutually exclusive: $P(\text{red or blue}) = \frac{5+3}{10} = \frac{4}{5}$.

ANSWER $\frac{1}{2}, \frac{1}{2}, \frac{1}{2}, \frac{4}{5}$.

GUIDED EXAMPLE · PRIME OR SIX

Problem. A fair die is rolled. Find the probability that the score is prime, the probability that it is not prime, and the probability that it is prime or 6.

Step 1. The prime scores are 2, 3 and 5. Remember that 1 is not prime.

Step 2. The event "prime or 6" contains 2, 3, 5 and 6. These outcomes do not overlap.

Now complete. Calculate all three probabilities, simplify each fraction, and check that the first two add to 1.

COMMON ERROR

The eleven possible sums of two dice are not eleven equally likely outcomes. Sum 7 has six ways and sum 12 has one, so $P(\text{sum} = 7) = \frac{6}{36}$, not $\frac{1}{11}$. Always count outcomes that are equally likely before dividing. A second error is to add probabilities of events that can happen together; add counts only when the events are mutually exclusive.

PRACTICE 10.2 · PROBABILITY AND COMPLEMENTS

SUPPORT 1–4 · CORE 1–8 · EXTENSION 9–12

State $n(S)$ before each calculation. Give each answer as a simplified fraction, and use a complement when it shortens the work.

1. A fair die is rolled. Find $P(4)$, $P(\text{odd})$ and $P(\text{score greater than } 4)$. F
2. Two fair coins are tossed. Find $P(HH)$, $P(\text{exactly one head})$ and $P(\text{at least one head})$. F
3. A fair five-sector spinner is numbered 1 to 5. Find $P(3)$, $P(\text{not } 3)$ and $P(\text{even})$. F
4. A bag contains 4 red, 5 blue and 3 green counters. One counter is taken. Find $P(\text{red})$, $P(\text{not blue})$ and $P(\text{red or green})$. P
5. Two fair dice are rolled. Find $P(\text{sum} = 8)$, $P(\text{sum} = 12)$ and $P(\text{sum at least } 11)$. P
6. Two fair dice are rolled. Find the probability that the two scores are the same, and the probability that they are different. P
7. Which of these values cannot be the probability of an event? F

   A 0 B 0.45 C 1 D 1.25

8. A bag contains 7 red and 3 yellow counters. Find $P(\text{red})$ and $P(\text{yellow})$, and explain why they add to 1. In 20 draws with replacement, how many red counters would you expect? P
9. A student says that the probability of obtaining a sum of 7 with two dice is $\frac{1}{11}$, because there are eleven possible sums. Identify the error and give the correct probability. R
10. Events A and B cannot happen at the same time. $P(A) = 0.35$ and $P(B) = 0.45$. Find $P(A \text{ or } B)$ and $P(\text{neither } A \text{ nor } B)$. R
11. Two fair dice are rolled. Find the probability that the sum is a multiple of 3. E
12. Design an eight-sector spinner for which $P(\text{red}) = \frac{1}{2}$, $P(\text{blue}) = \frac{1}{4}$ and $P(\text{green}) = \frac{1}{4}$. State how many sectors of each colour you need, and justify your design. E

10.3 Experimental probability and simulation

CRITERION A · III ◆ CRITERION C · III ◆ CHANGE

A theoretical probability describes a model. Data describe what happened. Comparing the two is how we decide whether the model is credible.

DEFINITION · RELATIVE FREQUENCY AND SIMULATION

The **frequency** of an event is the number of times it occurs in a set of trials. Its **relative frequency**, also called the **experimental probability**, is

$$\text{relative frequency} = \frac{\text{number of successes}}{\text{number of trials}}.$$

A **simulation** models a real situation by generating many random outcomes. A **random digit** is one of 0 to 9, each equally likely. The **expected frequency** of an event in N trials is $P(A) \times N$.

Suppose a die is rolled 60 times and the results in Table 10.2 are recorded. Each face would be expected about 10 times, because $P(\text{face}) = \frac{1}{6}$ and $60 \times \frac{1}{6} = 10$. The observed frequencies are close to 10 but not equal. That variation is normal: a fair die does not "remember" how often each face has appeared.

TABLE 10.2 · RESULTS OF 60 ROLLS OF A DIE, WITH RELATIVE FREQUENCIES ROUNDED TO THREE DECIMAL PLACES

Face	1	2	3	4	5	6	Total
Frequency	9	11	8	12	10	10	60
Relative frequency	0.150	0.183	0.133	0.200	0.167	0.167	1.000

The relative frequencies vary around $\frac{1}{6} \approx 0.167$. Face 4 appeared 20% of the time, while face 3 appeared 13.3% of the time. With only 60 rolls, differences of this size are not surprising. A much larger number of rolls would usually produce relative frequencies closer to $\frac{1}{6}$, though no number of rolls guarantees an exact match.

REAL-WORLD LINK · FORECASTS AND QUALITY CONTROL

A weather forecast of "70% chance of rain" is a probability model based on many earlier weather patterns. A factory that finds 4% of items faulty uses that relative frequency to estimate how many faults to expect in a large order. In both cases, the number is useful only if the model and the sample are representative.

METHOD · RUNNING A SIMULATION

Step 1. Define one trial and the event you are counting.

Step 2. Choose a random device: a die, a coin, a calculator or a table of random digits.

Step 3. Assign outcomes so that the proportion of digits or results that mean "success" equals the probability in the model. For 70%, use seven of the ten digits, such as 0 to 6.

Step 4. Generate many trials, record the number of successes and calculate the relative frequency.

Step 5. Compare with the theoretical probability, and state whether the model is supported and what its limitations are.

WORKED EXAMPLE 10.3.1 — COMPARING FREQUENCIES

PROBLEM

A basketball player is said to score 40% of her free throws. In a practice session she scores 28 of 80 free throws. Calculate the relative frequency and compare it with the model.

1. Relative frequency $= \frac{28}{80} = 0.35$.
2. The model predicts $0.40 \times 80 = 32$ successes.
3. She scored $32 - 28 = 4$ fewer than predicted, a difference of 0.05 in relative frequency.
4. The result is lower than the model, but 80 trials are few enough that random variation could explain the gap. The data do not prove that the model is wrong.

ANSWER **Relative frequency 0.35; expected frequency 32. The observed rate is lower, but the evidence is not decisive.**

WORKED EXAMPLE 10.3.2 — A DIGIT SIMULATION

PROBLEM

A free-throw shooter succeeds 70% of the time. Use the digits 0 to 6 for a success and 7 to 9 for a miss. The 25 digits below represent 25 simulated free throws. Count the successes, calculate the relative frequency and compare it with 70%.

Digits: 7 2 9 4 1 8 3 6 0 5 2 7 9 1 4 6 3 8 0 5 2 7 4 8 9

1. Successes are digits 0 to 6. Count them: 7, 2, 4, 1, 3, 6, 0, 5, 2, 7, 1, 4, 6, 3, 0, 5, 2, 7, 4 give 19 successes. The digits 9, 8, 9, 8, 9 are misses.
2. Relative frequency $= \frac{19}{25} = 0.76$, or 76%.
3. The model predicts $0.70 \times 25 = 17.5$ successes. The simulation produced 19, which is 1.5 more than expected.
4. With only 25 trials, a difference of 1.5 successes is ordinary. Repeating the simulation with 250 or 1000 trials would give a more stable estimate.

ANSWER **19 successes; relative frequency 0.76. The result is close to 70% but the sample is small.**

WORKED EXAMPLE 10.3.3 — EXPECTED FREQUENCY FROM A PERCENTAGE

PROBLEM

A quality check finds that 4% of items are faulty. A simulation uses two-digit random numbers, with 00, 01, 02 and 03 representing a faulty item. Explain why this is a suitable assignment, and find the expected number of faulty items in 600 items.

1. There are 100 two-digit numbers from 00 to 99, and each is equally likely.
2. The four codes 00, 01, 02 and 03 represent 4 out of 100, which is $0.04 = 4\%$.
3. The expected number is $0.04 \times 600 = 24$.
4. In a simulation, the observed number would vary around 24. A result far from 24 would suggest that the 4% model is not appropriate.

ANSWER **Four of the 100 two-digit codes represent a fault; expected faulty items $= 24$.**

GUIDED EXAMPLE · A SUSPICIOUS SPINNER

Problem. A four-sector spinner is spun 25 times. Red appears 9 times. The spinner is supposed to be fair. Calculate the relative frequency of red, the expected frequency under a fair model, and comment on whether the evidence is strong.

Step 1. The fair probability of red is $\frac{1}{4} = 0.25$.

Step 2. The expected number of reds in 25 spins is $0.25 \times 25 = 6.25$.

Now complete. Calculate the relative frequency, compare it with 0.25, and write two sentences explaining why 25 spins are not enough to conclude that the spinner is biased.

COMMON ERROR

Do not say that a coin is "due" to show tails after several heads. Each toss is independent, so the probability of tails remains $\frac{1}{2}$. Also check digit assignments carefully: digits 0 to 6 are seven digits and model 70%, while digits 0 to 7 are eight digits and model 80%.

PRACTICE 10.3 · EXPERIMENTAL PROBABILITY

SUPPORT 1–4 · CORE 1–8 · EXTENSION 9–10

Give relative frequencies as decimals or percentages, and compare them with the theoretical value. State the expected frequency before commenting on the result.

1. A coin is flipped 50 times and lands on heads 22 times. Find the relative frequency of heads, the expected frequency if the coin is fair, and the difference. F
2. A die is rolled 120 times and shows a six 18 times. Find the relative frequency of a six, the theoretical probability, and the expected frequency. F
3. A basketball player succeeds 90% of the time. Use digits 0 to 8 for a success. In 25 simulated throws the digits are
   8, 4, 9, 1, 6, 3, 7, 2, 5, 0, 8, 4, 6, 9, 2, 7, 5, 3, 1, 8, 0, 6, 4, 9, 7.
   Count the successes, calculate the relative frequency and compare it with the model. P
4. A weather model gives a 30% chance of rain each day. Describe how to simulate 40 days using random digits. State the expected number of rainy days. P
5. A four-sector spinner is spun 200 times and red appears 62 times. Find the relative frequency, the expected frequency for a fair spinner, and comment on whether the result proves bias. P
6. A factory finds that 4% of items are faulty. Describe a simulation using two-digit random numbers, and find the expected number of faulty items in 600. P
7. A student says, "The last five tosses were heads, so the next toss is more likely to be tails." Identify the error and explain the correct reasoning. R
8. Explain why 1000 trials usually give a better estimate of a probability than 20 trials, but why 1000 trials still do not guarantee an exact match with the theoretical value. R
9. In 80 trials an event occurs 24 times. What is its relative frequency? F

   A 0.24 B 0.30 C 0.80 D 3
10. Design a simulation for the probability that a family with three children has exactly two boys. Assume each child is equally likely to be a boy or a girl, and state the expected number of such families in 40 simulated families. E

10.4 Investigation · Is the game fair?

CRITERION B · I–III ◆ CRITERION C · IV ◆ MODELS

A game is fair when no player has an expected advantage. Probability lets us move beyond intuition and compare the expected number of wins for each player.

In the **prime-sum game**, two fair dice are rolled. Player A scores a point if the sum is prime; Player B scores a point otherwise. The possible prime sums are 2, 3, 5, 7 and 11. Table 10.3 groups the 36 outcomes.

TABLE 10.3 · OUTCOMES IN THE PRIME-SUM GAME

Result	Winning sums	Number of outcomes
Player A wins	2, 3, 5, 7, 11	15
Player B wins	4, 6, 8, 9, 10, 12	21
Total	All sums from 2 to 12	36

The prime-sum game

CRITERION B · 30 MINUTES · PAIRS

The situation. A and B each put a counter on the table. They roll two dice. If the sum is prime, A takes both counters; otherwise B takes both. They play 120 games. Before playing, decide whether you think the game is fair. Then use the table and a simulation to test your conjecture.

Maya's simulation. Maya used a random-number generator to simulate 30 games. Her recorded sums were:

Sums: 2, 6, 6, 3, 6, 10, 4, 5, 9, 7, 7, 10, 9, 7, 6, 4, 11, 4, 8, 7, 8, 5, 8, 8, 3, 11, 10, 6, 7, 7.

1. Explain why the 36 ordered outcomes are equally likely, but the eleven possible sums are not.
2. Use Table 10.3 to calculate $P(A)$ and $P(B)$. Simplify both fractions.
3. In 120 games, how many wins would you expect each player to have?
4. Which player has the advantage? How many expected wins separate them?
5. Count the prime sums in Maya's 30 games. Calculate the relative frequency of A winning.
6. Compare Maya's relative frequency with the theoretical probability from Question 2. Does the simulation support the model? Evaluate the strength of the evidence and state one limitation.
7. Change the rules so that the game is fair by moving exactly one sum from B's list to A's list. Which sums could you move? Explain.
8. A different game charges \$3 to play and pays \$12 to a winner. What probability of winning makes the game fair? Design a spinner or die rule that produces that probability.

Criterion task · The school fair game

ASSESSED CRITERIA
A i–iii · C i–iv · D i–v

A school fair plans a game called **One and Six**. A player spins a fair four-sector spinner numbered 1 to 4 and rolls a fair six-sided die. The player wins if the spinner shows 1 *and* the die shows 6. The game costs \$1 to play and pays a \$10 prize to each winner. The organisers expect 240 plays during the day.

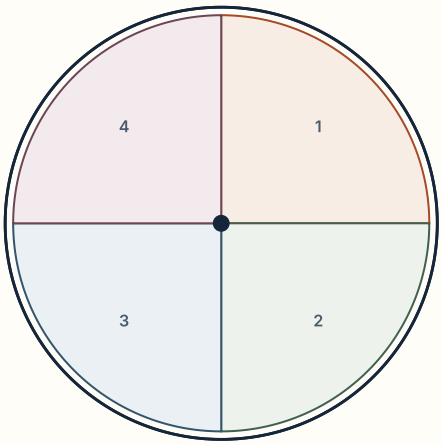


Figure 10.4 The spinner used in One and Six. A win requires a 1 on the spinner and a 6 on the die.

1. How many equally likely outcomes are there? Give one winning outcome and two losing outcomes, and explain why the outcomes are equally likely if both devices are fair.
2. Calculate $P(\text{win})$ and $P(\text{lose})$.
3. Find the expected number of winners and non-winners in 240 plays.
4. Calculate the expected takings, the expected prize payout and the expected profit. Find the expected profit per play, correct to the nearest cent.
5. Find the prize that would make the game fair, and compare it with the \$10 offered. Is the game fair to players? Justify your answer.
6. The organisers want an expected profit of \$100. Keeping the \$1 entry fee, find the prize they should offer.
7. The organisers also want players to have a more realistic chance of winning. Change the winning rule so that $P(\text{win}) = \frac{1}{12}$ with the same \$10 prize. Calculate the new expected profit from 240 plays and comment on the trade-off.
8. Write a note of 80–100 words for the organisers. State two assumptions behind your calculations, one limitation of the model, and your recommendation. Use probability evidence to support the recommendation.

CRITERION MAPPING Parts 1–4 · A i–iii, C i–ii | Parts 5–7 · D i–v, C iii–iv | Part 8 · C iv, D iii–v

FORMAT CALCULATIONS, TABLE AND WRITTEN RECOMMENDATION

TIME 50–60 MINUTES

RESOURCES CALCULATOR, SPINNER AND DIE OPTIONAL

10.5 Summary, revision and reflection

KEY IDEAS

- A sample space is the complete set of equally likely outcomes. It can be listed with a table, a tree diagram or an ordered list.
- Theoretical probability is $P(A) = \frac{n(A)}{n(S)}$. Every probability lies between 0 and 1, and the probabilities of all outcomes add to 1.
- The complement rule is $P(A') = 1 - P(A)$. It is useful when "not A" is easier to count than A.
- Relative frequency compares successes with trials. It is an estimate of probability, not a guarantee, and it usually stabilises as the number of trials increases.
- A simulation assigns random digits to outcomes so that the proportion of successes matches the model. Its results must be compared with theory and interpreted with variation in mind.
- A game is fair when the expected payout equals the cost of playing. Expected frequency is probability multiplied by the number of trials.

$$P(A) = \frac{n(A)}{n(S)} \cdot P(A') = 1 - P(A)$$

$$\text{relative frequency} = \frac{\text{successes}}{\text{trials}}$$

$$\text{expected frequency} = P(A) \times N$$

fair game: expected payout = cost of one play

two-stage sample space: $m \times n$ outcomes

Connections. Fractions and percentages from earlier chapters are the language of probability. The systematic listing strategies here prepare you for tree diagrams in Year 3, and the comparison of model with data leads directly into the statistical work of Chapter 11.

REVISION CHECKLIST

- ☐ I can construct a sample space using a list, table or tree diagram.
- ☐ I can count favourable outcomes and calculate a theoretical probability.
- ☐ I can simplify probabilities and check that they lie between 0 and 1.
- ☐ I can use the complement rule.
- ☐ I can calculate relative frequency and expected frequency.
- ☐ I can design a simulation with a valid digit assignment.
- ☐ I can compare experimental results with a theoretical model.
- ☐ I can test whether a game is fair and suggest a justified change.

REFLECTION

A simulation produces a relative frequency close to the theoretical probability. Does that prove the model is correct? Explain what else you would need to know before trusting the simulation for a real decision.

KEY VOCABULARY

trial
one performance of a chance experiment

outcome
one possible result of a trial

sample space
the set of all possible outcomes

event
a set of outcomes

equally likely
having the same chance

theoretical probability
favourable outcomes divided by all outcomes

complement
the event that A does not happen

relative frequency
successes divided by trials

experimental probability
a probability estimated from data

simulation
a random model of a real process

expected frequency
probability times number of trials

fair game
a game with equal expected gains

Mixed practice · Consolidate and extend

Decide first whether the question asks for a count, a theoretical probability, a relative frequency or an expected frequency. Name the sample space before you calculate.

Fluency and accuracy

1. List the sample space for one roll of a fair six-sided die. F
2. Two fair coins are tossed. Find the probability of exactly one tail. F
3. Two fair dice are rolled. Find the probability that the sum is 9. F
4. A bag contains 3 red and 7 blue counters. Find $P(\text{blue})$ and $P(\text{not blue})$. F
5. A fair die is rolled. Find the probability that the score is prime. F
6. A fair four-sector spinner is spun 200 times. How many times would you expect each sector to appear? F
7. An event occurs 18 times in 40 trials. Find its relative frequency. F
8. A game costs \$2 to play and pays \$8 to a winner. What probability of winning makes the game fair? F

Reasoning and problem solving

9. Two fair dice are rolled. Find $P(\text{sum even})$ and $P(\text{sum odd})$. Explain why your two answers are equal. P
10. A spinner has six equal sectors labelled $1, 1, 2, 3, 3, 3$. Find $P(3)$, $P(\text{not } 3)$ and $P(1)$. P
11. A bag contains 5 red, 4 blue and 3 green counters. Find $P(\text{red or blue})$ and $P(\text{not green})$. P
12. Two fair dice are rolled. Find $P(\text{sum at least } 10)$, $P(\text{sum less than } 4)$ and $P(\text{sum from } 4 \text{ to } 6 \text{ inclusive})$. P
13. A simulation of 60 trials models an event with probability 25%. The event occurs 18 times. Find the expected frequency and the relative frequency, and comment on the result. P
14. A student says, "The probability of rain is 0.7, so the probability of no rain is also 0.7." Identify the error and give the correct probability. R
15. Two fair dice are rolled. Find the probability that both scores are odd, and the probability that exactly one score is odd. P
16. Design a simulation for an event with probability 20%, using digits 0 to 9. Describe the digit assignment, the number of trials and the expected number of successes in 50 trials. P
17. A game costs \$4 to play and pays \$20 to a winner, with $P(\text{win}) = \frac{1}{5}$. Is the game fair? Explain what happens if the probability rises to $\frac{1}{4}$ or falls to $\frac{1}{10}$. R
18. A bag contains 4 red and 6 blue counters. Two red counters are added. Find the new probability of red. How many further red counters must be added to make $P(\text{red}) = \frac{2}{3}$? P

Extension and inquiry

19. Three fair coins are tossed. Find the probability of exactly two heads and the probability of at least two heads. E

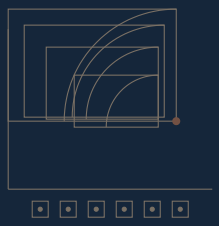
20. In a two-dice game, A wins if the sum is a multiple of 3 and B wins otherwise. Show that the game is not fair. Suggest one sum to move from B's outcomes to A's so that the game becomes fair, and verify the new probabilities. E

21. A simulation uses digits 0 to 8 to represent success, but the real success rate is 80%. Explain how the simulation misrepresents the situation and give a correct assignment. E

22. A raffle sells 500 tickets at $2 each and offers one prize of $600. Find the expected profit for the organiser, the expected profit per ticket, and the prize that would make the raffle fair. E

23. A student tosses a coin 10 times, obtains 7 heads and concludes that the coin must be biased. Evaluate the claim. What evidence would you collect before deciding? E

24. Design a fair game with a $3 entry fee and a $15 prize. State the required probability of winning and describe a device that produces it. Check the expected payout. E



11

PROBABILITY AND DATA

Univariate data

A queue is timed for twelve days, a class records its reaction times and a school audits its water use. A single number can hide the story; a well-chosen summary and display can reveal it.

KEY CONCEPT · RELATIONSHIPS

RELATED · PATTERN

RELATED · REPRESENTATION

CONTEXT · FAIRNESS AND DEVELOPMENT

INQUIRY

How can one set of measurements be summarised, displayed and compared without hiding the variation that matters?

MYP UNIT PLANNER

Framing the inquiry

UNIT 11 · APPROX. 10 HOURS
ATL FOCUS · INFORMATION LITERACY

KEY AND RELATED CONCEPTS

Relationships. The centre, spread and shape of a distribution are related: an outlier can pull the mean away from the median, and a long tail changes which summary is most representative.

Pattern. Repeated measurements vary. Patterns in that variation can be described, compared and used to make decisions.

Representation. A frequency table, dot plot or box plot highlights different features. The choice of representation shapes the claim a reader can make.

GLOBAL CONTEXT AND EXPLORATION

Fairness and development. Exploration: access to resources and equitable decisions. Data about waiting times, water use or travel can reveal unequal experiences and support fairer choices.

Evidence is only useful when the sample is representative, the measurements are comparable and the summary is appropriate to the shape of the data.

STATEMENT OF INQUIRY

Summarising a distribution by its centre, spread and shape — and representing it appropriately — allows fair comparisons and evidence-based decisions about real communities.

FACTUAL

How are the mean, median, mode, range, quartiles and interquartile range calculated from a data set?

CONCEPTUAL

How does the shape of a distribution, including outliers, affect which summary is most representative?

DEBATABLE

When does a comparison between two groups support a fair decision, and when is more evidence needed?

ATL · INFORMATION LITERACY IN ACTION

Check the source, the sample and the units before calculating. Record data in a consistent order, verify totals, and keep the raw values so that every summary can be checked. Ask who collected the data, who is missing from the sample and what the numbers do not measure.

INQUIRY PATHWAY FOR THIS CHAPTER

1	2	3	4	5	6
CONTEXT	**QUESTION**	**CONCEPT**	**INVESTIGATION**	**COMMUNICATION**	**REFLECTION**
A school compares waiting times at two canteen queues.	Which queue is slower, and which is more consistent?	Centre, spread, shape and representation.	Do dominant and non-dominant hands have different reaction times?	Tables, dot plots, box plots and evidence-based recommendations.	What can a summary hide, and when is more evidence needed?

Learning intentions and success criteria

YOU ARE LEARNING TO

— **Understand** a data set as a collection of measurements of one variable.

— **Know** how to calculate and interpret the mean, median, mode and range.

— **Know** how to find quartiles, the interquartile range and a five-number summary.

— **Be able to** construct and interpret dot plots and box plots.

— **Be able to** compare distributions using centre, spread and shape.

— **Be able to** evaluate a claim and recommend an evidence-based action.

SUCCESS CRITERIA

- [ ] I can order data and calculate the mean, median, mode and range.
- [ ] I can find quartiles, the IQR and the five-number summary.
- [ ] I can identify an outlier using the $1.5 \times$ IQR rule.
- [ ] I can draw and read a box plot.
- [ ] I can compare two distributions using more than one summary.
- [ ] I can explain why a chosen average or display is appropriate.
- [ ] I can state the limitations of a data-based claim.

PREREQUISITE CHECK · 10 MINUTES, NO CALCULATOR

1. Write these numbers in order: $7, 3, 9, 5, 1$.
2. Find the mean of $4, 6, 8, 10$.
3. Find the median of $3, 5, 8, 10, 12$.
4. Find the mode of $2, 3, 3, 4, 5, 3$.
5. Find the range of $12, 7, 15, 9$.
6. Calculate $\frac{1}{4}$ of 20.
7. Calculate $360 \div 4$.
8. A table shows 4 students have 0 pets, 5 have 1 pet and 3 have 2 pets. How many students are there?

Answers are in the Chapter 11 answer section. If more than two are incorrect, review ordering values and finding a simple average before continuing.

YEAR 1
Read pictograms and bar charts; find the mode and range; calculate a simple mean.

YEAR 2 · THIS CHAPTER
Frequency tables, mean/median/mode, quartiles, IQR, box plots and comparing distributions.

YEAR 3
Standard deviation, comparing larger data sets, bivariate data and lines of best fit.

CRITERION	STRANDS	EVIDENCE IN THIS CHAPTER
A · Knowing and understanding	i, ii, iii	Calculate and interpret summaries; find quartiles, IQR and outliers; read box plots.
B · Investigating patterns	i, ii, iii	Investigate reaction-time data; compare distributions; test a conjecture.
C · Communicating	i, ii, iii, iv	Use tables, dot plots and box plots; justify comparisons and limitations.
D · Applying mathematics	i, ii, iii, iv, v	Audit queue data; evaluate claims; recommend a fairer staffing decision.

11.1 Organising and summarising data

CRITERION A · I ◆ CRITERION C · I ◆ REPRESENTATION

A list of measurements is hard to read. Ordering it, counting repeats and choosing a summary turns the list into evidence.

DEFINITION · DATA AND MEASURES OF CENTRE

A **variable** is a quantity or characteristic that can vary. A **data set** is a collection of measurements of one variable; each measurement is a **value**. The **frequency** of a value is the number of times it occurs.

The **mean** is the total of all values divided by the number of values. The **median** is the middle value when the data are ordered. The **mode** is the most frequent value. The **range** is the largest value minus the smallest value.

Mean, median and mode are **measures of centre**; the range is a simple **measure of spread**.

For a frequency table, multiply each value by its frequency before adding. This is the same as adding every value one by one. Table 11.1 records the number of goals scored by a team in 20 matches.

TABLE 11.1 · GOALS SCORED IN 20 MATCHES

Goals	0	1	2	3	4
Frequency	3	5	6	4	2

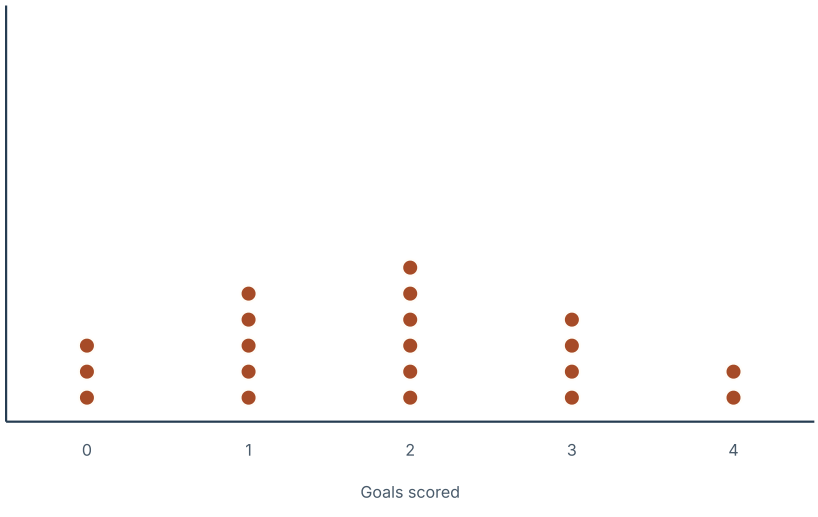


Figure 11.1 A dot plot of the goals data. Each dot is one match. The cluster at 2 and the long right tail are easier to see than in a frequency table.

KEY IDEA · CHOOSING AND CHECKING A SUMMARY

The mean uses every value and is affected by outliers. The median depends on position and is more resistant to outliers. The mode describes the most common value but may not exist or may not be unique. The range is quick but uses only two values. Always report the centre with a measure of spread.

WORKED EXAMPLE 11.1.1 FROM A FREQUENCY TABLE

PROBLEM

Using Table 11.1, find the mean, median, mode and range of the number of goals.

1. The total number of goals is $0(3) + 1(5) + 2(6) + 3(4) + 4(2) = 0 + 5 + 12 + 12 + 8 = 37$.
2. There are 20 matches, so the mean is $\frac{37}{20} = 1.85$ goals.
3. There are 20 values, so the median is between the 10th and 11th values. The cumulative frequencies are 3, 8, 14, 18 and 20, so both middle values are 2. The median is 2 goals.
4. The highest frequency is 6, for 2 goals, so the mode is 2 goals.
5. The largest value is 4 and the smallest is 0, so the range is $4 - 0 = 4$ goals.

ANSWER **Mean 1.85, median 2, mode 2, range 4 goals.**

The next worked example uses the daily screen times of 15 students, in minutes: $10, 12, 15, 18, 20, 22, 24, 26, 26, 28, 30, 32, 35, 38, 70$. The value 70 is much larger than the rest and will be important again in Section 11.2.

WORKED EXAMPLE 11.1.2 AN OUTLIER AT WORK

PROBLEM

For the 15 screen times above, calculate the mean and the median. Compare them and explain the difference.

1. The values are already ordered. The median is the 8th value, which is 26 minutes.
2. Add the values: $10 + 12 + 15 + 18 + 20 + 22 + 24 + 26 + 26 + 28 + 30 + 32 + 35 + 38 + 70 = 406$.
3. The mean is $\frac{406}{15} = 27.07$ minutes, correct to two decimal places.
4. The mean is greater than the median because the single value 70 pulls the total upward. The median, based on position, stays at 26.

ANSWER **Mean 27.07 minutes; median 26 minutes. The median is more representative of a typical student in this data set.**

GUIDED EXAMPLE · PETS PER STUDENT

Problem. A survey of 14 students records the number of pets: 4 students have 0, 5 have 1, 3 have 2 and 2 have 3.

Step 1. The total number of pets is $0(4) + 1(5) + 2(3) + 3(2) = 17$.

Step 2. The cumulative frequencies are 4, 9, 12, 14, so the median lies between the 7th and 8th values.

Now complete. Calculate the mean, median, mode and range. Explain why the mean is not a whole number and what that means in context.

COMMON ERROR

Always order the data before finding the median. For n values, the median is at position $\frac{n+1}{2}$ when n is odd, and is the mean of the two middle values when n is even. The mode is not necessarily unique, and a data set may have no mode. Do not use the range as a measure of centre.

PRACTICE 11.1 · SUMMARISING DATA

SUPPORT 1–4 · CORE 1–8 · EXTENSION 9–10

Order the values first. State which average you are calculating, and give units where they apply.

1. Find the mean of $6, 8, 10, 12, 14$. F
2. Find the median of $4, 7, 9, 12, 15, 18$. F
3. Find the mode of $3, 5, 5, 6, 8, 8, 9$. F
4. Find the range of $14, 9, 21, 17, 6$. F
5. A shoe shop records sizes sold: size 5, 2 pairs; size 6, 5 pairs; size 7, 8 pairs; size 8, 3 pairs; size 9, 2 pairs. Find the mean, median, mode and range. P
6. A data set is $12, 14, 15, 16, 17, 18, 45$. Calculate the mean and the median. Which is more representative, and why? P
7. The mean of five numbers is 12. Four of the numbers are $8, 10, 14, 16$. Find the fifth number. P
8. A student finds the median of $4, 9, 2, 7, 11$ by taking the middle number without ordering and writes 2. Identify the error and give the correct median. R
9. Data set A is $5, 6, 7, 8, 9$. Data set B is $2, 4, 7, 10, 12$. Show that the means are equal, then compare the spreads using the range. E
10. A data set $4, 6, 7, 9, 10$ has five values. What value must be added so that the mean of the six values is 8? E

11.2 Quartiles, IQR and box plots

CRITERION A · II ◆ CRITERION C · II ◆ PATTERN

Two data sets can share a median yet have very different spreads. Quartiles divide the ordered data into four parts and give a compact picture of that spread.

DEFINITION · QUARTILES AND THE FIVE-NUMBER SUMMARY

Order the data. The **median** is the second quartile, Q_2. The **lower quartile** Q_1 is the median of the lower half of the data, and the **upper quartile** Q_3 is the median of the upper half. For an odd number of values, exclude the median position when forming the two halves.

The **interquartile range** is $\text{IQR} = Q_3 - Q_1$. It measures the spread of the middle half of the data. The **five-number summary** is the minimum, Q_1, median, Q_3 and maximum. A **box plot** displays these five values on a number line.

A value is an **outlier** if it lies more than $1.5 \times \text{IQR}$ below Q_1 or above Q_3. Whiskers extend to the smallest and largest values that are not outliers; outliers are plotted separately.

For the 15 screen times in Section 11.1, the median is 26. The seven values below the median have median 18, and the seven values above it have median 32. So $Q_1 = 18$, $Q_2 = 26$, $Q_3 = 32$, and the IQR is $32 - 18 = 14$ minutes. The upper fence is $32 + 1.5(14) = 53$, so 70 is an outlier. The whiskers reach 10 and 38, the largest value that is not an outlier.

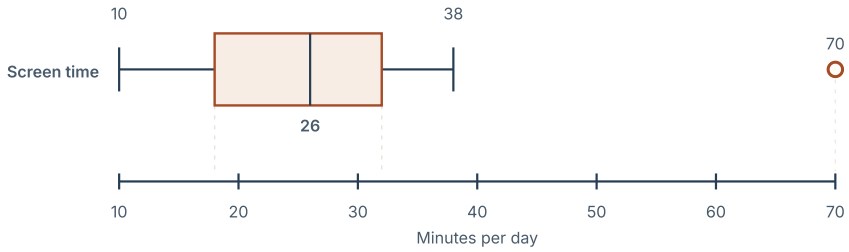


Figure 11.2 A box plot of the screen-time data. The box contains the middle half of the data; the line inside is the median. The value 70 is plotted separately because it lies beyond the upper fence of 53.

METHOD · DRAWING A BOX PLOT

Step 1. Order the data and find the five-number summary.

Step 2. Check for outliers using the $1.5 \times \text{IQR}$ fences, and adjust the whiskers.

Step 3. Draw a number line with a sensible scale and mark the five values.

Step 4. Draw the box from Q_1 to Q_3, the median line inside it, and whiskers to the minimum and maximum non-outliers. Plot any outliers as separate points.

Step 5. Label the axis with units and explain what the box and whiskers show.

WORKED EXAMPLE 11.2.1 — FIVE-NUMBER SUMMARY

PROBLEM

The data $3, 5, 6, 8, 9, 11, 12, 14, 15, 17, 20$ are ordered. Find the five-number summary, the IQR and any outliers.

1. There are 11 values, so the median is the 6th value: $Q_2 = 11$.
2. The lower half is $3, 5, 6, 8, 9$, whose median is 6, so $Q_1 = 6$.
3. The upper half is $12, 14, 15, 17, 20$, whose median is 15, so $Q_3 = 15$.
4. $\text{IQR} = 15 - 6 = 9$. The fences are $6 - 1.5(9) = -7.5$ and $15 + 1.5(9) = 28.5$.
5. The minimum 3 and maximum 20 lie inside the fences, so there are no outliers. The five-number summary is $3, 6, 11, 15, 20$.

ANSWER $3, 6, 11, 15, 20$**; IQR** $= 9$**; no outliers.**

WORKED EXAMPLE 11.2.2 — SCREEN TIME IN FULL

PROBLEM

Find the five-number summary and IQR of the 15 screen times, identify any outliers, and interpret the box plot.

1. The data are already ordered. The median is the 8th value, $Q_2 = 26$.
2. The lower seven values have median $Q_1 = 18$; the upper seven have median $Q_3 = 32$.
3. $\text{IQR} = 32 - 18 = 14$. The upper fence is $32 + 1.5(14) = 53$, so 70 is an outlier. The lower fence is $18 - 21 = -3$, so there are no low outliers.
4. The whiskers reach 10 and 38. The five-number summary is $10, 18, 26, 32, 38$, with an outlier at 70.
5. The middle half of the screen times lies between 18 and 32 minutes. The upper whisker is longer than the lower whisker, showing a longer right tail; the outlier at 70 is unusually high.

ANSWER **Five-number summary** $10, 18, 26, 32, 38$**; IQR** $= 14$**; outlier** 70**.**

GUIDED EXAMPLE · TWELVE VALUES

Problem. Find the median, quartiles, IQR and five-number summary of $2, 4, 4, 5, 7, 9, 10, 12, 13, 15, 17, 20$.

Step 1. There are 12 values, so the median is the mean of the 6th and 7th values: $\frac{9+10}{2} = 9.5$.

Step 2. The lower half is the first six values and the upper half is the last six.

Now complete. Find Q_1, Q_3, the IQR and the five-number summary. Check whether any value is an outlier.

COMMON ERROR

Do not use $Q_3 - Q_1$ as the range: the range is the maximum minus the minimum. For an odd number of values, do not include the median in both halves when finding quartiles; use the values on either side of the median position. A box plot is not a bar chart: the width of the box shows spread, not frequency.

PRACTICE 11.2 · QUARTILES AND BOX PLOTS

SUPPORT 1–4 · CORE 1–8 · EXTENSION 9–10

Order the data and show the median position. Check whether the whiskers reach the extreme values or stop at an outlier fence.

1. Find the median, Q_1, Q_3 and IQR of $5, 6, 7, 8, 9, 10, 11$. F
2. Find the median, Q_1, Q_3 and IQR of $12, 15, 18, 20, 22, 25, 28, 30$. F
3. Find the median, Q_1, Q_3 and IQR of the integers from 1 to 10. P
4. A data set has $Q_1 = 12$ and IQR $= 8$. Find Q_3 and the upper fence. Is 35 an outlier? P
5. A five-number summary is $4, 7, 9, 13, 18$. Find the range, IQR and median. P
6. Box plot A has median 10 and IQR 4. Box plot B has median 12 and IQR 8. Compare their centres and spreads. R
7. A student says the range is $Q_3 - Q_1$. Identify the error and state the correct definitions of range and IQR. R
8. For $2, 5, 6, 6, 7, 8, 9, 10, 50$, find the median, quartiles and IQR, and identify any outlier. P
9. Class A has median 65 and IQR 8; Class B has median 70 and IQR 15. Which class is more consistent, and which has the higher typical score? R
10. Explain why the IQR is less affected by a single extreme value than the range. Use an example to support your answer. E

11.3 Comparing distributions and making claims

CRITERION C · IV ◆ CRITERION D · I ◆ RELATIONSHIPS

A comparison is an argument. It needs a measure of centre, a measure of spread and a reason to believe the data are comparable.

Two water fountains were monitored for ten days. Fountain A supplied $120, 132, 140, 145, 150, 152, 160, 165, 170, 180$ litres per day. Fountain B supplied $90, 100, 110, 120, 130, 150, 170, 190, 210, 230$ litres per day. Their summaries are in Table 11.2 and their box plots are in Figure 11.3.

TABLE 11.2 · DAILY WATER USE AT TWO FOUNTAINS, IN LITRES

Fountain	Mean	Median	Range	IQR
A	151.4	151	60	25
B	150.0	140	140	80

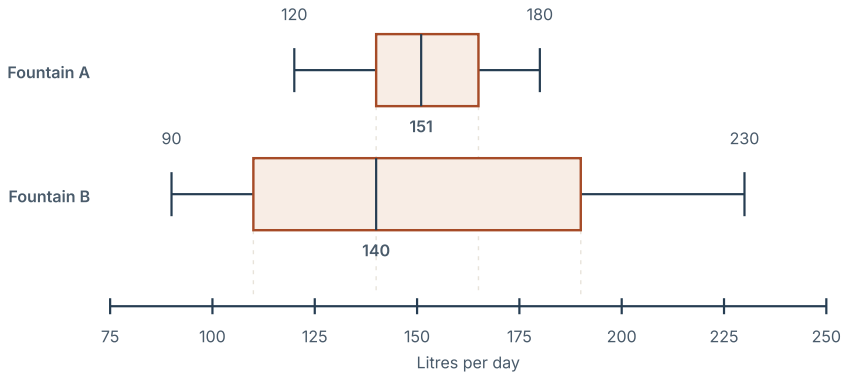


Figure 11.3 Box plots for the two fountains. Both have similar means, but the box and whiskers for Fountain B are much longer, showing greater day-to-day variation.

KEY IDEA · A FAIR COMPARISON

Compare **centre** (median or mean), **spread** (IQR or range) and **shape** (skew, outliers, clusters). Use the same measures and units for both groups. Refer to the context, and state the sample size and any limitation. A difference in one summary is evidence, not proof of a cause.

WORKED EXAMPLE 11.3.1 COMPARING FOUNTAINS

PROBLEM

Using Table 11.2 and Figure 11.3, compare the water use at the two fountains. Which fountain is more consistent?

1. The means are almost equal: 151.4 L and 150.0 L. The medians differ more: 151 L for A and 140 L for B.
2. Fountain B's range is 140 L, compared with 60 L for A. Its IQR is 80 L, compared with 25 L for A.
3. B's box is much longer and its whiskers extend further, so its daily use is more variable. A's middle half is tightly clustered between 140 L and 165 L.
4. Although the means are similar, A is more consistent and its median is higher. B has more low days and more high days, so it is less predictable.

ANSWER **Fountain A is more consistent: smaller range and IQR. B has a similar mean but much greater spread.**

WORKED EXAMPLE 11.3.2 — WHICH AVERAGE?

PROBLEM

A company reports that the mean salary is $52,000, but the median salary is $38,000. Explain what this suggests and which average better describes a typical employee.

1. The mean is greater than the median, which suggests a right-skewed distribution: a small number of very high salaries pull the mean upward.
2. The median is the middle salary, so half of the employees earn below $38,000. That is a better description of a typical employee.
3. The mean is still useful for the total wage bill, but it is not representative of most employees.

ANSWER **The distribution is likely right-skewed by high earners. The median, $38,000, is more representative of a typical employee; the mean describes the total payroll better.**

GUIDED EXAMPLE · A MISLEADING GRAPH

Problem. A poster compares two fountains. Its vertical axis starts at 140 L, so the bar for 160 L looks twice as tall as the bar for 150 L. Explain why the graph is misleading and describe a fairer display.

Step 1. On a full scale from 0, the heights would be in the ratio 160 : 150, not 2 : 1.

Now complete. Explain how the truncated axis exaggerates the difference, suggest a corrected graph, and state one summary that would give a fairer comparison.

COMMON ERROR

Do not compare means without checking spread, sample size and shape. A higher mean may come from one extreme value, and a small sample may give a misleading result. Also avoid claiming that one variable causes another simply because the groups differ.

PRACTICE 11.3 · COMPARING AND EVALUATING

SUPPORT 1–4 · CORE 1–8 · EXTENSION 9–10

Refer to centre, spread and context. State what the data do not show before making a recommendation.

1. Class A has mean score 72 and range 8; Class B has mean 75 and range 20. Which class is more consistent? Which has the higher average? [P]
2. Distribution A has median 20 and IQR 5; distribution B has median 20 and IQR 12. Compare their centres and spreads. [P]
3. A bar chart's vertical axis starts at 50. The bars for 55 and 60 appear in the ratio 2 : 1. Explain why this is misleading and how to fix it. [R]
4. A data set has mean 30 and median 22. What does this suggest about its shape? [R]
5. House prices in a street include one very expensive house. Would you report the mean or the median as typical? Explain. [R]
6. Two fountains have similar means, but A has IQR 25 L and B has IQR 80 L. Which is more reliable? Justify. [P]
7. A survey uses 5 students; another uses 200 students from the same school. Which is more likely to represent the school? Explain. [R]

8. A student claims, “The mean is always the best average.” Evaluate the claim with an example. R
9. Box plot A has median 14 and IQR 6; box plot B has median 18 and IQR 10. What claims are supported, and what can you not conclude? E
10. Construct a data set of seven values with median 10, range 14 and IQR 4. Show that it satisfies all three conditions. E

11.4 Investigation · Reaction times

CRITERION B · I–III ◆ CRITERION C · IV ◆ PATTERN

Is your dominant hand faster? A simple reaction-time test can produce data, but deciding what the data show requires care.

Twelve students caught a falling ruler with each hand. The catch distance in centimetres was converted to a reaction time in milliseconds. Each value below is the mean of three trials for that hand.

TABLE 11.3 · REACTION TIMES FOR DOMINANT AND NON-DOMINANT HANDS, IN MILLISECONDS

Student	1	2	3	4	5	6	7	8	9	10	11	12
Dominant	210	220	225	230	235	240	245	250	255	260	270	290
Non-dominant	230	240	245	250	255	260	265	270	280	290	300	320

Which hand is faster?

CRITERION B · 30 MINUTES · PAIRS

The situation. A group of students claims that the dominant hand has a faster reaction time. Use Table 11.3 to investigate the claim. Before calculating, write a conjecture and say what evidence would support it.

1. Calculate the mean and median reaction time for each hand.
2. Calculate the range and IQR for each hand. Show the quartiles you use.
3. Draw a box plot for each hand on the same scale, or use the five-number summaries to describe the boxes.
4. Compare the centre and spread of the two distributions. Does the dominant hand tend to be faster?
5. How large is the difference between the medians? Is that difference meaningful in the context of a reaction-time test?
6. Do the distributions overlap? What does the overlap imply about individual students?
7. Check for outliers using the $1.5 \times$ IQR rule. Evaluate the measurement method and the sample.
8. Write a conclusion of 60–80 words. State whether the claim is supported, and include one limitation and one improvement.

Criterion task · The canteen queue audit

ASSESSED CRITERIA
A I–III · C I–IV · D I–V

A school records the waiting time, in minutes, at two canteen queues over 12 school days.

Queue A: 4, 5, 5, 6, 6, 7, 7, 7, 8, 8, 9, 12.
Queue B: 2, 3, 4, 5, 6, 8, 9, 10, 11, 12, 14, 15.

1. Order each data set. Calculate the mean, median, mode and range of each queue.
2. Calculate the lower and upper quartiles and the IQR of each queue.
3. Write the five-number summary for each queue and draw both box plots on the same scale.
4. Compare the centre and spread of the two queues. Which queue is more consistent? Justify using at least two summary measures.
5. Does the data support the claim "Queue B is typically slower"? Use the medians and the overlap between the data sets in your answer.
6. The canteen manager can assign one extra staff member to one queue. Which queue would you choose, and why? Consider both the typical wait and the variability.
7. State two assumptions or limitations of the data. Suggest one change that would make the comparison fairer.
8. Write a note of 80–100 words for the canteen manager. Summarise the evidence, state your recommendation and acknowledge one limitation.

CRITERION MAPPING Parts 1–3 · A i–iii, C i–ii | Parts 4–6 · D i–v, C iii–iv | Parts 7–8 · C iv, D iii–v

FORMAT TABLE, BOX PLOTS AND WRITTEN RECOMMENDATION

TIME 50–60 MINUTES

RESOURCES CALCULATOR, RULER, GRID PAPER

11.5 Summary, revision and reflection

KEY IDEAS

- The mean uses every value; the median is the middle value; the mode is the most frequent value; the range is the difference between the largest and smallest values.
- Order the data before finding the median or quartiles. For an odd number of values, exclude the median position when forming the two halves.
- The IQR $= Q_3 - Q_1$ describes the spread of the middle half and is resistant to outliers.
- A box plot shows the five-number summary. Whiskers stop at the most extreme non-outliers; outliers are plotted separately.
- Compare distributions using centre, spread and shape, with the same measures and units.
- The mean is pulled by skew and outliers; the median is usually more representative when the data are skewed.

$$\text{mean} = \frac{\text{sum of all values}}{\text{number of values}}$$

$$\text{median position} = \frac{n+1}{2} \text{ (odd } n\text{)}$$

$$\text{IQR} = Q_3 - Q_1$$

outlier fences: $Q_1 - 1.5\text{IQR}$ and $Q_3 + 1.5\text{IQR}$

five-number summary: min, Q_1, median, Q_3, max

Connections. The ordering and fraction work from earlier chapters supports quartiles and the median. The comparison of model with data in Chapter 10 becomes the comparison of groups here, and Year 3 extends spread to standard deviation and comparisons to bivariate data.

REVISION CHECKLIST

- ☐ I can order data and calculate the mean, median, mode and range.
- ☐ I can find quartiles and the IQR.
- ☐ I can use the $1.5 \times$ IQR rule to identify outliers.
- ☐ I can construct and interpret a box plot.
- ☐ I can compare two distributions using centre, spread and shape.
- ☐ I can choose an appropriate average for skewed data.
- ☐ I can evaluate a data-based claim and state its limitations.
- ☐ I can make an evidence-based recommendation.

REFLECTION

Two groups have the same median but different box plots. What else would you ask for before deciding which group is more consistent, and what could a summary hide?

KEY VOCABULARY

variable
a quantity that can vary

frequency
the number of times a value occurs

mean
total divided by the number of values

median
the middle value when ordered

mode
the most frequent value

range
largest minus smallest

quartile
a value dividing ordered data into four parts

interquartile range
the spread of the middle half, $Q_3 - Q_1$

five-number summary
min, quartiles, median and max

box plot
a display of the five-number summary

outlier
a value far from the rest of the data

skew
asymmetry in a distribution

Mixed practice · Consolidate and extend

Order the data first. Decide whether the question asks for a measure of centre, a measure of spread, or a comparison, and give units where they apply.

Fluency and accuracy

1. Find the mean of $7, 9, 11, 13, 15$. F
2. Find the median of $3, 4, 6, 9, 12, 15$. F
3. Find the mode of $2, 4, 4, 5, 7, 7, 7, 9$. F
4. Find the range of $18, 12, 25, 14, 9$. F
5. Find Q_1, Q_3 and the IQR of $2, 4, 6, 8, 10, 12, 14, 16$. F
6. A five-number summary is $5, 8, 10, 15, 22$. Find the range and IQR. F
7. A data set has 6 values and mean 8. Find the total of the values. F
8. A data set has $Q_1 = 10$, $Q_3 = 18$. Find the upper fence and decide whether 32 is an outlier. F

Reasoning and problem solving

9. Distribution A has mean 20 and median 20; distribution B has mean 30 and median 22. Which is likely to be right-skewed? Explain. P
10. The data $4, 5, 6, 7, 8, 9, 10$ have mean 7 and median 7. Add the value 20 and recalculate both. Which measure changes less? P
11. Two classes have the same median score, but Class A has IQR 4 and Class B has IQR 10. Which class is more consistent? Explain. P
12. A graph of waiting times has a vertical axis starting at 90 minutes. Explain how this can exaggerate differences. R
13. Which average would you report for house prices when one house costs much more than all the others? Justify your choice. R
14. A frequency table records 0 goals 3 times, 1 goal 5 times, 2 goals 4 times and 3 goals 2 times. Find the mean, median, mode and range. P
15. A sample of 10 students is compared with a sample of 500 students. Which is more likely to represent the whole school? Explain. P
16. Data set A is $3, 4, 5, 6, 7$; data set B is $1, 3, 5, 7, 9$. Show that the means are equal and compare the spreads. P
17. A student finds the median of $2, 9, 4, 7, 5$ by taking the middle number without ordering and writes 4. Identify the error and give the correct median. R
18. A box plot has whiskers at 10 and 40, a box from 20 to 30 and a median line at 25. State Q_1, Q_3, the IQR and the range. P

Extension and inquiry

19. Construct a data set of nine values with median 15, $Q_1 = 10$ and $Q_3 = 20$. Show that your values satisfy all three conditions. E
20. A data set has mean 50 and median 40. A very large value is added. Describe what happens to the mean, the median and the range. E

21. Box plot A has median 60 and IQR 10; box plot B has median 70 and IQR 25. What claims are supported, and what can you not conclude about individual values? E

22. Design a fair comparison of two revision methods for a class test. State what you would measure, how you would assign students and one condition you would keep the same. E

23. A graph shows only the mean rainfall for each month. What variation could be hidden, and what display would reveal it? E

24. Return to the canteen audit. Which queue would you staff more, and what additional data would you collect before making a final decision? Use centre and spread in your answer. E

Answers and worked solutions

COMPLETE SOLUTIONS FOR EVERY EXERCISE

An answer ends a calculation; a solution records a decision. The entries below show the reasoning behind each result, so a difference between your work and mine becomes a precise question.

HOW EACH ENTRY IS ARRANGED

Exercise numbers match the chapter. Part letters follow the number, for example **3c**. Working is shown where it helps you locate a disagreement; routine arithmetic is compressed. Final answers are in bold, with units where required.

Some questions invite more than one valid route. The key gives one efficient method and, where useful, a note beginning *Alternatively*. A different correct method is not an error. Validity, accuracy and a conclusion that answers the question are what count.

INVESTIGATIONS AND CRITERION TASKS

Open tasks receive *sample responses*, not a single script: representative data, a conjecture, a verification and a limitation. Your data may differ. Use the sample to judge the completeness of your table, the quality of your reasoning and the care of your conclusion.

Academic honesty. These solutions support your attempt; they do not replace it. If you use a sample response, rewrite the reasoning in your own words and explain the choice of method. That is the evidence a teacher can assess.

ACCURACY AND NOTATION

Unless a question states otherwise, exact answers are given for integers, fractions and terminating decimals. Measurements are rounded to a sensible degree of accuracy, generally three significant figures. Never round intermediate values; rounding early is a common source of "correct method, wrong answer".

- **Fractions** in simplest form; **ratios** with whole numbers and no common factor.
- **Probability** as a fraction, decimal or percentage, with the form stated.
- **Angles** with the degree symbol; lengths, areas and volumes with units.
- **Money** to two decimal places where the context uses currency.
- **Data summaries** with the unit and number of values.

Equivalent fractions, ratios and expressions can look unlike each other. Test numerical equivalence before assuming an error.

Criterion tasks use course-specific rubrics that paraphrase the MYP criteria; they are not official IB descriptors, and sample responses show one level of performance.

Chapter 1 · Ratios and equivalent ratios

PREREQUISITE CHECK

1 $\frac{12}{18} = \frac{2}{3}$.

2 $60 \div 4 = \mathbf{15}$.

3 $1.5 \times 1000 = \mathbf{1500\ g}$.

4 $3 \times 7 + 4 \times 7 = 21 + 28 = \mathbf{49}$, or $7 \times 7 = 49$.

5 $5x = 45$, so $x = \mathbf{9}$.

6 $0.25 = \frac{25}{100} = \frac{1}{4}$.

PRACTICE 1.1 · READING AND WRITING RATIOS

1 $6 : 9 = \mathbf{2} : \mathbf{3}$.

2 Walk : bus $= 10 : 14 = \mathbf{5} : \mathbf{7}$. Walk : total $= 10 : 24 = \mathbf{5} : \mathbf{12}$.

3 **18 : 7.**

4 45 min : 120 min $= 45 : 120 = \mathbf{3} : \mathbf{8}$.

5 4 parts $= 12$, so 1 part $= 3$; dogs $= 7 \times 3 = \mathbf{21}$.

6 8 parts $= 320$, so 1 part $= 40$; fiction $= 5 \times 40 = \mathbf{200}$.

7 3 parts $= 15$, so 1 part $= 5$. Oranges $= 2 \times 5 = 10$; total $= 15 + 10 = \mathbf{25}$ pieces.

8 **False.** Order changes the comparison. For example, 2 : 3 means 2 of the first quantity for every 3 of the second; 3 : 2 reverses the relationship, and $\frac{2}{3} \neq \frac{3}{2}$.

9 **C.** $8 : 12 = 2 : 3$, and $4 : 6 = 2 : 3$. The others are $1 : 2$, $3 : 4$ and $1 : 2$.

10 Sam divided the two terms by different numbers. Both terms must be divided by the same non-zero factor. $5 : 10 = \mathbf{1} : \mathbf{2}$.

PRACTICE 1.2 · EQUIVALENT RATIOS

1 Divide by 6: **2 : 3.**

2 Divide by 5: **5 : 8.**

3 Multiply by 10: $6 : 9$, then divide by 3: **2 : 3.**

4 Multiply by 12: $9 : 10$; no common factor, so **9 : 10.**

5 Scale factor $15 \div 3 = 5$, so $? = 7 \times 5 = \mathbf{35}$.

6 Scale factor $24 \div 8 = 3$, so $? = 9 \div 3 = \mathbf{3}$.

7 Any three of **10 : 8**, **15 : 12**, **20 : 16**, **25 : 20** (multiplying both terms by 2, 3, 4, 5).

8 **Yes.** $8 : 12 = 2 : 3$ and $12 : 18 = 2 : 3$; both simplify to the same ratio. Cross products: $8 \times 18 = 144 = 12 \times 12$.

9 $14 : 21 = 2 : 3$, so $x : 3 = 2 : 3$ and $x = \mathbf{2}$.

10 $2 : 5 = 6 : 15$, so **6 : 15** (15 parts water).

11 Subtracting 3 from both terms does not preserve a ratio. $6 : 9 = 2 : 3$, while $3 : 4$ is not equivalent to $2 : 3$. Correct answer: **2 : 3.**

12 The total number of parts is $2 + 3 = 5$. One part is $40 \div 5 = 8$, so the ratio is **16 : 24.**

PRACTICE 1.3 · TABLES AND LINES

1 The second quantities are **10**, **15**, **20** (multiply by 2, 3, 4).

2 Missing cost $= \$60$. Cost : pens $= 12 : 3 = \mathbf{4} : \mathbf{1}$, or \$4 per pen.

3 $3 : 5 = 12 : 20$, so **20** cups of water.

4 **Table A.** Its ratios $2 : 5$, $4 : 10$ and $6 : 15$ all equal $2 : 5$. In Table B, $4 : 9 \neq 2 : 5$.

5 6 cups flour needs 8 cups milk; 9 needs 12; 12 needs **16** cups milk.

6 4 cm $\rightarrow$ 10 km; 6 cm $\rightarrow$ 15 km; 9 cm $\rightarrow$ **22.5** km.

7 10 minutes is five 2-minute intervals, so $5 \times 3 = \mathbf{15\ km}$.

8 **Yes.** $2 : 3 = 4 : 6 = 6 : 9$. Cross products: $2 \times 6 = 12 = 3 \times 4$, and $4 \times 9 = 36 = 6 \times 6$. The points lie on the same straight line through the origin.

9 **C.** $12 : 15 = 4 : 5$, not $3 : 4$. The others simplify to $3 : 4$.

10 **No.** $5 : 8 = 10 : 16$, but $15 : 23 \neq 5 : 8$. Change 23 to **24**, because $15 : 24 = 5 : 8$.

PRACTICE 1.4 · SHARING AND COMPARING

1 $5 + 3 = 8$ parts; $72 \div 8 = 9$. Ali **45**, Bea **27**. *Alternatively,* Ali receives $\frac{5}{8}(72) = 45$ and Bea receives $\frac{3}{8}(72) = 27$.

2 $2 + 3 + 4 = 9$ parts; $180 \div 9 = 20$. **40**, **60**, **80**.

3 $3 + 5 + 4 = 12$ parts; $96 \div 12 = 8$. **24**, **40**, **32**.

4 4 parts $= 36$, so 1 part $= 9$. Larger $= 7 \times 9 = \mathbf{63}$; total $= \mathbf{99}$.

5 Difference $= 8 - 3 = 5$ parts $= 45$, so 1 part $= 9$. Amounts: **27** and **72**.

6 $7 + 3 + 2 = 12$ parts; $120 \div 12 = 10$. **70**, **30**, **20**.

7 Length + width $= 60 \div 2 = 30$ cm. $7 + 3 = 10$ parts, so 1 part $= 3$ cm. Length **21 cm**, width **9 cm**.

8 $2 + 3 + 4 = 9$ parts; $180° \div 9 = 20°$. Angles: **40°**, **60°**, **80°**.

9 $\frac{4}{5} = 0.8$ and $\frac{7}{9} \approx 0.778$, so **4 : 5** is greater. Cross products: $4 \times 9 = 36 > 35 = 5 \times 7$.

10 5 parts $= 350$ g, so 1 part $= 70$ g. Sugar $= 2 \times 70 = \mathbf{140\ g}$; total $= \mathbf{490\ g}$.

11 $5 + 6 + 7 = 18$ parts; $360 \div 18 = 20$. **100**, **120**, **140** students.

12 Difference $= 5 - 3 = 2$ parts $= \$120$, so 1 part $= \$60$. Total $= 8 \times 60 = \mathbf{\$480}$.

INVESTIGATION · THE LEMONADE STAND

1 A: $2 : 5$ (already simplest). B: $4 : 10 = 2 : 5$. C: $6 : 15 = 2 : 5$. D: $3 : 8$ (already simplest).

2 A, B and C have the same simplified ratio $2:5$. If the other ingredients and preparation are the same, they should have the same lemon concentration and therefore a very similar taste. This is an inference from concentration, not a proof about flavour.

3 For 30 cups of water: A needs $30 \times \frac{2}{5} = 12$ cups lemon; B needs 12; C needs 12; D needs $30 \times \frac{3}{8} = 11.25$ cups (11¼ cups). A, B and C are the same scaled recipe; D is slightly less concentrated.

4 The points are $(2, 5)$, $(4, 10)$, $(6, 15)$, $(8, 20)$. They lie on a straight line through the origin. The graph shows a constant multiplicative relationship: doubling both quantities stays on the line.

5 **Sample conjecture:** If two lemonade recipes have the same simplified ratio, they have the same relative concentration and should taste the same when other factors are equal. Test: $3:7$ and $9:21$ both simplify to $3:7$, so the conjecture predicts the same concentration. The example is consistent with the conjecture, but a pattern cannot prove a claim about subjective taste.

6 Recipe A has $\frac{2}{5} = 0.4$ cups lemon per cup water. Recipe D has $\frac{3}{8} = 0.375$. Recipe A is more lemony because $0.4 > 0.375$.

7 The ratio does not account for the type of lemon, the amount of sugar, the temperature, the accuracy of measuring, or other ingredients. Two mixtures with the same lemon-to-water ratio can still taste different.

REAL-LIFE APPLICATION · COOKING FOR 45

1 Scale factor $= 45 \div 12 =$ **3.75**.

2 Rice: $600 \times 3.75 = 2250$ g $=$ **2.25 kg**. Lentils: $450 \times 3.75 = 1687.5$ g, which rounds to **1688 g** (or 1.69 kg to three significant figures). Oil: $90 \times 3.75 = 337.5$ mL, which rounds to **338 mL**.

3 $3.75 = 3 + \frac{3}{4}$. Scaling by 3 gives three copies of the original; scaling by $\frac{3}{4}$ gives three quarters of the original. Adding these amounts is the distributive property: 3.75 copies equal 3 copies plus $\frac{3}{4}$ of a copy.

4 2.4 kg $= 2400$ g. The recipe needs 2250 g, so there is 150 g extra. **Yes, enough.** The calculation is accurate to the nearest gram, but the recipe is a model: some rice may be lost in cooking, and measuring is approximate. A 150 g margin is about 6.7% of the required amount, so the conclusion is reasonable but should be checked against actual serving sizes.

CRITERION TASK · THE COMMUNITY GARDEN HARVEST

1 $120 \div 3 = 40$ kg per class.

2 Total parts $= 5 + 3 + 2 = 10$. One part $= 120 \div 10 = 12$ kg. 7A: $5 \times 12 = 60$ kg; 7B: $3 \times 12 = 36$ kg; 8A: $2 \times 12 = 24$ kg. Check: $60 + 36 + 24 = 120$.

3 **Sample comparison.** Equal share: 40 kg each. Proportional share: 60, 36, 24 kg. Relative to an equal share, 7A gains 20 kg, 7B loses 4 kg and 8A loses 16 kg. The proportional plan is more defensible if the purpose is to reward contribution, because the hours ratio $5:3:2$ is preserved. The equal plan is more defensible if the purpose is to meet equal need. A limitation is that hours may not measure effort, difficulty or need; the committee may combine the two principles, for example by guaranteeing a minimum share. No single answer is correct without stating the criterion for fairness.

4 $90 \div 10 = 9$ kg per part. 7A: **45 kg**; 7B: **27 kg**; 8A: **18 kg**.

5 15% of $120 = 0.15 \times 120 = 18$ kg reserved. Remainder $= 120 - 18 = 102$ kg. One part $= 102 \div 10 = 10.2$ kg. Shares: 7A $= 5 \times 10.2 =$ **51.0 kg**; 7B $=$ **30.6 kg**; 8A $=$ **20.4 kg**. Check: $51.0 + 30.6 + 20.4 = 102.0$ kg.

6 **Sample proposal.** "To the garden committee: We recommend sharing the 120 kg harvest in the ratio $5:3:2$, because it reflects the hours each class contributed. There are 10 parts, so one part is 12 kg: 7A receives 60 kg, 7B 36 kg and 8A 24 kg. The method is exact for the stated ratio, but the weights are rounded to the nearest kilogram, so the total may differ by a small amount. Our main assumption is that hours are a fair measure of contribution. This is a limitation: a class may have worked fewer hours but in harder conditions, or may have greater need. We therefore recommend the proportional share with a minimum allocation for any class whose share falls below 20 kg." The response should include the table, the method, a clear recommendation, a statement of accuracy and a limitation.

MIXED PRACTICE

1 **2 : 3**.

2 $0.8 : 1.2 = 8 : 12 =$ **2 : 3**.

3 Scale factor 4, so $? = 9 \times 4 =$ **36**.

4 $3 + 4 = 7$ parts; $84 \div 7 = 12$. **36** and **48**.

5 $2 + 5 + 7 = 14$ parts; $210 \div 14 = 15$. **30**, **75**, **105**.

6 40 min : 180 min $= 40 : 180 =$ **2 : 9**.

7 7 parts $= 21$ L, so 1 part $= 3$ L. Syrup $= 2 \times 3 =$ **6 L**.

8 Scale factor from 2 to 6 is 3, so $5 \times 3 = 15$. From 2 to 10 is 5, so $5 \times 5 = 25$. Missing values: **15**, **25**.

9 $5 : 8 = 0.625$ and $7 : 11 \approx 0.636$, so **7 : 11** is larger.

10 Length + width $= 78 \div 2 = 39$ cm. $8 + 5 = 13$ parts, so 1 part $= 3$ cm. Length **24 cm**, width **15 cm**.

11 $6 : 9$ still has common factor 3. $6 : 9 = \mathbf{2 : 3}$, which is simplest.

12 $3 : 4 = 21 : 28$, so milk $= \mathbf{28}$ cups. Total $= 21 + 28 = \mathbf{49}$ cups.

13 5 parts $= 40$, so 1 part $= 8$. Blue $= 3 \times 8 = 24$. Total $= 40 + 24 = \mathbf{64}$.

14 $3 + 4 + 5 = 12$ parts; $180° \div 12 = 15°$. Angles: $\mathbf{45°}$, $\mathbf{60°}$, $\mathbf{75°}$.

15 Largest share $= 5$ parts $= \$250$, so 1 part $= \$50$. Total $= 10 \times 50 = \mathbf{\$500}$. Shares: $\mathbf{\$100}$, $\mathbf{\$150}$, $\mathbf{\$250}$.

16 $\frac{3}{4} = 0.75$ and $\frac{5}{7} \approx 0.714$, so $\mathbf{3 : 4}$ is greater. Cross products: $3 \times 7 = 21 > 20 = 4 \times 5$.

17 $3\text{ cm} : 8\text{ km} = 12\text{ cm} : 32\text{ km}$, so the distance is $\mathbf{32\text{ km}}$.

18 1 part $= 2.5$ L, so water $= 4 \times 2.5 = \mathbf{10\text{ L}}$. Total $= 2.5 + 10 = \mathbf{12.5\text{ L}}$.

19 $7 + 5 = 12$ parts; $480 \div 12 = 40$. Girls $= \mathbf{280}$; boys $= \mathbf{200}$.

20 $5 + 4 = 9$ parts; $72 \div 9 = 8$. Chocolate $= 5 \times 8 = \mathbf{40}$; fruit $= 4 \times 8 = 32$. After 8 chocolate are eaten, chocolate $= 40 - 8 = 32$, so the new ratio is $32 : 32 = \mathbf{1 : 1}$.

21 Scale factor $= 18 \div 12 = 1.5$, so the longer side is $18 \times 1.5 = \mathbf{27\text{ cm}}$. The new dimensions are 18×27 cm; each dimension is multiplied by 1.5, so the scale ratio is $\mathbf{3 : 2}$ (new : original).

22 Make the shared quantity b the same: $a : b = 2 : 5 = 6 : 15$ and $b : c = 3 : 4 = 15 : 20$. Therefore $a : b : c = \mathbf{6 : 15 : 20}$.

23 Difference $= 5 - 4 = 1$ part $= \$36$, so 1 part $= \$36$. Total $= 9 \times 36 = \mathbf{\$324}$.

24 Scale factor $= 20 \div 8 = 2.5$. Flour $= 3 \times 2.5 = \mathbf{7.5}$ cups; milk $= 2 \times 2.5 = \mathbf{5}$ cups. The amounts are not whole numbers because the scale factor is not an integer; half-cups can be measured, so this is not a problem in a kitchen. If only whole cups can be measured, the cook must round and accept a slightly different ratio, which should be stated.

25 **Sample response.** Context: 600 mL of juice is shared among three bottles in the ratio $2 : 3 : 5$. Total parts $= 10$, so one part $= 60$ mL: 120 mL, 180 mL and 300 mL. A fair question is whether the bottle sizes differ; if the largest bottle is for a group of four and the smallest for one person, equal volumes may not be fair. The solution assumes the juice is shared by groups of equal size. The judgement should state this assumption and one alternative, such as sharing per person instead.

Chapter 2 · Proportional reasoning

PREREQUISITE CHECK

1 Divide both terms by 6: $\frac{18}{24} = \frac{3}{4}$.

2 $7 \times 12 = \mathbf{84}$.

3 $3.5 \times 100 = \mathbf{350\text{ cm}}$.

4 $4x = 52$, so $x = 52 \div 4 = \mathbf{13}$.

5 $0.4 = \frac{4}{10} = \frac{2}{5}$.

6 $15 \div 0.5 = 15 \div \frac{1}{2} = \mathbf{30}$.

PRACTICE 2.1 · PROPORTIONAL OR NOT?

1 **A is proportional:** $\frac{3}{2} = \frac{6}{4} = \frac{9}{6} = 1.5$. **B is not:** $\frac{4}{1} = 4$, $\frac{6}{2} = 3$, $\frac{8}{3} \approx 2.67$.

2 $k = \frac{10}{4} = \frac{15}{6} = \frac{20}{8} = \mathbf{2.5}$, so $\mathbf{y = 2.5x}$.

3 $x = 5 \Rightarrow y = 35$; $y = 56 \Rightarrow x = 8$. Missing values: **35** and **8**.

4 **No.** At $x = 1$, $y = 5$ and $\frac{y}{x} = 5$; at $x = 2$, $y = 9$ and $\frac{y}{x} = 4.5$. The ratios differ. Also $y = 1$ when $x = 0$, not 0.

5 $k = \frac{10}{4} = \mathbf{2.5}$; when $x = 7$, $y = 2.5 \times 7 = \mathbf{17.5}$.

6 **No.** $C = 5 + 0.10m$. At 0 minutes the cost is \$5, and the ratios $\frac{C}{m}$ decrease as m increases (for example $\frac{5.10}{1} = 5.10$ but $\frac{6.10}{2} = 3.05$). The graph has intercept 5.

7 **No.** $\frac{A}{s} = s$, which changes with s. For $s = 2$, $A = 4$ and $\frac{A}{s} = 2$; for $s = 3$, $A = 9$ and $\frac{A}{s} = 3$. Area is proportional to s^2, not to s.

8 $k = \frac{12}{3} = 4$; check $\frac{32}{8} = 4$, so the constant is consistent. Missing value $= 4 \times 5 = \mathbf{20}$.

9 The table is $x = 1, 3, 5, 7$ and $y = 3, 10, 17, 24$. The ratios are 3, 3.33, 3.4, 3.43, which are not equal. **The claim is false.** Equal differences do not imply proportionality; here $y = 3.5x - 0.5$.

10 When x increases by 2, y increases by $3.5 \times 2 = \mathbf{7}$, because $\Delta y = k\,\Delta x$. From $x = 6$ to $x = 11$ the increase is 5, so y increases by $3.5 \times 5 = 17.5$ and $y = 21 + 17.5 = \mathbf{38.5}$. Check: $3.5 \times 11 = 38.5$.

PRACTICE 2.2 · PROPORTIONAL PROBLEMS

1 One pen costs $6.25 \div 5 = \$1.25$. Eight pens cost $8 \times 1.25 = \mathbf{\$10.00}$.

2 One kilogram costs $4.80 \div 3 = \$1.60$. Seven kilograms cost $7 \times 1.60 = \mathbf{\$11.20}$.

3 $96 \div 12 = 8\text{ m}^2$ per litre, so 5 L cover $5 \times 8 = \mathbf{40\text{ m}^2}$.

4 One serving uses $600 \div 4 = 150$ g. Ten servings need $10 \times 150 = \mathbf{1500\text{ g} = 1.5\text{ kg}}$.

5 $\frac{3}{5} = \frac{x}{20}$, so $5x = 60$ and $x = \mathbf{12}$.

6 $\frac{8}{15} = \frac{24}{y}$, so $8y = 360$ and $y = \mathbf{45}$. Equivalently, the numerator is multiplied by 3, so the denominator is $15 \times 3 = 45$.

7 The speed is $150 \div 2 = 75$ km/h. In 5 hours the car travels $5 \times 75 = \mathbf{375\text{ km}}$.

8 Scale factor $= 7.5 \div 3 = 2.5$. Flour $= 2 \times 2.5 =$ **5** cups.

9 Shop A: $\$4.20 \div 3 = \$1.40$ per notebook. Shop B: $\$6.75 \div 5 = \$1.35$ per notebook. **Shop B is better value** by $\$0.05$ per notebook.

10 Adding 7 to both terms gives $\frac{11}{12}$, which is not equivalent to $\frac{4}{5}$. Adding the same number to numerator and denominator does not preserve a ratio. Correctly, $5x = 48$, so $x =$ **9.6**.

11 Coverage is $28 \div 3.5 = 8$ m^2 per litre. For 50 m^2: $50 \div 8 =$ **6.25 L**. To a sensible accuracy, about **6.3 L**. If paint is sold only in whole litres, buy 7 L.

12 Total parts $= 3 + 5 = 8$; one part $= 24 \div 8 = 3$ L. Red $= 3 \times 3 =$ **9 L**; white $= 5 \times 3 =$ **15 L**. Check: $9 + 15 = 24$ L.

PRACTICE 2.3 · SCALE AND SIMILARITY

1 Real length $= 4.5 \times 100 = 450$ cm $=$ **4.5 m**. *Alternatively,* solve the proportion $\frac{1}{100} = \frac{4.5}{x}$, giving $x = 450$.

2 8 m $= 800$ cm; plan length $= 800 \div 200 =$ **4 cm**.

3 $7.2 \times 50\,000 = 360\,000$ cm $= 3600$ m $=$ **3.6 km**.

4 3 cm : 12 km $= 3$ cm : 1 200 000 cm, so the scale is **1 : 400 000**.

5 Scale factor $= 6 \div 4 = 1.5$ (also $9 \div 6 = 1.5$ and $12 \div 8 = 1.5$). Ratios: $\frac{4}{6} = \frac{6}{9} = \frac{8}{12} = \frac{2}{3}$, so corresponding sides are in the same ratio.

6 New dimensions **15 cm** × **24 cm**. Perimeter: original $2(5 + 8) = 26$ cm, new $2(15 + 24) =$ **78 cm** $= 3 \times 26$. Area: original 40 cm^2, new $15 \times 24 =$ **360 cm^2** $= 9 \times 40$.

7 Real length $= 18 \times 24 = 432$ cm $=$ **4.32 m**.

8 Scale factor $= 25 \div 10 = 2.5$. New height $= 15 \times 2.5 =$ **37.5 cm**. Area $= 25 \times 37.5 =$ **937.5 cm^2** (the original area 150 cm^2 is multiplied by $2.5^2 = 6.25$).

9 The comparison is reversed. At 1 : 100, the drawing is 100 times smaller than the real object; the real object is 100 times larger than the drawing. Correct statement: **1 cm on the drawing represents 100 cm in reality.**

10 2 km $= 200\,000$ cm. At 1 : 25 000: $200\,000 \div 25\,000 =$ **8 cm**. At 1 : 50 000: $200\,000 \div 50\,000 =$ **4 cm**. The 1 : 25 000 map shows more detail, because each centimetre represents 250 m rather than 500 m.

INVESTIGATION · A PLAN THAT FITS

1 The garden is 15 m × 9 m $= 1500$ cm × 900 cm. At 1 : 50: **30 cm** × **18 cm**. At 1 : 100: **15 cm** × **9 cm**. At 1 : 200: **7.5 cm** × **4.5 cm**.

2 The usable area is 17 cm × 25.7 cm. The 1 : 50 plan is 30 cm long, which exceeds 25.7 cm, so it does not fit in either orientation. The 1 : 100 plan (15 × 9) and the 1 : 200 plan (7.5 × 4.5) both fit. The 1 : 100 plan uses the paper well without crowding the border.

3 At 1 : 100, the shed is 3 m × 2 m $= 300$ cm × 200 cm, so its plan dimensions are **3 cm** × **2 cm**.

4 At 1 : 100, the bed is 4.5 m × 1.8 m $= 450$ cm × 180 cm, so its plan dimensions are **4.5 cm** × **1.8 cm**.

5 At 1 : 100: 7.2 cm → 720 cm $=$ **7.2 m**; 4.8 cm → 480 cm $=$ **4.8 m**.

6 For a scale of 1 : n, with both lengths in the same unit: plan length $= \frac{\text{real length}}{n}$ and real length $= n \times$ plan length. Equivalently, $\frac{\text{plan length}}{\text{real length}} = \frac{1}{n}$.

7 **Sample answer.** At 1 : 200, the garden is only 7.5 cm × 4.5 cm, the shed is 1.5 cm × 1 cm and a bed is 2.25 cm × 0.9 cm. Small features are hard to draw accurately and hard to label without the labels overwhelming the plan. A 1 mm error on the plan represents 20 cm in reality at 1 : 200, but only 10 cm at 1 : 100, so the larger-scale plan is both easier to read and more precise.

REAL-LIFE APPLICATION · A MODEL SPORTS HALL

1 Divide each real length in centimetres by 200: length $3600 \div 200 =$ **18 cm**; width $2400 \div 200 =$ **12 cm**; height $800 \div 200 =$ **4 cm**.

2 **Yes.** The model's base is 18 cm × 12 cm, and the model is 4 cm tall. It fits on a 50 cm × 40 cm board with room to spare: $18 < 50$, $12 < 40$, and the height does not affect the board area.

3 Model base area $= 18 \times 12 = 216$ cm^2. Real floor area $= 36 \times 24 = 864$ m^2. The length scale factor is 200, so the area factor is $200^2 = 40\,000$. Check: 216 cm^2 × 40 000 $= 8\,640\,000$ cm^2. Since 1 m^2 $= 10\,000$ cm^2, this is $8\,640\,000 \div 10\,000 =$ **864 m^2**, as required.

4 **Sample answer.** The model cannot show internal features such as changing rooms, equipment, lighting, surface texture or people. At 1 : 200, 1 cm represents 2 m, so any feature smaller than about 2 m is either invisible or must be exaggerated. The scale controls the level of detail: the smaller the model, the more features must be simplified or omitted, and the easier it is for the model to give a misleading impression of space.

CRITERION TASK · THE NATURE RESERVE MAP

1 Multiply each map length by 10 000: 8.4 cm → 84 000 cm = **840 m**; 6.7 cm → 67 000 cm = **670 m**; 4.2 cm → 42 000 cm = **420 m**; 5.1 cm → 51 000 cm = **510 m**.

2 $840 + 670 + 420 + 510 = 2440$ m = **2.44 km**.

3 1 : 10 000 means 1 unit on the map represents 10 000 of the same units in reality. In centimetres, 1 cm on the map represents 10 000 cm = 100 m in reality.

4 **Sample choice:** 1 : 12 000. The reserve is 240 000 cm × 180 000 cm. At 1 : 12 000 the plan is $240\,000 \div 12\,000 =$ **20 cm** by $180\,000 \div 12\,000 =$ **15 cm**. This fits the 38 cm × 26 cm usable area and leaves a border of 9 cm horizontally and 5.5 cm vertically. 1 : 10 000 would also fit (24 × 18 cm), but 1 : 12 000 leaves more room for a title, key and labels while keeping both dimensions comfortably inside the paper. Accept any scale whose plan fits both dimensions and whose border is reasonable, with a correct justification.

5 The 4.2 cm section is 420 m = 42 000 cm in reality. At 1 : 12 000, its map length is $42\,000 \div 12\,000 =$ **3.5 cm**. It is shorter than the original 4.2 cm because the new scale is smaller: each centimetre now represents 120 m instead of 100 m, so the same real distance needs fewer centimetres.

6 **Sample proposal.** "Nature reserve trail map — scale 1 : 12 000 (1 cm represents 120 m). The reserve measures 20 cm by 15 cm on the plan, leaving a clear border. The four trail sections have real lengths 840 m, 670 m, 420 m and 510 m, giving a total trail length of 2.44 km. Map lengths are measured to the nearest 0.1 cm, so each represents about 12 m; the total therefore has an uncertainty of roughly ±48 m (four measurements of ±12 m). The map shows horizontal distances and the positions of the trails; it does not show changes in height, surface conditions or seasonal closures, so walkers should not use it as a substitute for a topographic map." The response should include the scale statement, the table of real lengths, a comment on accuracy, a limitation and a clear recommendation. A correct table and scale with a reasonable limitation should be accepted even if the chosen scale differs.

MIXED PRACTICE

1 $\frac{12.5}{5} = \frac{20}{8} = \frac{30}{12} =$ **2.5**.

2 $y = 2.5x$; when $x = 15$, $y =$ **37.5**.

3 $\frac{9}{4} = \frac{x}{10}$, so $4x = 90$ and $x =$ **22.5**.

4 $\frac{6}{11} = \frac{18}{y}$, so $6y = 198$ and $y =$ **33**.

5 Unit price = $9.50 \div 5 =$ \$1.90 per kg. Eight kilograms cost $8 \times 1.90 =$ **\$15.20**.

6 Fuel per kilometre = $7 \div 84 = \frac{1}{12}$ L. For 156 km: $156 \div 12 =$ **13 L**.

7 6 m = 600 cm; plan length = $600 \div 150 =$ **4 cm**.

8 $9.5 \times 40\,000 = 380\,000$ cm = 3800 m = **3.8 km**.

9 **Yes.** $\frac{7}{2} = \frac{14}{4} = \frac{24.5}{7} =$ **3.5**, so the ratios are equal and $k = 3.5$.

10 Area = $7 \times 11 = 77$ cm^2. A length factor of 4 multiplies area by $4^2 = 16$, so the new area is $77 \times 16 =$ **1232 cm²**. Check: $28 \times 44 = 1232$.

11 **No.** $C = 8 + 0.05m$. The fixed \$8 means $C = 8$ when $m = 0$, so the graph is a straight line with intercept 8, not through the origin. The ratios also change: $\frac{13}{100} = 0.13$ but $\frac{18}{200} = 0.09$.

12 Scale factor = $15 \div 6 = 2.5$. Pasta = $450 \times 2.5 =$ **1125 g**; sauce = $300 \times 2.5 =$ **750 g**.

13 One litre fills $10 \div 2.5 = 4$ glasses, so 6 L fill $6 \times 4 =$ **24** glasses.

14 Shop A: \$7.60 ÷ 4 = \$1.90 per kg. Shop B: \$11.10 ÷ 6 = \$1.85 per kg. **Shop B is better value** by \$0.05 per kg.

15 Scale factor = $30 \div 12 = 2.5$. New width = $8 \times 2.5 =$ **20 cm**. New area = $20 \times 30 =$ **600 cm²**; the area factor is $2.5^2 =$ **6.25**, and $96 \times 6.25 = 600$.

16 Real length = $45 \times 80 = 3600$ cm = **36 m**.

17 A 3 cm square has area 9 cm^2. Doubling the side to 6 cm gives area 36 cm^2, which is 4 times as large, not 2 times. **The claim is false:** when lengths are multiplied by k, areas are multiplied by k^2.

18 Real distance = $3 \times 20\,000 = 60\,000$ cm = 600 m. On Map B: $60\,000 \div 50\,000 =$ **1.2 cm**.

19 $k = \frac{10}{4} = 2.5$, so $y = 2.5 \times 14 =$ **35**. **Method 2:** the scale factor from 4 to 14 is 3.5, so $y = 10 \times 3.5 = 35$. **Method 3:** a unit increase of 1 in x gives 2.5 in y; 14 units give 35.

20 Real dimensions = $12 \times 50 = 600$ cm = **6 m** and $9 \times 50 = 450$ cm = **4.5 m**. Real area = $6 \times 4.5 =$ **27 m²**.

21 $k = \frac{12.5}{5} = \frac{22.5}{9} =$ **2.5**. Equivalently, the change in y is $22.5 - 12.5 = 10$ and the change in x is $9 - 5 = 4$, so $k = \frac{10}{4} = 2.5$; for a proportional relationship, $\frac{\Delta y}{\Delta x} = k$.

22 $4.8 \times 25\,000 = 120\,000$ cm = 1200 m = **1.2 km**. $3.6 \times 25\,000 = 90\,000$ cm = 900 m = **0.9 km**. Real area = $1.2 \times 0.9 =$ **1.08 km²**. Check with areas: $4.8 \times 3.6 = 17.28$ cm^2, and $17.28 \times 25\,000^2 = 10\,800\,000\,000$ cm^2 = 1.08 km^2.

23 The combined length factor is $2.5 \times 0.8 =$ **2**. **No,** the final image is not the same size: each length is twice the original, so the area is $2^2 = 4$ times the original.

24 Real height = $1.2 \times 150 = 180$ m. After the increase, the real height is $180 + 6 = 186$ m. The model height should be $186 \div 150 =$ **1.24 m**. Equivalently, add $6 \div 150 = 0.04$ m to the original 1.2 m.

25 **Sample response.** Room: 5 m × 4 m, scale 1 : 50, giving a plan 10 cm × 8 cm. Real lengths from the plan: door 4.2 cm → 2.1 m; window 3.2 cm → 1.6 m; bed 4.0 cm → 2.0 m; wardrobe 2.4 cm → 1.2 m. The plan cannot show ceiling height, so it cannot tell us whether the wardrobe fits under a sloped ceiling. A measurement to the nearest 0.1 cm corresponds to 5 cm in reality, so I would state lengths to the nearest 5 cm. Include a labelled sketch, the scale statement and an accuracy statement. Accept any consistent plan with correct conversions and a genuine limitation.

Chapter 3 · Rates and unit rates

PREREQUISITE CHECK

1 $240 \div 8 = \mathbf{30}$.

2 Divide both terms by 15: $45 : 60 = \mathbf{3 : 4}$.

3 $2.5 \times 60 = \mathbf{150}$ minutes.

4 $0.75 \times 1000 = \mathbf{750\ g}$.

5 $3.6 \times 4 = \mathbf{14.4}$.

6 The third decimal digit is 6, so $7.846 \approx \mathbf{7.85}$.

PRACTICE 3.1 · RATES AND UNIT RATES

1 (a) $240 \div 4 = \mathbf{60\ km/h}$. (b) $18 \div 6 = \mathbf{\$3/kg}$. (c) $96 \div 3 = \mathbf{32\ words/min}$. (d) $150 \div 5 = \mathbf{30\ mL/dose}$.

2 $42 \div 6 = \mathbf{7\ L/min}$. Time $= 105 \div 7 = \mathbf{15\ min}$.

3 $72.50 \div 5 = \mathbf{\$14.50}$ per shirt. Three shirts cost $3 \times 14.50 = \mathbf{\$43.50}$.

4 $320 \div 40 = \mathbf{8\ km/L}$. Fuel per 100 km: $100 \div 8 = \mathbf{12.5\ L/100\ km}$.

5 $750 \div 15 = \mathbf{50\ g}$ per muffin. For 24 muffins: $24 \times 50 = \mathbf{1200\ g = 1.2\ kg}$.

6 $72 \div 60 = 1.2$ beats per second, so the rate is **72 beats/min**. In 15 s: $15 \times 1.2 = \mathbf{18\ beats}$.

7 $31.50 \div 3.5 = \mathbf{\$9.00/kg}$. For 2.2 kg: $2.2 \times 9.00 = \mathbf{\$19.80}$.

8 $400 \div 80 = \mathbf{5\ m/s}$. Three minutes is 180 s, so the distance is $5 \times 180 = \mathbf{900\ m}$.

9 The division is correct, but the unit is wrong: $12 \div 15 = 0.8$ km per **minute** , not per hour. In km/h, $0.8 \times 60 = \mathbf{48\ km/h}$. Equivalently, 15 min $= 0.25$ h and $12 \div 0.25 = 48$ km/h.

10 Four bottles hold $4 \times 500 = 2000$ mL $= 2$ L. The cost is $5.60 \div 2 = \mathbf{\$2.80/L}$. The number of bottles says nothing about volume: 4 small bottles may hold less than 2 large ones, so compare total volume and price.

PRACTICE 3.2 · BEST BUYS

1 $3.60 \div 2 = \$1.80/kg$; $8.50 \div 5 = \$1.70/kg$. **The 5 kg bag is better value.** *Alternatively,* compare 10 kg: five 2 kg bags cost \$18.00, while two 5 kg bags cost \$17.00.

2 $4.20 \div 0.5 = \$8.40/kg$; $6.00 \div 0.75 = \$8.00/kg$. **The 750 g pack is better value.**

3 Six bottles: $6 \times 0.5 = 3$ L, so $3.00 \div 3 = \$1.00/L$. Twelve bottles: $12 \times 0.5 = 6$ L, so $5.40 \div 6 = \$0.90/L$. **The 12-pack is better value.**

4 $6.50 \div 0.25 = \$26.00/kg$; $23.00 \div 1 = \$23.00/kg$. **The 1 kg pack is better value.**

5 $7.20 \div 3 = \$2.40/L$; $11.50 \div 5 = \$2.30/L$. **The 5 L bottle is better value.**

6 4-pack: $4 \times 200 = 800$ sheets, so $3.20 \div 8 = \$0.40$ per 100 sheets. 9-pack: $9 \times 180 = 1620$ sheets, so $6.30 \div 16.2 \approx \$0.389$, or about 38.9 cents per 100 sheets. **The 9-pack is slightly better value.**

7 "Buy 2, get 1 free": 3 bars cost $2 \times 2.40 = \$4.80$, so \$1.60 per bar. The other shop charges $6.60 \div 3 = \$2.20$ per bar. **The first offer is much better value.**

8 $10.50 \div 1.5 = \$7.00/kg$; $14.96 \div 2.2 = \$6.80/kg$. **The 2.2 kg pack is better value.**

9 $10.50 \div 5 = \$2.10/kg$; $21.50 \div 10 = \$2.15/kg$. **The 5 kg bag is better value.** A larger bag can cost more per kilogram because of packaging, transport, storage or marketing; size alone is not evidence of value.

10 $30 \div 100 = \$0.30$ per Mbps; $60 \div 250 = \$0.24$ per Mbps. The 250 Mbps plan has the lower cost per Mbps, but it costs twice as much in total and may provide more speed than needed. **The lower unit rate is not automatically the better choice.**

11 The student compared totals, not unit prices. Six pens: $4.80 \div 6 = \$0.80$ each. Ten pens: $7.50 \div 10 = \$0.75$ each. **The 10-pack is better value** by \$0.05 per pen.

12 **Sample response.** 500 g of pasta for \$2.40 has unit price $2.40 \div 0.5 = \$4.80/kg$. 1 kg for \$4.50 has unit price $\$4.50/kg$, so the 1 kg pack is better value. A buyer living alone might still choose the 500 g pack to avoid waste or a larger upfront cost. The recommendation must state which criterion — price per kilogram or household need — is being prioritised.

PRACTICE 3.3 · SPEED, DENSITY AND FLOW

1 Speed $= 150 \div 2 =$ **75 km/h**. In 3.5 h: $3.5 \times 75 =$ **262.5 km**.

2 $240 \div 40 =$ **6 m/s**. In km/h: $6 \times 3.6 =$ **21.6 km/h**.

3 45 min $= 0.75$ h, so distance $= 72 \times 0.75 =$ **54 km**.

4 2 min $= 120$ s, so distance $= 5 \times 120 =$ **600 m**.

5 Speed $= 1800 \div 3 =$ **600 km/h**. Time for 2400 km: $2400 \div 600 =$ **4 h**.

6 1.2 kg $= 1200$ g, so density $= 1200 \div 150 =$ **8 g/cm³**. Mass of 80 cm³: $8 \times 80 =$ **640 g**.

7 Mass $= 0.9 \times 250 =$ **225 g**.

8 Time $= 90 \div 3.6 =$ **25 min**. Volume in 12 min: $3.6 \times 12 =$ **43.2 L**.

9 $90 \div 3.6 =$ **25 m/s**; $15 \times 3.6 =$ **54 km/h**.

10 The speed is $300 \div 4 = 75$ km/h, but that does not mean 75 minutes for 100 km. Time $= 100 \div 75 = \frac{4}{3}$ h $=$ **1 h 20 min**. The student confused the numerical value of the speed with a time in minutes.

INVESTIGATION · THE UNIT-PRICE TRAP

1 1 kg: $\$2.40/kg$. 5 kg: $10.50 \div 5 = \$2.10/kg$. 10 kg: $21.50 \div 10 = \$2.15/kg$.

2 The **5 kg package** is cheapest per kilogram at \$2.10/kg; the 1 kg package is \$2.40/kg and the 10 kg package is \$2.15/kg.

3 30 kg is six 5 kg packages: $6 \times 10.50 =$ **\$63.00**. (Three 10 kg packages would cost \$64.50, and thirty 1 kg packages would cost \$72.00.)

4 At the listed prices, **no** : the 10 kg package is \$0.05/kg more expensive than the 5 kg package. It does mean fewer bags to carry and store, so a buyer may value the convenience, but the mathematical comparison still favours the 5 kg package.

5 New price $= 21.50 \times 0.85 = \$18.275$, which rounds to **\$18.28** . New unit price $= 18.275 \div 10 = \$1.8275 \approx$ **\$1.83/kg**. Yes, it becomes the cheapest option, below \$2.10/kg.

6 **Rule:** divide the total price by the quantity, in the same unit, to get the unit price. Apply any offer to the price first, then divide. Compare unit prices, not totals, and check that the quantities are equivalent. The lowest unit price is the best value only when the products and the quantities needed are comparable.

7 **Sample response.** A family might choose a smaller package with a higher unit price because it fits the storage space, avoids waste, or costs less to buy today. The unit-price comparison remains correct; the decision also weighs budget, storage and how quickly the food will be used.

REAL-LIFE APPLICATION · WATER FOR THE GARDEN

1 Volume $= 1.2 \times 0.8 \times 0.9 = 0.864$ m³. Since 1 m³ $= 1000$ L, the capacity is **864 L**.

2 $864 \div 12 =$ **72 min**, or **1 h 12 min** .

3 The net rate is $12 - 18 = -6$ L/min. Water leaves faster than it enters, so the volume decreases by 6 L each minute; the tank will empty rather than fill.

4 The inlet must be **greater than 18 L/min** . At exactly 18 L/min the net rate is zero; any rate above 18 L/min produces a positive net rate.

5 Net rate $= 24 - 18 = 6$ L/min. Time $= 864 \div 6 =$ **144 min**, or **2 h 24 min** .

6 864 L $= 0.864$ thousand litres. Cost $= 0.864 \times 2.10 = \$1.8144 \approx$ **\$1.81**. **Assumption:** the tank starts empty, the rates stay constant, and there are no leaks or evaporation. If water is billed in whole thousands of litres, the charge may be higher.

CRITERION TASK · THE BREAKFAST CLUB CONTRACT

1 Supplier A: $18 + 2.10 \times 20 = \$60$; $18 + 2.10 \times 40 = \$102$; $18 + 2.10 \times 60 = \$144$. Supplier B: $6 + 2.40 \times 20 = \$54$; $6 + 2.40 \times 40 = \$102$; $6 + 2.40 \times 60 = \$150$.

2 At 40 L, both suppliers charge **\$102** .

3 Supplier A: $102 \div 40 =$ **\$2.55/L**. Supplier B: $102 \div 40 =$ **\$2.55/L**.

4 At 80 L: A costs $18 + 2.10 \times 80 = \$186$; B costs $6 + 2.40 \times 80 = \$198$. **Supplier A is cheaper by \$12.**

5 Let x be the litres. Solve $18 + 2.10x = 6 + 2.40x$. Then $12 = 0.30x$, so $x =$ **40 L**. Check: each charges $18 + 84 = \$102$ and $6 + 96 = \$102$. Below 40 L, B is cheaper; above 40 L, A is cheaper.

6 **Sample recommendation.** At the planned 40 L per week, both suppliers cost \$102, or \$2.55/L. The choice should therefore use non-price evidence: delivery reliability, refrigeration, milk quality, notice periods and whether the contract can change if demand rises. If the club expects demand to grow above 40 L, Supplier A is cheaper because its per-litre rate is lower; if demand may fall below 40 L, Supplier B is cheaper. **Limitation:** the comparison assumes the fees and per-litre prices stay fixed and ignores storage, spoilage and delivery timing. The break-even quantity is 40 L.

MIXED PRACTICE · CONSOLIDATE AND EXTEND

1 $180 \div 3 =$ **60 km/h**.

2 $12.60 \div 4 =$ **\$3.15/kg**.

3 $75 \div 5 =$ **15 L/min**.

4 $54 \div 3.6 =$ **15 m/s**.

5 $8 \times 3.6 =$ **28.8 km/h**.

6 $7 \times 2.80 =$ **\$19.60**.

7 $65 \times 4 = \mathbf{260\ km}$.

8 $450 \div 50 = \mathbf{9\ g/cm^3}$.

9 $2.5 \times 8 = \mathbf{20\ L}$.

10 $7.20 \div 3 = \$2.40/kg$; $11.50 \div 5 = \$2.30/kg$. **The 5 kg bag is better value.**

11 $2.70 \div 1.5 = \$1.80/L$; $3.40 \div 2 = \$1.70/L$. **The 2 L bottle is better value.**

12 $900 \div 12 = 75$ g per serving; $20 \times 75 = \mathbf{1500\ g = 1.5\ kg}$.

13 1 h 36 min $= 1.6$ h, so speed $= 240 \div 1.6 = \mathbf{150\ km/h}$.

14 $5 \times 7.5 = \mathbf{37.5\ min}$, or **37 min 30 s** .

15 $15.36 \div 2.4 = \mathbf{\$6.40/kg}$. For 1.5 kg: $1.5 \times 6.40 = \mathbf{\$9.60}$.

16 20 m/s $= 20 \times 3.6 = 72$ km/h, so 20 m/s is **faster** than 70 km/h. Comparing 20 with 70 without converting units is invalid.

17 $3.75 \div 0.75 = \$5.00/L$; $6.90 \div 1.5 = \$4.60/L$. **The 1.5 L bottle is better value.**

18 $6.5 \times 3.2 = \mathbf{20.8\ L}$. (The journey is 3.2 lots of 100 km.)

19 $12.50 \div 5 = \$2.50/kg$; $5.20 \div 2 = \$2.60/kg$. **The 5 kg bag is better value.** For 10 kg: two 5 kg bags cost \$25.00; five 2 kg bags cost \$26.00, so the saving is **\$1.00** .

20 Number of bottles $= 48 \div 2 = 24$. Total cost $= 24 \times 2.10 = \mathbf{\$50.40}$. Cost per litre $= 50.40 \div 48 = \mathbf{\$1.05/L}$.

21 Outward time $= 150 \div 60 = 2.5$ h; return time $= 150 \div 40 = 3.75$ h. Total time $= 6.25$ h; total distance $= 300$ km. Average speed $= 300 \div 6.25 = \mathbf{48\ km/h}$. It is not 50 km/h because more time is spent at the slower speed.

22 Combined rate $= 12 + 18 = 30$ L/min, so time $= 900 \div 30 = \mathbf{30\ min}$. With the outlet, net rate $= 30 - 6 = 24$ L/min, so time $= 900 \div 24 = \mathbf{37.5\ min}$ or **37 min 30 s** .

23 $8.50 \div 0.25 = \$34.00/kg$; $30.00 \div 1 = \$30.00/kg$. **The 1 kg pack is better value.** At \$30/kg, 2.5 kg costs $2.5 \times 30 = \mathbf{\$75.00}$.

24 Slower runner: $5 \div 8 = 0.625$ h $= \mathbf{37.5\ min}$. Faster runner: $5 \div 12 \approx 0.4167$ h $= \mathbf{25\ min}$. The slower runner finishes $37.5 - 25 = \mathbf{12.5\ min}$ later.

25 **Sample response.** 500 g for \$2.40 has unit price \$4.80/kg; 1 kg for \$4.50 has unit price \$4.50/kg, so the 1 kg pack is the better buy. A buyer might choose the 500 g pack if storage is limited, if the product may spoil, or if the smaller upfront cost matters more. The poster should show both unit prices, the saving per kilogram, and the non-price condition.

Chapter 4 · Integers and operations

PREREQUISITE CHECK

1 $47 + 28 = \mathbf{75}$.

2 $90 - 64 = \mathbf{26}$.

3 $7 \times 8 = \mathbf{56}$.

4 $72 \div 9 = \mathbf{8}$.

5 $2^4 = 2 \times 2 \times 2 \times 2 = \mathbf{16}$.

6 Multiplication before addition: $5 + 12 = \mathbf{17}$.

PRACTICE 4.1 · ORDER AND ABSOLUTE VALUE

1 (a) -5 (b) -120 (c) $+35$ (d) $+12$ (e) -7.

2 $-6, -4, -1, 0, 2, 5$, so $\mathbf{-6 < -4 < -1 < 0 < 2 < 5}$.

3 (a) $-4 < 1$ (b) $-9 < -2$ (c) $0 > -3$ (d) $-7 > -12$.

4 (a) opposite 11; $|-11| = 11$. (b) opposite -8; $|8| = 8$. (c) opposite 0; $|0| = 0$. (d) opposite 4; $|-4| = 4$.

5 (a) 15 (b) 7 (c) 6 (d) 0.

6 **9** and **-9**.

7 The negative sign means the account is overdrawn: \$45 is owed. The absolute value is **45**, which represents the **amount owed** , not a positive balance.

8 4°C is warmer. The difference is $4 - (-7) = 4 + 7 = \mathbf{11}$°C.

9 $-5 - (-18) = -5 + 18 = \mathbf{13}$ m. The diver ascends 13 m.

10 The student compared absolute values instead of positions. On the number line -12 is farther left than -5, so $\mathbf{-12 < -5}$. A larger absolute value makes a negative integer smaller, not greater.

11 The integers are **6** and **-6**. Since $6 > -6$, 6 is greater. Their sum is **0**: they are equal distances from zero on opposite sides, so they form a zero pair.

12 Order: $\mathbf{-8 < -3 < 0 < 2}$. The difference between highest and lowest is $2 - (-8) = \mathbf{10}$°C. Zero is the freezing point of water and the reference between negative and positive readings; it is a temperature, not the absence of temperature.

PRACTICE 4.2 · ADDING AND SUBTRACTING

1 (a) 4 (b) 4 (c) -10 (d) -6.

2 (a) -4 (b) -10 (c) 8 (d) 5.

3 (a) 8 because $-6 + 8 = 2$. (b) -3 because $-3 - 8 = -11$. (c) 8 because $5 - 8 = -3$.

4 $-9 + 4 + 6 = \mathbf{1}$.

5 $-4 + 9 - 6 = \mathbf{-1}$°C.

6 $80 - 125 + 60 = \mathbf{15}$. The balance is **\$15**, a positive balance.

7 Final depth: $-250 + 80 - 35 = \mathbf{-205}$ m. Net change: $80 - 35 = +45$ m, so the submarine rises 45 m overall.

8 $-3 - 7 + 4 = \mathbf{-6}$.

9 Subtracting -3 is the same as adding its opposite, 3: $7-(-3)=7+3=10$. On the number line, start at 7 and move 3 units right. The direction reverses because the change being removed was itself a move left.

10 Both terms are negative, so the result must be negative: $-2-5=-2+(-5)=\mathbf{-7}$, not 3. The student subtracted the magnitudes but changed the sign incorrectly.

11 The other integer is $-5-3=\mathbf{-8}$. Check: $3+(-8)=-5$.

12 Net change: $8-12+5-3=\mathbf{-2}$ yards. Total distance moved: $8+12+5+3=\mathbf{28}$ yards. The net change compares the final position with the start; total distance counts every yard moved, whatever the direction.

PRACTICE 4.3 · PRODUCTS, QUOTIENTS AND POWERS

1 (a) -42 (b) 40 (c) -36 (d) 12.

2 (a) -7 (b) 8 (c) -9 (d) 0.

3 (a) -32 (b) 16 (c) -16 (d) 1.

4 (a) $5+(-12)=\mathbf{-7}$. (b) $6+2=\mathbf{8}$. (c) $-8-5=\mathbf{-13}$.

5 $(-3)\times(-4)=12$ and $20\div(-5)=-4$, so $12+(-4)=\mathbf{8}$.

6 $3+7\times(-2)=3-14=\mathbf{-11}$°C.

7 $8+5\times(-3)=8-15=\mathbf{-7}$, so the lift finishes at floor -7.

8 (a) -8 (b) 24 (c) 3.

9 Continue the pattern: $4\times(-3)=-12$, $3\times(-3)=-9$, $2\times(-3)=-6$, $1\times(-3)=-3$, $0\times(-3)=0$, $(-1)\times(-3)=3$. Each time the first factor decreases by 1, the product increases by 3. Consistency therefore forces $(-4)\times(-3)=12$. Distributive check: $(-4)\times(-3)+(-4)\times 3=(-4)\times 0=0$, so $(-4)\times(-3)=12$.

10 $(-6)^2=(-6)(-6)=\mathbf{36}$, because the brackets include the sign in the base. $-6^2=-(6^2)=\mathbf{-36}$, because the power applies to 6 before the negative sign.

11 $-2^2+(-3)^2\times(-1)^3=-4+9\times(-1)=-4-9=\mathbf{-13}$.

12 The monthly changes are -4, so after 6 months the total change is $6\times(-4)=\mathbf{-\$24}$. Final balance: $30+(-24)=\mathbf{\$6}$. The negative change reduces the balance; the final positive value means \$6 remains.

REAL-LIFE APPLICATION · A DAY AT THE MOUNTAIN STATION

1 $\mathbf{-9<-8<-7<-5<-2<0<3}$.

2 $3-(-9)=3+9=\mathbf{12}$°C.

3 $-8-3=\mathbf{-11}$°C. The temperature falls by 11°C.

4 $3-(-9)=3+9=\mathbf{12}$°C, a rise of 12°C.

5 Sum: $-9-5+0+3-2-7-8=-28$. Mean: $-28\div 7=\mathbf{-4}$°C.

6 0°C is the freezing point of water and the boundary between the negative and positive readings in the table. It is a measured temperature, not "no temperature"; a reading of 0°C is colder than 3°C but warmer than -2°C.

7 Height difference: $2400-1600=800$ m $=0.8$ km. Predicted fall: $0.8\times 6=4.8$°C. Predicted station temperature: $6-4.8=\mathbf{1.2}$°C. The recorded mean is -4°C, which is 5.2°C colder. **Possible reasons:** the lapse rate is only an average; local exposure, snow cover, cold air sinking into the valley or the time of the readings can change the actual temperature.

INVESTIGATION · WHY IS NEGATIVE TIMES NEGATIVE POSITIVE?

1 Row -2: $4, 2, 0, -2, -4$. Row -1: $2, 1, 0, -1, -2$. Row 0: $0, 0, 0, 0, 0$. Row 1: $-2, -1, 0, 1, 2$. Row 2: $-4, -2, 0, 2, 4$.

2 As the column value increases by 1, the product decreases by 2: $4, 2, 0, -2, -4$.

3 As the row value increases by 1, the product in the column for -2 also decreases by 2: $4, 2, 0, -2, -4$.

4 The products increase by 4 each time the first factor decreases by 1: $-12, -8, -4, 0$. The next product must be **4**, so $(-1)\times(-4)=4$.

5 $0=(-1)\times 0=(-1)\times(1+(-1))=(-1)\times 1+(-1)\times(-1)=-1+?$. For the total to be 0, $?=\mathbf{1}$. Therefore $(-1)\times(-1)=1$.

6 **Multiplication:** same signs give a positive product, different signs give a negative product; any product with zero is zero. **Division:** same signs give a positive quotient, different signs give a negative quotient; $0\div a=0$ for $a\neq 0$, and division by zero is undefined. Examples: $(-5)(-3)=15$; $(-5)(3)=-15$; $-20\div(-4)=5$; $-20\div 4=-5$; $0\div(-6)=0$.

7 **Yes.** $(-2)^2=4$ (positive), $(-2)^3=-8$ (negative), $(-2)^4=16$ (positive). Each pair of negative factors gives a positive product; an odd exponent leaves one negative factor, so the result is negative, while an even exponent leaves none.

CRITERION TASK · THE CAVE RESEARCH STATION

1 Station 1: $1240 - 85 = \mathbf{1155}$ m; Station 2: $1240 - 160 = \mathbf{1080}$ m; Station 3: $1240 - 240 = \mathbf{1000}$ m; Station 4: $1240 - 310 = \mathbf{930}$ m.

2 Station 1 (1155 m) > Station 2 (1080 m) > Station 3 (1000 m) > Station 4 (930 m).

3 Entrance to Station 4: $1240 - 930 = \mathbf{310}$ m. Station 1 to Station 3: $1155 - 1000 = \mathbf{155}$ m.

4 Entrance to Station 4: $3 - 11 = -\mathbf{8}$°C, a fall of 8°C. Station 2 to Station 4: $3 - 6 = -\mathbf{3}$°C, a fall of 3°C.

5 $-2 - 5 + 3 = -\mathbf{4}$°C.

6 Charge after 3 trips and 5 hours: $100 - 3 \times 12 + 5 \times 7 = 100 - 36 + 35 = \mathbf{99}\%$. Two more trips use $2 \times 12 = 24\%$, leaving $99 - 24 = \mathbf{75}\%$. This is at least 70%, so **yes, two more trips are possible** on the stated figures, with 5 percentage points to spare. **Recommendation:** allow the two trips, but monitor the charge and keep a reserve for the return journey. **Assumption:** every trip uses exactly 12% and the panel restores exactly 7% per hour. **Limitation:** cold conditions, battery age, lights or other equipment could increase consumption, so the 75% estimate is not a guarantee.

MIXED PRACTICE · CONSOLIDATE AND EXTEND

1 $-\mathbf{9} < -\mathbf{6} < -\mathbf{2} < \mathbf{0} < \mathbf{4} < \mathbf{7}$.

2 (a) $-5 > -11$ (b) $3 > -4$ (c) $-8 < 0$.

3 (a) 14 (b) -9 (c) 0.

4 (a) 5 (b) -15 (c) -7 (d) 8.

5 (a) -42 (b) 32 (c) -7 (d) 9.

6 (a) -27 (b) 16 (c) -16.

7 $8 - 3 \times (-2) = 8 + 6 = \mathbf{14}$.

8 $-20 \div (-4) + 7 = 5 + 7 = \mathbf{12}$.

9 $(-4)^2 - 3 \times (-5) = 16 + 15 = \mathbf{31}$.

10 $-6 + 2 \times (-3) - (-8) \div 4 = -6 - 6 + 2 = -\mathbf{10}$.

11 $-5 - 4 + 7 = -\mathbf{2}$°C.

12 $60 - 95 + 40 = \mathbf{\$5}$.

13 Final depth: $-320 + 75 - 40 = -\mathbf{285}$ m. Net change: $75 - 40 = +35$ m, a rise of 35 m.

14 $6 - 9 + 4 = \mathbf{1}$, so the lift finishes at floor 1.

15 -12°C is colder. The difference is $-5 - (-12) = 7$ °C.

16 The signs differ, so subtract the magnitudes and keep the sign of the larger absolute value: $7 - 2 = 5$ and -7 is farther from zero, so $-7 + 2 = -\mathbf{5}$. On the number line, start at -7 and move 2 units right to -5; the result cannot be farther left than -7.

17 $(-5)^2 = (-5)(-5) = \mathbf{25}$. The bracket includes the negative sign in the base. $-5^2 = -(5^2) = -\mathbf{25}$, because the power applies to 5 before the negative sign.

18 Ascent $= 3 \times 6 = 18$ m. Final depth: $-24 + 18 = -\mathbf{6}$ m.

19 Final temperature: $-18 + 5 \times 4 = -18 + 20 = \mathbf{2}$°C. Overall change: $2 - (-18) = +\mathbf{20}$°C.

20 $850 - 1200 = -\mathbf{350}$ m. The hiker finishes 350 m below sea level.

21 Continuing: $1 \times (-3) = -3$, $0 \times (-3) = 0$. The products increase by 3 as the first factor decreases by 1. **Rule:** a positive integer times a negative integer is negative, with magnitude equal to the product of the magnitudes: $a \times (-b) = -(ab)$ for $a, b > 0$.

22 -7 is 7 units left of zero. Its opposite is the same distance on the other side, at 7. Taking the opposite of -7 therefore gives **7**, which is the same as $-(-7) = 7$.

23 Factor pairs with product -24: $(1, -24), (2, -12), (3, -8), (4, -6), (6, -4), (8, -3), (12, -2), (24, -1)$. Their sums are $-23, -10, -5, -2, 2, 5, 10, 23$. The pair with sum 2 is **6** and $-\mathbf{4}$. Check: $6 \times (-4) = -24$, $6 + (-4) = 2$.

24 Evaluate powers first: $-3^2 = -9$ and $(-2)^3 = -8$. Then multiply: $(-4) \times 2 = -8$. Expression: $-9 + (-8) - (-8) = -9 - 8 + 8 = -\mathbf{9}$.

25 Position: $-2 + 7 - 3 + 5 = \mathbf{7}$, so the lift finishes at floor 7. Total floors travelled: $|7| + |-3| + |5| = 7 + 3 + 5 = \mathbf{15}$ floors. The final floor compares the finishing position with the starting floor; total travel counts every floor moved, in both directions.

Chapter 5 · Algebraic expressions and equations

PREREQUISITE CHECK

1 $8 \times 7 = \mathbf{56}$.

2 $52 - 19 = \mathbf{33}$.

3 $96 \div 6 = \mathbf{16}$.

4 Multiplication before addition: $4 + 15 = \mathbf{19}$.

5 $2^4 = 2 \times 2 \times 2 \times 2 = \mathbf{16}$.

6 Subtract 9 from both sides: $x = \mathbf{6}$.

5.1 THE LANGUAGE OF ALGEBRA

1 (a) Terms $6x$, 11; coefficient 6; constant 11. (b) Terms $4y^2$, $-9y$, 2; coefficients 4 and -9; constant 2. (c) Terms 15, $-3d$; coefficient -3; constant 15.

2 (a) $n + 7$ (b) $4(x + 3)$ (c) $2m - 10$ (d) $\frac{p}{5} + 2$.

3 (a) $5(6) + 3 = \mathbf{33}$ (b) $5(0.5) + 3 = \mathbf{5.5}$ (c) $5(-4) + 3 = -20 + 3 = -\mathbf{17}$.

4 (a) $2(3)^2 - 3(3) = 18 - 9 = \mathbf{9}$ (b) $2(-2)^2 - 3(-2) = 8 + 6 = \mathbf{14}$.

5 $4(5 + 1) - 7 = 24 - 7 = \mathbf{17}$.

6 **B** $3x - 5$. "Five less than three times x" subtracts 5 from $3x$.

7 (a) $45 + 30h$ (b) $45 + 30(3.5) = 45 + 105 = \mathbf{\$150}$.

8 $2(2x + 1) + 2x$, which simplifies to $4x + 2 + 2x = \mathbf{6x + 2}$.

9 $4x$ at $x = -3$ is $4(-3) = \mathbf{-12}$, not 7. x^2 at $x = -3$ is $(-3)^2 = \mathbf{9}$, not -9. In the first case the student did not multiply: adding 4 and 3 gives 7, but $4x$ means $4 \times (-3)$. In the second, the student squared 3 and attached a negative sign; the negative sign is part of the base, so $(-3)^2 = 9$.

10 $2n$ means $2 \times n$, while $n + 2$ means add 2. At $n = 5$, $2n = 10$ but $n + 2 = 7$. The expressions have different values, so they are not equivalent.

11 (a) $150 + 12n$ (b) $150 + 12(40) = 150 + 480 = \mathbf{\$630}$ (c) 150 is the fixed hall-hire charge, paid once regardless of the number of people.

12 $3(x + 1) = 3x + 3$, so the second and third expressions are equivalent. At $x = 2$, $3x + 1 = 7$ while $3(x + 1) = 3x + 3 = 9$. The first expression is not equivalent to the other two.

Guided 5.1 $3(4) + 5 = 12 + 5 = \mathbf{\$17}$. For 10 smoothies: $3(10) + 5 = \mathbf{\$35}$. The 5 is the fixed delivery fee.

5.2 EQUIVALENT EXPRESSIONS

1 (a) $10x$ (b) $6a$ (c) $4m + 2$ (d) $2y + 11$.

2 (a) $5x - 3$ (b) $3a + 7$ (c) $-3b + 5$.

3 (a) $2x + 12$ (b) $10y - 15$ (c) $-x - 4$ (d) $-6a + 15$.

4 (a) $3x + 6 + 4x + 4 = \mathbf{7x + 10}$ (b) $10x - 5 - 2x - 6 = \mathbf{8x - 11}$.

5 (a) $3(2x + 3)$ (b) $5(2y - 3)$ (c) $4(2a + 3b)$.

6 **B** $10x - 15$.

7 $4(x + 3) = 4x + 12$, not $4x + 3$; the 4 multiplies both terms. $7x - 2x = 5x$, not $5x^2$; subtracting like terms changes the coefficient, not the power.

8 $2(2x + 5) + 2x = 4x + 10 + 2x = \mathbf{6x + 10}$. Also $2(3x + 5) = 6x + 10$. These are equivalent because both describe two lengths and two widths, and expanding the second gives the first.

9 $3x$ has variable part x, while $3x^2$ has variable part x^2. At $x = 2$, $3x = 6$ but $3x^2 = 12$. Their variable parts differ, so they are not like terms.

10 Yes, both are correct: $6(2x - 3) = 12x - 18$, and $-6(-2x + 3) = 12x - 18$. Taking out a factor of -6 is equivalent to taking out 6 and changing the signs inside the brackets.

11 At $x = 0$, $2(x + 4) = 8$ but $2x + 4 = 4$. One matching value is not enough because two different expressions can agree at a single point. A counterexample at $x = 0$ proves they are not equivalent.

12 A rectangle of height 4 and width $2x + 1$ splits into a region of area $4(2x) = 8x$ and a region of area $4(1) = 4$. The total area is $8x + 4$, so $4(2x + 1) = 8x + 4$ by the distributive property.

Guided 5.2 $-5(-4) = \mathbf{20}$, so the expansion is $-10x + 20$. At $x = 3$: $-30 + 20 = \mathbf{-10}$; the original is $-5(6 - 4) = -5(2) = \mathbf{-10}$.

5.3 SOLVING LINEAR EQUATIONS

1 (a) $x = 8$ (b) $y = 17$ (c) $m = 9$ (d) $n = 28$. Checks: $8 + 9 = 17$; $17 - 6 = 11$; $5(9) = 45$; $28 \div 4 = 7$.

2 (a) $2x = 10$, $x = 5$ (b) $4x = 24$, $x = 6$ (c) $3y = 15$, $y = 5$ (d) $-2a = -6$, $a = 3$.

3 (a) $x + 2 = 6$, $x = 4$ (b) $x - 1 = 5$, $x = 6$ (c) $4x + 2 = 14$, $4x = 12$, $x = 3$ (d) $12x - 8 = 40$, $12x = 48$, $x = 4$.

4 (a) $2x = 10$, $x = 5$ (b) $3x = 12$, $x = 4$ (c) $3y = 21$, $y = 7$.

5 $2x + 6 = 3x - 3$, so $6 + 3 = 3x - 2x$, giving $x = \mathbf{9}$. Check: left $2(12) = 24$; right $3(8) = 24$. *Alternatively,* subtract $2x$ first: $6 = x - 3$, so $x = 9$.

6 $4 + 2.50d = 34$. Subtract 4: $2.50d = 30$. Divide by 2.50: $d = \mathbf{12}$ km.

7 $2b + 3 = 15$. Subtract 3: $2b = 12$. Divide by 2: $b = \mathbf{6}$. Her brother is 6 years old.

8 **B** $x = -3$. $5 - 2(-3) = 5 + 6 = 11$.

9 The student added 4 to both sides instead of subtracting 4. Correctly: $3x + 4 = 19$, so $3x = 15$ and $x = \mathbf{5}$. Check: $3(5) + 4 = 19$.

10 The 2 was not distributed to -3. Correctly: $2x - 6 = 10$, so $2x = 16$ and $x = \mathbf{8}$. Check: $2(8 - 3) = 2(5) = 10$.

11 Substituting into the original equation tests the equality as it was stated, so it catches arithmetic and sign errors made during solving. For $x = 4$: left $5(4) + 2 = 22$, right $3(4) + 10 = 22$. Both sides agree, so $x = 4$ is correct.

12 $6x - 3 = 4x + 9$, so $2x = 12$ and $x = \mathbf{6}$. Check: left $3(11) = 33$; right $24 + 9 = 33$.

13 $18 + 0.10m = 32$. Subtract 18: $0.10m = 14$. Divide by 0.10: $m = \mathbf{140}$ minutes.

14 $2(x + 3) = 2x + 6$ expands to $2x + 6 = 2x + 6$, which is true for every value of x: it is an identity with infinitely many solutions. $2(x + 3) = 2x + 1$ expands to $2x + 6 = 2x + 1$, which would require $6 = 1$; this is false for every x, so there is no solution.

Guided 5.3 From $4x = 20$, divide both sides by 4: $x = \mathbf{5}$. Check: $4(5) - 3 = 20 - 3 = 17$.

REAL-LIFE APPLICATION · CHOOSING A REPAIR PLAN

1 Plan A: 12, 28, 44, 60, 76. Plan B: 20, 32, 44, 56, 68. Each row increases by the hourly rate: \$8 per 2 hours for A gives +16; \$6 per 2 hours for B gives +12. Extended: at 1, 3, 5 and 10 hours, A costs \$20, \$36, \$52 and \$92; B costs \$26, \$38, \$50 and \$80.

2 Plan A: $12 + 8h$, fixed fee \$12, hourly rate \$8. Plan B: $20 + 6h$, fixed fee \$20, hourly rate \$6.

3 At $h = 3$: A $= 12 + 24 = \$36$; B $= 20 + 18 = \$38$. Plan A is cheaper by \$2.

4 $12 + 8h = 20 + 6h$. Subtract $6h$: $12 + 2h = 20$. Subtract 12: $2h = 8$, so $h = \mathbf{4}$. Both plans cost $\$12 + 8(4) = \$20 + 6(4) = \mathbf{\$44}$. This is the break-even time.

5 For $h < 4$, Plan A is cheaper; for $h > 4$, Plan B is cheaper; at $h = 4$ the costs are equal. From the table, at 2 hours A costs \$28 and B \$32; at 6 hours A costs \$60 and B \$56.

6 At 12 hours: A $= 12 + 96 = \$108$; B $= 20 + 72 = \$92$. Plan B is cheaper by **\$16**, so the workshop should choose Plan B if the hire lasts 12 hours.

7 **Assumption:** the hourly rates are constant and no other charges apply. **Limitation:** the workshop may be billed for part-hours differently, may need parts or travel, or may finish early; any of these could change the total. The recommendation should be checked against the suppliers' actual terms.

INVESTIGATION · THE NUMBER TRICK

1 Start 4: $9, 18, 14, 7, \mathbf{3}$. Start 10: $15, 30, 26, 13, \mathbf{3}$. Start -3: $2, 4, 0, 0, \mathbf{3}$.

2 n; $n + 5$; $2(n + 5)$; $2(n + 5) - 4$; $\dfrac{2(n + 5) - 4}{2}$; $\dfrac{2(n + 5) - 4}{2} - n$.

3 $\dfrac{2(n + 5) - 4}{2} = \dfrac{2n + 10 - 4}{2} = \dfrac{2n + 6}{2} = n + 3$. Then $n + 3 - n = \mathbf{3}$. The starting number cancels, so the result is always 3.

4 $n = -2.5$: $-2.5 + 5 = 2.5$; doubled $= 5$; subtract 4 $= 1$; halved $= 0.5$; subtract -2.5: $0.5 - (-2.5) = \mathbf{3}$. The trick works because the algebra holds for all numbers, including negative and fractional values.

5 **Sample trick.** Think of a number; double it; add 14; halve the result; subtract the original number. Expressions: n, $2n$, $2n + 14$, $\dfrac{2n + 14}{2} = n + 7$, then $n + 7 - n = \mathbf{7}$. Other valid designs are possible if the final simplification gives 7.

6 Testing examples confirms only the cases tried. The simplification treats n as any number and shows that the starting value cancels identically, so the conclusion covers every possible input. This is a general argument rather than a finite check.

CRITERION TASK · THE FUN-RUN T-SHIRT ORDER

1 Print shop: $n = 5 : 85$; $10 : 125$; $20 : 205$; $30 : 285$. Online: $5 : 75$; $10 : 135$; $20 : 255$; $30 : 375$.

2 Print shop: $45 + 8n$, with fixed setup \$45 and variable cost $8n$. Online: $12n + 15$, with variable cost $12n$ and fixed delivery \$15.

3 For 10 shirts: print shop $45 + 80 = \$125$; online $120 + 15 = \$135$. The print shop is cheaper by **\$10**.

4 $45 + 8n = 12n + 15$. Subtract $8n$: $45 = 4n + 15$. Subtract 15: $30 = 4n$, so $n = 7.5$. The costs are equal at 7.5 shirts, which is not a whole number of shirts. At 7 shirts the online supplier is cheaper (\$99 vs \$101); at 8 shirts the print shop is cheaper (\$109 vs \$111). The practical break-even is therefore at 8 shirts.

5 Profit $= 15n - (45 + 8n) = 15n - 45 - 8n = \mathbf{7n - 45}$. A positive profit requires $7n > 45$, so $n > 6.43$. The smallest whole-number order is **7** shirts, giving $49 - 45 = \$4$ profit. Check revenue $7 \times 15 = \$105$, cost $45 + 56 = \$101$.

6 **Sample recommendation.** If the council expects to order at least 8 shirts and sell them all at \$15, it should use the print shop, which becomes cheaper from 8 shirts and leaves a larger profit as the order grows. For 7 or fewer shirts, the online supplier is cheaper. **Assumption:** every shirt is sold at \$15 and there are no returns or extra fees. **Limitation:** the model ignores minimum order quantities, delivery delays, quality differences, unsold stock and the risk that demand is lower than expected; these should be checked before ordering.

MIXED PRACTICE · CONSOLIDATE AND EXTEND

1 (a) $11x$ (b) $7a$ (c) $3m + 9$.

2 (a) $3x + 21$ (b) $12y - 6$ (c) $-4a - 12$.

3 (a) $x = 13$ (b) $x = 7$ (c) $x = 24$.

4 (a) $2x = 12$, $x = 6$ (b) $5x = 25$, $x = 5$.

5 (a) $24 - 3 = \mathbf{21}$ (b) $-8 - 3 = \mathbf{-11}$.

6 **B** $6x + 15$.

7 **C** $x = 6$. $4(6) - 3 = 21$ and $2(6) + 9 = 21$.

8 The 5 was multiplied by x only. Correctly, $5(x + 2) = \mathbf{5x + 10}$.

9 The student subtracted 3 instead of adding 3. Correctly, $2x = 14$, so $x = \mathbf{7}$.

10 $4x = 4 \cdot x$, while $x^4 = x \cdot x \cdot x \cdot x$. At $x = 2$, $4x = 8$ but $x^4 = 16$. The variable parts are different, so the terms are not like terms.

11 Perimeter $= 2(3x + 1) + 2(x - 2) = 6x + 2 + 2x - 4 = \mathbf{8x - 2}$.

12 Let the integers be n, $n + 1$, $n + 2$. Then $n + (n + 1) + (n + 2) = 42$, so $3n + 3 = 42$, $3n = 39$, $n = 13$. The integers are **13**, **14**, **15**.

13 $15 + 0.08m = 27$. Subtract 15: $0.08m = 12$. Divide by 0.08: $m = \mathbf{150}$ minutes.

14 $3x - 6 = 2x + 8$, so $x = \mathbf{14}$. Check: left $3(12) = 36$; right $2(18) = 36$.

15 $5 - 2x = 13$, so $-2x = 8$ and $x = \mathbf{-4}$. Check: $5 - 2(-4) = 13$.

16 $5 + 2.20d = 38$. Subtract 5: $2.20d = 33$. Divide by 2.20: $d = \mathbf{15}$ km.

17 $20 + 3h = 35 + 1.5h$, so $1.5h = 15$ and $h = \mathbf{10}$. The equal cost is $20 + 30 = \$50$. For $h > 10$, the second plan is cheaper because its hourly rate is lower; for example, at 12 hours: 56 versus 53.

18 $2(3x + 4) = 6x + 8$ and $-3(x - 2) = -3x + 6$. Adding: $6x + 8 - 3x + 6 = \mathbf{3x + 14}$.

19 The claim that $x = 2$ makes both sides 11 is correct: $3(2) + 5 = 11$ and $2(2) + 7 = 11$. Solving $3x + 5 = 2x + 7$ gives $x = 2$, so it is the only solution. The check confirms the solution but does not by itself prove uniqueness; the solving steps do.

20 $2x + 6 = 2x + 6$ is true for every x. Both sides are the same expression, so any value of x satisfies it; there are infinitely many solutions.

21 $2x + 6 = 2x + 1$ simplifies to $6 = 1$, which is false. No value of x can make it true, so there is no solution.

22 **Sample answer.** $4x + 3 = 2x + 13$. At $x = 5$: left 23, right 23. Solving: $2x = 10$, $x = 5$. Other equations are possible if the two sides are equal at $x = 5$ but not identical.

23 (a) $10x - 25 = 5(2x - 5)$ (b) $8a + 12b = 4(2a + 3b)$. Expanding each result returns the original expression.

24 $3(2n + 1) - 6n + 4 = 6n + 3 - 6n + 4 = \mathbf{7}$. The $6n$ terms cancel, so the value is 7 for every n.

25 $2(x + 3) + 2x = 30$, so $4x + 6 = 30$, $4x = 24$, $x = \mathbf{6}$. The dimensions are 9 cm by 6 cm, and the area is $9 \times 6 = \mathbf{54\ cm^2}$.

Chapter 6 · Numerical patterns and generalisation

PREREQUISITE CHECK

1 $7 \times 6 = \mathbf{42}$.

2 $4 \times 9 + 3 = 36 + 3 = \mathbf{39}$.

3 Multiplication first: $50 - 42 = \mathbf{8}$.

4 The rule is "add 3", so the next term is **15**.

5 The x-coordinate is the first number, **4**.

6 Subtract 1 from both sides: $2n = 16$. Divide by 2: $n = \mathbf{8}$.

6.1 SEQUENCES AND TERM-TO-TERM RULES

1 (a) 17, 20, 23 (b) 14, 10, 6 (c) 32, 64, 128 (d) 60, 50, 40.

2 (a) Add 5; 27, 32. (b) Subtract 6; 36, 30. (c) Multiply by 5; 625, 3125.

3 (a) 4, 13, 22, 31, 40 (b) 3, 6, 12, 24, 48 (c) 100, 88, 76, 64, 52.

4 Differences: 3, 5, 7, 9. Next terms: 36, 49. The differences increase by 2 each time, so they are not constant. A constant difference would produce a linear sequence.

5 (a) 40, 55, 70, 85, 100 (b) Week 5, because the total reaches \$100 exactly.

6 **B** Multiply by 3: $2 \times 3 = 6$, $6 \times 3 = 18$, $18 \times 3 = 54$.

7 (a) 0, −3 (b) Term $n = 12 - 3(n - 1) = 15 - 3n$. Solve $15 - 3n = -30$: $-3n = -45$, so $n = \mathbf{15}$. The 15th term is −30.

8 The differences are 3, 4, 5, not a constant 4. The next difference is 6, so the next term is $14 + 6 = \mathbf{20}$, not 18. The student used the first difference as if it continued unchanged.

9 The differences are 2, 4, 8, which are not constant. The rule multiplies each term by 2. A linear sequence must add the same amount at every step.

10 6, 2, −2, −6, −10, −14, −18, −22. The term-to-term rule is "subtract 4"; the 8th term is **−22**.

11 Extend both: A 17, 20, 23, 26, . . .; B 18, 22, 26, ... The next shared term is **26**. After the shared 14, A lists four further terms to reach 26 while B lists three.

12 (a) Each term is the sum of the two previous terms. (b) 13, 21, 34. (c) The differences are 0, 1, 1, 2, 3, 5, which change, so the sequence is not linear. The rule depends on two previous terms, not on adding a fixed amount.

Guided 6.1 Next terms 27, 31. Term 7 is six steps after term 1: $11 + 6(4) = \mathbf{35}$. Check: 11, 15, 19, 23, 27, 31, 35.

6.2 THE NTH TERM OF A LINEAR SEQUENCE

1 (a) $3n + 1$ (b) $5n + 1$ (c) $-2n + 11$ (d) $4n - 3$.

2 (a) $7n - 5$ (b) $-5n + 35$ (c) $n - 0.5$.

3 $4(10) + 1 = \mathbf{41}$; $4(50) + 1 = \mathbf{201}$.

4 (a) $7(12) - 2 = \mathbf{82}$. (b) Solve $7n - 2 = 61$: $7n = 63$, so $n = \mathbf{9}$.

5 $3n + 2 = 100$ gives $3n = 98$, so $n = 32\frac{2}{3}$. This is not a whole number, so 100 is not a term. $3n + 2 = 98$ gives $n = 32$, so 98 is the 32nd term.

6 (a) $d = -6$, so the rule is $-6n + 86$. Check: $n = 1$ gives 80. (b) $86 - 60 = \mathbf{26}$. (c) $86 - 6n = 2$, so $6n = 84$ and $n = \mathbf{14}$.

7 **A** $5n + 2$. Check: $n = 1$ gives 7, $n = 2$ gives 12.

8 At $n = 1$, $4n + 5 = 9$, but the first term is 5. The coefficient 4 is correct, but the constant must be 1: the rule is $\mathbf{4n + 1}$. Check $n = 2$: $8 + 1 = 9$.

9 At $n = 2$, $4(2) + 21 = 29$, but the second term is 21. The sequence decreases, so $d = -4$ and the zeroth term is $25 - (-4) = 29$. The rule is $\mathbf{-4n + 29}$. Check: $n = 1$ gives 25, $n = 2$ gives 21.

10 (a) $4n + 2$ (b) $4(25) + 2 = \mathbf{102}$ tiles.

11 (a) The nth term is $3n + 2$, so the 40th term is $120 + 2 = \mathbf{122}$. (b) $3n + 2 = 92$, so $3n = 90$ and $n = \mathbf{30}$.

12 At $n = 2$, $3n + 7 = 13$, but the second term is 10. The expression $3n + 4$ gives $6 + 4 = 10$. The constant in $dn + c$ is the zeroth term, found by subtracting d from the first term: $7 - 3 = 4$, not 7.

13 (a) $2n + 5 = 3n + 2$, so $n = 3$. Both give $2(3) + 5 = 11$. (b) At $n = 10$: A $= 25$, B $= 32$, so B is larger. B has the larger coefficient of n, so it grows faster.

14 For whole-number n, $2n$ is even. Adding 1 gives an odd number, so every term is odd. Examples: $n = 1$ gives 3, $n = 2$ gives 5, $n = 7$ gives 15. The structure $2n + 1$ applies to every position, so the conclusion is general.

Guided 6.2 The 15th term is $7(15) - 3 = 105 - 3 = \mathbf{102}$. Solving $7n - 3 = 95$: $7n = 98$, so $n = 14$. Since 14 is a whole number, 95 is the 14th term.

6.3 FUNCTION MACHINES AND LINEAR GRAPHS

1 $4(0) + 1 = 1$, then 5, 9, 13, 17. Outputs: 1, 5, 9, 13, 17.

2 (a) 3, 8, 13, 18, 23 (b) $5(12) - 2 = \mathbf{58}$.

3 $3n + 4 = 31$, so $3n = 27$ and $n = \mathbf{9}$.

4 $6n - 5 = 43$, so $6n = 48$ and $n = \mathbf{8}$.

5 $y = 3x - 1$ gives 2, 5, 8, 11, 14. Each increase of 1 in x increases y by 3, so the points lie on a straight line.

6 **B** $(2, 7)$, because $2(2) + 3 = 7$. The point $(2, 5)$ gives $2(2) + 3 = 7$, not 5.

7 (a) $4(0) - 2 = \mathbf{-2}$ (b) $4(3) - 2 = \mathbf{10}$ (c) $4x - 2 = 18$, so $4x = 20$ and $x = \mathbf{5}$.

8 (a) 3.5, 7.5, 11.5, 15.5 (b) $3.5 + 2(10) = \mathbf{23.5}$ dollars. (c) 3.5 is the fixed charge before travelling; 2 is the cost in dollars per kilometre.

9 Reverse: $26 - 5 = 21$, then $21 \div 3 = \mathbf{7}$. Check: $3(7) + 5 = 26$.

10 (a) $2n + 9 = 4n + 1$, so $2n = 8$ and $n = \mathbf{4}$. The common output is 17. (b) At $n = 10$: A $= 29$, B $= 41$; B is larger.

11 The student multiplied the sum $5 + 3$ instead of applying the operations in order. The rule is "multiply by 2, then add 3": $2(5) + 3 = \mathbf{13}$, not 16.

12 (a) The change in y is 3 for each increase of 1 in x, and $y = 2$ when $x = 0$, so $y = 3x + 2$. (b) $3(10) + 2 = \mathbf{32}$. (c) Yes: $3(5) + 2 = 17$, so $(5, 17)$ lies on the line.

13 (a) $5x - 4 = 81$, so $5x = 85$ and $x = \mathbf{17}$. (b) Yes: $5(10) - 4 = 50 - 4 = 46$, so the point $(10, 46)$ satisfies the rule.

Guided 6.3 (a) $5(4) - 3 = 20 - 3 = \mathbf{17}$. (b) Add 3 to 42 to get 45, then divide by 5: $45 \div 5 = \mathbf{9}$. Check: $5(9) - 3 = 42$.

REAL-LIFE APPLICATION · THE LANTERN ARCH

1 The table values are correct: A 4, 7, 10, 13, 16, 19; B 7, 9, 11, 13, 15, 17. Extended: at 8 panels, A $= 25$, B $= 21$; at 10 panels, A $= 31$, B $= 25$.

2 Design A: coefficient 3, meaning 3 lanterns per panel; constant 1, the crown lantern. Design B: coefficient 2, meaning 2 lanterns per panel; constant 5, the five lanterns on the base and crown.

3 At $n = 6$: A $= 3(6) + 1 = 19$; B $= 2(6) + 5 = 17$. Design B uses **2** fewer lanterns.

4 $3n + 1 = 2n + 5$, so $n = 4$. Both designs use $3(4) + 1 = 2(4) + 5 = \mathbf{13}$ lanterns. At 4 panels the designs are equal; for fewer than 4, A uses fewer, and for more than 4, B uses fewer.

5 Design A has $3n + 1$ lanterns, so the cost is $6(3n + 1) + 45 = 18n + 6 + 45 = \mathbf{18n + 51}$ dollars. Solve $18n + 51 \leq 300$: $18n \leq 249$, so $n \leq 13.83$. The largest whole number is **13** panels. Cost: $18(13) + 51 = 234 + 51 = \$285$. The next size, 14 panels, costs $252 + 51 = \$303$, which is over budget.

6 For 61 lanterns, design A: $3n + 1 = 61$, so $n = 20$ panels. Design B: $2n + 5 = 61$, so $n = 28$ panels. Both use 61 lanterns, so the materials cost $6(61) + 45 = \mathbf{\$411}$ for either design. A needs fewer, denser panels; B needs more, sparser panels. If both fit, the choice is aesthetic: A gives a denser arch with fewer joins, while B spreads the lanterns across more panels and may suit a wider courtyard.

7 **Assumption:** every lantern costs \$6, the frame and wiring cost exactly \$45, and all panels are the same size. **Limitation:** bulk prices, labour, electrical fittings, permits, weatherproofing or a larger frame for more panels could change the total. A more detailed quote could shift the affordable panel count, so the workshop should confirm costs before ordering.

INVESTIGATION · THE GROWING STAIRCASE

1 Totals for $n = 1$ to 6: 1, 3, 6, 10, 15, 21. Changes: 2, 3, 4, 5, 6.

2 Staircase n adds n tiles, so the term-to-term rule is "add 2, then 3, then 4, and so on". The amount added is not constant; it equals the new step number.

3 $T_{10} = 1 + 2 + \cdots + 10 = 55$; $T_{20} = 1 + 2 + \cdots + 20 = 210$. Pairing the first and last terms gives $5(11) = 55$ and $10(21) = 210$. No drawing is needed because the pairing uses the structure of the sum.

4 The sequence is not linear. Its first differences are $2, 3, 4, 5, 6$, which increase rather than staying constant. A linear sequence would add the same number of tiles at every step.

5 At $n = 4$, $2n - 1 = 7$, but staircase 4 has 10 tiles. The claim is false. It appears to work for $n = 1$ (1 tile) and $n = 2$ (3 tiles), but it fails at $n = 3$: $2(3) - 1 = 5$, while the staircase has 6 tiles. Two matching cases are not enough.

6 Write $T = 1 + 2 + \cdots + n$ and also $T = n + (n - 1) + \cdots + 1$. Adding the two lines pairs each term to $n + 1$, giving $2T = n(n + 1)$, so $T = \dfrac{n(n+1)}{2}$. Check: $n = 4$ gives $\frac{4\times5}{2} = 10$; $n = 10$ gives $\frac{10\times11}{2} = 55$.

7 Solve $\dfrac{n(n+1)}{2} = 78$, so $n(n + 1) = 156$. Since $12 \times 13 = 156$, $n = \mathbf{12}$. Check: $T_{12} = \frac{12\times13}{2} = 78$.

8 Comparing values: at $n = 6$, $T_6 = 21$ and $4n - 1 = 23$, so the linear sequence is larger; at $n = 7$, $T_7 = 28$ and $4n - 1 = 27$, so the staircase is larger. The staircase first overtakes at $n = \mathbf{7}$. Its increase at step n is n, which eventually exceeds the constant increase of 4, so it will overtake any linear sequence.

CRITERION TASK · THE MOSAIC TILE BORDER

1 For $n = 3, 5, 8, 10$, the border counts are $8, 16, 28, 36$. The expression is $B = 4n - 4$.

2 Four sides of length n give $4n$ tiles, but each of the four corner tiles has been counted twice, once for each side. Subtracting the four duplicated corners gives $4n - 4$. For $n = 5$: $4(5) - 4 = 16$.

3 $4n - 4 = 60$, so $4n = 64$ and $n = \mathbf{16}$. For 100: $4n - 4 = 100$, so $4n = 104$ and $n = \mathbf{26}$. Both are whole numbers, so both panels are possible.

4 Cost $= 2.50(4n - 4) + 12 = 10n - 10 + 12 = \mathbf{10n + 2}$ dollars.

5 Solve $10n + 2 \leq 200$: $10n \leq 198$, so $n \leq 19.8$. The largest whole-number side length is **19**, costing $10(19) + 2 = \$192$. The next size, $n = 20$, costs $200 + 2 = \$202$, which is over budget.

6 $4n - 4 = 3n + 6$, so $n = 10$. At $n = 10$, both designs use 36 border tiles. For $n > 10$, $3n + 6$ is smaller: at $n = 12$, design 1 uses 44 tiles and design 2 uses 42, so design 2 uses 2 fewer. Design 1 uses fewer for $n < 10$: at $n = 8$, 28 versus 30.

7 **Sample recommendation.** If the project wants the largest possible panels within \$200, it should choose design 1 with side length 19, which costs \$192. If more panels are planned and the visual difference is acceptable, design 2 becomes cheaper once the side length exceeds 10; for example, at side 12 it saves 2 border tiles per panel. **Assumption:** the \$2.50 price per tile and the \$12 grout charge remain fixed, and every panel needs only one border. **Limitation:** cutting waste, breakages, labour and a possible bulk discount are not included; these could change the cost ranking.

MIXED PRACTICE · CONSOLIDATE AND EXTEND

1 $27, 32, 37$.

2 $5n + 3$.

3 $d = -4$, so the rule is $-4n + 44$. Check: $n = 1$ gives 40.

4 $6(10) - 5 = \mathbf{55}$. Solve $6n - 5 = 79$: $6n = 84$, so $n = \mathbf{14}$.

5 $4(6) - 3 = \mathbf{21}$. Solve $4n - 3 = 41$: $4n = 44$, so $n = \mathbf{11}$.

6 $1, 4, 7, 10, 13$.

7 **A** $5n - 3$.

8 At $n = 1$, $5n + 9 = 14$, not 9. The coefficient 5 is correct, but the constant is 4: the rule is $\mathbf{5n + 4}$.

9 The differences are $4, 5, 6$, so the next difference is 7, not 6. The next term is $18 + 7 = \mathbf{25}$. The student assumed the difference stayed at 6.

10 $3n + 1$ has a constant term of 1, while $3(n + 1) = 3n + 3$. At $n = 2$: $3n + 1 = 7$, but $3(n + 1) = 9$. Different values mean the expressions are not equivalent.

11 The nth term is $3n + 1$, so the 30th pattern has $3(30) + 1 = \mathbf{91}$ tiles.

12 $4n + 2 = 80$ gives $4n = 78$ and $n = 19.5$, not a whole number: 80 is not a term. $4n + 2 = 82$ gives $n = 20$, so 82 is the 20th term.

13 $3n + 5 = 5n - 1$, so $2n = 6$ and $n = \mathbf{3}$. The common term is $3(3) + 5 = 14$. At $n = 10$: A $= 35$, B $= 49$.

14 **Counterexamples:** $3, 6, 9, \ldots$ has nth term $3n$, and $4, 7, 10, \ldots$ has nth term $3n + 1$. Both have first difference 3, but their constants differ. The first term determines the constant, so "difference 3" alone does not fix the rule as $3n + 2$.

15 The next difference is 12, so the next term is $30 + 12 = \mathbf{42}$. The differences $4, 6, 8, 10$ increase by 2 and are not constant, so the sequence is not linear. Its terms fit $n(n + 1)$.

16 After n weeks: $25 + 12(n - 1) = \mathbf{12n + 13}$. Solve $12n + 13 \geq 150$: $12n \geq 137$, so $n \geq 11.42$. Week 12 is the first whole-number week, with total $144 + 13 = \mathbf{\$157}$. Week 11 has $132 + 13 = \$145$.

17 The change is 3 for each 1 in x, and the value at $x = 0$ is 4, so $y = 3x + 4$. At $x = 12$: $36 + 4 = \mathbf{40}$. Solving $3x + 4 = 61$: $3x = 57$, so $x = \mathbf{19}$.

18 $5n + 2 = 3n + 12$, so $2n = 10$ and $n = \mathbf{5}$. Both outputs are 27. At $n = 10$: A $= 52$, B $= 42$, so A is larger. At $n = 0$: A $= 2$, B $= 12$, so A is smaller.

19 Step n adds $T_n - T_{n-1} = n$ tiles. Since $T_{12} - T_{11} = 78 - 66 = 12$, step **12** adds exactly 12 tiles.

20 First term $= 23 - 4(4) = 7$. Sequence: $7, 11, 15, 19, 23$. The nth term is $4n + 3$.

21 $6n - 1 = 100$ gives $n = \frac{101}{6}$, not a whole number, so 100 is not a term. $6n - 1 = 101$ gives $n = 17$, so 101 is the 17th term.

22 Set $4n + 3 = 7n - 6$: $9 = 3n$, so $n = 3$. Both give $4(3) + 3 = 7(3) - 6 = \mathbf{15}$. The common term is 15.

23 **Sample context.** A craft stall charges a fixed \$2 packing fee plus \$5 for each notebook. For n notebooks, the total cost in dollars is $5n + 2$. The 5 is the cost of each notebook and the 2 is the once-only packing fee. At $n = 8$: $5(8) + 2 = \mathbf{\$42}$. Other consistent contexts are acceptable.

24 Divide staircase n into L-shaped layers containing $1, 3, 5, \ldots, 2n - 1$ tiles. The layers stack to fill an $n \times n$ square, whose area is n^2. For $n = 4$: $1 + 3 + 5 + 7 = 16 = 4^2$. Therefore the sum of the first n odd numbers is n^2.

Chapter 7 · 2D geometry: angles, perimeter and area

PREREQUISITE CHECK

1 A right angle.

2 $180 - 47 = \mathbf{133}$.

3 $14 \times 6 = \mathbf{84}$.

4 $\frac{1}{2} \times 9 \times 8 = 36$.

5 $5^2 = 25$, so $3.14 \times 25 = \mathbf{78.5}$.

6 Subtract 30: $2x = 150$. Divide by 2: $x = \mathbf{75}$.

7.1 ANGLE RELATIONSHIPS

1 (a) Acute (b) right (c) obtuse (d) reflex (e) straight.

2 (a) $180 - 124 = \mathbf{56}^\circ$ (b) $180 - 38 - 67 = \mathbf{75}^\circ$.

3 (a) $360 - 120 - 85 - 95 = \mathbf{60}^\circ$ (b) $90 + 90 + 48 = 228$, so $y = 360 - 228 = \mathbf{132}^\circ$.

4 The vertically opposite angle is 68°. The two adjacent angles are $180 - 68 = \mathbf{112}^\circ$ each. The four angles are $68^\circ, 112^\circ, 68^\circ, 112^\circ$.

5 (a) Corresponding: $\mathbf{73}^\circ$. (b) Co-interior: $180 - 108 = \mathbf{72}^\circ$. (c) Alternate: $\mathbf{41}^\circ$.

6 $(3x + 15) + (2x + 10) = 180$, so $5x + 25 = 180$, $5x = 155$ and $x = \mathbf{31}$. The angles are $3(31) + 15 = \mathbf{108}^\circ$ and $2(31) + 10 = \mathbf{72}^\circ$. Check: $108 + 72 = 180$.

7 $2x + 3x + 4x + 90 = 360$, so $9x = 270$ and $x = \mathbf{30}$. The angles are $\mathbf{60}^\circ, \mathbf{90}^\circ, \mathbf{120}^\circ, \mathbf{90}^\circ$.

8 Alternate angles are equal: $5x - 10 = 3x + 40$, so $2x = 50$ and $x = \mathbf{25}$. Each angle is $5(25) - 10 = \mathbf{115}^\circ$.

9 Co-interior angles are supplementary: $(2x + 20) + (3x + 30) = 180$, so $5x + 50 = 180$, $5x = 130$ and $x = \mathbf{26}$. The angles are $\mathbf{82}^\circ$ and $\mathbf{98}^\circ$.

10 Co-interior angles are supplementary, not equal. The statement should say $108^\circ + 72^\circ = 180^\circ$, which is consistent with the co-interior rule. The student confused co-interior angles with alternate (or corresponding) angles.

11 **B** Vertically opposite angles are equal.

12 The vertically opposite angle is 63°; the two adjacent angles are $180 - 63 = \mathbf{117}^\circ$ each. A painter can measure one angle, set the opposite angle equal to it, and set the adjacent angles to 117°; the four angles must total 360°.

13 The angles a and b lie on a straight line, so $a + b = 180$. The angles b and c also lie on a straight line, so $b + c = 180$. Therefore $a = 180 - b = c$, so $a = c$. Vertically opposite angles are equal.

Guided 7.1 $6x - 10 = 4x + 30$, so $2x = 40$ and $x = \mathbf{20}$. Each alternate angle is $6(20) - 10 = 4(20) + 30 = \mathbf{110}^\circ$. The adjacent angle on the straight line is $180 - 110 = \mathbf{70}^\circ$.

7.2 TRIANGLES

1 (a) Equilateral and acute. (b) Scalene and right-angled. (c) Isosceles and obtuse. (d) Scalene and acute.

2 (a) $180 - 52 - 71 = \mathbf{57}^\circ$ (b) $180 - 90 - 34 = \mathbf{56}^\circ$ (c) $180 - 25 - 25 = \mathbf{130}^\circ$.

3 (a) $(180 - 84) \div 2 = \mathbf{48}^\circ$ each. (b) $180 - 2(63) = \mathbf{54}^\circ$.

4 $x + (x + 20) + 2x = 180$, so $4x + 20 = 180$, $4x = 160$ and $x = \mathbf{40}$. The angles are $\mathbf{40}^\circ, \mathbf{60}^\circ, \mathbf{80}^\circ$.

5 (a) The exterior angle equals the sum of the two opposite interior angles: $115 = 40 + y$, so $y = \mathbf{75}^\circ$. The adjacent interior angle is $180 - 115 = 65^\circ$. (b) $3x = x + 80$, so $2x = 80$, $x = \mathbf{40}$, and the exterior angle is $\mathbf{120}^\circ$.

6 $180 - 90 - 29 = \mathbf{61}^\circ$.

7 $70 + 60 + 60 = 190^\circ$, but a triangle's angles must add to 180°. No such triangle exists.

8 **C** The sum of the two opposite interior angles.

9 The interior apex angle is $180 - 130 = \mathbf{50}^\circ$. The base angles are $(180 - 50) \div 2 = \mathbf{65}^\circ$ each.

10 The side view is a right triangle: one angle is 12°, one is 90°, so the third is $180 - 90 - 12 = \mathbf{78}^\circ$.

11 Two right angles already sum to 180°, leaving 0° for the third angle. A triangle must have a positive third angle, so it cannot contain two right angles.

12 Exterior angles: $180 - 50 = \mathbf{130}^\circ$, $180 - 60 = \mathbf{120}^\circ$, $180 - 70 = \mathbf{110}^\circ$. Their sum is $\mathbf{360}^\circ$. The three exterior angles and three interior angles together make three straight lines: $3 \times 180 = 540^\circ$. Subtracting the interior sum 180° leaves $540 - 180 = \mathbf{360}^\circ$.

13 $(2x - 10) + (x + 30) + (3x - 20) = 180$. The constants give $-10 + 30 - 20 = 0$, so $6x = 180$ and $x = \mathbf{30}$. The angles are $2(30) - 10 = \mathbf{50}^\circ$, $30 + 30 = \mathbf{60}^\circ$ and $3(30) - 20 = \mathbf{70}^\circ$. All three are acute, so it is an acute triangle.

Guided 7.2 $6x + 30 = 180$, so $6x = 150$ and $x = \mathbf{25}$. The angles are $2(25) + 10 = \mathbf{60}^\circ$, $3(25) - 20 = \mathbf{55}^\circ$ and $25 + 40 = \mathbf{65}^\circ$. Check: $60 + 55 + 65 = 180$.

7.3 QUADRILATERALS

1 (a) $360 - 95 - 85 - 100 = \mathbf{80}^\circ$ (b) $360 - 120 - 60 - 60 = \mathbf{120}^\circ$.

2 The adjacent angles are $180 - 68 = \mathbf{112}^\circ$. The four angles are $\mathbf{68}^\circ, \mathbf{112}^\circ, \mathbf{68}^\circ, \mathbf{112}^\circ$.

3 Adjacent angles are supplementary: $(3x + 20) + (2x + 30) = 180$, so $5x + 50 = 180$, $5x = 130$ and $x = \mathbf{26}$. The angles are $\mathbf{98}^\circ$ and $\mathbf{82}^\circ$; the four angles are $\mathbf{98}^\circ, \mathbf{82}^\circ, \mathbf{98}^\circ, \mathbf{82}^\circ$.

4 $2x + 120 + 80 = 360$, so $2x = 160$ and $x = \mathbf{80}^\circ$.

5 The angles between parallel sides are co-interior: $180 - 72 = \mathbf{108}^\circ$.

6 $x + 2x + 3x + 4x = 360$, so $10x = 360$ and $x = \mathbf{36}$. The angles are $\mathbf{36}^\circ, \mathbf{72}^\circ, \mathbf{108}^\circ, \mathbf{144}^\circ$.

7 $(x + 20) + (2x - 10) + (x + 40) + (2x + 10) = 360$. The constants give $20 - 10 + 40 + 10 = 60$, so $6x + 60 = 360$, $6x = 300$ and $x = \mathbf{50}$. The angles are $\mathbf{70}^\circ, \mathbf{90}^\circ, \mathbf{90}^\circ, \mathbf{110}^\circ$. Check: $70 + 90 + 90 + 110 = 360$.

8 The student used the triangle angle sum. A quadrilateral's angles add to 360°, and $100 + 80 + 90 + 90 = 360$, so the quadrilateral is possible.

9 **C** A square.

10 **C** Opposite angles are equal.

11 The corner of the rectangle is 90°. The diagonal splits it into two angles, so the angle with the side edge is $90 - 58 = \mathbf{32}^\circ$. (The triangle containing the diagonal has angles $90^\circ, 58^\circ$ and 32°.)

12 If all four angles were acute, each would be less than 90°, so their sum would be less than 360°. But the angles of a quadrilateral must add to exactly 360°. Therefore not all four can be acute.

13 A diagonal joins two opposite vertices and divides the quadrilateral into two triangles. Each triangle has angle sum 180°, and the two triangles use all the interior angles of the quadrilateral exactly once. Therefore the quadrilateral's angle sum is $2 \times 180 = \mathbf{360}^\circ$.

Guided 7.3 $4x - 20 = 2x + 40$, so $2x = 60$ and $x = \mathbf{30}$. The opposite angles are $4(30) - 20 = 2(30) + 40 = \mathbf{100}^\circ$. The adjacent angle is $180 - 100 = \mathbf{80}^\circ$. The four angles are $\mathbf{100}^\circ, \mathbf{80}^\circ, \mathbf{100}^\circ, \mathbf{80}^\circ$.

7.4 PERIMETER AND AREA

1 (a) $2(12 + 7) = \mathbf{38}$ cm (b) $4(9) = \mathbf{36}$ cm (c) $6 + 8 + 10 = \mathbf{24}$ cm.

2 $P = 10 + 5 + 4 + 3 + 6 + 8 = \mathbf{36}$ m. Area $= 10 \times 8 - 4 \times 3 = 80 - 12 = \mathbf{68}$ m^2.

3 (a) $14 \times 9 = \mathbf{126}$ cm^2 (b) $11^2 = \mathbf{121}$ m^2.

4 (a) $\frac{1}{2}(12)(7) = \mathbf{42}$ cm^2 (b) $\frac{1}{2}(8.5)(4) = \mathbf{17}$ m^2.

5 (a) $9 \times 6 = \mathbf{54}$ cm^2 (b) $12.5 \times 4.8 = \mathbf{60}$ m^2.

6 (a) $\frac{1}{2}(6 + 10)(5) = \mathbf{40}$ cm^2 (b) $\frac{1}{2}(3.5 + 7.5)(4) = \mathbf{22}$ m^2.

7 (a) $C = 2(3.14)(7) = \mathbf{43.96}$ cm; $A = 3.14(49) = \mathbf{153.86}$ cm^2. (b) $r = 9$ m; $C = 2(3.14)(9) = \mathbf{56.52}$ m; $A = 3.14(81) = \mathbf{254.34}$ m^2.

8 Curved edge $= 3.14(5) = 15.7$ cm. Perimeter $= 15.7 + 10 = \mathbf{25.7}$ cm. Area $= \frac{1}{2}(3.14)(25) = \mathbf{39.25}$ cm^2.

9 Area $= 15 \times 10 - 6 \times 4 = 150 - 24 = \mathbf{126}$ m^2. The outer boundary is $15 + 6 + 6 + 4 + 9 + 10 = \mathbf{50}$ m.

10 $24 \times 35 = \mathbf{\$840}$.

11 The height must be perpendicular to the base; 7 cm is the slant side. The area is $9 \times 5 = \mathbf{45}$ cm^2, not 63 cm^2.

12 **C** $\frac{1}{2}bh$.

13 $C = 3.14(1.2) = \mathbf{3.768}$ m; $A = 3.14(0.6^2) = 3.14(0.36) = \mathbf{1.1304}$ m^2. Edging $= 3.768 \times 18 = \$67.824 \approx \mathbf{\$67.82}$.

14 (a) $2.5 \times 10\,000 = \mathbf{25\,000}$ cm^2 (b) $4\,500 \div 10\,000 = \mathbf{0.45}$ m^2.

15 $P = 2(l + w)$, so $36 = 2(12 + w)$, $18 = 12 + w$ and $w = \mathbf{6}$ cm. Area $= 12 \times 6 = \mathbf{72}$ cm^2.

16 The 1×9 rectangle has area 9 cm^2; the 5×5 square has area 25 cm^2. Both have perimeter 20 cm, so the same perimeter does not fix the area. Among rectangles with a given perimeter, the square has the largest area.

Guided 7.4 $P = 8 + 14 + 7 + 9 = \mathbf{38}$ cm. $A = \frac{1}{2}(8 + 14)(6) = \frac{1}{2}(22)(6) = \mathbf{66}$ cm^2.

REAL-LIFE APPLICATION · THE COURTYARD BUDGET

1 Courtyard $12 \times 8 = 96$ m^2; lawn $6 \times 3.5 = 21$ m^2; fountain $3.14(1^2) = 3.14$ m^2. Paved area $= 96 - 21 - 3.14 = \mathbf{71.86}$ m^2.

2 8% of $71.86 = 0.08(71.86) = 5.7488$. Area to order $= 71.86 + 5.7488 = 77.6088 \approx \mathbf{77.61}$ m^2. Cost $= 77.61 \times 45 = \mathbf{\$3\,492.45}$.

3 Subtotal $= 3\,492.45 + 880 = \mathbf{\$4\,372.45}$. Budget remaining $= 4\,500 - 4\,372.45 = \mathbf{\$127.55}$.

4 Rim $= 2(3.14)(1) = \mathbf{6.28}$ m. Cost $= 6.28 \times 30 = \mathbf{\$188.40}$. Total $= 4\,372.45 + 188.40 = \mathbf{\$4\,560.85}$, which is **\$60.85** over budget.

5 One valid change is to find pavers at \$42/m^2: $77.61 \times 42 = \$3\,259.62$, so the total becomes $3\,259.62 + 880 + 188.40 = \mathbf{\$4\,328.02}$, under budget by **\$171.98**. Other valid changes include reducing the paved area by enlarging the lawn or fountain; this saves paving but may change the design's use. The answer should state the change, show the new total and comment on whether the design's purpose is preserved.

6 **Assumptions:** the 8% allowance covers cuts and breakages; quoted prices include delivery and tax; the courtyard is flat and the shapes fit without overlap; labour is not charged separately. **Limitation:** cutting paving around circles can waste more than 8%, and a real quote may charge for labour, drainage or disposal. A higher wastage rate or an extra labour charge could push the total above budget, so the group should confirm the quote before ordering.

INVESTIGATION · ANGLE SUMS OF POLYGONS

1 Triangles: $1, 2, 3, 4, 5, 6$ for $n = 3, 4, 5, 6, 7, 8$. Angle sums: $180°, 360°, 540°, 720°, 900°, 1080°$. Regular interior angles: $60°, 90°, 108°, 120°, \approx 128.6°, 135°$.

2 The number of triangles is always 2 fewer than the number of sides: $n - 2$.

3 Each triangle contributes $180°$, so $S = 180(n - 2)°$. Triangle: $180(1) = 180°$; quadrilateral: $180(2) = 360°$; hexagon: $180(4) = 720°$.

4 From one vertex, diagonals can be drawn to every non-adjacent vertex. There are $n - 3$ such vertices, producing $n - 2$ triangles (the two sides adjacent to the chosen vertex form the first and last triangle). The triangles meet only along edges and use every part of the polygon's interior once, so there are no gaps or overlaps. Their angles at the boundary vertices are exactly the polygon's interior angles.

5 12-gon: $180(12 - 2) = 180(10) = \mathbf{1800}°$. 20-gon: $180(18) = \mathbf{3240}°$.

6 A regular polygon has n equal interior angles, and their sum is $180(n - 2)°$, so each is $180(n - 2) \div n$. Pentagon: $180(3) \div 5 = 108°$. Hexagon: $180(4) \div 6 = 120°$. Decagon: $180(8) \div 10 = 144°$.

7 Solve $180(n - 2) \div n = 150$: $180n - 360 = 150n$, so $30n = 360$ and $n = \mathbf{12}$. For $100°$: $180n - 360 = 100n$, so $80n = 360$ and $n = 4.5$. A polygon must have a whole number of sides, so $100°$ is impossible.

8 A regular pentagon has interior angle $108°$, so each exterior angle is $180 - 108 = \mathbf{72}°$. Five of them give $5 \times 72 = \mathbf{360}°$. In general, the exterior angles and interior angles form n straight lines: $n \times 180°$. Subtracting the interior sum $180(n - 2)°$ leaves $180n - 180n + 360 = \mathbf{360}°$.

9 The equilateral triangle ($60°$, six meet at a point), the square ($90°$, four meet) and the regular hexagon ($120°$, three meet) tile the plane. The regular pentagon does not: $108°$ does not divide $360°$ exactly ($3 \times 108 = 324$, leaving a $36°$ gap). In general, a regular polygon tiles only when 360 divided by its interior angle is a whole number.

Worked example 7.6.1 (a) $180(12 - 2) = 180(10) = \mathbf{1800}°$. (b) $1800 \div 12 = \mathbf{150}°$. (c) $180(n - 2) = 1260$, so $n - 2 = 7$ and $n = \mathbf{9}$; it is a nonagon.

PRACTICE 7.5 · BUDGET DECISIONS

1 Courtyard $= 10 \times 7 = 70$ m^2. Fountain $= 3.14(1.5^2) = 3.14(2.25) = 7.065$ m^2. Paved area $= 70 - 7.065 = 62.935 \approx \mathbf{62.94}$ m^2.

2 $62.94 \times 40 = \mathbf{\$2\,517.60}$.

3 $34 \times 25 = \mathbf{\$850}$.

4 Area to order $= 62.94 \times 1.10 = 69.234 \approx \mathbf{69.23}$ m^2. Paving cost $= 69.23 \times 40 = \mathbf{\$2\,769.20}$. Total with edging $= 2\,769.20 + 850 = \mathbf{\$3\,619.20}$.

5 Yes. $3\,800 - 3\,619.20 = \mathbf{\$180.80}$ remains. An assumption is that 10% extra paving covers cuts and breakages, the courtyard is flat, and the quoted prices include delivery; if wastage or labour is higher, the total could rise.

PRACTICE 7.6 · GENERALISING POLYGON ANGLES

1 $180(15 - 2) = 180(13) = \mathbf{2340}°$.

2 Octagon: $180(8 - 2) = 1080°$, so each interior angle is $1080 \div 8 = \mathbf{135}°$.

3 $360 \div 12 = \mathbf{30}°$.

4 $180(n - 2) = 1620$, so $n - 2 = 9$ and $n = \mathbf{11}$.

5 $360 \div 40 = \mathbf{9}$ sides. Interior angle $= 180 - 40 = \mathbf{140}°$; check: $180(7) \div 9 = 1260 \div 9 = 140°$.

6 Solve $180(n - 2) \div n = 172$: $180n - 360 = 172n$, so $8n = 360$ and $n = \mathbf{45}$. A regular 45-gon would have an interior angle of $172°$, so the claim is possible; the angle is very close to a straight line because the polygon has many sides.

CRITERION TASK · THE COMMUNITY COURTYARD

1 Courtyard $= 14 \times 9 = 126$ m^2. Stage $= \frac{1}{2}(4)(3) = 6$ m^2. Each tree bed $= 3.14(1.2^2) = 3.14(1.44) = 4.5216$ m^2; three beds $= 13.5648 \approx \mathbf{13.56}$ m^2. Paved area $= 126 - 6 - 13.56 = \mathbf{106.44}$ m^2.

2 Outer boundary $= 2(14 + 9) = 46$ m. Stage $= 4 + 3 + 5 = 12$ m. Tree-bed rims $= 3 \times 2(3.14)(1.2) = 3(7.536) = 22.608 \approx 22.61$ m. Total edging $= 46 + 12 + 22.61 = \mathbf{80.61}$ m.

3 Paving $= 106.44 \times 38 =$ **\$4 044.72**. Edging $= 80.61 \times 15 =$ **\$1 209.15**. Subtotal $= 4\,044.72 + 1\,209.15 =$ **\$5 253.87**.

4 Contingency $= 0.05(5\,253.87) =$ **\$262.69**. Total $= 5\,253.87 + 262.69 =$ **\$5 516.56**. This is **\$16.56** over the \$5,500 budget.

5 With $r = 1.5$ m, each bed has area $3.14(2.25) = 7.065$ m^2, so three beds have area $21.195 \approx$ **21.20 m^2**. Paved area $= 126 - 6 - 21.20 =$ **98.80 m^2**. Paving $= 98.80 \times 38 =$ \$3 754.40. Rims $= 3 \times 2(3.14)(1.5) = 28.26$ m, so total edging $= 46 + 12 + 28.26 =$ **86.26** m; edging cost $= 86.26 \times 15 =$ \$1 293.90. Subtotal $= 3\,754.40 + 1\,293.90 =$ \$5 048.30. Contingency $= 0.05(5\,048.30) =$ \$252.42. Total = **\$5 300.72**, which is **\$199.28** under budget. Enlarging the beds lowers the total because the extra bed area replaces paving at \$38/m^2 with rim edging at only \$15/m; the saving on paving is larger than the extra edging cost.

6 A justified recommendation is to use the 1.5 m beds if the courtyard layout allows: the total is \$5,300.72, under budget by \$199.28, and the larger beds provide more shade. **Assumption:** the three larger beds fit without overlapping the stage or paths, and the 5% contingency is sufficient. **Limitation:** larger beds may reduce circulation space, require root barriers, or change the paving cut pattern and wastage; a supplier's quote could differ from the model. The recommendation should refer to both the budget and the courtyard's use.

7 A complete response uses metres for lengths, square metres for areas, rounds each measurement to 2 decimal places before costing, rounds money to the nearest cent, and shows the formula before each substitution. All three areas, both perimeters and the contingency should be visible in the working.

MIXED PRACTICE

1 118° obtuse; 90° right; 247° reflex.

2 $180 - 133 =$ **47°**.

3 $360 - 85 - 100 - 95 =$ **80°**.

4 $180 - 42 - 63 =$ **75°**.

5 The other angles are **106°**, **74°**, **106°**.

6 $\frac{1}{2}(11)(6) =$ **33 cm^2**.

7 $C = 2(3.14)(4) =$ **25.12** m; $A = 3.14(16) =$ **50.24 m^2**.

8 $\frac{1}{2}(5 + 9)(4) =$ **28 cm^2**.

9 $(3x + 10) + (2x - 5) = 180$, so $5x + 5 = 180$, $5x = 175$ and $x =$ **35**. The angles are $3(35) + 10 =$ **115°** and $2(35) - 5 =$ **65°**.

10 $7x - 20 = 4x + 40$, so $3x = 60$ and $x =$ **20**. Each angle is $7(20) - 20 =$ **120°**.

11 The sum is $8x + 40 = 360$, so $x =$ **40**. The angles are **90°**, **110°**, **60°**, **100°**.

12 The other opposite interior angle is $110 - 45 =$ **65°**. The adjacent interior angle is 70°, and $70 + 45 + 65 = 180$.

13 The student used the quadrilateral angle sum. A triangle's angles add to 180°, and $80 + 70 + 30 = 180$, so such a triangle is possible.

14 Area $= 12 \times 9 - 5 \times 4 = 108 - 20 =$ **88 m^2**. The outer boundary is $12 + 5 + 5 + 4 + 7 + 9 =$ **42** m.

15 $C = 3.14(10) =$ **31.4** m; $r = 5$ m, so $A = 3.14(25) =$ **78.5 m^2**.

16 $18 \times 42.50 =$ **\$765**.

17 $P = 2(15 + 9) = 48$ m; cost $= 48 \times 14 =$ **\$672**.

18 The 6×4 rectangle has area 24 cm^2; the 5×5 square has area 25 cm^2. Both perimeters are 20 cm, so perimeter does not determine area. With a fixed perimeter, the square encloses the larger area.

19 Interior sum $= 180(12 - 2) = 180(10) =$ **1800°**. Each interior angle of a regular 12-gon is $1800 \div 12 =$ **150°**.

20 $n = 360 \div 24 =$ **15** sides. Interior angle $= 180 - 24 =$ **156°**; check: $180(15 - 2) \div 15 = 2340 \div 15 = 156°$.

21 $2 + 3 + 4 = 9$ parts, so one part is $180 \div 9 = 20°$. The angles are **40°**, **60°**, **80°**.

22 $1 + 2 + 3 + 4 = 10$ parts, so one part is $360 \div 10 = 36°$. The angles are **36°**, **72°**, **108°**, **144°**.

23 Paved area $= 20 \times 15 - 8 \times 6 = 300 - 48 =$ **252 m^2**. Paving $= 252 \times 35 =$ **\$8 820**. Outer perimeter $= 2(20 + 15) = 70$ m, so edging $= 70 \times 20 =$ **\$1 400**. Total $= 8\,820 + 1\,400 =$ **\$10 220**.

24 The original rectangle has area $2 \times 3 = 6$ cm^2 and perimeter $2(2 + 3) = 10$ cm. Doubling the sides gives 4×6, area 24 cm^2 and perimeter 20 cm. The perimeter doubles, but the area becomes four times as large, not twice. In general, doubling every length multiplies area by $2^2 = 4$.

Chapter 8 · 3D geometry: solids, surface area and volume

PREREQUISITE CHECK

1 $7 \times 4 =$ **28 cm^2**.

2 $\frac{1}{2}(6)(5) =$ **15 cm^2**.

3 $2(3.14)(3) =$ **18.84** cm.

4 $2.5 \times 2.5 =$ **6.25**.

5 $4 \times 4 \times 4 =$ **64**.

6 $3 \times 4 = 12$, then $12 \times 5 =$ **60**.

7 $100 \times 100 =$ **10 000 cm^2**.

8.1 SOLIDS, FACES AND NETS

1 (a) Cuboid (b) cylinder (c) triangular prism.

2 6 faces, 12 edges, 8 vertices.

3 5 faces (1 square and 4 triangles), 8 edges, 5 vertices.

4 Faces $= 2 + 5 =$ **7**; edges $= 2(5) + 5 =$ **15**; vertices $= 2(5) =$ **10**.

5 Four squares in a row wrap around the four side faces. The fifth square overlaps the first and the sixth overlaps the second, so there is no square left for the top or bottom. The net cannot close into a cube.

6 A cuboid with two opposite square faces, for example a square prism $3\text{ cm} \times 3\text{ cm} \times 5\text{ cm}$.

7 One possible net: a row of four rectangles $3 \times 2, 6 \times 2, 3 \times 2, 6 \times 2$, with a 6×3 rectangle above one 6×2 rectangle and a matching 6×3 rectangle below it. The six faces are two 6×3, two 6×2 and two 3×2. Surface area $= 2(18 + 12 + 6) =$ **72** cm^2.

8 A cube net has six equal squares: $6s^2 = 54$, so $s^2 = 9$ and $s =$ **3** cm. Surface area $=$ **54** cm^2.

9 Diameter 10 cm gives $r =$ **5** cm. The circumference $2\pi r = 31.4$ cm matches the rectangle's length, so $h =$ **12** cm. Rectangle area $= 31.4 \times 12 = 376.8\ \text{cm}^2$; circles $= 2(3.14)(25) = 157\ \text{cm}^2$; total $=$ **533.8** cm^2.

10 A face is any surface of a solid, including a curved one. A cylinder has **3 faces** (2 circular and 1 curved), **2 edges** (the circular rims) and **0 vertices**. Saying it has 2 faces counts only the flat circles and ignores the curved surface.

Guided 8.1 $6.28r = 18.84$, so $r =$ **3** cm. The solid is a cylinder of radius 3 cm and height 10 cm. Circles: $2(3.14)(9) = 56.52\ \text{cm}^2$; rectangle: $18.84 \times 10 = 188.4\ \text{cm}^2$; surface area $=$ **244.92** cm^2.

8.2 SURFACE AREA OF PRISMS

1 $SA = 2(10 \times 4 + 10 \times 6 + 4 \times 6) = 2(40 + 60 + 24) =$ **248** cm^2.

2 $SA = 6(7^2) = 6(49) =$ **294** cm^2.

3 $B = \frac{1}{2}(5)(12) = 30\ \text{cm}^2$; $P = 5 + 12 + 13 = 30$ cm; $SA = 2(30) + 30(20) = 60 + 600 =$ **660** cm^2.

4 $SA = 2(2.5 \times 1.2 + 2.5 \times 0.8 + 1.2 \times 0.8) = 2(3 + 2 + 0.96) =$ **11.92** m^2.

5 $SA = 2(30 \times 20 + 30 \times 15 + 20 \times 15) = 2(600 + 450 + 300) = 2700\ \text{cm}^2$. Extra 10% is 270 cm^2, so order **2970** cm^2.

6 The equal sides of each triangular end are $\sqrt{1.2^2 + 1.6^2} = 2.0$ m. Each end has area $\frac{1}{2}(2.4)(1.6) = 1.92\ \text{m}^2$. The two sloping rectangles each have area $2 \times 3.5 = 7\ \text{m}^2$. Canvas $= 2(1.92) + 2(7) = 3.84 + 14 =$ **17.84** m^2. The floor (8.4 m^2) is not canvas.

7 Open top: base $40 \times 25 = 1000$, two 40×20 sides $= 1600$, two 25×20 sides $= 1000$; total **3600** cm^2. Closed box: add the 1000 cm^2 top, total 4600 cm^2. The closed box needs **1000** cm^2 more card.

8 $SA = 2(1.5 \times 0.8 + 1.5 \times 0.6 + 0.8 \times 0.6) = 2(1.2 + 0.9 + 0.48) = 5.16\ \text{m}^2$. Paint needed $= 5.16 \div 12 =$ **0.43** L. Cost $= 0.43 \times 18 =$ **$7.74**.

9 $SA_1 = 2(16 + 40 + 40) = 192\ \text{cm}^2$; $SA_2 = 2(16 + 80 + 20) = 232\ \text{cm}^2$. Both volumes are 160 cm^3, so the same volume can require different amounts of material. The more compact $4 \times 4 \times 10$ cuboid uses less cardboard.

10 $5 \times 3 \times 2 = 30$ is the **volume** in cm^3, not the surface area. Correct surface area: $2(15 + 10 + 6) =$ **62** cm^2. The two quantities have different units and describe different properties.

8.3 SURFACE AREA OF CYLINDERS

1 Bases $= 2(3.14)(9) = 56.52\ \text{cm}^2$; curved $= 2(3.14)(3)(7) = 131.88\ \text{cm}^2$; total **188.4** cm^2.

2 $d = 10$, so $r = 5$. Bases $= 157\ \text{cm}^2$; curved $= 2(3.14)(5)(12) = 376.8\ \text{cm}^2$; total **533.8** cm^2.

3 Open top: one base $= 3.14(16) = 50.24\ \text{cm}^2$; curved $= 2(3.14)(4)(9) = 226.08\ \text{cm}^2$; total **276.32** cm^2.

4 A pipe has no end circles: $SA = 2\pi rh = 2(3.14)(0.5)(8) =$ **25.12** m^2.

5 Label area is the curved surface: $2(3.14)(3.5)(11) =$ **241.78** cm^2.

6 $628 = 2(3.14)(25) + 2(3.14)(5)h = 157 + 31.4h$. So $31.4h = 471$ and $h =$ **15** cm.

7 $V_A = 3.14(9)(4) = 113.04\ \text{cm}^3$; $V_B = 3.14(36)(1) = 113.04\ \text{cm}^3$. The volumes are equal. $SA_A = 2(3.14)(9) + 2(3.14)(3)(4) = 56.52 + 75.36 =$ **131.88** cm^2; $SA_B = 2(3.14)(36) + 2(3.14)(6)(1) = 226.08 + 37.68 =$ **263.76** cm^2. Equal volume does not mean equal surface area; the taller, narrower cylinder A uses less material.

8 $SA = 2(3.14)(1.44) + 2(3.14)(1.2)(2) = 9.0432 + 15.072 = 24.1152\ \text{m}^2 \approx$ Cost $= 24.1152 \times 7 =$ **$168.81** (to the nearest cent).

9 The lateral term $2\pi dh$ uses the diameter where the radius is needed, so it is twice as large as it should be. Correct formula with $d = 2r$: $SA = 2\pi r^2 + 2\pi rh = 2\pi(d/2)^2 + 2\pi(d/2)h =$ $\pi\mathbf{d^2/2} + \pi\mathbf{dh}$.

8.4 VOLUME AND CAPACITY

1 $V = 9 \times 4 \times 6 =$ **216** cm^3.

2 $V = 5^3 =$ **125** cm^3.

3 $V = B\ell = 30 \times 12 =$ **360** cm^3.

4 $V = 3.14(3^2)(8) = 3.14(9)(8) =$ **226.08** cm^3.

5 $r = 5$ cm; $V = 3.14(25)(14) = \mathbf{1099}$ cm^3.

6 $V = 20 \times 8 \times 6 = 960$ cm^3 $= 960$ mL. The label overstates the volume by $1000 - 960 = \mathbf{40}$ mL; it is not accurate.

7 $V = 2 \times 1.5 \times 1.2 = 3.6$ m^3 $= 3.6 \times 1000 = \mathbf{3600}$ L.

8 $V = 3.14(1.5^2)(3) = 3.14(2.25)(3) = 21.195$ m^3 $= \mathbf{21\,195}$ L.

9 $240 = 30\ell$, so $\ell = \mathbf{8}$ cm.

10 Box: $V = 20 \times 10 \times 10 = 2000$ cm^3, $SA = 2(200 + 200 + 100) = 1000$ cm^2. Cylinder: $V = 3.14(36)(18) = 2034.72$ cm^3, $SA = 2(3.14)(36) + 2(3.14)(6)(18) = 226.08 + 678.24 = 904.32$ cm^2. The cylinder holds 34.72 cm^3 more and uses 95.68 cm^2 less material.

8.5 COMPOSITE SOLIDS AND PACKAGING

1 Base volume $= 10 \times 6 \times 3 = 180$ cm^3; top volume $= 4 \times 6 \times 4 = 96$ cm^3; total **276** cm^3. Base SA $= 2(60 + 30 + 18) = 216$ cm^2; top SA $= 2(24 + 16 + 24) = 128$ cm^2; contact $= 4 \times 6 = 24$ cm^2, hidden on both, so subtract 48. Exposed SA $= 216 + 128 - 48 = \mathbf{296}$ cm^2.

2 $SA = 2(2 \times 1.5 + 2 \times 1 + 1.5 \times 1) = 2(3 + 2 + 1.5) = 13$ m^2. Cost $= 13 \times 9.50 = \mathbf{\$123.50}$.

3 $V = 3.14(0.64)(1.5) = 3.0144$ m^3 $= \mathbf{3014.4}$ L. $SA = 2(3.14)(0.64) + 2(3.14)(0.8)(1.5) = 4.0192 + 7.536 = \mathbf{11.5552}$ m^2.

4 One can $= 3.14(12.25)(11) = 423.115$ cm^3; six cans $= 2538.69$ cm^3. Carton $= 21 \times 14 \times 11 = 3234$ cm^3. Empty $= 3234 - 2538.69 = 695.31$ cm^3, which is $695.31 \div 3234 \approx \mathbf{21.5\%}$.

5 First prism $= 24 \times 5 = 120$ cm^3; second $= 500 - 120 = \mathbf{380}$ cm^3.

6 When two parts are joined, each contact face is hidden on both parts. Example: two $2 \times 2 \times 2$ cubes joined face to face. Their separate surface areas sum to $2(24) = 48$ cm^2, but the $2 \times 2 = 4$ cm^2 contact is hidden twice. Exposed SA $= 48 - 2(4) = \mathbf{40}$ cm^2, not 48.

7 Small cube: $SA = 6(9) = 54$ cm^2, $V = 27$ cm^3. Large cube: $SA = 6(36) = 216$ cm^2, $V = 216$ cm^3. Surface area has multiplied by 4; volume has multiplied by 8.

8 Each dimension is multiplied by 0.9, so surface area is multiplied by $0.9^2 = 0.81$. New SA $\approx 2400 \times 0.81 = 1944$ cm^2, a reduction of about 19%, not 10%. Area scales by the square of the length factor.

8.5 INVESTIGATION · HOW MUCH OF THE CARTON IS AIR?

1 One can: $V = 3.14(3.5^2)(11) = 3.14(12.25)(11) = 423.115$ cm^3. Twelve cans: $12(423.115) = \mathbf{5077.38}$ cm^3.

2 Carton volume $= 28 \times 21 \times 11 = 6468$ cm^3. Empty space $= 6468 - 5077.38 = 1390.62$ cm^3. Percentage $= 1390.62 \div 6468 \times 100 \approx \mathbf{21.5\%}$.

3 $SA = 2(588 + 308 + 231) = 2254$ cm^2 $= 0.2254$ m^2. Cost for 1000 cartons $= 1000(0.2254)(2.80) = \mathbf{\$631.12}$.

4 A 6×2 layout has length $6(7) = 42$ cm, width $2(7) = 14$ cm, height 11 cm. $SA = 2(42 \times 14 + 42 \times 11 + 14 \times 11) = 2(588 + 462 + 154) = \mathbf{2408}$ cm^2.

5 The 4×3 layout uses less cardboard: $2408 - 2254 = \mathbf{154}$ cm^2 less. For 1000 cartons, the saving is $1000(0.0154)(2.80) = \mathbf{\$43.12}$.

6 Any reasonable answer, such as: the long 6×2 carton may be harder to stack and handle, may need thicker board for strength, or may not fit a standard shelf. The least-cardboard design is not automatically the best design.

7 A tighter packing pattern, such as staggering the cans in a hexagonal arrangement, can reduce the gaps between cans. It may allow a shorter or narrower carton, but it complicates filling and may leave different corner gaps. Accept any calculation consistent with the proposed dimensions.

8 Assumptions: the cans are exact cylinders, the carton dimensions are exact, and there is no clearance or board thickness. Limitation: real cans need clearance, cardboard has thickness, and the packing machine may require a less efficient layout. These effects can change the material saving.

8.6 CRITERION TASK · THE WATER TANK

1 Curved area $= 2(3.14)(1.2)(2.5) = 18.84$ m^2. Top $= 3.14(1.2^2) = 3.14(1.44) = 4.5216$ m^2. Painted tank area $= 18.84 + 4.5216 = \mathbf{23.3616}$ m^2.

2 Base top $= 3 \times 3 = 9$ m^2. The tank covers $3.14(1.2^2) = 4.5216$ m^2 of it, so exposed top $= 9 - 4.5216 = 4.4784$ m^2. Four sides $= 4(3)(0.15) = 1.8$ m^2. Exposed base area $= 4.4784 + 1.8 = \mathbf{6.2784}$ m^2. Total paint area $= 23.3616 + 6.2784 = \mathbf{29.64}$ m^2.

3 Cost $= 29.64 \times 6.50 = 192.66$, so **\$192.66** to the nearest cent.

4 Capacity $= 3.14(1.44)(2.5) = 11.304$ m^3 $= \mathbf{11\,304}$ L. Concrete $= 9 \times 0.15 = 1.35$ m^3; cost $= 1.35 \times 180 = \mathbf{\$243.00}$. Total $= 192.66 + 243.00 = \mathbf{\$435.66}$, which is \$64.34 under the \$500 budget.

5 Cuboid capacity $= 2.4 \times 2.4 \times 2 = 11.52\ \text{m}^3 =$ **11 520** L. Painted area $= 2.4 \times 2.4 + 4(2.4)(2) = 5.76 + 19.2 = 24.96\ \text{m}^2$. The cuboid covers $5.76\ \text{m}^2$ of the base, so exposed base $= (9 - 5.76) + 1.8 = 3.24 + 1.8 = 5.04\ \text{m}^2$. Total paint area $= 24.96 + 5.04 = 30.00\ \text{m}^2$. Paint cost $= 30.00 \times 6.50 =$ **$195.00**. Total $= 195.00 + 243.00 =$ **$438.00**, which is $62.00 under budget.

6 Design B holds $11\,520 - 11\,304 =$ **216** L more water. Design A costs $438.00 - 435.66 =$ **$2.34** less and uses $30.00 - 29.64 = 0.36\ \text{m}^2$ less paint. Recommend A if 11,304 L is enough; recommend B if the extra capacity is needed. The concrete base and labour are the same for both, so the comparison depends mainly on capacity and paint. A also has a smaller footprint (a 2.4 m diameter circle rather than a 2.4×2.4 m square).

7 Assumption: one coat of paint with no waste, and the given areas are exact. Limitation: wall thickness, fittings, access ladders, overlapping paint and labour costs are ignored; a real quote may differ. Communication check: lengths in m, areas in m^2, volumes in m^3 or litres, and money to the nearest cent.

MIXED PRACTICE · CONSOLIDATE AND EXTEND

1 A cube.

2 5 faces, 9 edges, 6 vertices.

3 $6(8^2) = 6(64) =$ **384** cm^2.

4 $2(45 + 36 + 20) =$ **202** cm^2.

5 $7 \times 6 \times 3 =$ **126** cm^3.

6 $3.14(4)(9) =$ **113.04** cm^3.

7 $2(3.14)(9) + 2(3.14)(3)(5) = 56.52 + 94.2 =$ **150.72** cm^2.

8 $2.5\ \text{L} = 2.5 \times 1000 =$ **2500** cm^3.

9 $240 = 8(5)h = 40h$, so $h =$ **6** cm. $SA = 2(8 \times 5 + 8 \times 6 + 5 \times 6) = 2(40 + 48 + 30) =$ **236** cm^2.

10 $628 = 3.14(25)h = 78.5h$, so $h =$ **8** cm. $SA = 2(3.14)(25) + 2(3.14)(5)(8) = 157 + 251.2 =$ **408.2** cm^2.

11 Base $= 50 \times 30 = 1500$; two 50×20 sides $= 2000$; two 30×20 sides $= 1200$; total **4700** cm^2.

12 $6s^2 = 294$, so $s^2 = 49$ and $s =$ **7** cm. $V = 7^3 =$ **343** cm^3.

13 Curved area $= 2(3.14)(1.5)(4) = 37.68\ \text{m}^2$; cost $= 37.68 \times 6 =$ **$226.08**.

14 $SA_1 = 2(24 + 12 + 8) = 88\ \text{cm}^2$; $SA_2 = 2(24 + 16 + 6) = 92\ \text{cm}^2$. Both volumes are $48\ \text{cm}^3$; the $6 \times 4 \times 2$ cuboid uses less cardboard.

15 $V = 1.2 \times 0.8 \times 0.5 = 0.48\ \text{m}^3 =$ **480** L.

16 $r = 1$ m; $V = 3.14(1^2)(3) = 9.42\ \text{m}^3 =$ **9420** L.

17 Doubling every dimension multiplies volume by $2^3 = 8$, not 2. For a cube of edge 3 cm, $V = 27\ \text{cm}^3$; for edge 6 cm, $V = 216\ \text{cm}^3$. Surface area is multiplied by $2^2 = 4$: $54 \to 216\ \text{cm}^2$.

18 Volumes $= 5 \times 4 \times 3 + 2 \times 4 \times 2 = 60 + 16 =$ **76** cm^3. Base SA $= 2(20 + 15 + 12) = 94$; top SA $= 2(8 + 4 + 8) = 40$; contact $= 2 \times 4 = 8$, subtract 16. Exposed SA $= 94 + 40 - 16 =$ **118** cm^2.

19 A cylinder. The rectangle is $2\pi r = 2(3.14)(4) = 25.12$ cm long and $h = 10$ cm high. Total area $= 25.12 \times 10 + 2(3.14)(16) = 251.2 + 100.48 =$ **351.68** cm^2.

20 Cube: $V = 1000\ \text{cm}^3$, $SA = 600\ \text{cm}^2$. Cuboid: $V = 20 \times 5 \times 10 = 1000\ \text{cm}^3$, $SA = 2(100 + 200 + 50) = 700\ \text{cm}^2$. Both have the same volume, but the cube uses $100\ \text{cm}^2$ less material because it is more compact.

21 $720 = 12(10)h = 120h$, so $h =$ **6** cm. $SA = 2(120 + 72 + 60) =$ **504** cm^2.

22 Cube: $V = 1000\ \text{cm}^3$, $SA = 600\ \text{cm}^2$. Box: $V = 20 \times 20 \times 2 = 800\ \text{cm}^3$, $SA = 2(400 + 40 + 40) = 960\ \text{cm}^2$. The cube has the larger volume but uses less cardboard, so the claim is false. A compact shape can enclose more space with less surface.

23 Volumes 27 and 216 give side lengths $s = \sqrt[3]{27} = 3$ cm and $s = \sqrt[3]{216} = 6$ cm. SA $= 6(9) = 54\ \text{cm}^2$ and $6(36) = 216\ \text{cm}^2$. The side length has doubled, the surface area has multiplied by 4, and the volume has multiplied by 8.

24 Original: $V = 3.14(9)(8) = 226.08\ \text{cm}^3$; $SA = 2(3.14)(9) + 2(3.14)(3)(8) = 56.52 + 150.72 = 207.24\ \text{cm}^2$. New: $r = 6$, $h = 4$; $V = 3.14(36)(4) = 452.16\ \text{cm}^3$, so the volume has doubled. $SA = 2(3.14)(36) + 2(3.14)(6)(4) = 226.08 + 150.72 = 376.8\ \text{cm}^2$, an increase of $169.56\ \text{cm}^2$. Surface area has not doubled.

25 Tank $= 3.14(1^2)(3) = 9.42\ \text{m}^3$; base $= 2 \times 2 \times 1 = 4\ \text{m}^3$; total $= 13.42\ \text{m}^3 =$ **13 420** L. The surface areas cannot simply be added: the tank's circular base and the part of the cuboid's top beneath it are in contact, so $2(3.14) =$ **6.28** m^2 is hidden. The cuboid's bottom may also be on the ground.

26 One possible change: replace the full top with a shrink-wrap lid. The top face has area $28 \times 21 = 588\ \text{cm}^2$, so the cardboard falls from 2254 to $2254 - 588 = 1666\ \text{cm}^2$, a 26.1% reduction. It might fail because the cans lose protection from stacking and dust, and the carton becomes weaker. Accept any proposal with a correct calculation and a realistic limitation.

Chapter 9 · Transformations and symmetry

PREREQUISITE CHECK

1 Quadrant IV: x is positive and y is negative.

2 $(1+4,\ 3-2) = (\mathbf{5}, \mathbf{1})$.

3 The x-axis keeps x and changes the sign of y: $(\mathbf{4}, \mathbf{-2})$.

4 $(2\times 3,\ -3\times 3) = (\mathbf{6}, \mathbf{-9})$.

5 A quarter turn is $\mathbf{90}^\circ$.

6 $5^2 = \mathbf{25}\ \text{cm}^2$.

7 $x = 18 \div 2 = \mathbf{9}$.

9.1 COORDINATES AND TRANSLATIONS

1 $(2+3,\ 5-7) = (\mathbf{5}, \mathbf{-2})$.

2 $(1-2, 1+3) = (\mathbf{-1}, \mathbf{4})$; $(4-2, 1+3) = (\mathbf{2}, \mathbf{4})$; $(1-2, 2+3) = (\mathbf{-1}, \mathbf{5})$.

3 $(5-4,\ -3-2) = (\mathbf{1}, \mathbf{-5})$.

4 $\begin{pmatrix} 3-(-2) \\ -1-4 \end{pmatrix} = \begin{pmatrix} 5 \\ -5 \end{pmatrix}$: 5 right and 5 down.

5 Subtract the vector: $(2-3,\ 1-(-4)) = (\mathbf{-1}, \mathbf{5})$.

6 $(4, -1) \to (5, 1) \to (\mathbf{2}, \mathbf{6})$. The single vector is $\begin{pmatrix} 1-3 \\ 2+5 \end{pmatrix} = \begin{pmatrix} -2 \\ 7 \end{pmatrix}$.

7 $\begin{pmatrix} 8-2 \\ -1-3 \end{pmatrix} = \begin{pmatrix} 6 \\ -4 \end{pmatrix}$: the robot moves 6 units right and 4 units down.

8 The student added 5 to the y-coordinate instead of subtracting 5, writing $4+5=9$. Correct image: $(3+2,\ 4-5) = (\mathbf{5}, \mathbf{-1})$.

9 For any point (x, y), the first order gives $(x+a+c,\ y+b+d)$ and the reverse order gives $(x+c+a,\ y+d+b)$. These are equal because addition is commutative. Example with $(1, 2)$, $\begin{pmatrix} 3 \\ -1 \end{pmatrix}$ and $\begin{pmatrix} -2 \\ 4 \end{pmatrix}$: both orders give $(\mathbf{2}, \mathbf{5})$.

10 The vectors must be opposites, so their sum is $\begin{pmatrix} 0 \\ 0 \end{pmatrix}$. Example: $\begin{pmatrix} 3 \\ -2 \end{pmatrix}$ followed by $\begin{pmatrix} -3 \\ 2 \end{pmatrix}$ returns every point to its start, because $(x+3-3,\ y-2+2) = (x, y)$.

Guided 9.1 $D'(2, -1)$, $E'(4, -1)$, $F'(2, 2)$. Both triangles are right-angled with legs 2 and 3, so each area is $\frac{1}{2}(2)(3) = \mathbf{3}$ square units. The areas are equal because a translation changes only position, not lengths or angles, so the image is congruent.

9.2 REFLECTION AND LINE SYMMETRY

1 $(\mathbf{3}, \mathbf{-4})$.

2 $(\mathbf{2}, \mathbf{5})$.

3 In $y = x$, swap coordinates: $(\mathbf{-1}, \mathbf{4})$.

4 In $y = -x$, $(x, y) \to (-y, -x)$: $(\mathbf{2}, \mathbf{-6})$.

5 $x' = 2(2) - 3 = 1$, so $(\mathbf{1}, \mathbf{1})$.

6 $y' = 2(3) - 4 = 2$, so $(\mathbf{-1}, \mathbf{2})$.

7 Change the sign of each x: $(\mathbf{-1}, \mathbf{1})$, $(\mathbf{-4}, \mathbf{1})$, $(\mathbf{-1}, \mathbf{3})$.

8 The midpoint of P and P' is $\left(\frac{2+6}{2}, 1\right) = (4, 1)$. The mirror is vertical, so it is $\mathbf{x = 4}$.

9 The reflection in the x-axis should change the sign of y, not x. The student's $(-2, 5)$ is the reflection in the y-axis. Correct image: $(\mathbf{2}, \mathbf{-5})$.

10 Square **4**; rectangle **2**; equilateral triangle **3**; regular hexagon **6**.

11 The motif extends $3+5=8$ cm from the mirror. Its reflection extends 8 cm on the other side, so the logo is $2\times 8 = \mathbf{16}$ cm wide. The two motifs are separated by $2\times 3 = 6$ cm around the mirror.

12 Reflection in $x = 1$: $x' = 2 - x$. Reflection in $x = 4$: $x'' = 8 - x' = 8 - (2 - x) = x + 6$. The combined map is $(x, y) \to (x+6, y)$, a translation by $\begin{pmatrix} 6 \\ 0 \end{pmatrix}$. Example: $(2, 3) \to (0, 3) \to (8, 3)$.

9.3 ROTATION AND ROTATIONAL SYMMETRY

1 $(x, y) \to (-y, x)$: $(\mathbf{-1}, \mathbf{3})$.

2 $(x, y) \to (y, -x)$: $(\mathbf{4}, \mathbf{2})$.

3 $(x, y) \to (-x, -y)$: $(\mathbf{-5}, \mathbf{2})$.

4 $(x, y) \to (y, -x)$: $(1, 0) \to (\mathbf{0}, \mathbf{-1})$; $(3, 0) \to (\mathbf{0}, \mathbf{-3})$; $(1, 2) \to (\mathbf{2}, \mathbf{-1})$.

5 Relative to the centre: $(4-1, 2-1) = (3, 1) \to (-1, 3)$. Add the centre: $(-1+1, 3+1) = (\mathbf{0}, \mathbf{4})$. Check: both points are $\sqrt{10}$ from $(1, 1)$.

6 180° about (a, b): $(2a-x, 2b-y) = (8-2, 2-3) = (\mathbf{6}, \mathbf{-1})$.

7 Order **8**, angle $360^\circ \div 8 = \mathbf{45}^\circ$.

8 Angle $360^\circ \div 5 = \mathbf{72}^\circ$. It can also have line symmetry: a regular pentagon has order 5 and 5 mirror lines. Rotational symmetry does not rule out reflections.

9 $(-y, x)$ is the rule for 90° *anticlockwise*. Clockwise, $(x, y) \to (y, -x)$, so $(3, -1) \to (\mathbf{-1}, \mathbf{-3})$.

10 $(2, 1) \to (-1, 2) \to (-2, -1)$. This is $(x, y) \to (-x, -y)$, the 180° rule. Two 90° anticlockwise turns make a half-turn, because $90^\circ + 90^\circ = 180^\circ$.

9.4 ENLARGEMENT AND SIMILARITY

1 $(2\times 4, 3\times 4) = (\mathbf{8}, \mathbf{12})$.

2 $(\mathbf{-6}, \mathbf{4})$.

3 $\left(6\times\frac{1}{2}, -4\times\frac{1}{2}\right) = (\mathbf{3}, \mathbf{-2})$.

4 $(\mathbf{2}, \mathbf{2})$, $(\mathbf{4}, \mathbf{2})$, $(\mathbf{2}, \mathbf{6})$.

5 $(\mathbf{1},\mathbf{1}), (\mathbf{3},\mathbf{1}), (\mathbf{3},\mathbf{2}), (\mathbf{1},\mathbf{2})$.

6 Relative to $(2,2)$: $(3,4) - (2,2) = (1,2)$. Multiply by 3: $(3,6)$. Add the centre: $(3+2, 6+2) = (\mathbf{5},\mathbf{8})$.

7 $5 \times 4^2 = 5 \times 16 = \mathbf{80}$ cm$^\mathbf{2}$.

8 $6 \div 2 = 3$ and $9 \div 3 = 3$, so $k = \mathbf{3}$.

9 Areas scale by k^2, not k. For $k = 3$, the area factor is $3^2 = \mathbf{9}$: the area becomes nine times as large, not three times.

10 $18 \div 6 = 3$ and $12 \div 4 = 3$, so $k = \mathbf{3}$. Area ratio new : old $= 3^2 : 1 = \mathbf{9} : \mathbf{1}$ (old : new $= 1 : 9$).

11 Perimeter $= 24 \times 1.5 = \mathbf{36}$ cm. Area $= 32 \times 1.5^2 = 32 \times 2.25 = \mathbf{72}$ cm$^\mathbf{2}$. The area factor is $k^2 = 2.25$, an increase of 125%, not 50%; a 50% increase would apply only to lengths.

9.5 INVESTIGATION · SYMMETRY AND TESSELLATION

1 Nonagon: $\frac{7\times180^\circ}{9} = 140^\circ$, and $360 \div 140 \approx 2.57$. Dodecagon: $\frac{10\times180^\circ}{12} = 150^\circ$, and $360 \div 150 = 2.4$. Neither quotient is a whole number, so neither tessellates by itself.

2 Triangle (60°, quotient 6), square (90°, quotient 4) and hexagon (120°, quotient 3) tessellate. The pentagon, octagon, decagon, nonagon and dodecagon do not, because 360° is not a whole-number multiple of their interior angles.

3 $3 \times 108^\circ = 324^\circ$, so the gap is $360^\circ - 324^\circ = \mathbf{36}^\circ$. A fourth pentagon needs 108°, which exceeds the gap by $108^\circ - 36^\circ = 72^\circ$, so it would overlap its neighbours.

4 $2 \times 135^\circ + 90^\circ = 270^\circ + 90^\circ = 360^\circ$, so the angles fit exactly. The pattern uses two kinds of regular polygon, and the same arrangement (octagon, octagon, square) occurs at every vertex, so it is semi-regular.

5 Yes. Rotating a copy 180° about the midpoint of one side produces a congruent triangle on the other side; the pair forms a parallelogram. Repeating this across the plane covers it. At each vertex the three angles of the triangle appear twice, so the angles sum to $2(\alpha + \beta + \gamma) = 2 \times 180^\circ = 360^\circ$.

6 Each interior angle is 120°, and $3 \times 120^\circ = 360^\circ$, so three hexagons meet exactly. Corresponding sides coincide, leaving no gap, and the angles fill the space around each vertex exactly, so the tiles do not overlap.

7 The claim is false. A scalene triangle and a parallelogram are not regular, yet both tessellate (every triangle and every quadrilateral does). Corrected statement: every triangle and every quadrilateral tessellates; a *regular* polygon tessellates by itself exactly when 360° is a whole-number multiple of its interior angle.

8 Sample: a 2×1 rectangle has two mirror lines (vertical and horizontal) and rotational symmetry of order 2 (angle 180°). Repeat it by a translation of $\begin{pmatrix}3\\0\end{pmatrix}$, reflect every second copy in the vertical line $x = 4.5$, and rotate one copy 180° about its centre. A strip of identical, equally spaced rectangles has no rotational symmetry as a whole (order 1), although each tile has order 2. Accept any motif and description that are consistent.

9 Sample evaluation: the strip has no gaps if each translation equals the motif width plus the chosen spacing, and the repetition is exact if every vector and centre is stated from a labelled reference point. The rotation is hardest to describe precisely because both a centre and an angle are needed. Improvement: use only translations, or mark one reference vertex on every copy so a viewer can check each image. Accept any justified judgement.

CRITERION TASK · THE COMMUNITY BANNER

1 Area $= \frac{1}{2}(2)(2) = \mathbf{2}$ square units. One square unit is $15 \times 15 = 225$ cm^2, so the area is $2 \times 225 = \mathbf{450}$ cm$^\mathbf{2}$ $= 0.045$ m^2.

2 Add 6 to each x: $A'(7,1)$, $B'(9,1)$, $C'(7,3)$.

3 Use $x' = 12 - x$: $A'(11,1)$, $B'(9,1)$, $C'(11,3)$.

4 Use $(x,y) \to (12-x, 8-y)$: $A'(11,7)$, $B'(9,7)$, $C'(11,5)$.

5 Multiply by 2: $A'(2,2)$, $B'(6,2)$, $C'(2,6)$. Area $= \frac{1}{2}(4)(4) = \mathbf{8}$ square units, which is $2^2 \times 2 = 8$.

6 Copies 1, 2 and 3 are congruent to the original: translation, reflection and rotation preserve all lengths and angles. Copy 4 is similar but not congruent: its angles are the same ($90^\circ, 45^\circ, 45^\circ$) but every side is doubled, so it is twice as wide and twice as high.

7 Total $= 2 + 2 + 2 + 2 + 8 = \mathbf{16}$ square units. Since 1 unit $= 0.15$ m, 1 square unit $= 0.0225$ m^2, so the total is $16 \times 0.0225 = \mathbf{0.36}$ m$^\mathbf{2}$.

8 Fabric needed $= 0.36 \times 1.1 = \mathbf{0.396}$ m$^\mathbf{2}$. Cost $= 0.396 \times 12 = \$4.752$, which rounds to **\$4.75**.

9 Model note (77 words): “The translated and reflected copies create a steady horizontal rhythm, while the half-turned copy balances the upper field. The enlarged copy draws the eye and makes the motif the banner's focus. I assume the fabric costs \$12 per square metre with no minimum order, and I ignore cutting waste beyond 10%. A limitation is that the 15 cm grid may not suit the printer's repeat. Snapping the motif to whole-grid points would make the design easier to reproduce.” Award credit for a clear use of transformation vocabulary, a sensible assumption, a real limitation and a workable improvement.

MIXED PRACTICE · CONSOLIDATE AND EXTEND

1 $(-3+5, 4-2) = (\mathbf{2}, \mathbf{2})$.

2 $(\mathbf{-2}, \mathbf{-5})$.

3 $(x, y) \to (y, -x)$: $(\mathbf{1}, \mathbf{-4})$.

4 $(\mathbf{6}, \mathbf{-4})$.

5 **6** axes.

6 Order **3**, angle **120°**.

7 $x' = 2(2) - 5 = -1$: $(\mathbf{-1}, \mathbf{3})$.

8 $7 \times 2^2 = \mathbf{28}\ \mathbf{cm^2}$.

9 $A'(-2, 4)$, $B'(1, 4)$, $C'(-2, 7)$.

10 Swap coordinates: $(\mathbf{1}, \mathbf{-2})$, $(\mathbf{1}, \mathbf{1})$, $(\mathbf{4}, \mathbf{1})$, $(\mathbf{4}, \mathbf{-2})$.

11 $(x, y) \to (4 - x, 4 - y)$: $(\mathbf{3}, \mathbf{3})$, $(\mathbf{0}, \mathbf{3})$, $(\mathbf{3}, \mathbf{1})$.

12 Relative to $(1, 1)$, the object is $(2, 1) - (1, 1) = (1, 0)$ and the image is $(5, 1) - (1, 1) = (4, 0)$. The horizontal distance has multiplied by 4, so $k = \mathbf{4}$.

13 The joining segment is horizontal and its midpoint is $(3, 3)$, so the mirror is the vertical line $\mathbf{x = 3}$.

14 The final image is the original shape in its starting position: clockwise then anticlockwise turns of the same angle cancel. The combined transformation is the identity.

15 The sides are in the ratio $3 : 6 = 4 : 8 = 1 : 2$, so the rectangles are similar with $k = 2$. They are not congruent because their sizes differ. The areas are $12\ \text{cm}^2$ and $48\ \text{cm}^2$, a ratio of $\mathbf{4 : 1}$, which is $k^2 : 1$.

16 $3 \times 108° = 324°$, leaving a gap of $\mathbf{36}°$. A fourth pentagon would bring the total to $432°$, overlapping by $72°$, so three is the maximum.

17 Yes, the order matters. Reflect first, then translate: $(2, 1) \to (-2, 1) \to (2, 1)$. Translate first, then reflect: $(2, 1) \to (6, 1) \to (-6, 1)$. The results are different.

18 A square fits. Not every shape with order 4 has four mirror lines: a four-armed pinwheel can have order 4 and no mirror line. Rotational symmetry and line symmetry are independent properties.

19 The centre is $(2.5, 2.5)$. The rotation sends $(1, 1) \to (4, 1)$, $(4, 1) \to (4, 4)$, $(4, 4) \to (1, 4)$ and $(1, 4) \to (1, 1)$. The image coincides with the original square; the vertices are cycled, so the square has rotational symmetry of order 4 about its centre.

20 Original area $= 6$. After $k = 2$: $6 \times 4 = 24$. After $k = 3$: $24 \times 9 = 216$. The total area factor is $216 \div 6 = \mathbf{36} = 6^2$. Successive scale factors multiply $(2 \times 3 = 6)$, so the area factors multiply $(4 \times 9 = 36)$; a single enlargement with $k = 6$ would also give $k^2 = 36$.

21 Four copies meet at each vertex, so the interior angle is $360° \div 4 = 90°$. Then $\frac{(n-2)180°}{n} = 90°$ gives $n = \mathbf{4}$: the polygon is a square.

22 $60 \div 5 = \mathbf{12}$ copies fit. The twelfth copy starts at $11 \times 5 = 55$ and ends at $55 + 5 = \mathbf{60}$, so the last copy ends exactly at the far edge of the strip.

23 Reflection in $x = 3$: $x' = 6 - x$. Reflection in $x = 7$: $x'' = 14 - x' = 14 - (6 - x) = x + 8$. The result is the translation $\begin{pmatrix} 8 \\ 0 \end{pmatrix}$. Check: $(1, 2) \to (5, 2) \to (9, 2)$, a movement of 8 right.

24 The claim is false. An isosceles triangle has one mirror line but no rotational symmetry beyond order 1; a rectangle has two mirror lines and order 2. Correct statement: line symmetry does not guarantee rotational symmetry; shapes with two or more mirror lines do have rotational symmetry, but a shape with exactly one line may have none.

25 One quarter of the banner is $96 \div 4 = 24$ square units. The enlarged motif has area $2 \times 2^2 = 8$ square units, and $8 \leq 24$, so scale factor 2 is acceptable. It occupies $8 \div 96 = \frac{1}{12} \approx 8.3\%$ of the banner, well under the limit.

26 Sample: draw the motif with vertices $(1, 1), (3, 1), (1, 3)$. (1) Translate it by $\begin{pmatrix} 4 \\ 0 \end{pmatrix}$ twice to make a row. (2) Reflect the row in the vertical line $x = 10$. (3) Rotate the middle motif $180°$ about $(5, 2)$. Check: each image vertex must be the mirror image in $x = 10$, or the same distance from $(5, 2)$ on the opposite ray. A single misplotted point breaks the symmetry and is visible immediately. Accept any reproducible pattern with three named transformations and a valid check.

Chapter 10 · Probability and simulation

PREREQUISITE CHECK

1 Divide numerator and denominator by 6: $\frac{6}{36} = \frac{1}{6}$.

2 $0.35 = \frac{35}{100} = \frac{7}{20}$.

3 $1 - \frac{3}{8} = \frac{8}{8} - \frac{3}{8} = \frac{5}{8}$.

4 $0.15 \times 240 = \mathbf{36}$.

5 $2 \times 2 = \mathbf{4}$ outcomes: HH, HT, TH, TT.

6 **6** faces.

7 $\mathbf{2, 3, 5, 7, 11}$. (1 is not prime.)

8 $\frac{28}{80} = \mathbf{0.35}$.

10.1 OUTCOMES, SAMPLE SPACES AND EVENTS

1 $S = \{A, B, C, D\}$, so $n(S) = \mathbf{4}$.

2 $S = \{1, 2, 3, 4, 5, 6\}$; even $= \{2, 4, 6\}$, so **3** outcomes.

3 $S = \{HH, HT, TH, TT\}$. At least one tail: $\{HT, TH, TT\}$, so **3** outcomes.

4 $n(S) = \mathbf{8}$ sector outcomes. Distinct scores: $\{\mathbf{1}, \mathbf{2}, \mathbf{3}, \mathbf{4}\}$. The distinct scores are not equally likely: score 2 has 3 sectors, score 3 has 2 and score 4 has 1.

5 $6 \times 6 = \mathbf{36}$ outcomes. Sum 9: $(3, 6), (4, 5), (5, 4), (6, 3)$, so **4** outcomes.

6 $S = \{HHH, HHT, HTH, HTT, THH, THT, TTH, TTT\}$. Exactly two heads: HHT, HTH, THH, so **3**. At least two heads adds HHH, so **4**.

7 $3 \times 2 = \mathbf{6}$ meals: pasta–fruit, pasta–cake, curry–fruit, curry–cake, fish–fruit, fish–cake.

8 The student omitted TH. The coins are distinct, so HT (head then tail) and TH (tail then head) are different outcomes. Correct: $S = \{HH, HT, TH, TT\}$, so $n(S) = \mathbf{4}$.

9 Sum 7 occurs in 6 cells: $(1, 6), (2, 5), (3, 4), (4, 3), (5, 2), (6, 1)$. Sum 12 occurs in 1 cell: $(6, 6)$. The probabilities are $\frac{6}{36} = \frac{1}{6}$ and $\frac{1}{36}$, so sum 7 is six times as likely.

10 AB, AC, BA, BC, CA, CB, so **6** outcomes. There are 3 choices for the first letter and 2 for the second: $3 \times 2 = 6$.

Guided 10.1 $n(S) = 4 \times 3 = \mathbf{12}$. Sum 5: $(2, 3), (3, 2), (4, 1)$, so **3** outcomes. Other events also have several outcomes: sum 4 has $(1, 3), (2, 2), (3, 1)$, also 3 outcomes; sum 3 has 2 outcomes.

10.2 THEORETICAL PROBABILITY AND COMPLEMENTS

1 $P(4) = \frac{1}{6}$; $P(\text{odd}) = \frac{3}{6} = \frac{1}{2}$; $P(\text{greater than } 4) = \frac{2}{6} = \frac{1}{3}$.

2 $P(HH) = \frac{1}{4}$; $P(\text{exactly one head}) = \frac{2}{4} = \frac{1}{2}$; $P(\text{at least one head}) = \frac{3}{4}$. *Alternatively,* $P(\text{exactly one}) = 1 - P(0) - P(2) = 1 - \frac{1}{4} - \frac{1}{4} = \frac{1}{2}$.

3 $P(3) = \frac{1}{5}$; $P(\text{not } 3) = 1 - \frac{1}{5} = \frac{4}{5}$; $P(\text{even}) = \frac{2}{5}$.

4 There are $4 + 5 + 3 = 12$ counters. $P(\text{red}) = \frac{4}{12} = \frac{1}{3}$; $P(\text{not blue}) = \frac{4+3}{12} = \frac{7}{12}$; $P(\text{red or green}) = \frac{4+3}{12} = \frac{7}{12}$.

5 $P(\text{sum} = 8) = \frac{5}{36}$; $P(\text{sum} = 12) = \frac{1}{36}$; $P(\text{sum at least } 11) = \frac{2+1}{36} = \frac{1}{12}$.

6 Same: 6 outcomes, so $P = \frac{6}{36} = \frac{1}{6}$. Different: $36 - 6 = 30$ outcomes, so $P = \frac{30}{36} = \frac{5}{6}$.

7 **D.** A probability cannot exceed 1; 1.25 is greater than 1.

8 $P(\text{red}) = \frac{7}{10}$; $P(\text{yellow}) = \frac{3}{10}$. They add to 1 because every counter is either red or yellow. Expected red $= 0.7 \times 20 = \mathbf{14}$.

9 The eleven sums are not equally likely. Sum 7 has 6 outcomes out of 36, while some sums have fewer. Correct probability: $\frac{6}{36} = \frac{1}{6}$, not $\frac{1}{11}$.

10 Since the events are mutually exclusive, $P(A \text{ or } B) = 0.35 + 0.45 = \mathbf{0.80}$. Therefore $P(\text{neither}) = 1 - 0.80 = \mathbf{0.20}$.

11 Multiples of 3 are 3, 6, 9 and 12: $2 + 5 + 4 + 1 = 12$ outcomes. $P = \frac{12}{36} = \frac{1}{3}$.

12 Use **4** red sectors, **2** blue and **2** green, giving 8 sectors in total. Then $P(\text{red}) = \frac{4}{8} = \frac{1}{2}$, $P(\text{blue}) = \frac{2}{8} = \frac{1}{4}$ and $P(\text{green}) = \frac{2}{8} = \frac{1}{4}$.

Guided 10.2 Prime scores: 2, 3, 5, so $P(\text{prime}) = \frac{3}{6} = \frac{1}{2}$. Not prime: 1, 4, 6, so $P(\text{not prime}) = \frac{1}{2}$. Prime or 6: 2, 3, 5, 6, so $P = \frac{4}{6} = \frac{2}{3}$. The first two add to 1.

10.3 EXPERIMENTAL PROBABILITY AND SIMULATION

1 Relative frequency $= \frac{22}{50} = \mathbf{0.44}$. Expected heads $= 0.5 \times 50 = \mathbf{25}$. The result is 3 heads fewer than expected, a difference of 0.06.

2 Relative frequency $= \frac{18}{120} = \mathbf{0.15}$. Theoretical probability $= \frac{1}{6} \approx \mathbf{0.167}$. Expected sixes $= \frac{1}{6} \times 120 = \mathbf{20}$; the result is 2 fewer.

3 Successes are digits 0–8. The sequence has three 9s, so there are $25 - 3 = \mathbf{22}$ successes. Relative frequency $= \frac{22}{25} = \mathbf{0.88}$. Expected $= 0.9 \times 25 = \mathbf{22.5}$, so the result is close; 25 trials are a small sample.

4 Let digits 0, 1 and 2 mean rain and 3–9 mean no rain, since 3 of 10 digits represent 30%. Generate 40 digits, count the digits 0–2, divide by 40, and compare with 0.30. Expected rainy days $= 0.30 \times 40 = \mathbf{12}$.

5 Relative frequency $= \frac{62}{200} = \mathbf{0.31}$. Expected for a fair spinner $= \frac{1}{4} \times 200 = \mathbf{50}$. Red appeared 12 more times than expected. This is not proof of bias: random variation can produce such a result, although 0.31 is noticeably above 0.25. Repeat with more spins or inspect the spinner.

6 Two-digit numbers run from 00 to 99: 100 equally likely codes. Let 00, 01, 02 and 03 mean faulty; these are 4 of 100, or 4%. Expected faulty $= 0.04 \times 600 = \mathbf{24}$. Compare the simulated count with 24.

7 The tosses are independent; the coin has no memory. After five heads, $P(\text{tails})$ is still $\frac{1}{2}$. A run of heads is unusual-looking but not evidence that tails is "due".

8 More trials reduce the effect of random variation, so the relative frequency usually moves closer to the theoretical probability. But 1000 trials are still random: the result can differ from the theoretical value, and an exact match is not guaranteed.

9 **B.** $\frac{24}{80} = 0.30$.

10 Use three digits per family: 0–4 for a boy and 5–9 for a girl (or the reverse). Exactly two boys can occur in the orders BBG, BGB and GBB, so $P = \frac{3}{8}$. In 40 families, expected $= 40 \times \frac{3}{8} = \mathbf{15}$. Assumptions: each child is equally likely to be a boy or girl, and the three births are independent.

Guided 10.3 Relative frequency $= \frac{9}{25} = \mathbf{0.36}$. Expected $= 0.25 \times 25 = \mathbf{6.25}$. The observed 9 is 2.75 above expectation, but with only 25 spins this is plausible random variation. It is not strong evidence of bias; more spins are needed.

10.4 INVESTIGATION · IS THE GAME FAIR?

1 Each ordered pair of scores has probability $\frac{1}{36}$, so the 36 ordered outcomes are equally likely. A sum can occur in different numbers of ways: sum 7 has 6 outcomes and sum 12 has 1, so the eleven sums are not equally likely.

2 $P(A) = \frac{15}{36} = \mathbf{\frac{5}{12}}$; $P(B) = \frac{21}{36} = \mathbf{\frac{7}{12}}$. Check: $\frac{5}{12} + \frac{7}{12} = 1$.

3 A: $120 \times \frac{5}{12} = \mathbf{50}$ expected wins. B: $120 \times \frac{7}{12} = \mathbf{70}$ expected wins.

4 **B** has the advantage, with 20 more expected wins than A. B's probability $\frac{7}{12}$ is greater than A's $\frac{5}{12}$.

5 Prime sums in Maya's data: 2, 3, 5, 7, 7, 7, 7, 11, 5, 3, 11, 7, 7 — that is, **13** prime sums. Relative frequency $= \frac{13}{30} \approx \mathbf{0.433}$.

6 Theoretical $P(A) = \frac{5}{12} \approx 0.417$. Maya's 0.433 is close: the difference is about 0.016, roughly half a game in 30. The simulation supports the model. Limitations: only 30 games, random variation, and the dice may not be perfectly fair; a close result is evidence, not proof.

7 A needs 3 more outcomes to reach 18, because B has 21 and A has 15. Move sum **4** (3 outcomes) or sum **10** (3 outcomes) from B to A. Then A has 18 outcomes and B has 18, so both probabilities are $\frac{18}{36} = \frac{1}{2}$.

8 Fair means expected payout equals the cost: $P(\text{win}) \times 12 = 3$, so $P(\text{win}) = \frac{3}{12} = \frac{1}{4}$. A fair four-sector spinner with one winning sector, or an eight-sector spinner with two winning sectors, produces this probability. The expected payout is $12 \times \frac{1}{4} = \$3$, equal to the entry fee.

Evaluation guide A strong response lists or counts the outcomes accurately, computes both probabilities, interprets fairness in terms of expected wins or expected payout, and evaluates the simulation with reference to sample size and assumptions. Accept any valid fair redesign that produces 18 outcomes for each player, or an equivalent argument.

CRITERION TASK · THE SCHOOL FAIR GAME

1 $4 \times 6 = \mathbf{24}$ equally likely outcomes. One winning outcome is $(1, 6)$; losing examples are $(1, 5)$ and $(2, 6)$. If both devices are fair, each spinner sector has probability $\frac{1}{4}$, each die face has probability $\frac{1}{6}$, and each ordered pair has probability $\frac{1}{24}$.

2 $P(\text{win}) = \mathbf{\frac{1}{24}}$; $P(\text{lose}) = 1 - \frac{1}{24} = \mathbf{\frac{23}{24}}$.

3 Expected winners $= 240 \times \frac{1}{24} = \mathbf{10}$. Expected non-winners $= 240 \times \frac{23}{24} = \mathbf{230}$.

4 Takings $= 240 \times \$1 = \mathbf{\$240}$. Payout $= 10 \times \$10 = \mathbf{\$100}$. Expected profit $= \$240 - \$100 = \mathbf{\$140}$. Profit per play $= \frac{140}{240} \approx \mathbf{\$0.58}$.

5 For fairness, expected payout = takings: $10 \times \text{prize} = \$240$, so the fair prize is **\$24**. The offered prize is only \$10. The game is not fair to players: the expected return is $\frac{10}{24} \approx \$0.42$ per \$1 played, an expected loss of about \$0.58 per play.

6 For a profit of \$100, payout must be $\$240 - \$100 = \$140$. With 10 expected winners, prize $= \frac{\$140}{10} = \mathbf{\$14}$.

7 For example, win if the spinner shows 1 or 2 and the die shows 6. Then $P(\text{win}) = \frac{2}{4} \times \frac{1}{6} = \frac{2}{24} = \mathbf{\frac{1}{12}}$. Expected winners $= 240 \times \frac{1}{12} = \mathbf{20}$; payout $= 20 \times \$10 = \mathbf{\$200}$; profit $= \$240 - \$200 = \mathbf{\$40}$. The chance of winning doubles, but the expected profit falls from \$140 to \$40.

8 **Sample note.** "The model assumes a fair spinner and die, exactly 240 plays, independent plays and no costs other than prizes. In practice the number of winners will vary, and staffing or prize donations are ignored. At a \$10 prize the school expects \$140 profit, which is generous but leaves players with only a 1 in 24 chance. A \$14 prize meets the \$100 profit target, while \$24 would make the game fair. I recommend \$14 if fundraising is the priority, or the 1-in-12 rule if a more realistic chance matters more than profit." (94 words)

Marking guidance Criterion A: correct sample space and probabilities. Criterion C: clear notation, labelled outcomes and a structured note. Criterion D: correct financial model, interpretation of fairness and evaluation of assumptions. Accept any valid redesign with correct supporting calculations.

MIXED PRACTICE · CONSOLIDATE AND EXTEND

1 $S = \{1, 2, 3, 4, 5, 6\}$, $n(S) = 6$.

2 Exactly one tail: HT, TH, so $P = \frac{2}{4} = \mathbf{\frac{1}{2}}$.

3 Sum 9: 4 outcomes, so $P = \frac{4}{36} = \mathbf{\frac{1}{9}}$.

4 $P(\text{blue}) = \mathbf{\frac{7}{10}}$; $P(\text{not blue}) = \mathbf{\frac{3}{10}}$.

5 Prime scores 2, 3, 5: $P = \frac{3}{6} = \mathbf{\frac{1}{2}}$.

6 $200 \times \frac{1}{4} = \mathbf{50}$ times per sector.

7 $\frac{18}{40} = \mathbf{0.45}$.

8 Fair: $P(\text{win}) \times 8 = 2$, so $P(\text{win}) = \mathbf{\frac{1}{4}}$.

9 Even sums (2,4,6,8,10,12): $1 + 3 + 5 + 5 + 3 + 1 = 18$ outcomes. Odd sums: $36 - 18 = 18$. Both probabilities are $\frac{18}{36} = \frac{1}{2}$. The distribution is symmetric about 7, but the counts must still be checked.

10 $P(3) = \frac{3}{6} = \frac{1}{2}$; $P(\text{not } 3) = \frac{1}{2}$; $P(1) = \frac{2}{6} = \frac{1}{3}$.

11 12 counters. Red or blue: $\frac{5+4}{12} = \frac{3}{4}$; not green: $\frac{5+4}{12} = \frac{3}{4}$.

12 At least 10: $\frac{3+2+1}{36} = \mathbf{\frac{1}{6}}$. Less than 4: $\frac{1+2}{36} = \mathbf{\frac{1}{12}}$. From 4 to 6 inclusive: $\frac{3+4+5}{36} = \frac{12}{36} = \mathbf{\frac{1}{3}}$.

13 Expected $= 0.25 \times 60 =$ **15**. Relative frequency $= \frac{18}{60} =$ **0.30**. The event occurred 3 times more than expected, which is plausible random variation with 60 trials; more trials would clarify whether the model is appropriate.

14 Rain and no rain are complementary, so their probabilities must add to 1, not be equal. $P(\text{no rain}) = 1 - 0.7 =$ **0.3**.

15 Both odd: $3 \times 3 = 9$ outcomes, $P = \frac{9}{36} = \frac{1}{4}$. Exactly one odd: $3 \times 3 + 3 \times 3 = 18$ outcomes, $P = \frac{18}{36} = \frac{1}{2}$.

16 Let 0 and 1 mean success (2 of 10 digits, or 20%) and 2–9 mean failure. Run 50 trials, count successes, divide by 50. Expected successes $= 0.2 \times 50 =$ **10**.

17 Expected payout $= 20 \times \frac{1}{5} =$ \$4, equal to the \$4 cost, so the game is fair. If $P = \frac{1}{4}$, expected payout is \$5, which favours the player; if $P = \frac{1}{10}$, expected payout is \$2, which favours the game.

18 After adding 2 red: 6 red, 6 blue, so $P(\text{red}) = \frac{6}{12} = \frac{1}{2}$. Let x further red counters be added: $\frac{6+x}{12+x} = \frac{2}{3}$. Then $18 + 3x = 24 + 2x$, so $x =$ **6**. Check: 12 red out of 18 gives $\frac{2}{3}$.

19 Exactly two heads: HHT, HTH, THH, so $P = \frac{3}{8}$. At least two heads: those three plus HHH, so $P = \frac{4}{8} = \frac{1}{2}$.

20 Multiples of 3: sums 3, 6, 9, 12, with $2 + 5 + 4 + 1 = 12$ outcomes, so $P(A) = \frac{12}{36} = \frac{1}{3}$ and $P(B) = \frac{2}{3}$. Not fair. Move sum 7, which has 6 outcomes, from B to A: A then has $12 + 6 = 18$ outcomes and B has $36 - 18 = 18$, so both probabilities are $\frac{1}{2}$.

21 Digits 0–8 are 9 of 10 digits, so the simulation uses 90% success, not 80%. It will overestimate the success rate. Correct assignment: 0–7 mean success (8 of 10, or 80%) and 8–9 mean failure.

22 Takings $= 500 \times \$2 = \$1000$. Expected payout $= \$600$. Expected profit $= \$1000 - \$600 =$ **\$400**, or $\$400 \div 500 =$ **\$0.80** per ticket. For a fair raffle, expected payout equals takings, so the fair prize is **\$1000**.

23 Seven heads in ten tosses does not prove bias: a fair coin can produce this by chance, and ten tosses are a small sample. The claim needs much more data. Toss the coin many times and compare the relative frequency with 0.5; also check that the tosses are independent and the coin is physically symmetric.

24 Fair: $P(\text{win}) \times 15 = 3$, so $P(\text{win}) = \frac{3}{15} = \frac{1}{5}$. Use a fair five-sector spinner with one winning sector; the expected payout is $15 \times \frac{1}{5} = \$3$, equal to the entry fee.

Chapter 11 · Univariate data

PREREQUISITE CHECK

1 **1, 3, 5, 7, 9**.

2 $\frac{4+6+8+10}{4} = \frac{28}{4} =$ **7**.

3 The middle value of five ordered values is the third: **8**.

4 3 occurs three times, more than any other value, so the mode is **3**.

5 $15 - 7 =$ **8**.

6 $20 \div 4 =$ **5**.

7 $360 \div 4 =$ **90**.

8 $4 + 5 + 3 =$ **12** students.

11.1 ORGANISING AND SUMMARISING DATA

1 $6 + 8 + 10 + 12 + 14 = 50$, so the mean is $\frac{50}{5} =$ **10**.

2 There are six ordered values, so the median is the mean of the third and fourth: $\frac{9+12}{2} =$ **10.5**.

3 Both 5 and 8 occur twice, more often than 3, 6 or 9. The data are **bimodal** : the modes are **5** and **8**; there is no single mode.

4 The largest value is 21 and the smallest is 6, so the range is $21 - 6 =$ **15**.

5 Total pairs $= 2 + 5 + 8 + 3 + 2 = 20$. Total of sizes $= 5(2) + 6(5) + 7(8) + 8(3) + 9(2) = 10 + 30 + 56 + 24 + 18 = 138$. Mean $= \frac{138}{20} =$ **6.9**. The 10th and 11th values are both 7, so the median is **7**; the mode is also **7** because it has frequency 8. Range $= 9 - 5 =$ **4**.

6 Total $= 12 + 14 + 15 + 16 + 17 + 18 + 45 = 137$, so mean $= \frac{137}{7} \approx$ **19.57**. The median is the fourth value, **16**. The median is more representative because 45 is far above the rest and pulls the mean upward. (It is also beyond the upper fence $18 + 1.5(4) = 24$.)

7 The mean of five numbers is 12, so their total is $5(12) = 60$. The four given numbers total $8 + 10 + 14 + 16 = 48$, so the fifth is $60 - 48 =$ **12**.

8 The student did not order the data. In order the values are $2, 4, 7, 9, 11$, so the median is **7**, not 2. The value 2 is the minimum; its position in the unordered list happened to be third.

9 A: $5 + 6 + 7 + 8 + 9 = 35$, mean $\frac{35}{5} = 7$, range $9 - 5 = 4$. B: $2 + 4 + 7 + 10 + 12 = 35$, mean $\frac{35}{5} = 7$, range $12 - 2 = 10$. The means are equal, but B has the larger range, so B is more spread out. Range alone does not describe the middle of the data.

10 The five values total $4 + 6 + 7 + 9 + 10 = 36$. For six values to have mean 8, their total must be $6(8) = 48$. The added value must be $48 - 36 =$ **12**.

Guided 11.1 Total pets $= 0(4) + 1(5) + 2(3) + 3(2) = 17$, so mean $= \frac{17}{14} \approx \mathbf{1.21}$ pets. Cumulative frequencies are 4, 9, 12, 14; the 7th and 8th values are both 1, so median $= \mathbf{1}$. Mode $= \mathbf{1}$ (frequency 5). Range $= 3 - 0 = \mathbf{3}$. The mean is not whole because 17 is not divisible by 14. A mean number of pets need not be a possible count for one student; it is a summary of the group.

11.2 QUARTILES, IQR AND BOX PLOTS

1 There are seven values. Median $= \mathbf{8}$. Lower half $5, 6, 7$, so $Q_1 = \mathbf{6}$; upper half $9, 10, 11$, so $Q_3 = \mathbf{10}$. IQR $= 10 - 6 = \mathbf{4}$.

2 There are eight values, so the median is $\frac{20+22}{2} = \mathbf{21}$. Lower half $12, 15, 18, 20$ gives $Q_1 = \frac{15+18}{2} = \mathbf{16.5}$. Upper half $22, 25, 28, 30$ gives $Q_3 = \frac{25+28}{2} = \mathbf{26.5}$. IQR $= 26.5 - 16.5 = \mathbf{10}$.

3 The integers are $1, 2, \ldots, 10$. Median $= \frac{5+6}{2} = \mathbf{5.5}$. Lower half 1–5, so $Q_1 = \mathbf{3}$; upper half 6–10, so $Q_3 = \mathbf{8}$. IQR $= 8 - 3 = \mathbf{5}$.

4 $Q_3 = Q_1 + \text{IQR} = 12 + 8 = \mathbf{20}$. Upper fence $= 20 + 1.5(8) = \mathbf{32}$. Since 35 is greater than 32, 35 **is** an outlier.

5 Range $= 18 - 4 = \mathbf{14}$. IQR $= 13 - 7 = \mathbf{6}$. Median $= \mathbf{9}$. The median is the middle of the five-number summary, not the midpoint of the minimum and maximum.

6 B has the higher centre: median 12 versus 10. A is more consistent because its IQR is 4, compared with 8 for B. The IQR describes only the middle half; these summaries do not tell us the full ranges or whether every value in B is above every value in A.

7 The student has confused the range with the interquartile range. Range = maximum − minimum. IQR $= Q_3 - Q_1$. For example, in $2, 4, 5, 7, 9$, the range is $9 - 2 = 7$, while $Q_1 = 3$, $Q_3 = 8$ and the IQR is 5.

8 There are nine values, so median $= \mathbf{7}$. Lower half $2, 5, 6, 6$, so $Q_1 = \frac{5+6}{2} = \mathbf{5.5}$. Upper half $8, 9, 10, 50$, so $Q_3 = \frac{9+10}{2} = \mathbf{9.5}$. IQR $= 9.5 - 5.5 = \mathbf{4}$. Upper fence $= 9.5 + 1.5(4) = \mathbf{15.5}$, so **50** is an outlier. Lower fence $= 5.5 - 6 = -0.5$, so 2 is not an outlier.

9 Class A is more consistent because its IQR is smaller (8 versus 15). Class B has the higher typical score because its median is 70, compared with 65. The classes may still overlap, and IQR does not show the overall range.

10 The IQR depends on the positions of Q_1 and Q_3, which lie inside the ordered data. Changing one extreme value usually leaves those quartiles unchanged, while the range changes immediately because it uses the minimum and maximum. Example: $2, 5, 6, 7, 8, 9, 10$ has range 8 and IQR 4. Replace 10 by 100: the range becomes 98, but $Q_1 = 5$ and $Q_3 = 9$ are unchanged, so the IQR remains 4. The range is therefore much more sensitive to a single extreme value.

Guided 11.2 Median $= \frac{9+10}{2} = \mathbf{9.5}$. Lower six values $2, 4, 4, 5, 7, 9$ give $Q_1 = \frac{4+5}{2} = \mathbf{4.5}$. Upper six $10, 12, 13, 15, 17, 20$ give $Q_3 = \frac{13+15}{2} = \mathbf{14}$. IQR $= 14 - 4.5 = \mathbf{9.5}$. Five-number summary: **2**, **4.5**, **9.5**, **14**, **20**. Fences: $4.5 - 1.5(9.5) = -9.75$ and $14 + 1.5(9.5) = 28.25$. Both extremes lie inside the fences, so there are no outliers and the whiskers reach 2 and 20.

11.3 COMPARING DISTRIBUTIONS AND MAKING CLAIMS

1 Class A is more consistent because its range is smaller (8 versus 20). Class B has the higher average (mean 75 versus 72). A comparison of consistency should normally use the IQR as well as the range, because the range depends on only two values.

2 The distributions have the same centre (median 20), but B is more spread out: its IQR is 12, compared with 5 for A. A is more concentrated in its middle half. The IQR does not show the full range or the shape.

3 Starting the axis at 50 hides the first 50 units. The bars show visible heights $55 - 50 = 5$ and $60 - 50 = 10$, a ratio $2 : 1$, although the values are in the ratio $55 : 60 = 11 : 12$. The visual difference is exaggerated. Start the axis at 0, or use a dot plot or box plot and label the actual values.

4 The mean is greater than the median, so the distribution is likely **right-skewed** : a few high values pull the mean upward while the median stays near the centre of the bulk of the data. Check with a dot plot or box plot before concluding.

5 Report the **median** . One very expensive house is a high outlier, so it pulls the mean above what a typical house costs. The median is resistant to that extreme value. The mean may still be useful for the total value of the street's housing.

6 Fountain A is more reliable because its IQR is smaller (25 L versus 80 L): its daily use is more predictable. Both have similar means, but B varies much more from day to day. A smaller IQR does not guarantee that every day is close to the mean.

7 The sample of 200 is more likely to represent the school, because a larger sample usually reduces the effect of random variation. This is not automatic: the sample must still be selected without bias, and 200 students from one class or one year group would not represent the whole school.

8 The claim is false. The mean is sensitive to outliers and skew. Example: $10, 10, 10, 10, 100$ has mean 28, but the median is 10, which better describes a typical value. For house prices or salaries, the median is often more representative; the mean is useful when totals matter.

9 Supported: B has the higher typical value (median 18 versus 14) and greater spread (IQR 10 versus 6). Not supported: that every value in B is greater than every value in A, that B's overall range is larger, or that the difference has a particular cause. We do not know the sample sizes, the full data or the context.

10 One valid set is $\mathbf{4}, \mathbf{8}, \mathbf{9}, \mathbf{10}, \mathbf{11}, \mathbf{12}, \mathbf{18}$. There are seven values, so the median is the fourth, 10. The range is $18 - 4 = 14$. The lower half $4, 8, 9$ has median $Q_1 = 8$; the upper half $11, 12, 18$ has median $Q_3 = 12$; so the IQR is $12 - 8 = 4$. All three conditions are satisfied. Other sets with these properties are acceptable.

Guided 11.3 On a zero-based scale the bars would be in the ratio $160 : 150 = 16 : 15$, so the 160 L bar would be only about 6.7% taller, not twice as tall. The truncated axis makes the visible portions (10 L and 20 L) look like $2 : 1$. A corrected graph starts at 0 or uses a dot/box plot with the actual values. A fair comparison would report that the means are close but compare the medians and spreads, for example the IQR, before recommending either fountain.

11.4 INVESTIGATION · REACTION TIMES

1 Dominant: total $= 2930$, so mean $= \frac{2930}{12} \approx \mathbf{244.17}$ ms; median $= \frac{240+245}{2} = \mathbf{242.5}$ ms. Non-dominant: total $= 3205$, so mean $= \frac{3205}{12} \approx \mathbf{267.08}$ ms; median $= \frac{260+265}{2} = \mathbf{262.5}$ ms.

2 Dominant: lower six $210, \ldots, 240$ give $Q_1 = 227.5$; upper six $245, \ldots, 290$ give $Q_3 = 257.5$; IQR $= \mathbf{30}$ ms; range $= 290 - 210 = \mathbf{80}$ ms. Non-dominant: $Q_1 = 247.5$, $Q_3 = 285$, IQR $= \mathbf{37.5}$ ms; range $= 320 - 230 = \mathbf{90}$ ms.

3 Five-number summaries: dominant **210**, **227.5**, **242.5**, **257.5**, **290**; non-dominant **230**, **247.5**, **262.5**, **285**, **320**. On a common scale from about 200 to 330, the non-dominant box is shifted to the right and is slightly longer. Both whiskers are long, and the distributions overlap substantially.

4 The dominant hand has the lower centre: mean about 244.2 ms versus 267.1 ms, and median 242.5 ms versus 262.5 ms. Its spread is slightly smaller (IQR 30 versus 37.5; range 80 versus 90). The claim that the dominant hand tends to be faster is supported on average.

5 The medians differ by $262.5 - 242.5 = \mathbf{20}$ ms. This is about 8% of the dominant median and is small compared with the spread of either distribution (IQRs of 30 and 37.5 ms). It may be detectable in a careful experiment but is unlikely to be important in most everyday tasks. The size of a difference should be judged in context, not only by whether it exists.

6 The ranges overlap from 230 to 290 ms: some dominant-hand times are slower than some non-dominant-hand times, and vice versa. The overlap means hand dominance does not determine an individual's reaction time. A group difference in centre does not allow a prediction about one person.

7 Dominant fences: $227.5 - 45 = 182.5$ and $257.5 + 45 = 302.5$; non-dominant fences: $247.5 - 56.25 = 191.25$ and $285 + 56.25 = 341.25$. All values lie inside the fences, so there are no outliers. Limitations include only 12 volunteers, three trials per hand, unknown order of testing (practice or fatigue), and a conversion from catch distance that includes movement time. A stronger design would randomise the order of hands, use more trials and participants, and standardise the procedure.

8 Sample conclusion: "The dominant hand was faster on average: the median reaction time was 20 ms lower and the mean about 23 ms lower. The dominant hand was also slightly less variable. However, the distributions overlap from 230 to 290 ms, and 12 students are a small sample. The claim is supported on average, but it does not predict an individual's performance; a larger, randomised test is needed." A strong answer quotes both centres and a spread, notes the overlap, and includes a limitation and an improvement.

Evaluation guide Award credit for correct totals and quartiles, a sensible common scale, comparison of centre and spread rather than centre alone, recognition of overlap, correct fence calculations, and an evaluation that distinguishes a group tendency from an individual prediction. Accept any justified conclusion about practical importance; do not accept a claim that the dominant hand is faster for every student.

CRITERION TASK · THE CANTEEN QUEUE AUDIT

1 Queue A: total $= 84$, mean $= \frac{84}{12} =$ **7** minutes; median $= \frac{7+7}{2} =$ **7**; mode $=$ **7** (three days); range $= 12 - 4 =$ **8**. Queue B: total $= 99$, mean $= \frac{99}{12} =$ **8.25**; median $= \frac{8+9}{2} =$ **8.5**; no mode, because every value occurs once; range $= 15 - 2 =$ **13**.

2 Queue A: lower six $4, 5, 5, 6, 6, 7$ give $Q_1 = 5.5$; upper six $7, 7, 8, 8, 9, 12$ give $Q_3 = 8$; IQR $=$ **2.5**. Queue B: lower six $2, 3, 4, 5, 6, 8$ give $Q_1 = 4.5$; upper six $9, 10, 11, 12, 14, 15$ give $Q_3 = 11.5$; IQR $=$ **7**.

3 Five-number summaries: A **4**, **5.5**, **7**, **8**, **12**; B **2**, **4.5**, **8.5**, **11.5**, **15**. On a common scale, B's box is much longer and shifted right. For A the upper fence is $8 + 1.5(2.5) =$ **11.75**, so 12 is an outlier: draw the upper whisker to 9 and plot 12 separately. For B the fences are -6 and 22, so there are no outliers; whiskers reach 2 and 15.

4 Queue A is more consistent: IQR 2.5 versus 7 minutes, and range 8 versus 13 minutes. Queue B has the higher typical waiting time: mean 8.25 versus 7 minutes, median 8.5 versus 7 minutes. A is usually quicker and more predictable, although it has one unusual 12-minute wait.

5 Yes, the claim is supported at the centre: B's median is 1.5 minutes higher and its mean is 1.25 minutes higher. But the data overlap: A has waits of 7, 8, 9 and 12 minutes, while B has days of 2–6 minutes. So B is typically slower, but not slower on every day. With only 12 days, the difference is evidence rather than proof.

6 Assign the extra staff member to **Queue B** . It has the higher typical wait and much greater variability. Extra staff could reduce both the average wait and the long tail. However, A's 12-minute day is a warning: monitor A at peak times as well, because a single very long wait matters to the students experiencing it.

7 Assumptions/limitations: the 12 days represent normal conditions; waits were measured consistently; the queues had comparable staffing and demand; one school's canteen is typical. The data do not show arrival rates, queue lengths, time of day, day of week, staffing or menu. A fairer comparison would record the same days and times, more days, the number of students served and the staff on duty.

8 Sample note: "Queue B had the longer typical wait (median 8.5 minutes, mean 8.25) and was much more variable (IQR 7 minutes, range 13) than Queue A (median 7, IQR 2.5). I recommend assigning the extra staff member to Queue B. Queue A also had one 12-minute wait, so both queues should be monitored at peak times. The conclusion is based on 12 days and does not include staffing or arrival-rate data, so a longer audit would make the recommendation more reliable."

Marking guidance **A:** correct ordering, totals, averages, quartiles and fences. **C:** clear tables and box plots with a common scale, correct units and a coherent comparison. **D:** a justified staffing recommendation that uses centre and spread, acknowledges overlap and states limitations. For full D credit, the recommendation must follow from the evidence and consider the consequence of the choice.

MIXED PRACTICE · CONSOLIDATE AND EXTEND

1 $7 + 9 + 11 + 13 + 15 = 55$, so mean $= \frac{55}{5} =$ **11**.

2 For six ordered values the median is $\frac{6+9}{2} =$ **7.5**.

3 **7** occurs three times, more than any other value.

4 $25 - 9 =$ **16**.

5 There are eight values. $Q_1 = \frac{4+6}{2} =$ **5**; $Q_3 = \frac{12+14}{2} =$ **13**; IQR $= 13 - 5 =$ **8**.

6 Range $= 22 - 5 =$ **17**; IQR $= 15 - 8 =$ **7**.

7 Six values with mean 8 have total $6(8) =$ **48**.

8 IQR $= 18 - 10 = 8$, so the upper fence is $18 + 1.5(8) =$ **30**. Since 32 is greater than 30, **32** is an outlier.

9 Distribution B is likely right-skewed. Its mean (30) is well above its median (22), so high values are pulling the mean upward. Distribution A has mean equal to median, which is consistent with a roughly symmetric distribution.

10 Original data: total $= 49$, mean $= 7$, median $= 7$. After adding 20: total $= 69$ over 8 values, so mean $= 8.625$; the middle two values are 7 and 8, so median $= 7.5$. The median changes less (by 0.5) than the mean (by 1.625), because it is resistant to the added high value.

11 Class A is more consistent: its IQR is 4, compared with 10 for B. Same median means similar centre, but B's middle half is more spread out. This does not tell us the full ranges or the individual scores.

12 The axis hides the first 90 minutes. A 92-minute wait shown from 90 has visible height 2, and a 96-minute wait has height 6, so the second looks three times as tall, although the actual values are in the ratio $92 : 96 = 23 : 24$, only about 4% apart. Start the axis at 0 or show the actual values on a dot plot.

13 Report the median. One very expensive house is a high outlier; it pulls the mean above the price of a typical house. The median is resistant to that value. Report the mean as well if the total value of the housing stock is needed.

14 Total frequency $= 3 + 5 + 4 + 2 = 14$. Total goals $= 0(3) + 1(5) + 2(4) + 3(2) = 19$, so mean $= \frac{19}{14} \approx \mathbf{1.36}$. The 7th and 8th values are both 1, so median $= \mathbf{1}$. Mode $= \mathbf{1}$ (frequency 5). Range $= 3 - 0 = \mathbf{3}$.

15 The sample of 500 is more likely to represent the whole school, because a larger sample usually reduces random variation. The method still matters: a biased sample of 500 is not automatically representative. The sample of 10 is too small to support a confident claim about the school.

16 A: total $= 25$, mean $= 5$, range $= 7 - 3 = 4$. B: total $= 25$, mean $= 5$, range $= 9 - 1 = 8$. The means are equal, but B has the larger range, so its values are more spread out. The IQRs are 3 for A and 6 for B, which tells the same story about the middle half.

17 The student used the middle position of the unordered list. The ordered data are $2, 4, 5, 7, 9$, so the median is **5**, not 4. Ordering is essential because the median is defined by position in the ordered list.

18 $Q_1 = \mathbf{20}$, $Q_3 = \mathbf{30}$, IQR $= \mathbf{10}$, and the range shown by the whiskers is $40 - 10 = \mathbf{30}$. The median is 25. A box plot does not show frequencies, so we cannot infer how many values lie in each part.

19 One valid set is $\mathbf{6}, \mathbf{10}, \mathbf{10}, \mathbf{12}, \mathbf{15}, \mathbf{18}, \mathbf{20}, \mathbf{20}, \mathbf{24}$. The median (5th value) is 15. The lower four $6, 10, 10, 12$ have median $Q_1 = \frac{10+10}{2} = 10$; the upper four $18, 20, 20, 24$ have median $Q_3 = \frac{20+20}{2} = 20$. Both quartile conditions hold. Other valid sets are acceptable.

20 Adding a very large value increases the total and the count, so the mean increases, often substantially. The median is based on position, so it changes little or not at all. The range increases because the maximum increases. The mean is the summary most affected by the new extreme value.

21 Supported: B has the higher typical value (median 70 versus 60) and greater spread (IQR 25 versus 10). Not supported: every value in B is larger than every value in A; the full ranges or shapes; the reason for the difference; or a prediction about an individual. The sample sizes and context are unknown.

22 Measure a common test score, ideally the improvement from a pre-test to a post-test. Randomly assign students to the two revision methods so the groups are comparable. Keep the test, the time allowed, the teacher and the conditions the same. Compare medians and IQRs as well as means, and use enough students to reduce the effect of chance. A fair comparison also requires that students actually follow their assigned method.

23 A monthly mean can hide large day-to-day variation, dry spells, extreme storms, the number of rainy days and the shape of the distribution. Two months with the same mean can have very different experiences. A dot plot or box plot of daily rainfall, or a display of the number of rainy days alongside the total, would reveal the hidden variation.

24 Queue B should receive more staffing: its median is higher (8.5 versus 7 minutes), its mean is higher and its IQR is much larger (7 versus 2.5 minutes), so waits there are both longer and less predictable. Additional data: arrival rates and queue lengths at different times, staffing levels, day of the week, the number of students served and a longer run of days. Also monitor Queue A's occasional 12-minute wait.

REFERENCE · FORMULA BANK

Formula bank

A compact reminder of the relationships used in this course. A formula is a tool, not a substitute for reasoning: identify the quantities, check the units, and estimate whether the result is sensible.

RATIO, PROPORTION AND RATE

Equivalent ratios	$a : b = ka : kb$ for $k \neq 0$
Part-to-whole	$\frac{a}{a+b}$
Share of a in $a : b$	$T \times \frac{a}{a+b}$
Proportion	$\frac{a}{b} = \frac{c}{d} \iff ad = bc$
Scale drawing	actual = drawing × scale factor
Rate	$r = \frac{a}{b}$
Speed	$s = \frac{d}{t}, \quad d = st, \quad t = \frac{d}{s}$
Density	$\rho = \frac{m}{V}, \quad m = \rho V$
Flow rate	$Q = \frac{V}{t}, \quad V = Qt$

INTEGERS AND ALGEBRA

Subtracting a negative	$a - (-b) = a + b$
Products of signs	$(-a)(-b) = ab, \quad (-a)b = -ab$
Distributive law	$a(b + c) = ab + ac$
Linear equation	$ax + b = c \Rightarrow x = \frac{c-b}{a}$
Linear sequence	$u_n = dn + c$, d common difference
Function machine	$y = mx + c$
Check a solution	substitute and compare both sides

ANGLES AND PLANE SHAPES

Angles on a line	sum $= 180°$
Angles at a point	sum $= 360°$
Triangle	angle sum $= 180°$
Quadrilateral	angle sum $= 360°$
Polygon	angle sum $= (n - 2)180°$
Rectangle	$P = 2(l + w), \quad A = lw$
Parallelogram	$A = bh$
Triangle	$A = \frac{1}{2}bh$
Trapezium	$A = \frac{1}{2}(a + b)h$
Circle	$C = 2\pi r, \quad A = \pi r^2$

SOLIDS

Prism surface area	$S = 2B + Ph$
Prism volume	$V = Bh$
Cylinder	$S = 2\pi r^2 + 2\pi rh$
Cylinder volume	$V = \pi r^2 h$
Composite solids	add or subtract component volumes

TRANSFORMATIONS

Translation	$(x, y) \mapsto (x + a,\ y + b)$
Reflection in x-axis	$(x, y) \mapsto (x,\ -y)$
Reflection in y-axis	$(x, y) \mapsto (-x,\ y)$
Rotation $180°$	$(x, y) \mapsto (-x,\ -y)$
Enlargement, centre at origin	$(x, y) \mapsto (kx,\ ky)$
Similar shapes	lengths $\times k$, areas $\times k^2$, volumes $\times k^3$

PROBABILITY AND STATISTICS

Theoretical probability	$P(E) = \frac{n(E)}{n(S)}$
Complement	$P(\text{not } E) = 1 - P(E)$
Mutually exclusive events	$P(A \text{ or } B) = P(A) + P(B)$
Relative frequency	$\frac{\text{successes}}{\text{trials}}$
Expected count	$nP(E)$
Mean	$\bar{x} = \frac{\sum x}{n}$
Range	maximum − minimum
Interquartile range	$\text{IQR} = Q_3 - Q_1$
Outlier fences	$Q_1 - 1.5\text{IQR}$ and $Q_3 + 1.5\text{IQR}$

Notation. l length · w width · h height · r radius · b base · B area of a base · P perimeter · n number of terms or trials · k scale factor · ρ density · Q flow rate or quartile. In probability, $n(E)$ is the number of favourable outcomes and $n(S)$ the number of equally likely outcomes.

REFERENCE · VOCABULARY

Glossary

Definitions are written for this course. Where a word has a technical meaning, the definition names the quantities involved and the context in which the term is used.

absolute value — the distance of a number from zero; never negative.

alternate angles — equal angles on opposite sides of a transversal between parallel lines.

area — the surface enclosed by a two-dimensional shape, in square units.

balance method — solve an equation by applying the same operation to both sides.

best buy — the product with the lowest unit price.

bias — a systematic tendency to favour some outcomes or groups.

box plot — a five-number summary drawn on a number line.

centre of enlargement — the fixed point from which enlargement distances are multiplied.

circumference — the distance around a circle.

coefficient — the number multiplying a variable in a term.

common difference — the constant change between consecutive terms of a linear sequence.

complement — all outcomes not in a given event.

composite solid — a solid formed by joining or removing simpler solids.

constant — a quantity with a fixed value.

coordinates — an ordered pair locating a point on a grid.

corresponding angles — equal angles in matching positions at parallel lines.

density — mass per unit volume.

dependent variable — the output that responds to the independent variable.

diameter — the distance across a circle through its centre.

distribution — the pattern of a data set: centre, spread and shape.

dot plot — values marked by dots above a number line.

double number line — parallel number lines showing equivalent ratios.

enlargement — a transformation multiplying distances from a centre by a scale factor.

equation — a statement that two expressions are equal.

equivalent expressions — expressions equal for every permitted variable value.

equivalent ratios — ratios describing the same comparison.

evaluate — substitute values and calculate a result.

event — a set of outcomes from a sample space.

expected count — trials multiplied by the probability of success.

experimental probability — the relative frequency observed in an experiment.

expression — numbers, variables and operations without an equals sign.

fair game — a game in which every player has the same chance of winning.

five-number summary — minimum, lower quartile, median, upper quartile and maximum.

flow rate — the volume passing a point per unit time.

frequency — the number of times a value occurs.

function machine — a rule turning an input into an output.

generalisation — a statement describing a pattern for all its terms.

independent variable — the input that is chosen or changed.

integers — the whole numbers and their negatives, including zero.

interquartile range — upper quartile minus lower quartile; the middle-half spread.

inverse operation — an operation that undoes another.

like terms — terms with identical variables and powers.

linear equation — an equation whose variables have exponent 1.

linear sequence — a sequence with a constant difference.

line of symmetry — a line that reflects a figure onto itself.

mean — the sum of the values divided by their number.

median — the middle value of ordered data.

mode — the most frequent value.

mutually exclusive — events that cannot occur together.

net — a flat pattern that folds into a solid.

nth term — an expression giving any term from its position n.

order of operations — brackets, indices, multiplication and division, then addition and subtraction.

origin — the point $(0, 0)$.

outcome — a single possible result of an experiment.

outlier — a value far outside the rest of the data.

parallel lines — lines in the same plane that never meet.

parallelogram — a quadrilateral with two pairs of parallel sides.

perimeter — the distance around a two-dimensional shape.

pi — circumference divided by diameter of a circle, approximately 3.14159.

polygon — a closed plane figure made of straight sides.

prism — a solid with constant cross-section and parallel congruent bases.

probability — a number from 0 to 1 measuring likelihood.

proportion — an equation stating that two ratios are equal.

proportional — describing quantities whose ratio remains constant.

quadrilateral — a polygon with four sides.

quartile — a value dividing ordered data into quarters.

radius — the distance from a circle's centre to its circumference.

range — the maximum value minus the minimum value.

rate — a comparison of quantities with different units.

ratio — a comparison of quantities of the same kind.

reflection — a transformation flipping a figure across a mirror line.

relative frequency — successes divided by trials.

rotation — a transformation turning a figure about a centre.

rotational symmetry — a figure maps onto itself after a rotation under $360°$.

sample — a subset of a population used for information.

sample space — all possible outcomes of an experiment.

scale drawing — a drawing with all lengths in a fixed ratio to reality.

scale factor — the multiplier relating corresponding lengths.

sector — the circle region between two radii and an arc.

sequence — an ordered list of numbers formed by a rule.

similar figures — equal angles; corresponding lengths in a constant ratio.

simulation — a random model imitating a real process.

speed — the distance travelled per unit time.

spread — how far apart data values are.

substitution — replacing a variable with a value.

surface area — the total area of all faces and curved surfaces.

term — a number or product in an expression; a sequence element.

term-to-term rule — a rule for finding the next term from the previous one.

tessellation — shapes covering a plane without gaps or overlaps.

theoretical probability — favourable equally likely outcomes divided by all equally likely outcomes.

translation — a transformation moving every point by the same vector.

trapezium — a quadrilateral with exactly one pair of parallel sides.

tree diagram — branches listing outcomes of successive events.

triangle — a polygon with three sides; angles sum to $180°$.

unit price — the cost of one unit.

unit rate — a rate with denominator 1.

univariate data — one variable recorded for each item.

variable — a symbol for a value that can change.

vertically opposite angles — equal angles opposite where two lines cross.

volume — the space occupied by a solid, in cubic units.

whisker — a box-plot line to the most extreme non-outlier.

Notation at a glance

$=$	is equal to	$\parallel$	is parallel to
$\neq$	is not equal to	$\perp$	is perpendicular to
$\approx$	is approximately equal to	π	pi, about 3.14159
$<, >$	is less than, is greater than	$\bar{x}$	the mean of a data set
$a : b$	the ratio of a to b	$Q_1,\ Q_3$	lower and upper quartiles
$\frac{a}{b}$	a divided by b	IQR	interquartile range, $Q_3 - Q_1$
$a^2,\ a^3$	a squared, a cubed	$P(E)$	the probability of event E
$\sqrt{a}$	the square root of a	$P(\text{not } E)$	the complement of E
(x, y)	the coordinates of a point	$n(S)$	number of outcomes in the sample space
$y = mx + c$	a linear relationship	$n(E)$	number of favourable outcomes
u_n	the nth term of a sequence		
$\angle ABC$	the angle at vertex B	cm, cm^2, cm^3	length, area and volume units
$45°$	degrees		

Units matter. Write the unit with every measurement and read it as part of the answer. A number without its unit is often incomplete.

REFERENCE · INDEX

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REFERENCE · SOURCES AND CREDITS

References and acknowledgements

This coursebook is an original work. Public framework material was consulted to align the curriculum architecture; no third-party text, exercise, illustration, layout or page design has been reproduced.

FRAMEWORK AND CURRICULUM

- International Baccalaureate Organization. *Middle Years Programme: Mathematics subject brief.* IB Organization. Public information on aims, objectives and assessment criteria.
- International Baccalaureate Organization. *MYP: From principles into practice.* IB Organization. Public guidance on the inquiry cycle, concepts, global contexts and approaches to learning.
- International Baccalaureate Organization. *Mathematics guide: Middle Years Programme.* IB Organization. Public framework structure consulted for concept strands and criterion language; all wording here is independently written.

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The publicly listed contents of Oxford University Press's *MYP Mathematics 2* were consulted only to check the typical Year 2 scope and sequence. No proprietary wording, exercise, illustration, layout, page design or answer text was copied or adapted.

PEDAGOGY AND ASSESSMENT

- Wiggins, G., and McTighe, J. *Understanding by Design.* 2nd ed. ASCD, 2005.
- Black, P., and Wiliam, D. *Inside the Black Box.* King's College London, 1998.
- Boaler, J. *Mathematical Mindsets.* Jossey-Bass, 2016.
- National Council of Teachers of Mathematics. *Principles to Actions.* NCTM, 2014.

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MYP MATHEMATICS 2
STUDENT COURSEBOOK

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SUPPLEMENTARY SECTION

MYP Mathematics 2 — Student Coursebook

Spec-gap additions and verified corrections · 2026 verification pass

- How to use this section
- Chapters 1–11 — Try-it checks and Stretch & Apply boxes, with answers
- Answers-section entries to add
- Corrections applied in this edition

Appended to the coursebook — the original pagination is unchanged.

How to use this section

Same gap as MYP1: the book's "How the labels work" page (p10) lists Definition, Key idea, Common error, ATL and Criterion-strip boxes, but the course spec also requires **Try-it checks **and** Stretch & Apply (SaA)** boxes, of which the book has none. One of each per chapter (11 chapters), ready to drop in.

Placement convention

- **Try-it check**: after the first worked example of the chapter's central section

(section noted per chapter).

- **Stretch & Apply**: at the end of the concept sequence, just before the chapter's

investigation/criterion task.

Label-page text to add (p10, "How the labels work"):

> Try-it checks ask one or two rapid questions the moment a method is first shown. They test whether the new move is safe before you carry on; answers are given, so you can correct yourself in seconds. Stretch & Apply boxes move the method one step into a new context. They are deliberately short, sit at the end of a concept sequence, and reward connecting the idea to something you have already seen.

Chapter 1 — Ratios and equivalent ratios

Trial placement: after the first worked example in 1.2 (equivalent ratios); SaA before "The lemonade stand".

Try-it check

1. Write 15 : 25 in simplest form.
2. A recipe mixes flour and water in the ratio 3 : 2. For 18 cups of flour, how many cups

of water are needed?

Answers: (1) 3 : 5. (2) 12.

Stretch & Apply

A fruit punch mixes juice and soda in the ratio 2 : 3. A student mixes 4 cups of juice with 6 cups of soda and says the punch "tastes twice as strong because there is more of it". Explain, in ratio language, why the taste does not change.

Answer direction: 4 : 6 simplifies to 2 : 3 — an equivalent ratio; strength depends on the ratio, not on the total amount.

Chapter 2 — Proportional reasoning

Trial placement: after the first worked example in 2.1 (recognising proportional relationships); SaA before "A plan that fits".

Try-it check

1. Is the relationship $y = 4x$ proportional? What is the constant of proportionality?
2. Five pens cost £3. At the same price, what do 8 pens cost?

Answers: (1) Yes; $k = 4$. (2) £4.80.

Stretch & Apply

A map drawn at scale 1 : 100 000 shows two towns 4 cm apart. What is the real distance? If the map is then enlarged with scale factor 2 (every length doubles), what is the new scale?

Answer: 4 km; 1 : 50 000.

Chapter 3 — Rates and unit rates

Trial placement: after the first worked example in 3.2 (unit prices and best buys); SaA before "The unit-price trap".

Try-it check

1. 3 kg of apples cost £4.50. Find the unit price.
2. Which is the better buy: 500 g for £1.20, or 1 kg for £2.30?

Answers: (1) £1.50 per kg. (2) 500 g costs £2.40 per kg equivalent, so the 1 kg pack at £2.30 per kg is better.

Stretch & Apply

Printer A prints 24 pages in 6 minutes; printer B prints 36 pages in 10 minutes. Which is faster, and by how many pages per minute?

Answer: A does 4 pages/min; B does 3.6 pages/min — A is faster by 0.4 pages/min.

Chapter 4 — Integers and operations

Trial placement: after the first worked example in 4.2 (adding and subtracting integers); SaA before "A mountain weather station".

Try-it check

1. $(-7) + 12$
2. $(-5) - (-9)$

Answers: (1) 5. (2) 4.

Stretch & Apply

A hot-air balloon is at 300 m. It rises 240 m, then descends 520 m. Using signed numbers for each stage, what is its final height above the ground?

Answer: $+240$ then -520; $300 + 240 - 520 = 20$ m.

Chapter 5 — Algebraic expressions and equations

Trial placement: after the first worked example in 5.3 (solving linear equations); SaA before "Choosing a repair plan".

Try-it check

1. Solve $3x + 4 = 19$.
2. Evaluate $2n - 5$ when $n = 6$.

Answers: (1) $x = 5$. (2) 7.

Stretch & Apply

A repair plan charges £10 call-out fee plus £15 per hour of work. Write an expression for the cost C of h hours, then find h when the bill is £100.

Answer: $C = 10 + 15h$; $15h = 90$, so $h = 6$.

Chapter 6 — Numerical patterns and generalisation

Trial placement: after the first worked example in 6.2 (the nth term); SaA before "The lantern arch".

Try-it check

1. Find the nth term of the sequence 5, 8, 11, 14, …
2. Use your rule to find the 20th term.

Answers: (1) $3n + 2$. (2) 62.

Stretch & Apply

A function machine is "in → multiply by 2 → subtract 3 → out". Which input gives the output 17? Write the whole machine as one expression.

Answer: $2x - 3 = 17$, so $x = 10$.

Chapter 7 — 2D geometry: angles, perimeter and area

Trial placement: after the first worked example in 7.1 (angle relationships); SaA before "The courtyard budget".

Try-it check

1. The angles of a triangle are $x°$, $(x + 25)°$ and $(2x - 5)°$. Find x.
2. Find the area of a circle of radius 7 cm, leaving your answer in terms of π.

Answers: (1) $4x + 20° = 180°$, so $x = 40$ (angles 40°, 65°, 75°). (2) 49π cm².

Stretch & Apply

Floor tiles are 20 cm squares. A corridor 4 m long is tiled edge to edge along its length. How many tiles are needed, and what total area do they cover in m^2?

Answer: 4 m = 200 cm, so 10 tiles; $10 \times 0.04\ m^2 = 0.4\ m^2$.

Chapter 8 — 3D geometry: solids, surface area and volume

Trial placement: after the first worked example in 8.4 (volume of prisms and cylinders); SaA before "The water tank".

Try-it check

1. Find the volume of a cuboid measuring 5 cm by 3 cm by 4 cm.
2. Find the volume of a cylinder of radius 3 cm and height 10 cm, in terms of π.

Answers: (1) 60 cm^3. (2) 90π cm^3.

Stretch & Apply

A water tank 2 m by 1.5 m by 1.2 m is completely full. How many litres of water does it contain? (1 $m^3 = 1000$ L)

Answer: $2 \times 1.5 \times 1.2 = 3.6\ m^3 = 3600$ L.

Chapter 9 — Transformations and symmetry

Trial placement: after the first worked example in 9.1 (coordinates and translations); SaA before "Symmetry and tessellation".

Try-it check

1. Translate the point $(2, -3)$ by $(-4, +5)$.
2. Reflect the point (1, 4) in the x-axis.

Answers: (1) $(-2, 2)$. (2) $(1, -4)$.

Stretch & Apply

A flag on a community banner is at (3, 2). It is translated by $(5, -1)$, then its image is reflected in the y-axis. Where is the flag finally?

Answer: $(3, 2) \to (8, 1) \to (-8, 1)$.

Chapter 10 — Probability and simulation

Trial placement: after the first worked example in 10.2 (theoretical probability); SaA before "Is the game fair?".

Try-it check

1. A bag holds 3 red and 5 blue balls. Find P(red).
2. Find P(not red).

Answers: (1) 3/8. (2) 5/8 (the complement).

Stretch & Apply

A two-digit code uses the digits 1–9 for each digit. How many different codes are possible, and what is the probability that the two digits are the same?

Answer: $9 \times 9 = 81$ codes; 9 of them (11, 22, …, 99), so $9/81 = 1/9$.

Chapter 11 — Univariate data

Trial placement: after the first worked example in 11.1 (organising and summarising data); SaA before "Reaction times".

Try-it check

1. Find the mean, median and range of 4, 6, 6, 8, 10.
2. What is the mode of the data?

Answers: (1) Mean 6.8, median 6, range 6. (2) 6.

Stretch & Apply

Five students' puzzle times (seconds) are 12, 15, 15, 18, 40. Which average best represents a "typical" time, and why?

Answer: the median (15) — the 40 s outlier pulls the mean up to 20, so the median represents the typical performance better.

Answers-section entries to add

Append under a new heading in the answers part (page 222 onwards):

> TRY-IT & STRETCH & APPLY — CH 1: 1 (3 : 5; 12) 2 (4 : 6 = 2 : 3, equivalent ratios give the same taste). CH 2: 1 (yes, k = 4; £4.80) 2 (4 km; 1 : 50 000). CH 3: 1 (£1.50/kg; 1 kg pack) 2 (A, by 0.4 ppm). CH 4: 1 (5; 4) 2 (20 m). CH 5: 1 (x = 5; 7) 2 (C = 10 + 15h; h = 6). CH 6: 1 (3n + 2; 62) 2 (2x − 3; x = 10). CH 7: 1 (x = 40; 49π cm^2) 2 (10 tiles; 0.4 m^2). CH 8: 1 (60 cm^3; 90π cm^3) 2 (3600 L). CH 9: 1 ((−2, 2); (1, −4)) 2 ((−8, 1)). CH 10: 1 (3/8; 5/8) 2 (81; 1/9). CH 11: 1 (6.8; 6; 6; mode 6) 2 (median 15).

Corrections applied in this edition

Every answer-key entry and worked example in this book was independently recomputed during the verification pass. **No corrections were required.** The Try-it and Stretch & Apply boxes in this section were likewise recomputed before inclusion.

ABOUT THIS COURSEBOOK

A concept-driven inquiry course in ratio and proportion, algebra and pattern, geometry and space, probability and data — the second volume of a coherent five-year series.

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SADIKOU Moukoddim

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